

OHPE.

Manual

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PART 1

Getting started.

Quick start.

Ohpe is a small synth studio in one window: four instruments, a drum kit, sequencers, a tape recorder and a mixer. It runs as a standalone app or as a plug-in (VST3, Audio Unit) in your music software.



Engine keys on the left, the display with four encoders in the centre, mode keys on the right, the keyboard row below.

1. Parts: four instruments at once

Ohpe has **four parts (1–4).** Think of them as four separate synths living side by side: a bass, a pad, a lead and an arp, for example. Each part has its own sound, its own engine, its own effects and its own MIDI channel, and all four can play at the same time.

The **part selector** at the top picks the part you're editing right now (the edit part). Everything you touch (engine, knobs, effects, presets, the on-screen keyboard) follows that part. The display marks it with a yellow badge, P1–P4.

The **drum kit** sits next to the four parts as an extra instrument: 24 pads on MIDI channel 10.

Light colours everywhere: yellow = what you're editing · violet = in use elsewhere · green = playing · red = recording.

2. Engines: what kind of synth a part is

The **engine keys** on the left choose how the edit part makes sound. There are eight:

- **Analog:** classic oscillators and filter. Basses, leads, pads.
- **Wavetable:** morphing waveforms, and you can make your own from any sound.
- **Cluster:** up to seven detuned oscillators per key, for huge chords and supersaws.
- **Vocal:** a singing formant voice.
- **FM:** six operators. Bells, electric pianos, metallic sounds.
- **Pluck:** modelled strings and mallets.
- **Grain:** granular clouds from a sample, the tape or a built-in texture.
- **Sampler:** play your own audio files, or the built-in instruments.

Each part can use a different engine. The small light under every engine key shows which parts are using it: yellow for the part you're editing, violet for the others.

3. Pages: where you edit

The **mode keys** on the right choose what the display shows:

- **Synth:** the engine's sound controls
- **LFO:** movement and modulation
- **Effects:** the effect chain of each part and of the master
- **Tape:** an 8-track recorder for building whole songs
- **Envelope:** how notes start, hold and fade, and how they react to playing
- **Mixer:** levels, pan, EQ and sends for every channel
- **Sequencer:** steps, Euclid, arpeggiator and Orbit patterns
- **Drum:** the drum kit's pads

Each mode holds several **pages**. A page always shows **four values**, and those four values belong to the **four encoders** under the display. Turn the pages with the **arrows** in the display's header. You can also drag the bars on the display directly.

4. The basic workflow

1. Pick a part (1–4).
2. Pick an engine, or load a sound from the **LIBRARY** (430 factory presets).
3. Shape it on the Synth, Envelope, LFO and Effects pages, using four encoders per page.
4. Repeat for the other parts, then play them with the sequencers or record them to tape.
5. Balance everything in the Mixer.

5. Play, record, stop

- **Keyboard:** play with the mouse or any MIDI keyboard; the octave keys shift it.
- **Sequencer:** tap steps to switch them on, Shift-tap a step to edit it and lock parameters on it, press Play. In a host, the sequencer follows the host transport while Play is on.
- **Transport** (bottom left): **REC** records on the tape at once (it arms the selected track when none is armed), **PLAY** runs the sequencer and the tape, **STOP** stops (twice = back to the start), **|◀** and **LOOP**. Standalone keys: Space, R, Enter, L.
- Drop an audio file on the instrument to load it into the Sampler or Grain engine, or onto a drum pad.

6. Handy shortcuts

- **RANDOM** rolls a new playable sound for the edit part.
- **BREED** crosses two sounds into new ones.
- **PRO** shows every parameter of a page at once, as editable pictures.
- **Right-click** any control for MIDI Learn. Everything in Ohpe can be mapped.
- **Double-click** resets a value. Hold **Shift** while dragging for fine control.

Where to go next: Playing the interface explains every key and page, First recording in 30 seconds gets a song onto tape, and Installing Ohpe covers macOS and Windows.

Free & open source (AGPL v3) · macOS & Windows · Standalone · VST3 · AU · github.com/thegroovex/Ohpe

Feature overview.

Analog engine	two band-limited oscillators (shape morph, detune, octave, mix), sub, noise, ladder and state-variable filters (LP / BP / HP), drive, key tracking
FM engine	six operators, eight algorithms with smooth morphing between neighbours, feedback, per-operator ratio and level
Wavetable engine	six built-in tables of eight frames with smooth morphing, warp modes, unison, sub oscillator; own tables from any sound (audio to wavetable: pitch-synchronous analysis, 8–256 frames), single-cycle sets and a user library
Pluck engine	waveguide string and tuned mallet resonators for plucked and struck sounds
Cluster engine	up to seven detuned oscillators per key as unison or chord shapes, slow drift, resonant low-pass
Grain engine	granular synthesis from a loaded sample, the tape or a built-in texture
Vocal engine	formant voice with vowels, breath, vibrato and jitter; formant tracks analysed from your own sounds
Sampler engine	key zones or automatic slices from WAV / AIFF / FLAC files, loop modes, built-in keys sound and ten recorded factory instruments (kalimba, marimba, vibraphone, harp, harpsichord, recorder …)
Drum kit	24 pads on MIDI channel 10: five recorded factory kits (Studio, Tight, Room, Lo-Fi, Hybrid – CC0), own samples or synth models, choke groups
Factory sounds	412 part presets across all engines and categories, level-matched; 18 project presets using all four parts, the sequencers and the drum kit; six drum kits, ten recorded instruments and four keyboard instruments – see Factory sounds
Four parts	each with its own engine, envelopes, LFO, MIDI channel, scale / chord mode and MPE
Shared per voice	amp and modulation envelopes, LFO with several destinations, glide, 16-voice polyphony
Sequencers	step sequencer (up to 64 steps, probability, ratchets, parameter locks, 16 patterns, song mode), Euclid, arpeggiator and Orbit (polymeter), each with a target part – step, Euclid and Orbit can also play the drum kit
Tape machine	eight stereo tracks: record / overdub / punch from the input, parts or drums, varispeed, reverse, loop (set in / out by hand at the playhead, snapped to bar or beat, edges draggable – Setting the loop), tape character, undo, export as WAV master and stems; count-in of one or two bars (recording starts exactly on the bar) and a built-in metronome; the tempo is dragged, typed or tapped right on the display – see Tempo, metronome and count-in
Mixer	eight channels with EQ, compressor, volume, pan, mute / solo, two sends (A: delay, B: reverb), master compressor, tape saturation and limiter
FX racks	six serial slots per part and four on the master: hyper (unison widener), distortion (eight modes), flanger, compressor, multiband, EQ, filter, echo, hall, chorus, phaser, bitcrush, filter drive, delay, reverb, mid / side stereo, space 3D – reorder by dragging, each with mix and bypass – see FX rack
3D space	every part can move around your head – front, sides, behind, above, near and far – on tempo-synced motion paths (circle, figure-8, fly-by with Doppler, random orbit, pendulum, spiral); binaural on headphones with Ohpe's own parametric head model (no measured data set), mono-compatible on speakers; also as rack effect Space 3D; 75 "3D" presets (72 part presets, three projects) – see 3D Space
Life	one LIFE control per part keeps held notes evolving: every voice drifts on its own (a few cents of pitch, breathing filter, wandering engine colour, grain scatter) – 0 = static, on every engine, a modulation target – see Life
Resonator body	every part can play through a physically modelled body – coupled strings with sympathetic strings tuned to the note or the held chord, a modal plate, an open or closed tube; it follows the keyboard, pitch bend and MPE (pressure rings it, slide moves the excitation point); also as rack effect Resonator; 22 factory presets use it – see Resonator

Morph space	an XY pad per part between four complete sounds – also across different engines: parameters morph where the sounds share their structure, otherwise the sounds play in parallel and cross-fade at even loudness; play it by mouse, automation, MIDI, the modulation matrix, mod wheel or MPE pressure / slide; 7 morph presets – see Morph space
Voice Control	sing, hum or talk into an input and your voice moves the sound (not a vocoder – it only modulates): level, vowel (A-E-I-O-U map), brightness, consonants, pitch and syllable onsets as modulation sources, one-press Vowel → Filter + Morph – see Voice Control
Motion recorder	turn a part's knobs while the music runs and up to four movements per part loop as lanes with their own length (1 – 64 sixteenths, polymetric to the pattern); overdub by touching again, depth control, also modulation sources – see Motion recorder
Performance pads	four pads throw effects on the main mix for live sets – a quantised **stutter** (1/4 … 1/32, triplets), a **tape stop** over 1/8 … 2 bars, a resonant **filter sweep** (low-pass down / high-pass up) and a **reverb freeze** that holds the moment endlessly; hold = momentary, shift-click = latch, MIDI-learnable, matrix sources – see Performance pads
Harmony brain	Ohpe remembers the key and chords you play on any part; other parts follow (snap to the key, the chord tones or the root), the memory feeds three modulation sources, and an isomorphic grid keyboard with MPE (four layouts, one finger per member channel) plays it from the screen – see Harmony brain
SUB layer	every part gets a sub oscillator one or two octaves under each voice – Sine, Round or Square, into the engine's filter or direct, locked to the voice's pitch (glide, bend), band-limited and mono, level 0 = untouched sound – see SUB layer
Key Gravity	a spectral stage per part after the engines pulls every partial toward the key and the held chord – Scale, Chord or Chord+Scale, glide on chord changes, formant keep; at high amounts noise and drums become pitched texture in key; 0 = untouched sound – see Key Gravity
Expressive playing	Per part, velocity and aftertouch (channel, poly and MPE pressure) each drive two musical functions – cutoff, vibrato, glide time, attack, FM amount, position, drive and more – with amount and curve; the glide factory sounds stretch their glide under pressure – see Expressive playing
In-app updates	check for a new version from inside the standalone or the plug-in (by hand or, opt-in, once a day – never while playing), read its notes in English or German, download it in the background with resume and SHA-256 verification, then open the installer; stable / preview channel, skip or remind later, previous version as rollback, content packs without reinstalling; no telemetry – see Updates
Content packs	install preset packs from `.ohpepack` files (file dialog or drag & drop on the window): SHA-256 of every file checked before anything changes, one step of rollback, size limits; pack manager with version, size, license / signature (shown) and remove; filter the library by pack; presets point at pack content by `"<pack>/<item>"` and say which pack is missing – see Content packs
Settings and night skin	one settings view behind the gear key: language (EN / DE), tooltips and a help overlay, help pages, a low-glare dark skin (or auto with the system), display brightness, animations, UI size 75 – 200 %, MIDI setup (channels, MPE, bend ranges, inputs, clock / sync) and a MIDI learn list to edit or clear mappings, updates and about – saved per user, never in projects – see Settings
Accessibility and UI size	UI sizes 75 – 200 % (right-click on the housing), full keyboard operation with a focus ring (Tab, F6, arrows, Enter / Space), screen reader names and values in EN / DE, WCAG AA contrast in both skins – see Accessibility
Preset breeding	cross two sounds (also across engines – a hybrid keeps the other engine as its layer) with a chosen mutation, hear six children with waveform / brightness thumbnails at their parents' loudness, keep the best and breed further generations with undo; children save to the library as favourites – see Preset breeding
Bounce to engine	one BOUNCE key moves a tape track, the tape mix, the loop range or a few bars of the live input into the Grain or Sampler engine of a part – as a seamless endless loop (grains wrap round the seam, held sampler notes sustain it) or a one-shot; bounces are saved as WAV and come back with the session – see Bounce to engine
Tape memory	Ohpe always remembers the last 30 s of its own output; one CAPTURE key (MIDI-mappable, also on the Grid-Controller) turns it into a playable Sampler instrument – chromatic (pitch detected and tuned, root set) or sliced by onsets – saved inside the session, exportable as WAV – see Tape memory

Modulation matrix	16 routings per part from 20 sources (three LFOs, three envelopes, velocity, MPE, macros, a sequencer lane …) to pitch, level, pan, 24 engine parameters, the LFOs and every rack slot; rings on the encoders show what moves – see Modulation
Sync	host transport in the plug-in; MIDI clock in / out and network tempo sync (Link) in the standalone app
Multi-out	every channel can go to the main output or its own stereo output (8 aux outputs) – in the plug-in and in the standalone app
MIDI	per-part channels, MPE, every control is MIDI-learnable (right-click → MIDI Learn), maps can be exported / imported; all parameters are automatable
Standalone app	choose audio system, device, sample rate and buffer size, latency display, multi-out routing to device channels, MIDI inputs
Preset library	one LIBRARY key: categories (Bass, Keys, Pads, Leads, Plucks, Drums, FX, Arp, Textures, Vocal), engine filter, search, favourites, star ratings, sorting, factory and user banks, preview by selecting, part and whole-project presets, previous / next from MIDI
Display	a deep, layered display with parallax; stereo scope (goniometer, correlation, M / S levels), a moving 3D spectrum terrain, the head-centred 3D space view and the living display – the sound as a 3D object that breathes with the envelope, bulges with the LFOs and stretches with the stereo width – see Living display
Direct manipulation	drag the value bars on the display, Shift / Cmd / Ctrl for fine control, double-click to reset; pro view with every parameter of a mode at once – as editable pictures (envelopes, filter, LFO, FM graph, wavetable, grains, 3D) with violet modulation ranges, plus Discover: BRIGHT / HARD / ALIVE / WIDE, VARY, cross, dice, history
Languages	English and German, switchable at any time

Playing the interface.

Ohpe's surface: engine keys with the **LIFE** bar (left), the display (centre), mode keys (right), four encoders below the display, the **TRANSPORT** keys with the tape readout, the tempo, the metronome and the count-in (bottom left), the **OCTAVE** keys, **MIDI** and the keyboard readout (bottom right) and the keyboard row – **PITCH** and **MOD** wheels, a three-octave keyboard, **glide** mode and time. The header holds the **part selector**, the **preset strip**, the **LIBRARY**, **PRO** and **BREED** keys, **RANDOM** and **SAVE**, and the **settings** key (gear: language EN / DE, help, light / dark skin, MIDI, updates, about). Every function can be reached with the mouse and – through MIDI learn – from a controller.



The plate

Keyboard row: wheels, keyboard, glide

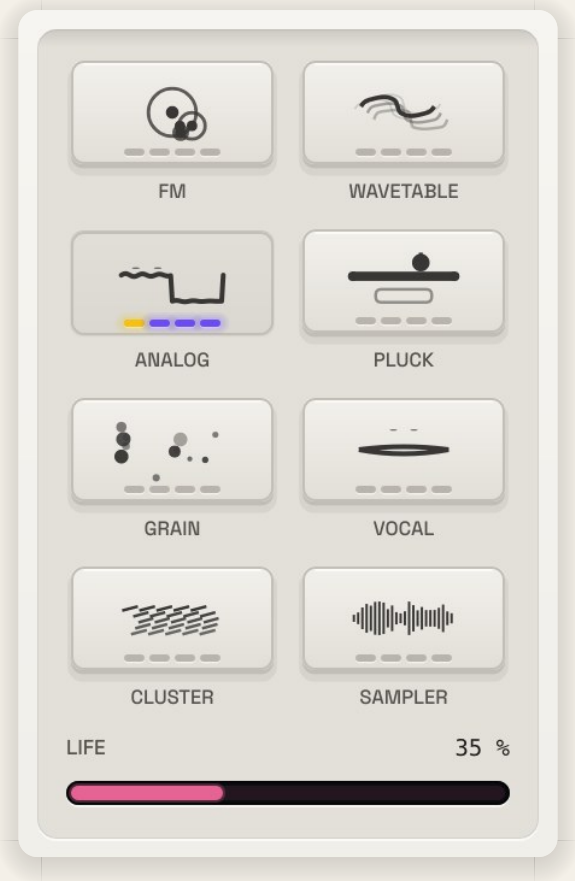


Keyboard row

Control	What it does
PITCH wheel	pitch bend for the edit part (its MIDI channel, like the on-screen keyboard); range = the part's Bend Range. Drag up / down; it springs back to the centre when you let go.
MOD wheel	the mod wheel (CC 1): a source in the modulation matrix; every factory preset answers it (factory sounds). Stays where you leave it; double-click = 0.
Keyboard	three octaves (C3–C6 by default); **OCTAVE −/+** moves it. **GRID** (between OCTAVE + and MIDI) swaps it for the isomorphic grid keyboard with MPE.
PORTAMENTO: AUTO / LEGATO / ALWAYS	glide mode of the edit part: Auto glides while the previous note still sounds, Legato only while another key is held, Always from every previous note.
GLIDE	glide time of the edit part (0 = off).

Both wheels are host parameters (Pitch Wheel `perf.bend`, Mod Wheel `perf.mod`): right-click for MIDI learn, Shift / Cmd / Ctrl for fine control. They are performance controls, not part of presets. The on-screen wheels also **follow your controller:** pitch bend and the mod wheel (CC 1) arriving from the host or a MIDI device on the edit part's channel (an omni part: any channel but the drum channel; an MPE part: channel 1) move them, as do host automation and MIDI learn of `perf.bend` / `perf.mod` – whichever moved last is shown.

Life



LIFE at the foot of the engine column

LIFE at the foot of the engine column makes the edit part's held notes keep evolving, the way a played instrument never sounds exactly the same twice. At **0** the sound is static and exactly as without it; turn it up and every voice starts to move on its own:

- a slow **pitch drift** of a few cents (at most about ±12 cents at 100 %; the voices of a chord drift apart),
- **filter breathing** – the brightness rises and falls slowly (filter engines: cutoff; FM: index; Vocal: vowel; Pluck: tone),
- **engine colour** – wavetable position and warp, FM index, formant, grain density wander a little,
- **grain scatter** – the Grain engine's grains start at slightly scattered positions,
- a hint of **pan** and **level** motion (about ±0.25 dB – breathing, never a jump).

The motion comes from slow random drifts and free "chaos" LFOs at uneven rates, different for every voice (seeded per part, voice and note, so a bounce sounds the same every time). Around 15 – 35 % brings pads and keys to life without sounding detuned; higher values give lush, drifting textures. Changes glide over about 60 ms, so you can ride it while notes hold.

Drag the bar (Shift / Cmd / Ctrl = fine), double-click = 0, right-click = MIDI learn. While the part plays, the lit bar shimmers slowly – faster with more life; a fine light tick shows where the modulation matrix moves it (target **Life**, MODULATION.md). LIFE is also on the Envelope pages (after Poly Glide) and is part of part presets (`pN.life`, PARAMETERS.md).

Light bars

Engine, mode, transport, octave and MIDI keys carry a short **light bar** at the bottom edge of the key, under the pictogram. One colour language: **yellow** = what you are editing, **violet** = running or in use elsewhere, **green** = playing, **red** = recording. A light that means "playing" or "recording" **breathes** gently (a soft pulse of about 1.8 s, never fully off).

Keys	Light bar
Engine	four segments, one per part (1 – 4, left to right): a segment lights when that part plays the engine – **yellow** for the edit part, **violet** for another part.
Mode	**yellow** = the page on the display. **Violet:** SEQUENCER = the sequencer runs; EFFECTS = a slot of any FX rack (a part or the master) is on; DRUM = a running sequencer plays the drum kit; LFO = a part's LFO has depth or a modulation routing is set. **Green** on TAPE = the tape plays, **red** on TAPE = recording (both breathing). The status colour wins; the shown page keeps its sunk key.
Transport	PLAY **green** while running (the triangle glows too), REC **red** (steady = ready; the circle glows), LOOP **yellow** while on. While PLAY runs and REC records, bar and sign breathe.
OCTAVE − / + · MIDI · GRID	OCTAVE **yellow** while the keyboard is shifted that way; MIDI **red** during MIDI learn; GRID **yellow** while the grid keyboard is shown.

Hover a key: its tooltip says what the bar means (engine keys also list the parts that use them).



Transport, keyboard row and octave block while recording

The **readouts** next to the keys (p39: the tape readout shares its row with the editable tempo, the metronome and the count-in, see Tempo, metronome and count-in): the tape readout above the transport keys shows the state (**REC** red, **PLAY** green – both breathing while running – TAPE), the tape time (m:ss.t) and bar.beat; the keyboard readout in the octave block shows the keyboard's range (C3–C6) and its octave shift, and **MIDI LEARN** in red while a control waits for a MIDI message.



Light bars: several parts with different engines

RANDOM and SAVE

RANDOM gives the edit part's engine new sound values (its Synth pages; level, tuning, voices, the sample, envelopes, effects and modulation stay). **SAVE** opens the library's save panel for the edit part.

Every rolled sound is meant to be playable straight away:

- **Musical ranges:** each engine parameter is rolled inside a range where the engine still sounds like an instrument – the filter never closes, resonance, FM index / feedback, drive and grain spray stay moderate, FM ratios and grain pitches land on musical intervals.
- **Little noise:** noise, crackle and breath are off in most rolls and low when they come; a roll that still sounds like noise is rolled again.
- **No slide:** portamento and pitch envelope are reset – glide is off (a short legato glide in a few rolls only, never on the first note), a short pitch blip on the attack comes in about one roll in 25; glide pressure is neutral.
- **Level-matched:** Ohpe plays the new sound offline (C4 + G4), rolls again if it is silent or broken, and sets the part **Gain** so it plays as loud as the factory presets (the median of their loudest moment, K-weighted – plucks and pads are compared by how loud a note sounds, not by how long it lasts) with its true peak at or below -1 dBTP. The part volume is not touched. When the part's own effects keep the level down (a compressor first in its rack, say), the new sound plays about as loud as the sound it was rolled from.

The **BREED** key (next to RANDOM; a click opens / closes the view, its light bar is lit while it is open) opens breeding: cross two sounds, hear six children, keep the best and breed further. RANDOM only rolls – holding or right-clicking it no longer opens breeding. While the breeding view is open, RANDOM rolls a new parent B and SAVE stores the selected child in the library as a favourite. Right-click BREED for MIDI learn of `lib.breed`. The breeding learns locally what you like (learning breed: LEARN, CURIOSITY, WHY, RESET in its foot row).



The breeding view

Display: drag the bars

The four rows on the display show what the four encoders control. The rows are controls too:

Action	Result
Drag a row (right or up = more)	changes the value relative to where it was
Click on the bar	jumps to that position, then keep dragging
Shift, **Cmd** (macOS) or **Ctrl** while dragging	fine control: a tenth of the speed. You can press or release the key during a drag; the value never jumps.
Double-click	back to the default value
Right-click	MIDI Learn / clear mapping / reset

The encoders behave the same: drag (fine with Shift / Cmd / Ctrl held at the start of the drag), double-click resets, right-click opens MIDI learn. The small caption above the rows names the page (Macros, Amp Envelope, Track …); the arrows in the header turn the pages.

Display views: character, stereo scope, spectrum terrain, living object

The display is built in layers like a real instrument screen: the dot grid lies behind the deep glass, the values float in front of it, the character casts a faint shadow and the glass itself carries a soft inner shadow and sheen. When you move the mouse over the display the back layers shift a little against the front (parallax).

The left half of the display shows one of five **views** – click it to cycle, right-click for the view menu:

View	What it shows
Character (default)	the animated character of the engine / mode. On the Synth and Mixer pages it stands on a slowly moving **spectrum terrain** of its sound (the edit part, the master).
Stereo scope	the master output: goniometer (mid up, side across, L upper left, R upper right), correlation meter (−1 … +1; red = out of phase) and the **M** / **S** level bars.

View	What it shows
Spectrum terrain	the master spectrum as a perspective landscape: new rows rise at the front and recede to the horizon. It comes to rest when the sound stops.
3D space (1.0)	the head-centred view of the 3D stage: every part that uses the space as a dot with its trail around the listener's head, the edit part labelled.
Living object	the sound as a 3D object: it breathes with the amp envelope, bulges with the LFOs, ripples with the mod envelope and the matrix, stretches with the stereo width and splits in two when the channels drift out of phase. The edit part (the master on the Mixer page); it comes to rest when the sound stops – see Living display.

The **stereo scope appears by itself** whenever the encoders show a **Mid/Side** effect in an FX rack slot: in a part's rack it shows that part after its rack (marked P1–P4), in the master rack (MFX 1–MFX 4) the master output (marked Master). The M and S bars use the colours of the Mid Level and Side Level encoders. In the rack view of the FX page the strip below the effect chain shows the stereo scope when a Mid/Side slot is selected there.



Mid/Side FX slot with the stereo scope



Master scope with Mid/Side in the master rack



Spectrum terrain



Living object

Animated 3D visuals (view menu) switches the terrain and the parallax off – the display then stays still, the scope keeps working. The setting belongs to the computer (it applies to every project); the chosen view is saved with the project. The visuals read the sound through a lock-free buffer: nothing waits or allocates on the audio thread, the analysis runs on the screen at 30 frames per second.

Parts

Ohpe has four parts, each with its own engine and sound. The **part selector** (1–4, header) chooses the **edit part**: the engine keys, Synth, Envelope, LFO and the FX slot pages, the library (part presets load there) and the on-screen keyboard (it plays on the part's MIDI channel) follow it. The display shows the edit part as a yellow badge (P1–P4). MIDI: Edit Part (`ui.part`).

Synth mode (0.6): after the engine's own pages come two pages every engine shares – **Glide** (time, mode, timing, poly glide) and **Pitch Envelope** (amount, time, curve, voices). See SEQUENCERS.md. Then the page **Sub** (p31: level, octave, shape, direct) – the part's SUB layer, a sub oscillator one or two octaves under every voice for extra low end.

Envelope mode: after the envelopes and the voice pages come the **Expression** pages – **Velocity** (target 1, amount 1, target 2, amount 2), **Aftertouch** (the same for channel / poly / MPE pressure) and **Expression Curves.** The display draws each slot's response with a dot that follows your playing; the pro view (Synth) has them as the group Expression. See Expressive playing.



The Expression page (Aftertouch)

Samples for the Sampler and Grain engines

With the Sampler or Grain engine on the edit part, the display shows **Load sample** and **Clear** and the name of the loaded file. You can also **drag an audio file** (WAV, AIFF, FLAC, Ogg) from your file manager onto the instrument: it goes to the edit part's Grain engine when Grain is selected, otherwise to its Sampler (the part switches to the Sampler). Each part has its own samples. Sample paths are saved with the project.

BOUNCE (third key, p23) puts tape or live-input audio into the engine you are looking at: a menu offers the edit track, Tape 1 … 8, the tape mix and the input (1–8 bars), plus Endless loop (on by default – the bounce loops seamlessly). See Bounce to engine. On the Sampler page a fourth key, **CAPTURE**, puts the tape memory into this part (Tape memory).

User wavetables and formants

With the Wavetable or Vocal engine on the edit part, the Synth page shows **IMPORT**, **USER** and **CLEAR**. Drag a sound or a wavetable file onto the instrument (or use IMPORT: file, tape track, live input, library; frames 8 … 256, smoothing): the sound is analysed in the background – a progress bar on the display – into a user wavetable (Wavetable engine) or formant tracks (Vocal engine), saved in the user wavetable library and switched on. The display then shows the frame stack (the frame at the current position in yellow) or the five formant tracks. **USER** switches between the user and the built-in material (`pN.usertable.wt` / `pN.usertable.vocal`), **CLEAR** removes it. Details: Wavetable engine.

Pro view



Pro view

PRO opens the pro view: **every parameter of the current mode at once**, in groups – in Synth mode the engine, its filter, the envelopes, LFO, pitch & level, voice, glide (p32: with Glide Pressure, MIDI.md), pitch envelope, sub, layer and MIDI of the part; in Tape mode the transport, loop, tape character and all eight tracks; in Drum mode the kit and all 24 pads. The tabs on top switch the mode. The layout uses one to four columns, as many as the window is wide, and scrolls when a mode has more than fits.

See – grab – hear (pro view 2.0). Every group card that has a picture shows it first, and the picture is the control:

Card	Picture	Grab
Amp / Mod envelope, Env 3	ADSR curve, a green dot follows the newest note	drag the four handles (peak = attack, knee = decay + sustain, sustain end, release end)
Filter	frequency response	X = cutoff, Y = resonance, wheel = drive
LFO, LFO 2 / 3	the wave on a tempo grid, a moving dot	up / down = depth, left / right = rate, a quick swipe or the wheel = shape
FM engine	operator graph: circles = operators (filled = heard), lines = modulation, thickness = index	drag a circle up / down = its ratio, wheel on a circle = its level, empty space = index, the badge = next algorithm
Wavetable engine	the frames as a stack, the glowing plane = position	up / down = position, left / right = warp, wheel = table
Grain / Sampler	the waveform; grains as a cloud around the position (green while playing)	grain: X = position, Y = size, wheel = spray; sampler: drag start / loop markers
Analog / Pluck / Cluster / Vocal	oscillator mix preview / string / detuned stack / formant curve	X and Y move the engine's first two macros
Pitch & Level	tuning ring and level meter	up / down on the ring = tune, wheel = fine, on the fader = level
Sub	the voice (grey) over its sub wave (yellow), level fader	up / down = level, swipe = shape, click the octave / route caption, wheel = level – see SUB layer

Card	Picture	Grab
FX slots, sends	the chain as cards with a mini picture of each effect	X = mix, Y = the effect's first parameter, wheel = type
3D Space	the head from above, the sound as a dot	drag the dot = direction and distance, wheel = elevation

Shift (Cmd / Ctrl) gives fine control, double-click resets, right-click opens MIDI learn – in the picture as on the bars. The bars stay below each picture as the precise row; the chevron in the card header folds them away. Hover over a bar or a picture for the value and one sentence on what it does.

Colours: yellow = what you edit, **violet = modulation** (every routed parameter shows the range its routings move it as a violet arc, band or shadow, on the bars as a violet band around the value – the filter shows the whole swept response), green = playing, red = recording.

Modulation by drag: the source chips on the DISCOVER line (LFO 1–3, Mod Env, Env 3, Mod Wheel) can be dragged onto any picture: the routing goes into the next free slot of the part's modulation matrix (amount 35 %), targeting the parameter under the drop point.

SIMPLE | ALL (top right): Simple shows the pictures and Discover only, All adds every bar. Modes without pictures (Mixer, Sequencer, Tape, Drum) always show their bars.

Discover



Discover with six variations

The strip at the top of the pro view (part modes) is for finding sounds rather than setting them:

- **BRIGHT · HARD · ALIVE · WIDE** – the character macros of the part (`pN.char.*`, -100 … +100 %, 0 = neutral). Each moves several real parameters musically: BRIGHT opens filter / tone / FM index, HARD gives a faster attack, drive and feedback, ALIVE drives the part's Life amount plus drift and jitter, WIDE widens unison, spread and the layer. They are host parameters (automation, MIDI learn) and matrix targets.
- **VARY** – six relatives of the current sound appear as numbered tiles. Each tile's wave is drawn from how that relative moved away from the starting sound (different changes = a different wave, shaped by its amp envelope); the dots top right show how far it went (1 = closest of the six, 3 = farthest). The tiles are cleared when the sound's parameters change (another engine). Click a tile to **audition** it (the sound is applied and a short note plays), **hold or double-click** to **keep** it. Drag one tile onto another to **cross** the two – the child replaces the target tile. Pressing VARY again without keeping returns to the sound you started from.
- **Dice** – rolls every unlocked group at once; the slider below sets the strength from **gentle** to **wild** (it also sets how far VARY goes). Each card of the sound has its own **dice** (roll only this group) and **lock** (VARY and the dice leave it alone); the lock in the strip unlocks all.
- **History** – the last 20 states as dots; click one to go back to it.

VARY, the dice and crossing use the same breeding core as the breeding view (mutation per gene group with safety caps on resonance, feedback and drive; crossover per group).

Sequencer views and parameter locks



Step locks

In Sequencer mode the tabs on the display choose the view: **Step**, **Euclid**, **Arp**, **Orbit** and **Sync**. Euclid and Orbit draw their rings / lanes; Arp and Sync show a summary line.

Step view:

- **Click** a step to switch it on or off (the four tabs on the left choose steps 1–16 … 49–64).
- **Shift-click** (or right-click) a step to **select it for editing.** The display shows LOCK STEP n and the encoders edit the step: **Note, Velocity, Gate, Probability** (page 1) and **Ratchet** (page 2).

- While a step is selected, switch to any other mode (Synth, Envelope, LFO, Mixer, Effects, Drum) and turn an encoder or drag a bar: the value is **locked on that step** instead of changing the parameter. Locked values show a square marker on the display and on the encoder. A step holds up to four locks; double-click a locked value to remove that lock.
- Steps with locks show small marks along their top edge (one per lock); the selected step is outlined.
- **Clear locks** removes all locks of the step; **Done**, a click on the LOCK badge, Esc or a second Shift-click ends the editing.

Transport keys

The **TRANSPORT** keys in the bottom-left panel (one row: TO START, STOP, PLAY, REC, LOOP under the tape readout) work on every page:

Key	Action
REC (red)	one-touch record on the tape: arms the selected track when no track is armed, switches recording on and starts the tape. Pressed while recording: punch out (the tape keeps playing). Lit while recording ready, breathing while it records.
PLAY	starts / stops the sequencer and the tape together (Play starts tape, on by default; off: PLAY runs the sequencer alone)
|◀	tape back to the start
STOP	stops tape, recording and sequencer; pressed again while stopped: back to the start
LOOP	tape loop on / off; without a loop range it loops what is recorded (or four bars)

While the tape runs, every page shows ● **REC 0:12.3** (red, breathing) or ▶ **TAPE 0:12.3** (green) in the display header. In the standalone app: **Space** = PLAY, **R** = REC, **Enter / Home** = to the start, **L** = loop (in a plug-in the host keeps its own keys). Right-click any transport key (or a button on the tape page) for MIDI learn – every transport function has a trigger, see MIDI.

Tempo, metronome and count-in

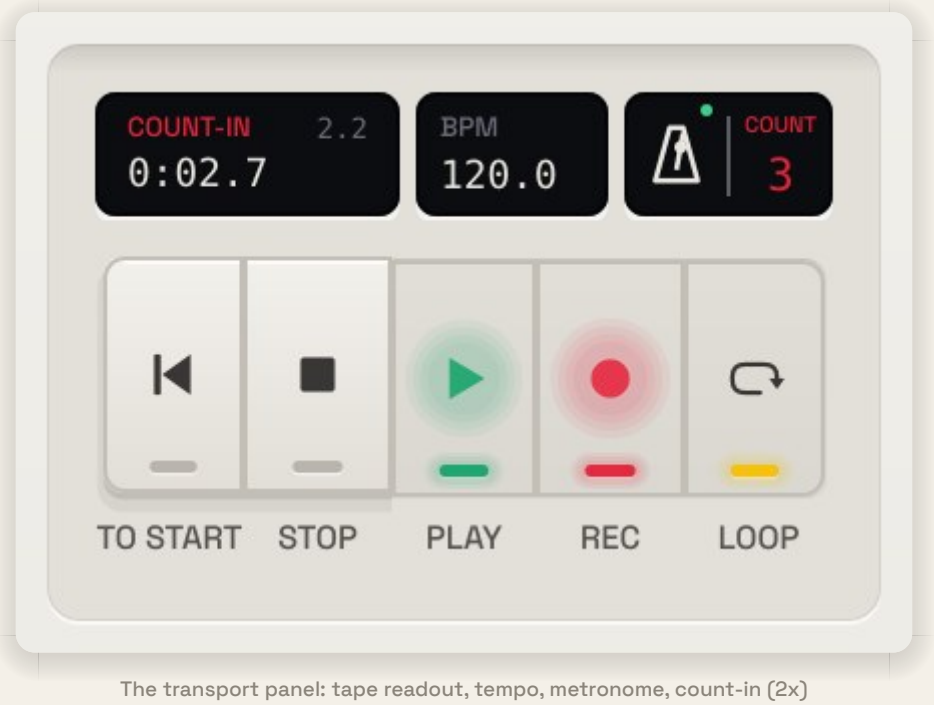


Tape page during a one-bar count-in: countdown, header tempo with TAP, the transport panel's tempo, click and count-in

Setting the tempo. The tempo is a control wherever it is shown – on the right of the display header (**120.0 BPM**, every page) and in the transport panel (**BPM 120.0**, next to the tape readout):

Gesture	Action
drag up / down	± 1 BPM per 4 pixels; with **Shift** (or Cmd / Ctrl) ± 0.1 BPM
mouse wheel	± 1 BPM (fine: ± 0.1)
double-click	type a value (20–300, a comma works too), **Return** sets it, **Esc** cancels
TAP (header, left of the tempo)	click in time: from the second tap on the tempo is the average of the last four intervals; a pause of more than 2 s starts over. Standalone key **T**.
right-click	MIDI learn / clear / reset of `tempo` (TAP: `metro.tap`)

While the tempo comes from outside – the plug-in host's transport, or an incoming **MIDI clock** (`sync.src`) – it is shown with **HOST** or **CLOCK** and cannot be edited there (TAP is hidden); change it in the host or on the clock source. With network sync (Link, standalone) Ohpe's tempo stays editable and sets the session tempo. The sequencer page keeps its Tempo encoder as before.



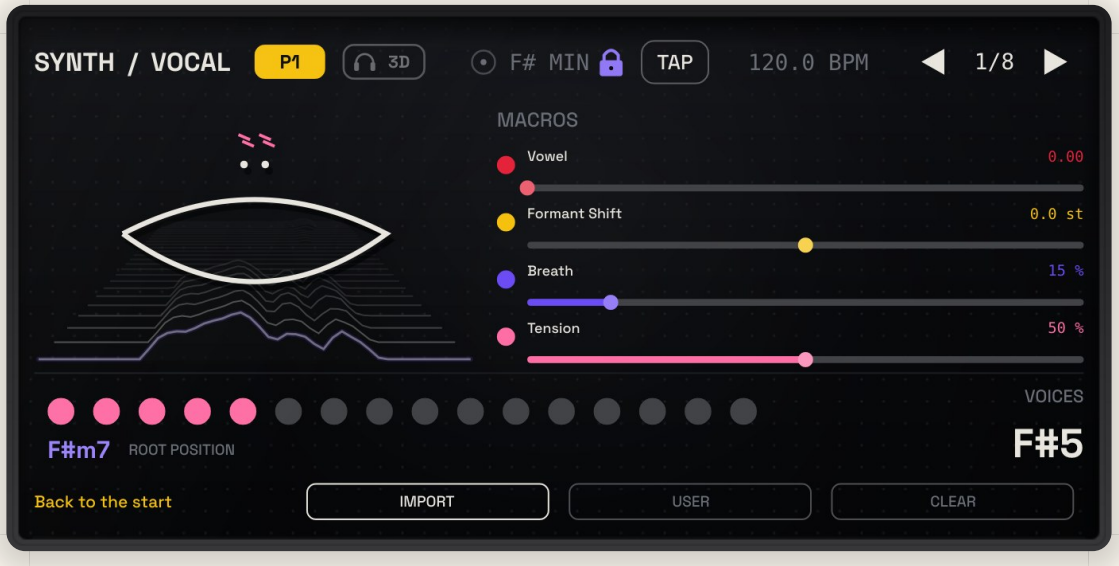
The transport panel: tape readout, tempo, metronome, count-in (2x)

Metronome. The metronome key (pyramid sign, right of the tempo in the transport panel) switches the **click** on and off (`metro.on`, standalone key **K**). It clicks on every beat while the **sequencer or the tape runs** – never while everything is stopped – and flashes green on each click; beat 1 of the bar is accented (higher and louder). The click is Ohpe's own synthesised sound (no samples), plays on the **main output** only and is **never recorded** onto the tape. Right-click the key: click level (25–100 %), accent on beat 1, Click in the plug-in too, MIDI learn. In a **plug-in** the click stays silent unless Click in the plug-in too (`metro.plugin`) is on – hosts have their own metronome; the standalone app clicks whenever the metronome is on. The click waits for Ohpe's reported latency (output limiter), so it sits exactly on the beat with the parts – and on the host's beat after its latency compensation.

Count-in. The count-in key (**COUNT**, `metro.countin`) cycles **– → 1 → 2** (off, one bar, two bars). When a recording starts from stop – **REC**, or PLAY with a track armed and recording on – the tape waits: the display shows the countdown (big beat number 4 – 3 – 2 – 1, the bar 1 / 2, red), the count-in key shows the beats left, and the click plays even with the metronome off (also in a plug-in). The tape starts recording **exactly on the first sample of the next bar**; a sequencer started with it (Play starts tape) starts on that same sample. If the sequencer is already running, the count-in counts on its grid and the recording starts on one of its bar lines (between one and the chosen number of bars later). **REC** or **STOP** during the count-in cancels it – nothing is recorded. Punching in while the tape already rolls has no count-in. Ohpe counts in 4/4.

The settings are host parameters (saved with the session / project, automatable, MIDI-mappable) and belong to the player: presets do not carry them. Parameters: PARAMETERS.md.

Key, chord and Scale Lock



The vocal page in F sharp minor: F#m7 under the voice dots, F#5 under VOICES, the key with Scale Lock on the header (2x)

The voice strip. Under the display's main area of the Synth, Envelope and LFO pages: sixteen voice dots (lit = voices sounding), on the right the small **VOICES** legend over the **last note** (big). The note's room is as wide as the longest note name (C#-1, Ab9), so the legend and the note never overlap in either language; the dots move closer on a narrow window instead. Notes are **spelled in the key** (Gb5 in E flat major, F#5 in F sharp minor; German: H, B).

The chord. Under the voice dots Ohpe names the **chord that sounds** – all four parts together, including the notes the one-finger chords and the harmony's following parts add (not the drum kit). Triads, sus2 / sus4, power chords, sixths, sevenths (maj7, 7, m7, m7b5, dim7, m(maj7), 7sus4; a seventh without its fifth too), add9 and ninths; the lowest note decides between two readings (Am7 over A, C6 over C) and makes a **slash chord** when it is not the root: C/E (1st inversion), G7/F (3rd inversion), or a bass outside the chord, C/D. Next to it the position (1ST INVERSION). Violet while it sounds, dim afterwards – the last chord stays until the next one. Chord roots are spelled in the key as well (Bbmaj7 in F major, Db in A flat major, C# in A major).

The key on the header. Left of **TAP** and the tempo: the key and scale (F# DOR, German FIS is written F#, H DUR) and a padlock for **Scale Lock**. The header is laid out so that every page title and engine name (English and German) is shown in full next to the whole key: on the Synth page a long title keeps the engine name (WAVETABLE rather than SYNTH / WAVETABLE), the mixer, tape and drum pages shorten MIXER / CHANNEL 3 to MIXER 3 only when space is short, and on a very full header (incoming clock) the tempo and key text get slightly smaller. A title is never cut off, and the key never covers the 3D badge. The same moves as the tempo:

Gesture	What it does
drag up / down	root by semitones (Shift / Cmd / Ctrl: the scale)
mouse wheel	root ± 1 (Shift: the scale)
click	menu: key, scale (all 14), Scale Lock on / off, Snap to the nearest note / White keys play the key, MIDI learn of the four parameters
click on the padlock	Scale Lock on / off
right-click on the padlock	MIDI learn / clear / reset of `key.lock`

This is the global key `key.root` / `key.scale` that the part's **Scale** quantise, the one-finger chords, the harmony brain (before it heard anything; Take Key writes into it) and the in-key grid already use – nothing new to keep in sync.

Key Gravity on the header. Right next to the key, left of it: a small **ring** for the edit part's Key Gravity (`pN.grav.amt`). It fills clockwise in the accent colour of its encoder with the amount and is empty at 0 % (the stage is off). Tooltip: Key Gravity – pulls the sound into key.

Gesture	What it does
click	opens the **Key Gravity** encoder page of the edit part (Effects mode, Sends tab)
drag up / down	the amount (Shift: fine)
mouse wheel	the amount ± 5 % (Shift: ± 1 %)
right-click	MIDI learn / clear / reset of `pN.grav.amt`

The two Key Gravity pages at the end of the Sends tab and the pro view's card stay where they were.

Scale Lock (`key.lock`, German In Tonart): every note that reaches a part plays in the key – the on-screen keyboard, MIDI from a keyboard or the host, the sequencers, the arpeggiator (it gets locked notes). Two modes (`key.lock.mode`):

Mode	What happens
Nearest (default)	every note snaps to the nearest note of the key (a tie goes down)
White Keys	the white keys play the key's degrees upwards: C4 plays the root nearest middle C, D4 the 2nd degree, … (pentatonic: five white keys per octave); black keys are silent. The grid keyboard is not remapped (it is in key already).

With Scale Lock on, the **transpose** of the step sequencer and of Orbit's lanes moves in **scale degrees** (+2 = two notes of the key up) instead of semitones; a sequencer playing the drum kit keeps semitones. Scale Lock off leaves everything exactly as before. The padlock is violet and closed while it is on; two small keys under it show White Keys.

From the host. The plug-in formats do not pass a key or scale to an instrument, so the key is Ohpe's own: `key.root` / `key.scale` are host parameters and can be automated from the host. Should a host key ever be passed in, the readout turns violet (the tooltip says the key comes from the host), root and scale are read-only and Scale Lock stays editable. Everything is MIDI-mappable. Parameters: PARAMETERS.md.

Tape page



Tape page

The left half of the display is the **transport panel:** the state (● **REC** breathing, REC READY, PLAY in green, STOP), the armed tracks, the time and bar.beat, the reels (they turn with the tape) and five buttons – |◀ · ■ · ▶ · ● · **LOOP** – plus one line that says what to do next (Pick a track, press REC). Below: the eight tracks with their content, the play position (red while recording), the loop region (yellow) and the **arm dots** – click a dot to arm or disarm the track, click a lane to select it.



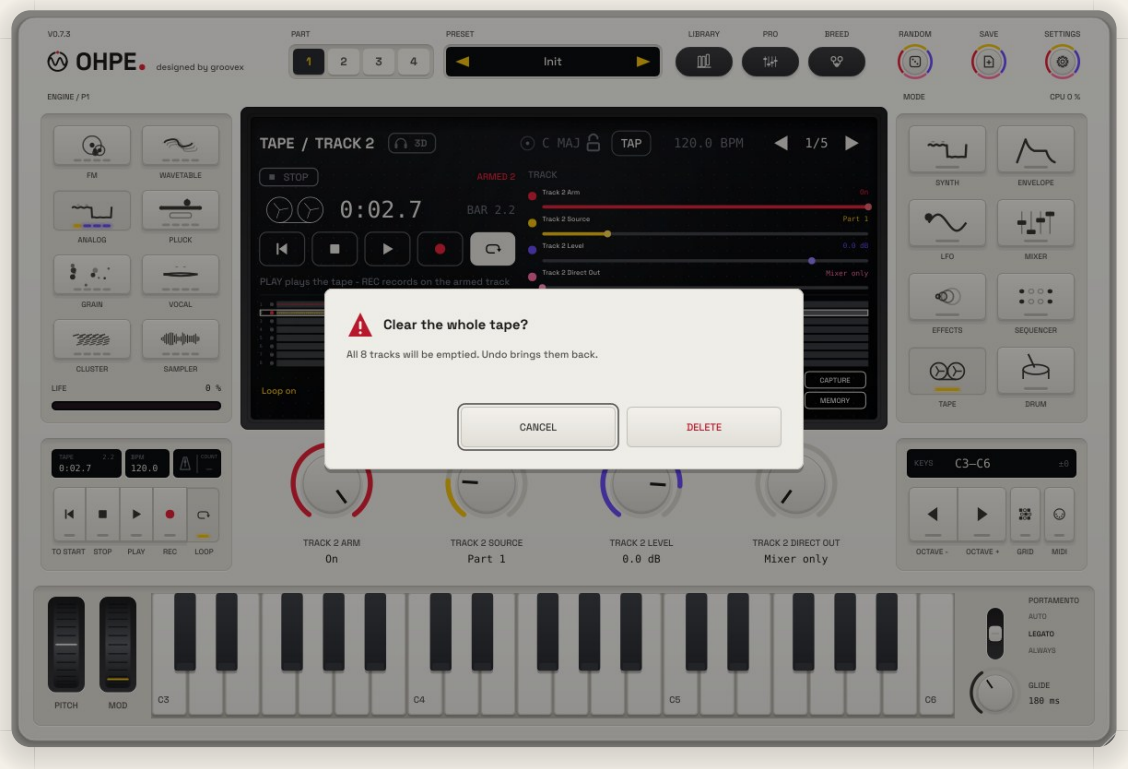
Tape page while recording

Encoder pages:

Page	Encoders
Track	Arm, Source (input, part 1–4, drum), Level, Direct out of the selected track
Transport	Overdub, Speed (varispeed 0.5–2×), Reverse, Play starts tape
Loop	Loop on/off, Loop in, Loop out, Snap (grid for setting the loop: Bar, Beat, Off – p42)
Tape character	Saturation, Wow, Flutter, Hiss
Bounce	Bounce source, Bounce target (Grain / Sampler), Endless loop, Input length

Soft keys on the display – two rows: the edits of the selected track above, the whole tape below:

Key	Action
Lift	copies loop in … loop out of the selected track to the clipboard and leaves silence
Cut	moves loop in … loop out of the selected track to the clipboard and closes the gap (later audio moves earlier)
Drop	places the clipboard at the play position (overwrites)
Splice	inserts the clipboard at the play position (the rest moves back)
Undo / Redo	unlimited undo of recordings and edits
Clear Track	empties the selected track at once – one undo step, so **Undo** brings it back
Clear All	empties all eight tracks after a confirmation (Clear the whole tape? – **Cancel** is the default, Return and Esc cancel; **Delete** in red clears). Undoable too
Save Tape	saves the whole recorder – all eight tracks, the tape length, the loop range and the edit track – as one `.ohpetape` file you can load again later, in any project
Load Tape	loads an `.ohpetape` file (or drop one on the window). When the tape holds material, Ohpe asks first (Replace the recorded tape?); the load is one undo step, so **Undo** brings the old tape back. A tape from another sample rate is converted and keeps its pitch
Export	renders master + the eight tracks as **24-bit WAV** files into a folder you choose (offline, faster than real time; the live audio is not interrupted)
Bounce	moves the edit track (or the source on the Bounce page; the loop range while the loop is on) into the Grain or Sampler engine of the edit part, as an endless loop by default – Bounce to engine
Capture	(end of the first row, p67) turns the tape memory – the last seconds of Ohpe's own output – into an instrument in the Sampler of the target part (Chromatic or Slices); dark while the memory is empty – Tape memory
Memory	(end of the second row) the memory menu: Chromatic / Slices, length (5–30 s), source (main out, part 1–4), target part, restore the last capture (after a preset change), export the last capture as WAV, clear the memory, MIDI learn for CAPTURE



Clear all asks first

Loop keys. The **loop readout** left of the soft keys shows the loop range that Lift and Cut use – in tape time and in bars, a lit dot while the tape loops, the pass while recording inside the loop. Click it to swap the keys for the loop keys (click again or **Back** to return); the encoders jump to the Loop page for fine adjustment:



The loop keys behind the loop readout

Key	Action
Set In / Set Out	loop in / out at the playhead – stopped or while the tape plays – on the nearest line of the grid; **Shift** = exactly at the playhead. Set Out switches the loop on at once
Loop	loop on / off

Key	Action
Snap Bar / Beat / Off	the grid for setting the loop (`tape.loopsnap`; 4/4 from the start of the tape at the project tempo)
To In	playhead to loop in
Back	back to the edit keys
Undo Loop / Redo Loop	undo / redo of the loop range (every key press, drag or encoder turn is one step; the tape's audio undo is separate)
Half / Double	the loop half / twice as long from loop in
Fit	the loop from the start around the recorded material (to the next grid line) and on

On the lanes the loop is a yellow band with a handle at each edge: drag an edge to move it (on the grid; Shift: free). Loop in always stays before loop out. Recording inside the loop layers every pass (Overdub on) or replaces it (Overdub off); the whole take is one undo step. Details: Setting the loop.

The four tape keys of the lower row are refused while recording. Every tape edit is MIDI-mappable: Tape Edit Track (`tape.edit.track`), Tape Lift / Cut / Drop / Splice / Undo / Redo / Return to Zero / Clear Track / Clear All (`tape.edit.*`); Tape Clear All from a controller opens the same confirmation on the screen. Saving and loading use the file dialog and have no MIDI trigger. Short guide: first recording in 30 seconds.

Drum kit page



Drum kit page

The Drum mode shows all 24 pads. Click a pad to play and select it; the encoders edit it – model, tune, decay, tone (page 1), level, pan, choke group, source (page 2) and the kit's volume and base note (page 3). **Load** puts an audio file on the selected pad (or drag a file onto a pad), **Clear** returns the pad to its factory sound.

Mod view (modulation matrix, 0.7)



Mod view

In **LFO** mode the tabs on the display switch between **LFO** (LFO 1, as before), **Mod**, the modulation matrix of the edit part, **Morph** (the morph pad), **Motion** (the motion recorder) **Harmony** (the harmony brain) and **Voice** (Voice Control). The Mod view lists all **16 routings** in two columns: source → target, a bar for the amount (to the right = positive, to the left = negative) and the routing's live value. Targets on an engine or an effect show the real parameter name ("Cutoff", "FX2 Hyper Detune"), so you see what moves without counting slots. Below the list a **trace** draws the value of the selected routing over the last three seconds.

Action	Result
Click a routing in the list	selects it; the encoders edit it
◀ Mod / Mod ▶ (soft keys)	previous / next routing
Clear (soft key)	sets the selected routing back to Off (source, via, target, amount, curve)
Page arrows	turn the encoder pages

Encoder pages of the Mod view: **source, via, target, amount** of the selected routing · **curve** and the **sequencer lane** · **macros 1–4** · **LFO 2** (rate, sync, shape, depth / mode, phase, fade) · **LFO 3** (the same) · **envelope 3** (delay, attack, hold, decay / sustain, release). Everything is in the pro view too (LFO tab).

Modulation rings. Wherever a parameter is the target of a routing – on the encoders and on the display rows of any page – a thin second ring (encoder) or line above the bar (display row) shows the **reach** of all its routings around the set value, and a dot or tick shows the **live position** while notes play. Turn the encoder to move the centre; the reach moves with it.

FX rack views



FX rack of a part

Key Gravity. The **Sends** tab of **Effects** mode ends with two encoder pages **Key Gravity** for the edit part: Gravity, Target, Glide, Curve, then Formant, Mix and Quality – the part's Key Gravity stage, which pulls every partial toward the key and the held chord (0 = off). In the pro view it is the card **Key Gravity.** Every control has its help line in both languages.

In **Effects** mode the tabs switch between **Sends** (as before), **Rack** – the six slots of the edit part's FX rack – and **Master**, the four slots of the master rack. The chain shows every slot as a block in signal order: effect name, mix bar, output level, **BYPASS** when bypassed; empty slots are dimmed. The selected slot is outlined, and the lower half of the display shows **what its effect does:** the transfer curve of the distortion, the response of the EQ and the filter, the static curve and live gain reduction of the compressor, the gain of each band of the multiband, the motion of the modulated effects.

Action	Result
Click a block	selects the slot; the encoders edit it
Drag a block onto another one	moves the slot there (the others close up); undoable, part of the preset
◀ Move / Move ▶ (soft keys)	moves the selected slot one place earlier / later
Clear (soft key)	empties the slot (type Off)
FX editor (soft key) or click on the slot's picture	opens the FX editor: the slot large, every parameter at once, points to drag

Encoder pages of a slot: **type, mix, bypass** · then the parameters of its effect, four per page. While an effect chosen for the first time is being prepared, the display says Preparing effect… (a moment; the slot passes the sound dry meanwhile). The arrows in the display header turn the pages – or open the **FX editor** to see every parameter of the slot at once.



Master rack

FX editor



FX editor: the EQ with band 2 selected

The **FX editor** shows the selected slot large, with **every parameter at once:** on the left a picture of what the effect does, edited directly with the mouse, on the right every parameter of the slot as a labelled bar (type, mix, bypass, then all of the effect's values). Open it with the **FX editor** soft key or a click on the slot's picture at the bottom of the display; **Close**, Esc or the soft key close it, and it closes by itself when you leave the Rack / Master view. It lies over the display and the two key columns – the **four encoders stay live below it.** The tabs at the top pick the slot (right-click: MIDI learn of its type).

In the picture	Result
Drag a point	moves its values (across / up), e.g. frequency and gain of an EQ band
Alt-drag or mouse wheel on a point	its second value: the band's Q, the compressor's knee, the LFO shape, the damping …
Shift (or Cmd / Ctrl) while dragging	fine
Double-click a point	resets it (Alt + double-click: its second value)
Click a point	selects it: its values go to the encoders (an extra encoder page right after Slot, named after the point)
Right-click a point	MIDI learn / automation menu of its parameter
Bars on the right	as in the pro view: drag, click to jump, double-click resets, right-click MIDI learn; a coloured dot marks the parameters on the encoders now

Every edit is a normal parameter change – the host records it as automation, MIDI learn and the host's undo work as everywhere, the modulation matrix shows its range as a violet bar on the point.

What the pictures show (yellow = what you edit):

- **EQ** – the response curve (20 Hz – 20 kHz, ±18 dB) with five points: low shelf, bands 1–3, high shelf; the selected band also alone (faint); the output level on its own track.
- **Compressor** – the transfer curve (input across, output up, -60 – 0 dB): the threshold point on the 1:1 line, the ratio point at the curve's end (lower = stronger), the knee as a light band (Alt / wheel); makeup on its own track, the live gain reduction (red); attack and release as an envelope with two points.
- **Multiband** – the three bands between their split frequencies: drag a band's level up / down, drag the split points across; the live gain of each band in violet; the output level on its own track.
- **Filter / Drive** – the response with one point: cutoff across, resonance up, drive on the wheel.
- **Distortion** – the transfer curve; the point moves bias (across) and drive (up), tone on the wheel; output level, pre-filter frequency (wheel: filter type). **Bitcrush** – a sine through the crusher: bits up, down-sampling across.
- **Chorus, phaser, flanger, hyper, tremolo** – the modulation as a wave: rate (or the tempo division when synced) across, depth up; the voices of chorus / hyper as several lines; extra values (spread, feedback, delay, voices, dimension, low cut) as slider rows below. **Wobble** – the filter sweep between its low and high frequency (two points), rate and drive below. **Formant** – the vowel (A E I O U across) and shift (up) with the two formant peaks, LFO rate and depth below. **Comb** – the comb response: frequency across, feedback up, damping on the wheel.
- **Delay / Echo** – the echo taps: time across, feedback up (the following taps decay with it); in ping-pong / tap mode the right channel's taps in pink (ratio below); the band between low cut and high cut below.
- **Reverb / Hall** – the dry hit, the pre-delay (point across) and the decaying tail: its length (size / decay) across, its brightness (damping) up; size and width below.
- **Mid/Side / Width** – the stereo picture (L / R axes, the image as wide as the width), balance; mid and side levels (Mid/Side) or the output (Width) on their own tracks.
- **Space 3D / Resonator** – their own pictures (the 3D position, the body); the bars edit them.



FX editor: the compressor of the master rack



FX editor: a modulation effect (Hyper)

3D Space view



The 3D Space view

The fourth tab of **Effects** mode, **3D Space**, edits the edit part's 3D stage: the **3D switch** (Off / Headphones / Speakers, a segmented switch on encoder 1 – turn the encoder or click a segment; p41), motion path, length and rate on the first page, then position (azimuth, elevation, distance, spread), the shape of the motion, room, air and distance, and on the last page the global **3D Monitoring** (Headphones / Speakers) and **Mono Check**. Instead of the effect chain the display shows the head-centred 3D view of all four parts, with the edit part's motion path drawn ahead of its dot as a dotted curve and its name bottom left; the strip below sums up the motion and the monitoring (while the part's 3D is off, the first line of the strip says Choose Headphones or Speakers (encoder 1) to place and move this part instead of the motion). The rack soft keys (move, clear) belong to the **Rack** and **Master** tabs only.

The **3D badge** in the display header (headphones pictogram + "3D") lights up in ultraviolet and pulses gently while the edit part plays binaural 3D; on speaker monitoring it is outlined in yellow with a pair of speakers, otherwise dim (with headphones or speakers – the monitoring a switch-on would use). Click it to jump to this view; hover it for Binaural 3D – best with headphones. The part selector shows the same pictogram per part, the library shows it after the name of every preset tagged **3D**.



The 3D badge in the display header

Body view



The Body view

The fifth tab of **Effects** mode, **Body**, edits the edit part's resonator body: on, body model, tuning (Fixed / Note / Chord) and mix on the first page, then tune, size, material and decay, then position, inharmonicity, sympathetic strings and width, and on the last page drive and the MPE amounts (pressure, slide). The display draws the body – the string pair with its sympathetic strings, the plate with its moving mode pattern or

the bore with its standing wave – with the excitation point as a yellow dot; the strip below names the model, the tuning and the pitch the body follows. A Resonator slot in the rack shows the same body in the Rack view.

Morph view



The Morph view

The third tab of **LFO** mode, **Morph**, shows the edit part's morph pad: four corner chips (A top left, B top right, C bottom left, D bottom right) in the encoder colours, the pad with the glided position, a soft glow from each corner as strong as its share, and the edges between the corners – solid where two corners morph their parameters (one sound), dotted where they cross-fade as parallel sounds (another engine or effect). The strip below says what the pad does right now (one sound, or how many sounds play in parallel).

Action	Result
Drag on the pad	moves the position (X, Y); right-click = MIDI learn
Click a corner chip, or **A … D** (soft keys)	menu: **Current sound**, **Preset** ▸ (any part preset of the library), **Clear corner**, **Clear all corners**
Encoders, page Morph	Morph on, X, Y, Glide
Encoders, page Control	X / Y Control (Off, Pressure, Slide, Mod Wheel), MPE, expression smoothing

Assigning a corner switches the morph on. An empty corner stands for the part's own sound.

Motion view



The Motion view

The fourth tab of **LFO** mode, **Motion**, shows the edit part's motion recorder: four lane rows with the recorded knob movements in the encoder colours over a sixteenth grid of the loop, and the playhead – green while playing, red while recording. A lane being touched has a breathing red rim; the strip below says what the recorder does right now.

Action	Result
Rec (soft key)	record: with the transport running, the knobs of the part you touch take lanes (up to four)
Play (soft key)	loop the lanes with the transport
Clear (soft key)	clear the part's lanes
Encoders, page Motion	Play, Rec, Length (1 – 64 sixteenths, polymetric), Depth

The lanes are also the matrix sources **Motion 1 … 4**.

Pads view



The Pads view

The sixth tab of **Effects** mode, **Pads**, holds the four performance pads – throws on the main mix for live sets: **Stutter**, **Tape Stop**, **Sweep** and **Freeze**. Each pad shows a small picture of what it does and its main setting; a pad waiting for its grid line blinks violet, an engaged pad lights green, a ringing freeze tail stays dim green. The strip below says which pads play.

Action	Result
press and hold a pad	the pad plays while held (engages on the next line of Quantise)
Shift-click a pad / soft key	latches the pad on or off
right-click a pad	MIDI learn (map a pad to a note-like CC button)
Encoders, page Pads	the four pads (turn = on / off)
Encoders, page Pad Timing	Quantise (Off, 1/16 … 1 bar), Stutter Length, Tape Stop Time, Sweep Time
Encoders, page Sweep & Freeze	Sweep Filter (low-pass / high-pass), Sweep Resonance, Freeze Level

The pads are also the matrix sources **Pad Sweep** and **Pads Held**.

Harmony view



The Harmony view and the grid keyboard

The fifth tab of **LFO** mode, **Harmony**, shows the harmony brain: the remembered key and chord, twelve bars for the pitch-class memory (key notes light, chord tones violet, held notes green, the key's root marked yellow) and the last eight chords. The strip says what the brain does and whether the edit part follows.

Action	Result
Take Key (soft key)	write the remembered key into the global Key / Scale
Hold (soft key)	freeze the memory
Forget (soft key)	forget key, chord and history
Encoders, page Harmony	Follow (edit part: Off / Scale / Chord / Root), Memory (bars), Hold, Grid Layout
Encoders, page Grid	Grid Layout (Fourths / Thirds / Whole Tones / Sequential), In Key / Chromatic, Fixed C (off = the key's root bottom left)

The **GRID** key (between OCTAVE + and MIDI) swaps the piano keys for the isomorphic grid keyboard: 4 × 24 cells in fourths, thirds, whole tones or sequential rows, in key or chromatic, the key's root bottom left (or C); chord tones carry a violet bar. With MPE on for the edit part each touch has its own channel – slide sideways to bend, up / down for slide, pressure from touch or pen. Every chord has the same shape in every key; the key's tooltip says so, and the first time the grid is switched on in a project the keyboard readout shows "Same chord shape in every key" for a few seconds.

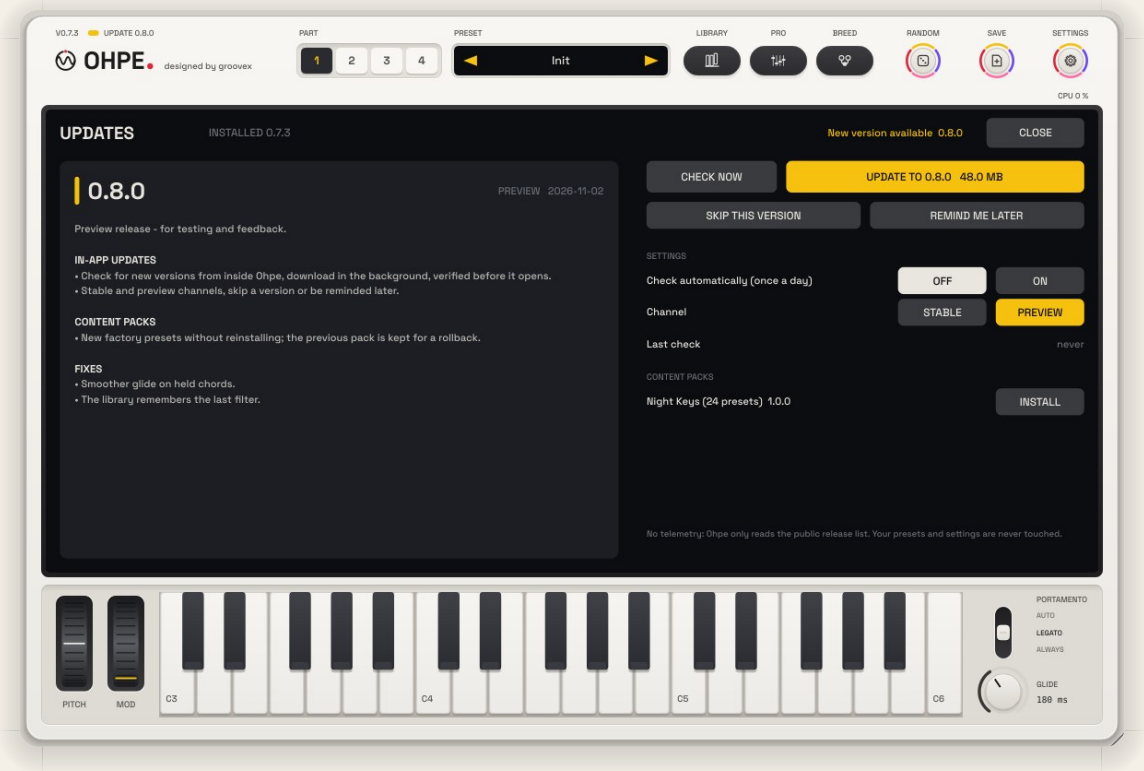
Voice view



The Voice view

The sixth tab of **LFO** mode, **Voice**, shows Voice Control: the vowel map (A E I O U, the sung vowel as a yellow dot, grey below the gate), the input meter with the gate line and bars for level, brightness, noisiness, pitch and onset. Encoder pages **Voice** (on, input gain, gate, smoothing) and **Voice Options** (pitch tracking, monitor). Soft keys: **On** and **Vowel → Filter + Morph** (three routings into the edit part's matrix). The input: standalone Settings > Audio > Input device, plug-ins the instrument's input / sidechain in the host.

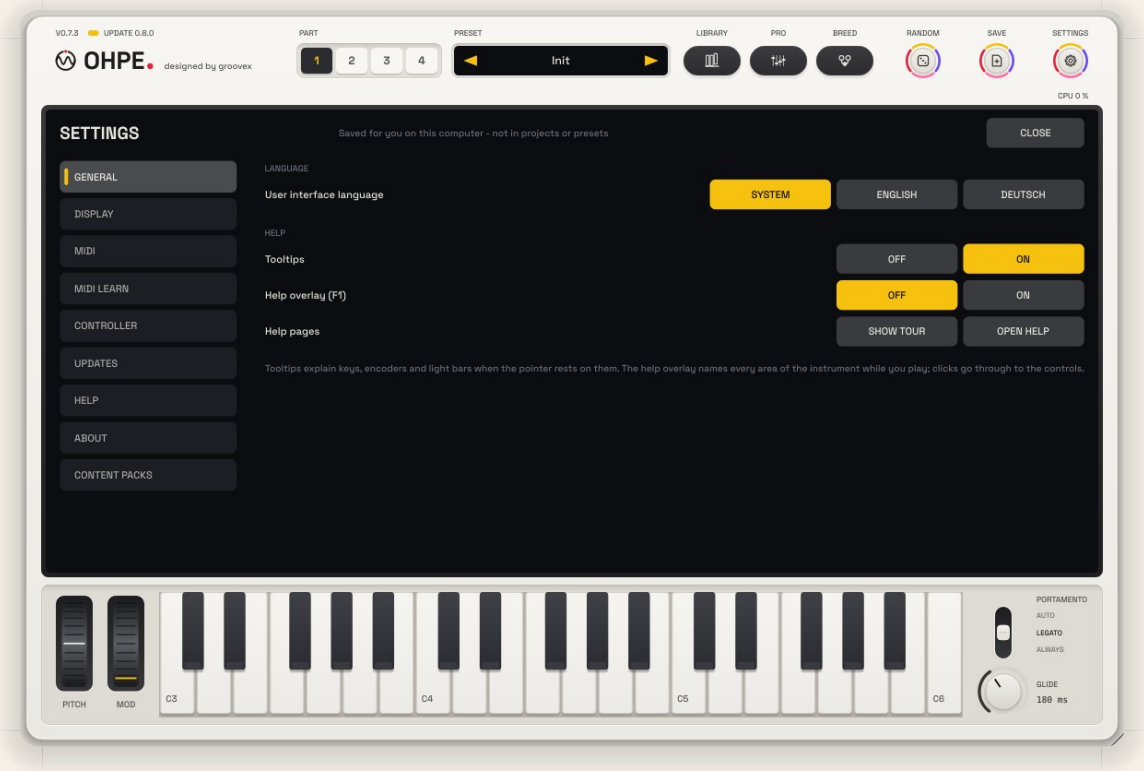
Updates



The update view

A click on the **version legend** above the wordmark (or the wordmark) opens the update view when an update is on offer, otherwise the **About** page of the settings. A small yellow light bar and `UPDATE x.y.z` next to the version say that a newer version is on offer. The view shows its release notes in the UI language, **CHECK NOW**, **UPDATE** (background download with a violet progress bar, **CANCEL**), **INSTALL** after the SHA-256 check, **SKIP THIS VERSION**, **REMIND ME LATER**, the settings (daily automatic check – off until switched on, never while playing –, channel Stable / Preview), content packs and **PREVIOUS VERSION**. Details: Updates.

Settings



The settings

The **gear key** at the right of the header opens the settings (the language pills moved in here). They are saved for you on this computer – not in projects or presets – and apply to every Ohpe you open: **General** (language, tooltips, help overlay, help pages), **Display** (light / dark / auto skin, display brightness, animations, window size), **MIDI** (channel, MPE and bend ranges of the four parts, the one-tap Fingerboard mode per part with its MPE HD status, MIDI inputs, clock and sync), **MIDI learn** (every mapping: edit, clear, clear all, map files), **Controller** (the grid controller: on by default and connected automatically, ports, status, controls, the live virtual controller), **Updates**, **Help** and **About**. **F1** switches the help overlay on and off.



The night skin

Details: Settings.

Tooltips and help search

Every control has a short **tooltip** in English and German (Settings > General > Tooltips): keys, encoders, wheels, the display's rows, page arrows, soft keys, tabs, steps, pads and tracks, the pro view's bars, pictures and card icons, the FX editor, the library, the breeding view, the settings and update pages. It says what the control does, and keys or values that learn MIDI with a right-click end with Right-click: MIDI learn. German texts keep the studio terms musicians use (Low / High Shelf, Q, Main, Tape-Memory).

The **help search** sits above the topics of Settings > Help; **Ctrl / Cmd + F** opens it from anywhere. Type English or German words – case, umlauts, word beginnings and small typos do not matter (`huellkurve`, `envelpoe`, `modrad`). The results are help topics, parameters (with where they are: Synth > Filter) and controls of the plate. **Return** or a click opens the best one: a topic shows on the page, a parameter's mode and page come up with its encoder ringed in yellow for a few seconds, a control is ringed on the plate; settings pages open directly. **Esc** clears the search.

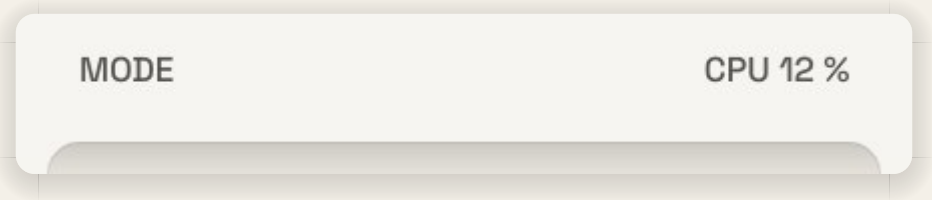
Library search

The library's search field matches every word (AND) against name, category, tags (**MPE**, **Fingerboard**, **Morph**), engine, author, pack, bank, kind, favourite and rating, in English and German, regardless of case and umlauts (`MPE`, `bass mpe`, `fläche`). The quick chips **MPE / Fingerboard / Morph** under the categories show presets that carry all selected ones. Details: Preset library.

Content packs

`.ohpepack` files add preset packs: **INSTALL PACK ...** (Settings > Updates, Settings > Content packs, the library's **PACKS** menu) or drop the file on the window. The library's **PACKS** chip filters by an installed pack; pack presets carry a **P**. Details: Content packs.

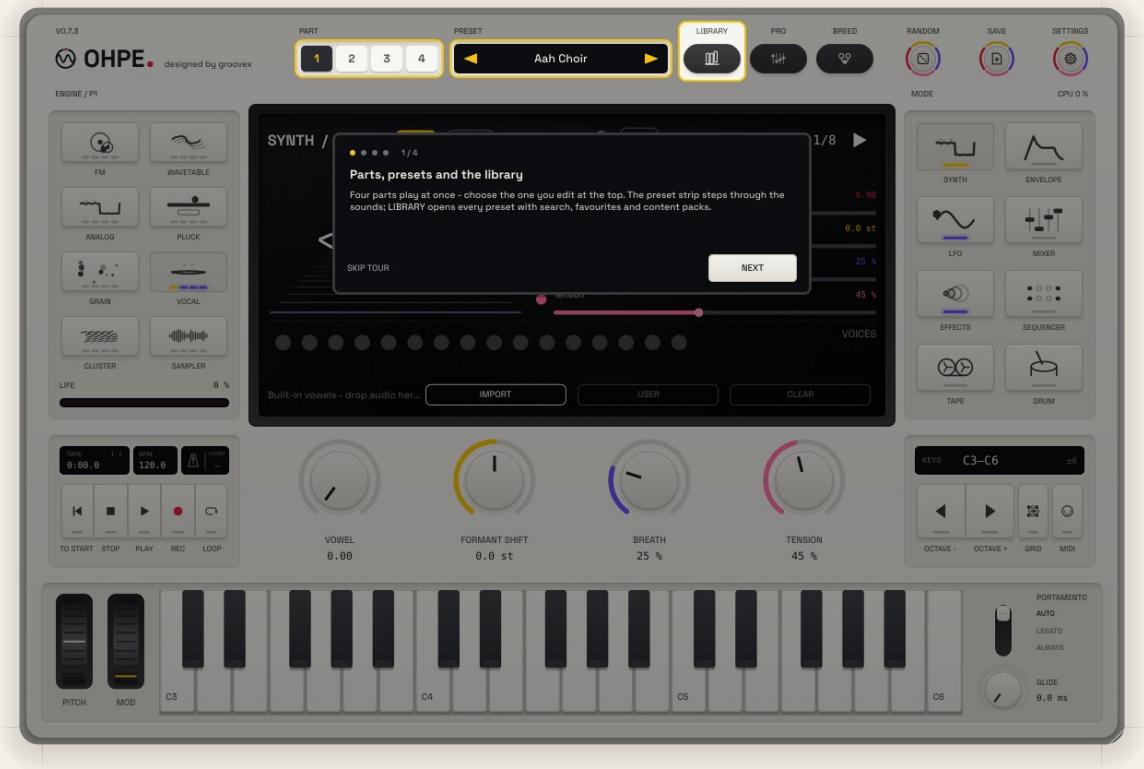
CPU meter



The CPU meter on the mode column's legend line

`CPU 12 %` on the legend line of the mode column (right, next to MODE) shows how much of each audio block's time Ohpe needs to compute its sound – averaged over about half a second; 100 % would be the whole time (the audio would break up). The tooltip adds the slowest block of the last two seconds. The legend turns red from 80 % on average or when a block took longer than its own time. It is a display only (no parameter, not saved). Parts that have been silent for a few seconds sleep and cost almost nothing until they play again. Details: Performance.

First-start tour



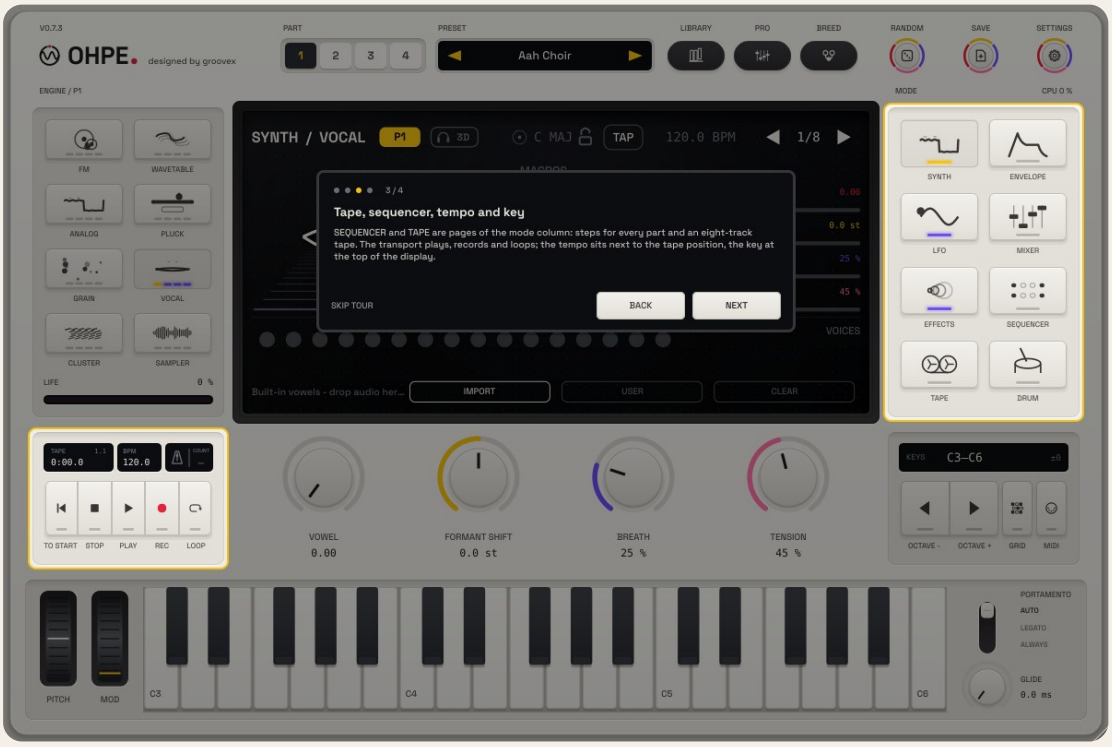
Step 1 of the first-start tour: parts, the preset strip and LIBRARY

The first time the standalone app starts, and the first time a plug-in window opens, a short tour of four steps shows the plate. The rest of the plate is dimmed; what a step talks about gets a yellow outline:

1. **Parts, presets and the library** – the part selector, the preset strip and **LIBRARY**.
2. **Encoders, soft keys and the FX editor** – the display with its soft keys and the four encoders below it; on the Effects page a click on the effect picture opens the FX editor.
3. **Tape, sequencer, tempo and key** – the mode column (SEQUENCER, TAPE) and the transport with the tape position and the tempo; the key sits at the top of the display.
4. **Help, settings and the Grid-Controller** – the gear key: language, night skin, help pages, help overlay, MIDI and the Grid-Controller.

NEXT / **BACK** (or → / ←, Return) step through it, **DONE** on the last step closes it, **SKIP TOUR** (or Esc) ends it at once. The plate waits while the tour is open. It shows once for the standalone app and once for the plug-in (stored in `settings.json`, key `"tour"`); **Settings > General > SHOW TOUR** and **SHOW TOUR** on the Help tab start it again at any time. It follows the language and the skin.


OHPE.
designed by groovex
Aah Choir
SYNTH / VOCAL
P1
TAP
120.0 BPM
1/8
MACROS
Vowel
Formant Shift
Breath
Tension
VOICES
IMPORT
USER
CLEAR
VOWEL
0.00
FORMANT SHIFT
0.0 st
BREATH
25 %
TENSION
45 %
2 / 4
Encoders, soft keys and the FX editor
The four encoders below the display set the four values shown above them. The soft keys at the foot of the display switch pages and functions. On the Effects page, click the effect picture to open the FX editor.
SKIP TOUR
BACK
NEXT


OHPE.
designed by groovex
Aah Choir
SYNTH / VOCAL
P1
TAP
120.0 BPM
1/8
3 / 4
Tape, sequencer, tempo and key
SEQUENCER and TAPE are pages of the mode column: steps for every part and an eight-track tape. The transport plays, records and loops; the tempo sits next to the tape position, the key at the top of the display.
SKIP TOUR
BACK
NEXT
VOWEL
0.00
FORMANT SHIFT
0.0 st
BREATH
25 %
TENSION
45 %



Crash-safe start (standalone, p63)



The restore notice after a session that did not end cleanly

The standalone app notes when a session is running and clears the note when you quit normally. If it finds the note at the next start (a crash, a power cut, the app was killed), it does not load the last session blindly: Ohpe starts with the factory start sound and shows a small notice at the top of the plate – **RESTORE** loads the last saved session after all, **START FRESH** keeps the start sound (the notice disappears either way). A saved state that cannot be loaded (damaged file) also starts with the start sound; the notice then only says so (**OK**). The plate stays playable while the notice is shown. A second running instance of the app never counts as a crash.

Ohpe saves the standalone session when you quit. RESTORE brings back that saved state, so changes made after the last save in a crashed session are not included. A damaged session, preset, tape, wavetable, content pack or settings file never crashes Ohpe: it is refused, or the parts that can be read are loaded and the rest falls back to defaults. In a plug-in host, a damaged host state is refused the same way and the instance keeps its sound.

Keys and shortcuts

Key	Action
Esc	closes the library (cancel), the pro view, the update view or the settings, or ends step editing
F1	help overlay on / off
Ctrl / Cmd + F	help search (Settings > Help)
Space (standalone)	PLAY / stop
R (standalone)	REC: record on the tape / punch out
Enter or Home (standalone)	tape to the start
L (standalone)	tape loop on / off
I / O (standalone)	tape loop in / out at the playhead (on the grid; with Shift: free)
T (standalone)	tap tempo
K (standalone)	metronome (click) on / off
Enter (library)	keep the selected preset and close
↑ / ↓ (library)	previous / next preset in the list
Tab / Shift+Tab, F6	keyboard operation: next / previous control, next area (focus ring) – arrows / Page change values, Enter / Space press keys, Esc gives the keys back to the notes; see accessibility
right-click on the housing	UI size 75 – 200 %

See also: preset library · sequencers · tape, mixer and effects · MIDI.

PART 2

Sound engines.

Grain, Vocal, Sampler and the drum kit.

Phase 2 adds three engines for every part (selected with **Engine**, `pN.engine`) and the **drum kit**, which has its own mixer channel. All of them sound right away: without loaded samples the Grain engine granulates a built-in evolving pad texture, the Sampler plays a built-in electric-piano-like keys tone, and the drum kit plays its factory recordings. Every parameter is listed in PARAMETERS.md.

Grain

Granular synthesis: every voice sprays short, softly windowed grains read from a source.

Control	What it does
Position (macro 1)	where in the source the grains are read (0 = start, 1 = end)
Grain Size (macro 2)	length of a grain, 5 – 500 ms
Density (macro 3)	grains started per second per voice, 1 – 100
Spray (macro 4)	random scatter of the position (up to ±0.5 s) and of the grain timing
Grain Pitch	transposes the grains (±24 semitones) on top of the played note
Reverse Chance	share of grains that play backwards
Stereo Spread	random panning of the grains
Scan Speed	moves the position while a note is held (1 = real time, negative = backwards)
Source	Sample (the loaded file or the built-in pad texture) or Tape (a tape region, see below)
Tone	low-pass brightness; the mod envelope and the LFO target Timbre move it

The played note transposes relative to **C4 (MIDI 60)**: C4 plays the source at its original pitch. With Source = Tape the engine granulates the tape: the **edit track** of the tape page (Track / `tape.edit.track`), or the **mix of all tracks** while the edit track is empty – its recorded length from the start, at most 60 s, normalised like a bounce. It follows the tape: after a recording pass, an edit, undo / redo or a change of the edit track the grains read the new material within a moment. With an empty tape the engine plays its sample (or built-in texture). To keep a take while the tape moves on, use **BOUNCE** instead.

Vocal

Formant synthesis of a singing voice: a glottal pulse plus breath noise drives five formant resonators.

Control	What it does
Vowel (macro 1)	morphs continuously A (0) – E (1) – I (2) – O (3) – U (4)
Formant Shift (macro 2)	moves all formants ±12 semitones (smaller or larger voice)
Breath (macro 3)	breath noise instead of a clear tone
Tension (macro 4)	lax and dark (0) to pressed and bright (1)
Vibrato Depth / Rate	pitch vibrato up to ±½ semitone, 0.5 – 10 Hz
Jitter	slow random pitch movement for a natural voice

The mod envelope and the LFO target Timbre move the vowel position.

User formants

The Vocal engine's Synth page has the soft keys **IMPORT**, **USER** and **CLEAR**, like the Wavetable engine (user wavetables). An imported sound (file, tape track, live input) is analysed by linear prediction: about every frame of the sound gives five formants (frequency, level, bandwidth – the five strongest peaks of the spectral envelope between 150 Hz and 5 kHz), up to 64 formant frames. With **USER** on (`pN.usertable.vocal`) Vowel 0 … 4 moves through these formant tracks from the start to the end of the sound instead of A – E – I – O – U; formant shift, breath, tension and vibrato work as before. The display draws the five tracks with the current vowel position. The material is saved like a user wavetable (library, presets, projects).

Sampler

Plays samples across the keyboard, either as **zones** (multisampling) or as **slices**.

Control	What it does
Start (macro 1)	start offset inside the sample or slice
Tone (macro 2)	low-pass brightness (mod envelope and LFO Timbre move it)
Loop Start / Loop End (macros 3, 4)	loop region as a share of the sample (when the file has no loop of its own)
Root Key	the note that plays a zone at its original pitch (zones may bring their own root key)
Loop	Off (one shot), Forward, Ping-Pong
Reverse	plays backwards
Mode	Zones or Slices
Slice Sensitivity	how small an attack still starts a new slice
First Slice Key	the key of slice 1 (default C2 = 36); each next key plays the next slice

Zones: each zone has a key range, a velocity range, a root key and optional loop points. Loop points stored in a WAV or AIFF file are used automatically.

Slices: Ohpe finds the attacks (transients) of the first sample and cuts it into slices, at most 64. Every slice plays on its own key at its original pitch and stops where the next slice begins. Changing Slice Sensitivity re-slices the sample.

Drum kit

24 pads, played on **MIDI channel 10** (pad 1 = `drum.base`, default C2 = 36, then one pad per semitone). The kit plays on mixer channel 5 (source Drum).

Real recordings first. The factory kit is a recorded acoustic kit (CC0 recordings of a real drum set, see THIRD-PARTY.md): kick, snare, rimshot, cross-stick, ghost note, closed / half-open / pedal / open hi-hat, three toms, ride, ride bell, crash, sizzle crash, cowbell, tambourine, shaker, claves, conga and bongo. Close, room-style and overhead microphones are mixed per piece, and every hit is humanised a little (±4 cents, ±0.4 dB); soft hits are darker, like a real drum. Load your own sample onto any pad to replace the recording; clearing it brings the recording back.

Synth models. Every pad also has a synthetic drum model, used when the pad has no sample (pad 5, the clap) or when **Source** is set to Synth:

Model	Sound
Kick	sine with a pitch sweep and a click
Snare	two tuned body tones plus filtered noise

Model	Sound
Hat	six inharmonic square oscillators and noise, high-passed
Clap	three noise bursts and a tail through a band-pass
Tom	sine with a short pitch drop and a stick noise
Perc	two-operator FM blip (rim, bell, woodblock)

Per pad: **Source** (Sample / Synth), **Synth Model**, **Tune** (±24 semitones), **Decay**, **Level**, **Pan**, **Tone** and **Choke Group** (0 = none, 1 – 8). Decay and Tone are centred: 0.5 is the natural sound; with samples, Decay below 0.5 shortens the hit and Tone below 0.5 darkens it. Every pad is monophonic, and pads of the same choke group cut each other – the closed, half-open, pedal and open hi-hats (pads 6, 7, 8, 17) share group 1. Note-offs are ignored.

Factory instruments and drum kits

- **Instrument** (`pN.sampler.inst`): the Sampler plays one of ten recorded instruments (kalimba, marimba, vibraphone, glockenspiel, harp, harpsichord, recorder, singing glass, zither, bowed metal) instead of its built-in keys tone while no sample of your own is loaded. Recorder, singing glass and bowed metal have loop points: set Loop to Forward for endless notes.
- **Instrument** (`pN.grain.inst`): the Grain engine granulates one note of the chosen instrument instead of its built-in texture (a loaded sample or the tape source still wins). For plucked and struck instruments keep Position near the start, where the note is loudest.
- **Keyboard Instrument** (0.8, `pN.sampler.inst2` / `pN.grain.inst2`): Grand Piano, Upright Piano (recordings, every few semitones, the grand with two velocity layers), Tine Piano and Reed Piano (Ohpe's own modelled electric pianos, two velocity layers: soft notes ring like bells, hard notes bark). When not Off it replaces Instrument. In the Grain engine the pianos make breathing piano clouds (keep Position near the start).
- **Drum Kit** (`drum.kit`): Studio (the original kit), Tight, Room, Lo-Fi, Hybrid or Boom (0.5: synthesised deep kick / sub kit). All kits use the same pad layout; your own pad samples always win.

Details and sources: FACTORY-SOUNDS.md, THIRD-PARTY.md.

Loading samples

Samples are loaded from disk in the plug-in (WAV, AIFF, FLAC and Ogg Vorbis): one sample for the whole keyboard, additional multisample zones, the grain source, or a sample per drum pad (it replaces the pad's factory recording). Ohpe saves the **paths** of the loaded files with the project, not the audio; when a project is opened again the files are reloaded. Files that were moved or deleted are skipped and that engine or pad uses its built-in sound or factory recording.

Bounce to engine: the BOUNCE key puts a tape track, the tape mix or a few bars of the live input straight into the Grain or Sampler engine – by default as a seamless endless loop (the grains wrap round the seam, a held Sampler note sustains it). Bounces are WAV files in the bounce library, so they are reloaded like any other sample. See Bounce to engine.

Wavetable engine.

Wavetable plays through a table of eight single-cycle frames and glides smoothly between them. Six tables are built in (generated when Ohpe starts, no sample files): **Classic** (sine → triangle → saw → square → narrowing pulse), **Harmonic** (an additive stack that grows frame by frame), **Formant** (vowel-like a – e – i – o – u), **Digital** (sparse, comb-like spectra), **Glass** (bell-like overtones) and **Grit** (seeded noisy spectra that get brighter).

Every table is stored as a set of band-limited copies, one per octave; the engine crossfades between them depending on the note and on how much the phase is warped, so high notes stay free of aliasing. **Warp** is a smooth phase distortion (Bend: one bulge per cycle, Twist: two), which brightens the wave without adding foldover; it eases off over the top octaves for that reason. Three unison oscillators, a sub sine and a resonant low-pass complete the voice.

Encoder	Parameter	Range	What it does
1	Position	0–1	position in the table (frame morph)
2	Warp	0–1	phase distortion depth
3	Tone	0–1	low-pass cutoff (≈ 40 Hz – 20 kHz); mod envelope and LFO Timbre move it
4	Unison	0–1	adds two detuned side oscillators, detune and stereo width
–	Table	list	Classic, Harmonic, Formant, Digital, Glass, Grit
–	Warp Mode	list	Bend, Twist
–	Env > Position	−1 – +1	how far the mod envelope sweeps the position
–	Resonance	0–1	filter resonance
–	Sub	0–1	sine one octave below
–	Key Track	0–1	cutoff follows the keyboard
–	Unison Voices (0.8)	1–8	3 = the classic centre + two sides; 1, 2, 4 … 8 = an evenly spread stack (reese, supersaw)
–	Unison Width (0.8)	0–1	stereo spread of the stack (not used by the classic three)
–	FM Amount (0.8)	0–1	phase modulation of the table by a sine (growls, metallic edges); fades out on the top notes so it never aliases
–	FM Ratio (0.8)	0.25–8	frequency of the FM sine relative to the note
–	LFO > Position (0.8)	−1 – +1	the part LFO moves the table position (slow scans, talking growls)
–	Env > FM (0.8)	−1 – +1	the mod envelope adds (or removes) FM

Parameter IDs:
`p1.wavetable.pos`, `.warp`, `.tone`, `.unison`, `.table`, `.warpmode`, `.envpos`, `.reso`, `.sub`, `.keytrack`, 0.8: `.univoices`, `.uniwidth`, `.fm`, `.fmratio`, `.lfopos`, `.envfm`.

Bass with a wide stack (0.8): the stack spreads the low end too – set the part's Bass Mono (`pN.monobass`, e.g. 150 Hz) or a Width slot with Mono Below to keep the bass mono-safe.

Starting points: pad – Formant, Position 0.3, Env > Position 0.4, slow amp attack; bass – Classic, Position 0.45, Unison 0, Sub 0.5, Tone 0.5; bright lead – Digital, Warp 0.4, Unison 0.5.

User wavetables: audio to wavetable



User wavetable on the display

On the Synth page of the Wavetable engine the display shows three soft keys: **IMPORT**, **USER** and **CLEAR**. Drag a sound (WAV, AIFF, FLAC, Ogg) or a wavetable file onto the instrument, or press **IMPORT**:

IMPORT menu	What it does
Audio or wavetable file …	file chooser (sounds, single-cycle sets, Ohpe tables `.ohpewt`)
From tape track > 1 … 8	analyses the recorded part of a tape track (up to 30 s)
Record input > 2 / 4 / 8 s	records the audio input (mono), then analyses it
Library > …	loads a table from the user wavetable library
Content packs > pack > …	uses a table of an installed wavetable bank (read-only, Wavetable engine)
Frames > 8 … 256	how many frames the analysis cuts (default 64)
Smoothing > Off / Light / Medium / Strong	spectral smoothing across neighbouring frames
Cancel import	while an import runs

How a sound becomes a table. The analysis runs on a worker thread – the audio never stops. It finds the pitch of the sound, then cuts exactly one period at evenly spaced points through the sound (period-synchronous slicing; the end of each period is crossfaded into the one before it, so the cycle loops without a click), resamples each period to 2048 samples, turns every frame so its fundamental starts as a rising sine (phase alignment: neighbouring frames morph without cancellation), optionally smooths the spectra across frames, removes DC, normalises every frame and builds the band-limited mip levels (one per octave, like the built-in tables), so a user table never aliases. Sounds without a clear pitch are cut into 2048-sample windows. Position then scans through the sound from its start (0) to its end (1).

Wavetable files. A WAV / AIFF / FLAC file that is a set of single cycles is read as it is: 2048 samples per frame (the file length is a multiple of 2048, up to 256 frames), or the frame size of a `clm` chunk or of evenly spaced cue points (other sizes are resampled to 2048). Ohpe's own format `.ohpewt` keeps the frames (and the formant tracks for the Vocal engine).

Library, presets, projects. Every analysed sound and imported set is saved into the user wavetable library – `…/groovex/Ohpe/Wavetables` in the user application data folder (macOS: `~/Library/Application Support/groovex/Ohpe/Wavetables`); the environment variable `OHPE_WAVETABLE_DIR` sets another folder. Part presets (block `"usertables"`), projects and the plug-in state (section `"usertables"`) refer to the table by its path, then by its file name in the library, and carry an embedded copy for tables up to 64 frames. When none of these is found the part plays its built-in table and the display says so – nothing else changes.

Wavetable banks (content packs). An installed content pack of kind `wavetables` adds its tables (`wavetables/*.wav`, single-cycle sets read like an imported set) to the menu under **Content packs >** . They are read-only: nothing is copied into the user library, and presets / projects store only the reference `"<pack>/<table>"` (`{"name": "Glass Choir", "content": "glass/glass-choir", "frames": 64}`), never the table. Loading re-solves the reference again; when the pack is not installed the display says **"Content pack missing – install it in Settings > Updates"** with the pack's id and the part plays its built-in table. The status line names the pack of a pack table.

USER switches between the user table and the built-in Table (host parameter `pN.usertable.wt`, MIDI-learnable; an import switches it on). **CLEAR** removes the part's user table. The display shows the frame stack, the frame at the current position lit in yellow; while an import runs, a progress bar.

Pluck engine.

Pluck models struck and plucked things. The **string** is a waveguide: a short delay line that recirculates a burst of shaped noise, losing a little energy and brightness on every pass. Its tuning compensates the delay of the loop filters exactly, so every note is in tune across the keyboard. The **mallet** is a set of five tuned resonators (like a struck wooden or metal bar) excited by a soft or hard mallet. **Material** blends between the two; a small resonant **body** colours both. The string rings by itself: while the key is held it decays naturally, the amp envelope only shapes the overall level and the release.

Encoder	Parameter	Range	What it does
1	Material	0–1	0 = string, 1 = mallet, in between a blend
2	Damping	0–1	decay time (long ringing → short thud)
3	Brightness	0–1	how fast the upper harmonics fade; mod envelope and LFO Timbre move it
4	Body	0–1	amount of body resonance
–	Excitation	0–1	soft → hard pluck / mallet (velocity adds to it)
–	Pick Position	0.02–0.5	where the string is plucked (near the end = thin and bright, middle = hollow)
–	Body Size	0–1	small → large body

Parameter IDs: `p1.pluck.material`, `.damping`, `.bright`, `.body`, `.excite`, `.pickpos`, `.bodysize`.

Starting points: nylon – Brightness 0.35, Body 0.5, Pick Position 0.25; harp – Damping 0.2, Brightness 0.6, long amp release; marimba-like – Material 1, Excitation 0.4, Damping 0.5.

Cluster engine.

Cluster plays a whole stack of oscillators for every key: up to seven band-limited saw / pulse oscillators, de-tuned against each other and spread across the stereo field. The stack can be a thick unison or a chord shape, so one finger plays octaves, fifths, triads, sevenths or stacked fourths. A slow random **drift** keeps the stack alive; a stereo resonant low-pass shapes it.

Encoder	Parameter	Range	What it does
1	Detune	0–1	detune between the stack voices (fine near 0, up to ± 55 cents)
2	Spread	0–1	stereo width of the stack
3	Tone	0–1	low-pass cutoff; mod envelope and LFO Timbre move it
4	Chord	list	Unison, Octave, Fifth, Major, Minor, Sus2, Sus4, Major 7, Minor 7, Dominant 7, Add 9, Fourths
–	Stack Voices	1–7	oscillators per note (the first ones of the chord shape)
–	Shape	0–1	saw → square → narrow pulse
–	Side Level	0–1	level of the stack voices against the centre voice
–	Resonance	0–1	filter resonance
–	Key Track	0–1	cutoff follows the keyboard
–	Drift	0–1	slow random pitch movement of the stack voices

Parameter IDs: `p1.cluster.detune`, `.spread`, `.tone`, `.chord`, `.stack`, `.shape`, `.side`, `.reso`, `.keytrack`, `.drift`.

Starting points: big lead – Unison, Detune 0.35, Spread 0.7; chord stab – Minor 7, Stack Voices 4, short amp decay, Tone 0.6 with some mod envelope; organ-like – Octave, Detune 0.05, Shape 0.5.

Layer engine and cross-modulation.

Every part can play a **second engine** with every note: the **layer.** Its sound uses the parameters that engine already has in the part (`pN.wavetable.*`, `pN.grain.*` …), so a part preset can combine, say, a plucked string with a grain cloud, a reese with a vocal, a piano with its own frozen recording. Presets that do this carry the category **Hybrid.**

Parameter	ID	Range	What it does
Layer Engine	`pN.layer.engine`	Off, FM … Sampler	the second engine (Off, or the part's own engine = no layer)
Layer Level	`pN.layer.level`	0–1	level of the layer
Layer Tune	`pN.layer.tune`	−24 … +24 semitones	e.g. an octave or a fifth above
Layer Fine	`pN.layer.fine`	±50 cents	detune against the main engine
Cross-Mod Mode	`pN.layer.xmode`	Ring, AM, Fold	how the layer modulates the main engine
Cross-Mod	`pN.layer.xmod`	0–1	amount of cross-modulation (0 = plain layering)
Layer Width	`pN.layer.width`	0–1	stereo width of the layer (0 mono, 0.5 as is, 1 doubled side)
Layer Own Envelope	`pN.layer.ownenv`	off / on	the layer uses its own amp envelope …
Layer Attack / Decay / Sustain / Release	`pN.layer.a .d .s .r`		… with these times (e.g. a pluck that blooms into a slow cloud)

Cross-modulation: Ring multiplies the main engine by the layer (metallic, inharmonic with detuned layers), AM lets the layer's envelope shape the main engine (a grain layer chops a pad), Fold folds the sum of both (harmonics that move with both engines). With MPE or aftertouch, the target Layer Blend moves the balance between main engine and layer.

The layer plays the same notes, glide, pitch envelope, LFO and expression as the main engine; its voices count against the same 16-voice limit per engine. With the layer off the part sounds exactly as before 0.8 (tested). The page **Layer** follows the engine pages in the Synth mode.

PART 3

Sequencing, tape and mixing.

Sequencers, parts and scale mode.

Ohpe has **four parts** that play at the same time, each with its own engine, envelopes, LFO and mixer channel (part 1–4 → mixer channels 1–4 by default, each routable to Main or to its own stereo output). Four sequencers drive them. Every sequencer has a **target part**, so a typical set-up is

Part	Engine (example)	Driven by
1	FM	Step sequencer (`seq.step.channel` = 1)
2	Analog	Euclid (`seq.euclid.part` = 2)
3	FM	Arpeggiator (`seq.arp.part` = 3), played from the keyboard
4	Analog	Orbit (`seq.orbit.part` = 4)

All sequencers run from the same transport (internal clock, host, MIDI clock or Link – see the MIDI guide) and start and stop with **Sequencer Run** (`seq.run`). Their steps sit on absolute positions of the transport, so they stay in phase with each other and with the host, also across loops and jumps.

Tempo: drag, type or tap it on the display header or in the transport panel (read-only while the host or a MIDI clock sets it); the metronome clicks on every beat while the sequencer runs, and a sequencer started together with a counted-in tape recording starts exactly on the recording's first sample – see Tempo, metronome and count-in. Ohpe has no step / live recording into the sequencer, so the count-in applies to tape recordings.

Step sequencer – patterns and song mode

The 64-step sequencer from 0.1 now holds **16 patterns.**

- **Pattern** (`seq.step.pattern`, 1–16) chooses the pattern that plays. A new choice waits until the playing pattern has reached its end, so changes always land on the bar.
- Each pattern can have its **own length** (or follow `seq.step.length`).
- **Song mode** (`seq.step.song`) plays the **chain** instead: up to 64 entries of pattern × repeats, looping. The song position is taken from the transport, so a host loop or a jump lands on the right pattern and step.
- Patterns, lengths and the chain are saved with the state. Patterns from 0.1 open as pattern 1.
- **Ties (0.5):** a step at gate 100 % ties into the next step when that step plays the same note – the note is held instead of struck again and ends with the next step's gate. Chains of 100 % steps hold a note (or a one-finger chord) for as long as you like. Not on the drum kit (pads are one-shots).

Euclid

Four rings, each spreading **pulses** as evenly as possible over **steps** (1–32) with Bjorklund's algorithm, turned by **rotation** and playing its own **note.** Rings of different lengths form a polymeter: rings of 3 and 4 steps meet every 12 steps.

Parameter	Range	
`seq.euclid.on`	off / on	
`seq.euclid.part`	1–4	target part (default 2)
`seq.euclid.rate`	1/4 … 1/32, triplets, dotted	shared grid of all rings
`seq.euclid.gate`, `.vel`	5–100 %, 1–127	note length, velocity
`seq.euclid.rN.steps / .pulses / .rot / .note`	per ring 1–4	

Parameter	Range	
`seq.euclid.rN.vel` (0.5)	0–127	velocity of one ring; 0 = `seq.euclid.vel` (accents: loud kick, soft hats)
`seq.euclid.swing`, `.human` (0.5)	0–75 %, 0–100 %	swing and humanise, see below

Examples: 3 pulses in 8 steps `x..x..x.`, 5 in 8 `x.xx.xx.`, 7 in 16 `x..x.x.x..x.x.x.`.

Arpeggiator

The notes played into the target part (keyboard, MIDI on the part's channel, one-finger chords) are held and played one after another.

- **Mode**: Up, Down, Up/Down (turning points are not repeated), Random, As Played, Chord (all held notes on every step).
- **Octaves** 1–4, **Gate** 5–100 %, **Rate** 1/4 … 1/32 including triplets (T) and dotted (.).
- By default the arpeggio plays **while keys are held** and stops when you let go. **Latch** (option, off by default) keeps the notes after the keys are released; the next key pressed with no key held starts a new set. Latch is your playing choice: loading a part preset never switches it on or off.
- While the transport runs the steps sit on its grid. When it is stopped the arpeggiator runs free at the current tempo and starts exactly on the first key.
- **Pedals**: the sustain pedal (CC 64, from any input that plays the part: MIDI keyboard, host, grid controller, an MPE controller's master channel) holds the arpeggio's notes: keys you let go stay in the pattern until the pedal comes up, keys pressed meanwhile are added. Sostenuto (CC 66) holds only the keys that were down when it went down. Pedal up removes every key you no longer hold; with none left the arpeggio stops cleanly. The pedal holds the arpeggio's key set, not its sounding notes, so each step stays as short as Gate. With **Latch** on the set stays after pedal up anyway; while the pedal is down a new key adds to the latched set instead of starting a new one. Switching the arpeggiator off or to another part releases everything it held. (An MPE part is played directly, without the arpeggiator.)

Orbit (polymeter)

Four lanes of up to 16 steps, each with its own **length**, **rate** and **transpose**, all playing into one part. Every lane has its own steps (note, velocity). Lanes drift against each other and meet again after the least common multiple of their lengths – long, evolving patterns from a few steps. With **Scale Lock** on (key readout on the display header) a lane's transpose – like the step sequencer's – counts degrees of the key instead of semitones.

- **Lane target** (`seq.orbit.lN.part`, 0.5): 0 = the Orbit target part (or the drum kit); 1–4 = this lane plays that part, never the kit. With the drums switched on, lanes 1–3 can play the kit while lane 4 plays a bass line – one sequencer for a whole groove.
- **Ties** (0.5): at **Gate** 100 % (`seq.orbit.gate`) a step that repeats the note of the step before it continues that note (held, not struck again) – sustained bass notes and chords. Lanes playing the drum kit are never tied.
- **Swing** and **Humanise** (`seq.orbit.swing`, `seq.orbit.human`, 0.5), see below.

Swing and humanise

- **Swing** (`seq.orbit.swing`, `seq.euclid.swing`, 0–75 %) moves every note on an off-beat sixteenth late by swing ÷ 2 of a sixteenth – the same amount as the step sequencer's `seq.step.swing` (30 % ≈ a 57 % shuffle, 75 % = 3/8 of a sixteenth). Lanes and grids at 1/8 or slower never land on an off-beat sixteenth, so all parts of a groove swing together.
- **Humanise** (`seq.orbit.human`, `seq.euclid.human`, 0–100 %): at 100 % a note starts up to 10 ms late and its velocity moves by up to ±16. The variation is repeatable (it starts the same way every time the sequencer is reset), so a bounce sounds like what you heard.

Sequencers playing the drum kit

The step sequencer, Euclid and Orbit can play the **drum kit** instead of their target part: `seq.step.drums`, `seq.euclid.drums`, `seq.orbit.drums` (off / on). Their notes then go to the pads (note – `drum.base` = pad, by default C2 = kick, D2 = snare, F2 = closed hi-hat …), as if played on MIDI channel 10; scale quantise and chords do not apply. Orbit makes a drum machine with one lane per drum and its own length per lane; Euclid plays four polyrhythmic drums. The factory project presets use both – see FACTORY-SOUNDS.md.

Poly glide

Glide (`pN.glide`) slides the pitch from note to note. With one voice (mono) every new note slides from the last. With **Poly Glide** (`pN.polyglide`) on and several voices, every note of a new chord takes over the voice of the nearest note just released and slides from there – chords glide into each other voice by voice (the sliding pads of the factory bank). Without released voices (the first chord, or notes added to held ones) notes start on their own pitch.

Portamento and pitch envelope

Every melodic engine shares the same portamento and pitch envelope; both sit on the last two pages of the **Synth** mode (Glide, Pitch Envelope) for every engine.

Parameter	What it does
Glide (`pN.glide`, 0 … 2 s)	Portamento time. 0 = off.
Glide Mode (`pN.glide.mode`)	**Auto** (default, as before 0.6): a note glides while the previous note still sounds (also in its release). **Legato:** a note glides only while another key is held; with one voice the envelopes keep running (true legato), detached notes start fresh. **Always:** every note glides from the last note played, even after silence.
Glide Timing (`pN.glide.rate`)	**Constant Time:** every glide takes the Glide time. **Constant Rate:** the Glide time is per octave – small steps arrive quickly, big jumps slide longer.
Poly Glide (`pN.polyglide`)	Chords glide voice by voice (see above).
Pitch Env Amount (`pN.pitchenv.amt`, −24 … +24 semitones)	Every new note starts this far away and slides into its pitch: negative = **rises** from below, positive = **falls** from above. 0 = off.
Pitch Env Time (`pN.pitchenv.time`, 5 ms … 4 s)	How long the slide takes.
Pitch Env Curve (`pN.pitchenv.curve`)	0 = linear, 1 = fast start and a slow arrival.

Mono note priority (0.6): with one voice, releasing the newest key returns to the previous key still held, legato and gliding back. **MIDI:** CC 65 switches portamento off / on, CC 5 sets the glide time (see MIDI.md). **Sequencers:** a step at gate 100 % (step sequencer, Orbit) now ends one sample after the next step starts, so a different next note overlaps it – mono parts in glide mode Legato glide exactly there.

The factory bank has rise and fall pads and synths (Rise Strings, Rise Choir, Rise Glass, Rise Warmth, Rise Bloom, Fall Haze, Rise Lead, Scoop Keys, Rise Bass, Zip Pluck, Fall Lead) and portamento leads and basses (Legato Glide Lead, Portamento Square, Glide Sine Bass, Glide Poly Synth).

Scale and chord mode

- **Key** (`key.root`) and **Scale** (`key.scale`: chromatic, major, minor, the church modes, harmonic and melodic minor, pentatonics, blues, whole tone) are global.
- **Scale Quantise** (`pN.quant`) moves every note going into a part to the nearest note of the scale – from the keyboard, MIDI and the sequencers.
- **Chord** (`pN.chord`): one key plays a chord – Triad, Seventh, Sus2, Sus4, Power, Octave, Sixth, Add9. In a scale the chords are diatonic (in C major a D plays D minor); with the chromatic scale they use fixed major intervals. Chords feed the arpeggiator, so one finger plays an arpeggiated chord.
- A key's release always ends exactly the notes it started, even if the settings changed in between.

Parameter IDs

All IDs are listed in PARAMETERS.md and by `ohpe_param_dump --markdown`.

Tape machine, mixer dynamics and FX rack.

../PARAMETERS.md*

Ohpe 0.2 adds an eight-track tape machine, dynamics in the mixer, an effect slot for the part, a stereo audio input in the plug-in and an offline bounce of the tape to WAV files.

```
input bus ┐                    ┌ tape track 1..8 ─┬ mixer channel (source "tape N") ┐
part 1 – FX slot ──────────────┤   (record from   │                                 ├ master
(parts 2-4, drum) │            │   input / part)  └ direct out → Out 1..8           │
                  └ mixer channel 1 ──────────────────────────────────────────────────┘
channel: source → EQ → compressor-lite → fader / pan → Main or Out N (+ sends A / B)
master:  sum + returns → compressor → tape saturation → volume → low-pass → limiter (ceiling) → Main
```

Tape machine

First recording in 30 seconds



Tape page while recording

1. Press **TAPE** (mode keys, right). The display shows the transport panel and the eight tracks.
2. **Pick a track:** click its lane (track 1 is selected at first). Its source is Part 1 – set another on encoder 2 (Source: input, part 1–4, drum).
3. Press **REC** (red key, bottom left – or **R** in the standalone app). The selected track is armed, recording starts and the tape and the sequencer run. The display shows ● **REC** (red, breathing) and the time.
4. **Play** – on the keyboard, a MIDI keyboard or with the sequencer.
5. Press **STOP** (or **Space**). Press **STOP** again: back to the start. Press **PLAY:** you hear your take. **Undo** on the display takes the whole pass back.

More takes: arm another track (click its dot, or Arm on encoder 1) and press **REC** again; with Overdub (Transport page) a take is added to the track instead of replacing it. **REC** while the tape plays = punch in, **REC** again = punch out.

Transport

Key / control	What it does	MIDI trigger
REC	one-touch record (arms the selected track when none is armed, starts the tape); while recording: punch out	`tape.tx.rec`
PLAY (body)	sequencer and tape start / stop together; with Play starts tape off only the sequencer	`tape.tx.play`
▶ (tape page)	the tape starts / stops (and the sequencer with it while Play starts tape is on)	`tape.tx.tplay`
STOP	everything stops (recording ends); when already stopped: to the start	`tape.tx.stop`
|◀	to the start	`tape.edit.rtz`
LOOP	loop on / off (without a range: the recorded length, or four bars)	`tape.tx.loop`
loop readout → **SET IN / SET OUT**	loop in / out at the playhead (on the grid; Shift: free) – see Setting the loop	`tape.edit.loopin` / `tape.edit.loopout`
arm dot / Arm	arms / disarms a track; the trigger works on the edit track	`tape.tx.arm`
Play starts tape	setting, on by default: PLAY runs sequencer and tape together	`tape.tx.link`

Triggers fire on a press (CC ≥ 64), so momentary controller buttons work as transport keys. They are not part of presets.

Eight stereo tracks, one shared transport. The tape is 10 minutes long by default; the length (10 s … 60 min) is a setting of the session (`TapeMachine::setLengthSeconds`, stored in the `tape` section of the state), not a parameter. Memory is only used where audio was recorded (blocks of 4096 frames), so a long, empty tape costs almost nothing.

What	How
Play / record	`tape.play`, `tape.rec` (punch in / out while playing). Recording needs at least one armed track (`tape.trN.arm`); the REC key arms the selected track by itself.
Source per track	`tape.trN.src`: Input (plug-in input bus), Part 1–4, Drum.
Overdub / replace	`tape.overdub` on: the new take is added; off: it replaces what was there (and the old material is not heard while it is replaced).
Monitoring	an armed track with source Input plays the input through the track (so it can be heard and routed); parts are heard on their own mixer channels.
Varispeed	`tape.speed` 0.5 … 2.0 (1.0 in the middle of the control), glides smoothly while rolling, 4-point interpolation. Pitch follows the speed.
Reverse	`tape.reverse` – the playhead runs backwards; at the start (or end) of the tape it stops and stays silent.
Loop	`tape.loop`, `tape.loopin`, `tape.loopout` (seconds, whole frames), `tape.loopsnap` (grid for setting it). See Setting the loop.
Locate	`TapeMachine::locate (seconds)` (applied at the next audio block).
Character	`tape.sat` (soft saturation, gentle top-end loss), `tape.wow`, `tape.flutter` (speed modulation shared by all tracks, as on one transport), `tape.hiss`. All at 0 = exact, untouched playback.
Track level	`tape.trN.level` (dB).
Direct out	`tape.trN.out`: Mixer only or Out 1–8. The track then also plays on that output (after level and character); if that output is not enabled, it plays on Main. Mixer channels can select tape tracks as source as well (channels 6–8 start on tracks 1–3).

Count-in and click. With a count-in (`metro.countin`: 1 or 2 bars) a recording started from stop waits for it and starts exactly on the first sample of the next bar (sample-accurate, also in the middle of an audio block); STOP or REC during the count-in cancels it. The metronome (`metro.on`) clicks while the tape or the

sequencer runs; the tape alone clicks on the tape's bar.beat grid. The click goes to the main output only and is never recorded (the tape records the parts, the drum kit and the input). See Tempo, metronome and count-in.

Setting the loop



Tape page with the loop keys

The loop can be set by ear while the tape plays: press **SET IN** where the phrase starts and **SET OUT** where it ends – the tape loops at once.

- **The loop readout** left of the tape page's keys shows LOOP (its dot lit while the tape loops), the range in tape time and in bars (BAR 3.1–5.1 2 bars) and, while recording inside the loop, the pass (PASS 3, red). Click it to switch the keys to the **loop keys**; click it again (or **BACK**) for the edit keys. The encoders jump to the Loop page: loop on, loop in, loop out, grid – for fine adjustment (Shift / Cmd / Ctrl: a tenth of the speed).
- **Loop keys** (two rows): **SET IN** · **SET OUT** · **LOOP** (on / off) · **SNAP** (Bar → Beat → Off) · **TO IN** (playhead to loop in) · **BACK**; **UNDO LOOP** · **REDO LOOP** · **HALF** · **DOUBLE** · **FIT** (loop from the start around the recorded material, to the next grid line).
- **At the playhead, while stopped or playing.** SET IN / SET OUT take the playhead position and snap it to the nearest line of the grid (`tape.loopsnap`: Bar by default, Beat, Off) – a key pressed a little late still lands on the downbeat. **Shift** (keys, shortcuts, dragging) sets it freely, exactly at the playhead. The grid is the tape's own: 4/4 from the start of the tape at the project tempo (the same bar.beat as the readout and the click).
- **SET OUT switches the loop on.** If the playhead is already past the new loop out (it was snapped back to the nearest bar), it jumps into the loop at once, at the same distance from loop in – no gap, no run-on.
- **In < out, always.** The shortest loop is one grid step (0.1 s off the grid). Setting an edge on the wrong side of the other one moves the loop along with its length (SET IN after loop out: the whole loop moves there); dragging an edge stops at the other one; an encoder, MIDI controller or host automation that turns one edge past the other pushes the other edge along (the edge that moved wins). Loop out never goes past the end of the tape.

- **Dragging on the lanes.** The loop is the yellow band over the track lanes (brighter while it loops); its edges are lines with a handle at the top. Drag an edge (the cursor turns into ↔) – on the grid, Shift: free. The time axis holds still during a drag; right-click on an edge: MIDI learn of loop in / out.
- **Undo.** Every change of the range is one step of the **loop history** (64 steps): each key press, each drag, each encoder turn once it rests (0.6 s). UNDO LOOP / REDO LOOP walk through it – separate from the tape's audio undo, so undoing a loop change never touches a recording. Loading a tape file starts a new loop history.
- **Recording inside the loop.** With the loop on, REC records pass after pass while the tape goes round: with **Overdub on** every pass is laid over the ones before (layering), with **Overdub off** every pass replaces the previous one (the last pass stays). The readout counts the passes. The whole take – from REC to punch-out or STOP – is **one undo step** of the tape (UNDO on the edit keys removes all passes at once; to keep only some, punch out after the pass you want and record the next take separately). Nothing is written outside the loop.
- **MIDI.** Every loop function is a host parameter and MIDI-mappable: `tape.edit.loopin` (set loop in), `tape.edit.loopout` (set loop out), `tape.edit.loophalf`, `tape.edit.loopdouble`, `tape.edit.loopundo`, `tape.edit.loopredo` (triggers: a press, CC ≥ 64; snapped to the grid), `tape.loopsnap` (grid), plus the existing `tape.tx.loop` (loop on / off), `tape.loop`, `tape.loopin`, `tape.loopout`. The display shows what a trigger did.
- **Shortcuts (standalone):** **I** set loop in, **O** set loop out (Shift: free), **L** loop on / off.

Editing

All edits work on one track, take times in seconds (or frames), and are one undo step each:

Edit	Result
lift (start, end)	copies the range to the clipboard and leaves silence
cut (start, end)	copies the range to the clipboard and closes the gap (later audio moves earlier)
drop (at)	puts the clipboard at `at`, over what is there
splice (at)	inserts the clipboard at `at`, later audio moves later (the tape end cuts it off)
clear track / clear all	silence (undoable; on the tape page Clear Track at once, Clear All after a confirmation)

Undo / redo are unlimited. One recording pass (from pressing record to releasing it) is one step. A new edit after undo drops the redo steps. Edits are refused (return `false`) while recording.

How it works (real time)

- Each track is a page table of 4096-frame blocks; empty pages are silence.
- The audio thread never allocates: the message thread keeps a pool of 256 empty blocks filled (`housekeeping()`, called by the plug-in at 20 Hz) and hands them over through a wait-free queue.
- Committed blocks are frozen. Recording into a frozen block first copies it into a fresh one (copy-on-write), so every undo state stays intact. Undo states share all unchanged blocks and all unchanged tracks; one step costs a page list for each changed track plus the blocks that really changed.
- Edits and undo swap page tables on the message thread under a two-flag exclusion: the message thread waits until the audio thread is outside the tape (at most one block); the audio thread never waits – during the few milliseconds of an edit the tape is silent.
- If recording ever runs out of fresh blocks, the samples are dropped and counted (`dropouts()`), never allocated on the audio thread.

Mixer dynamics

- **Compressor-lite** per channel (`mix.chN.comp`, 0 = off): one knob sets threshold (−4 … −30 dB), ratio (1.5 … 6) and half of the make-up gain; after the EQ, before the fader.
- **Master compressor** (`mix.master.comp` on / off, threshold, ratio, attack, release, make-up), stereo-linked, soft knee.
- **Master tape saturation** (`mix.master.tapesat`, 0 = off).
- **Master low-pass** (0.6, `mix.master.lowpass`, 500 Hz … 20 kHz, 12 dB/octave; 20 kHz = off): after the volume, before the limiter – darkens the whole mix, e.g. 4–5 kHz for lo-fi.
- **Limiter** (`mix.master.limit`) with an adjustable **ceiling** (`mix.master.ceiling`, default −0.3 dBFS as in 0.1). Since 0.8 it looks ahead 1.5 ms (83 samples at 48 kHz of latency with its true-peak interpolator, constant – also when off and on the aux outputs – and reported to the host), measures true peaks (inter-sample peaks of a 4× windowed-sinc interpolation, accurate up to near Nyquist, e.g. a saturated hi-hat) and ramps its gain down before a peak arrives instead of jumping – loud mixes stay clean, no clipping distortion. Its release is two-stage: fast after a single peak, gliding to slower while it works continuously (no pumping, and more drive never makes it quieter).
- **Compressor detector** (0.8): the master compressor, compressor-lite and the rack Compressor hold a peak for 12 ms before they let go, so fast settings no longer follow the waveform of a bass (no distortion on the sub).
- **Ducking** (0.8, sidechain): `mix.duck.src` picks the key channel (Off, Ch 1 … Ch 8), usually the drums (channel 5). Its low end (below about 150 Hz – the kick) pulls the other channels down by their own amount `mix.chN.duck` (0 … 1, up to −24 dB at a full hit); `mix.duck.release` (0.03 … 1 s) sets how fast they come back. The key follows its own recent peak, so every kick reaches the set depth whatever its level. The factory projects duck the basses most, pads less and leads / keys a touch.
- Gain reduction is readable for meters: `Mixer::gainReduction (channel)`, master = 8.
- **Delay sync** (0.5, `fxa.delay.sync`): Off, 1/16, 1/8T, 1/8, 1/4T, 1/8., 1/4, 1/4., 1/2. Any setting but Off sets the send delay's time from the tempo (host, Link or internal) and overrides `fxa.delay.time` (at most 2 s).

With every new control at its default the mixer is bit-identical to 0.1.

FX slot of the part

`p1.fx.type`: Off, Chorus, Phaser, Bitcrush, Filter Drive, Delay, Reverb, Hyper, Distortion, Flanger, Compressor, Multiband, EQ, Filter, Echo, Hall (0.7), Mid/Side (0.7) – an insert effect between part 1 and the mixer (and in front of the tape when a track records part 1). Each effect keeps its own parameters (`p1.fx.chorus.rate` …); chorus, phaser, bitcrush and filter drive have their own mix control, delay and reverb get one from the slot (`p1.fx.delay.mix`, `p1.fx.reverb.mix`). Off is an exact bypass. Since 0.7 this slot is slot 1 of the part's FX rack (below).

Chorus ensemble (0.8): the chorus has four more parameters (appended, in every slot of every rack): voices (1 … 4; 1 = the classic two-tap chorus, 2 … 4 = ensemble: staggered taps with evenly spread LFO phases, panned across the stereo field), drift (a slow random wander of every tap – less static, more like analog ensembles), spread (stereo spread of the ensemble voices) and low cut (20 Hz … 1 kHz, 12 dB/octave on the chorus' share only: below it the sound stays dry and mono, so a chorused bass keeps a solid sub). The defaults (1 voice, no drift, 20 Hz) give the 0.2 chorus. Since the 0.8 punch pass every factory chorus is an ensemble, and electric pianos, organs, pads, plucks, leads, arps, voices and the mid range of reese / neuro basses carry one.

FX rack

Every part has an **FX rack of six serial slots**, the master a rack of four:

```
part — FX 1 — FX 2 — FX 3 — FX 4 — FX 5 — FX 6 → mixer channel / tape
master: sum + returns — MFX 1 — MFX 2 — MFX 3 — MFX 4 — compressor — tape saturation — volume — low-pass — limiter
```

- Slot 1 is the FX slot above (`pN.fx.*`), slots 2 – 6 are `pN.fx2.*` … `pN.fx6.*`, the master slots `mfx1.*` … `mfx4.*`. Every slot has the same type list and its own parameter block per type (`p1.fx3.chorus.rate`), so two slots can hold the same effect with different settings.
- Per slot: **Mix** (`pN.fxK.mix`, dry / wet of the whole slot, 100 % = the effect's own output) and **Bypass** (`pN.fxK.bypass`). Mix and bypass glide over 10 ms – no clicks when switching.
- **Order:** the slots run 1, 2, … ; moving a slot rewrites the parameters of the slots in between (type, mix, bypass and every block), so the order is part of the preset, undoable and automatable like any other value.
- **Compatibility:** 0.6 presets and projects load into slot 1 unchanged and sound bit-identical (slot mix 100 %, the other slots Off). An empty or bypassed slot is an exact bypass.
- **Real time:** slot 1 has every type ready at once (as before). Slots 2 – 6 and the master slots allocate an effect's memory only when that type is first chosen (the plug-in does this within a moment on the message thread; until then the slot passes the signal dry).
- Host names tell the slots apart: "P2 FX3 Chorus Rate", "Master FX 1 Delay Mix" (the 0.2 slot of part 1 keeps its names).
- The modulation matrix can move each slot's mix and four parameters of its effect – for the 0.2 types the first four, for the rack types the four listed below as Mod 1–4 (MODULATION.md).
- On screen: the **Rack** and **Master** views of the Effects mode (USER-INTERFACE.md), and the graphical **FX editor** – the selected slot large with every parameter at once and points to drag in its picture: EQ bands on the response curve, threshold and ratio on the compressor's curve, rate and depth on a modulation wave, time and feedback as echo taps … (USER-INTERFACE.md).

Rack effect types

Besides the six 0.2 types every slot offers nine rack types. Everything that sweeps (cutoffs, times, gains, drive) glides sample by sample, so automation and modulation never zipper; the nonlinear stages run oversampled.

Type	What it does	Parameters	Mod 1–4
Hyper	unison widener: up to seven slowly drifting, detuned copies across the stereo field, then a dimension stage of cross-fed short delays; Retrigger restarts the motion with every note	rate, detune, voices, spread, dimension, retrigger, mix	rate, detune, spread, dimension
Distortion	eight modes – tube, soft clip, hard clip, diode, linear fold, sine fold, asymmetric, downsample – with bias, a pre-filter (low / band / high-pass) and a post tone control	mode, drive, bias, pre-filter, pre-filter freq, tone, output, mix	drive, bias, pre-filter freq, tone
Flanger	stereo flanger, free or tempo-synced, positive or negative feedback	rate, sync, depth, delay, feedback, stereo, mix	rate, depth, delay, feedback
Compressor	stereo-linked compressor with soft knee and makeup gain; mix gives parallel compression	threshold, ratio, attack, release, knee, makeup, mix	threshold, ratio, attack, makeup
Multiband	three bands with upward and downward compression; at depth 0 the crossovers sum flat	depth, upward, downward, time, low / mid / high gain, low / high split, output, mix	depth, low, mid, high gain
EQ	low shelf, three bells, high shelf	frequencies, gains, Q, output	band 1 freq / gain, band 2 freq / gain
Filter	low-pass / high-pass 12 or 24 dB, band-pass, notch, comb + / −, vowel formant; drive in front; the cutoff glides in octaves	type, cutoff, resonance, drive, mix	cutoff, resonance, drive, mix
Echo	stereo, ping-pong or tap delay (right tap at a ratio of the left), free or synced, filtered feedback	mode, sync, time, ratio, feedback, low cut, high cut, width, mix	feedback, time (free), high cut, width
Hall	plate or hall reverb: input diffusion and an eight-line feedback network with gentle modulation	mode, size, decay, damping, pre-delay, width, mix	size, decay, damping, pre-delay

IDs, ranges and defaults: PARAMETERS.md.

Mid/Side stereo

Mid/Side (type 16, appended to the type list after the nine 0.7 rack types) works on the stereo image. It is available in every slot of every rack – the six slots of each part and the four master slots – with its own parameter block per slot (`p1.fx.midside.width`, `p2.fx4.midside.width`, `mfx1.midside.width` …). It splits the signal into mid M = (L + R) / 2 and side S = (L − R) / 2, shapes both and puts them back together:

Parameter (`<slot>.midside.*`)	Range	What it does
Stereo Width `width`	0 … 200 %	side × width: 0 % = mono, 100 % = unchanged, 200 % = twice as wide
Mid Level `mid` / Side Level `side`	−24 … +12 dB	level of M and of S
Balance `balance`	−100 … +100 %	left / right balance
Bass Mono `bassmono` + Mono Below `monofreq`	on / off, 40 … 500 Hz	removes the side below the frequency (24 dB/oct.): tight, mono-safe lows, wide highs
Micro-Shift `haas` + Shift Time `haastime`	0 … 100 %, 1 … 30 ms	adds the mid, delayed by a few milliseconds, to the side: wider without changing the mono sum
Mid Tilt `midtilt` / Side Tilt `sidetilt`	−6 … +6 dB	a tilt EQ each for M and S around 650 Hz (+ = brighter, − = darker)

- **Mono-safe:** width, side level, bass mono, side tilt and micro-shift only touch S, and S cancels in the mono sum L + R = 2 M – folding to mono never loses or colours the mid.
- **Clean:** all gains glide (20 ms), so turning the encoders or modulating never clicks; no allocation after the slot is prepared. With every parameter at its default (width 100 %, 0 dB, balance centre, bass mono off, micro-shift 0) the effect passes the signal **bit-identically**.
- **Modulation:** the matrix targets FX k Param 1–4 are width, mid level, side level and balance.
- **In the master rack** (`mfx1` … `mfx4`, before the master compressor and limiter) it shapes the whole mix: Bass Mono keeps the low end of the mix centred, Width narrows or widens it.
- While the encoders show a Mid/Side slot the display shows the **stereo scope** of that part after its rack, or of the master output for a master slot (goniometer, correlation, M / S levels; see USER-INTERFACE.md).



Mid/Side FX slot with the stereo scope

More FX types

Five types are appended to the type list of every slot (`type` 17 – 21, after Mid/Side). Each is a complete insert with its own controls (`pN.fxK.<id>.*`, `mfxK.<id>.*`):

Type	What it does	Controls
Comb (`comb`)	modulated comb filter with damped feedback (negative = hollow, odd harmonics), stereo offset	`freq`, `feedback`, `damp`, `rate`, `depth`, `spread`, `mix`
Formant (`formant`)	three formant band-passes morphing a – e – i – o – u, shiftable, with an LFO on the vowel (free or tempo-synced, restarts with each note)	`vowel`, `shift`, `reso`, `rate`, `sync`, `depth`, `mix`
Wobble (`wobble`)	resonant low- / band-pass swept between two cutoffs by a tempo-synced LFO (sine, triangle, saws, square, S&H) that restarts with every note (`retrig`)	`rate`, `sync`, `shape`, `low`, `high`, `reso`, `mode`, `drive`, `retrig`, `mix`
Tremolo (`tremolo`)	tremolo or autopan, free or synced	`rate`, `sync`, `depth`, `mode`, `shape`
Width (`width`)	side gain 0 – 200 %, a Haas-style spread that cancels in the mono sum, mono below a frequency (4th order)	`width`, `spread`, `haas` (ms), `mono` (20 Hz = off), `output`
Resonator (`resonator`, p19)	the physically modelled body as an effect: string pair with sympathetic strings, modal plate, open / closed tube; in a part rack it follows that part's keyboard – see RESONATOR.md	`model`, `follow`, `tune`, `size`, `material`, `decay`, `position`, `inharm`, `sympathetic`, `width`, `drive`, `mix`, `pressamt`, `slideamt`

Sync divisions: Free (the rate in Hz), 1/1 … 1/32, dotted, triplets, then 2/1 and 4/1 – from the host, Link or the internal tempo. The effects are self-contained classes (`src/core/fx/InsertEffects2.h`).

Audio input (plug-in)

The plug-in has one audio input bus, **"Input"** (stereo or mono), off by default so that hosts see a plain instrument. Enable it in the host (side-chain / input routing) to record external audio with track source Input. The standalone app does not open an input device yet (that needs a microphone permission entry in the app bundle; planned).

Bounce / export

`tapeio::bounce (core, options)` (`src/plugin/TapeIO.h`, message thread) renders offline, faster than real time, while the live audio keeps running:

- a private copy of the instrument with the current parameters and the committed tape content, from the start of the tape to the end of the content plus a tail (default 2 s), speed 1 forward, no recording, sequencer stopped;
- `<name> Master.wav` – the Main bus (channels routed to Out 1–8 are not in it);
- `<name> Track 1.wav` … `Track 8.wav` – the eight tracks after level and tape character, before the mixer, all the same length so they line up in any editor;
- 24-bit stereo WAV at the current sample rate. Errors come back localised (`BounceResult`).

Tape files (.ohpetape)

Save Tape / Load Tape on the tape page keep the whole recorder in one file, independent of the project: all eight tracks, the tape length, the loop range (in, out, on) and the edit track (`src/core/tape/TapeFile.h`). Layout: the magic `OHPETPF1`, the length of a small UTF-8 JSON header (`{"format":"ohpe-tape","version":1,"sampleRate":…,"lengthSeconds":…,"loopIn":…,"loopOut":…,"loop":…,"editTrack":…}`), then the tape audio exactly as in a session (below: float32 blocks that hold audio, lossless). Loading sets the tape length and installs the content as **one undo step** – Undo brings the previous tape back; it asks first when the tape holds

material, and it is refused while recording. A tape saved at another sample rate is resampled on load (linear), so pitch and timing stay; the loop points are stored in seconds. **Export** (24-bit WAV stems, below) stays the way to take the audio into other software.

Tape audio in sessions

The plug-in state stays the 0.1 JSON document; when the tape holds audio, the state continues after a zero byte with the tape audio: a small header (`OHPETAPE`, version, sample rate, length) and every block that holds audio as float32 (lossless, little-endian). Silent blocks are not stored. A session is therefore self-contained – nothing breaks when a project is moved or archived – at a cost of about 23 MB per stereo track-minute of recorded audio (bounce long takes to WAV if a host struggles with large states). Not stored: the undo history, the clipboard, and a take that is still recording at the moment the host saves. A state without tape audio (0.1, or an empty tape) loads with an empty tape. Ohpe 0.1 cannot read states that contain tape audio.

Tape memory.

Ohpe never forgets the last half minute. While it runs it keeps a rolling memory of its own output, and one key – **CAPTURE** – turns what you just played into a new instrument. Played a phrase you liked by accident, a happy accident on the FX, a chord that rang out just right? Press CAPTURE and play it back across the keyboard.

What it keeps

- **Source** (`tapemem.src`): the main output (before the metronome click) or one part after its FX rack.
- **Length** (`tapemem.len`): how much a capture takes – 5, 10, 20 or 30 s. The memory always holds 30 s.
- The memory is a preallocated ring buffer with 16-bit frames (+6 dB headroom): about 6 MB at 48 kHz. Writing it costs the audio thread no allocation and no lock; nothing is written to disk until you capture.

CAPTURE

- Tape page: **CAPTURE** at the end of the first soft-key row, **MEMORY** (the settings) at the end of the second.
- Sampler page: **CAPTURE** (fourth key) captures into the part you are looking at.
- MIDI: `tapemem.go` (Capture) – MIDI learn from the MEMORY menu or the MIDI menu's remote controls.
- Grid-Controller: the Add Track button. It is dim while the memory holds sound and blinks while a capture is analysed.

The target part (`tapemem.to`) is the edit part unless you choose Part 1–4. The analysis – trimming the silence, finding the onsets and the pitch, retuning – runs on a worker thread; the display reports the result ("Captured into Part 2: chromatic, root A3"). The part switches to the Sampler.

Two modes (`tapemem.mode`)

- **Chromatic** – one playable sound: the strongest note of the memory (the stretch between two onsets with the most energy, at most 8 s; the whole memory when it has no onsets). Its pitch is detected (50 Hz – 2 kHz) and corrected to the nearest semitone, and the zone's root key is set to that note – so the keyboard plays it in tune. Sounds without a clear pitch (noise, drums) keep their pitch with root C4.
- **Slices** – the memory (silence trimmed, normalised) in the Sampler's slice mode: one slice per onset, from C2 upwards (the drum pads). Slice Sensitivity on the Sampler page re-slices it.

Saved with the session, exported as WAV

The last capture of each part is saved inside the session or project (state section `tapememory`, see file formats) as long as the part's Sampler still plays it – no external file is needed. MEMORY > Export last capture as WAV writes a 24-bit WAV (the root as its unity note) into the capture library: `<user application data>/groovex/Ohpe/Captures` (the environment variable `OHPE_CAPTURE_DIR` overrides it). MEMORY > Clear the memory forgets what is held.

Loading another sound after a capture

A preset chosen in the library or with the preset strip arrows loads completely into the part – engine, factory instrument or sample, all parameters – even while the part plays a capture: the capture leaves the part's Sampler, so it never covers the new sound. It is not lost: it stays in this session's memory, MEMORY > Restore last capture puts it back into its part (engine Sampler, mode and root as captured) and Export last capture

as WAV still writes it; the display says so when it happens. Browsing the library without loading brings the capture back when the library closes. A project loads its own sound into every part the same way; a project saved while a part plays a capture keeps that capture (audio inside) and brings it back. Part presets store the sound's settings, not the captured audio. RANDOM and BREED shape the captured sound (they never change the engine or the sample choice), so the capture stays. A capture belongs to its part only; other parts and presets never pick it up.

Parameters

`tapemem.len`, `tapemem.src`, `tapemem.mode`, `tapemem.to`, `tapemem.go` – see PARAMETERS.md. They are tools, not sound: presets do not store them.

Bounce to engine.



A tape loop bounced into the Grain engine

Bounce moves audio from the tape or the live input into the **Grain** or **Sampler** engine of a part with one button – resample a jam, turn a recorded phrase into a playable instrument, or freeze a few bars of a guitar into a granular pad. By default the bounce becomes an **endless loop**: its end flows into its start without a click, so a held key sustains it forever.

One button

Where	What happens
BOUNCE on the tape page (soft key)	bounces at once with the bounce settings (default: the edit track into the Grain engine of the edit part, endless loop). While the input records, the key lights and a second press finishes the recording early
BOUNCE on the Grain / Sampler page	a menu: Edit track, Tape 1 … 8 (empty tracks are greyed), Tape Mix, Input (n bars), the input length and Endless loop. Choosing a source bounces into the engine you are looking at
MIDI / automation: `bounce.go`	bounces with the bounce settings into the edit part (MIDI learn in the MIDI menu, Remote controls)

The part switches to the target engine, the bounce replaces its sample (the factory instrument choice stays and comes back when you clear the sample) and the display reports what happened, e.g. Bounced to Grain: Tape 2 · 4.0 s · endless loop. Undo is simple: **Clear** on the engine page goes back to the built-in sound, or load another sample.

Sources

Source	Material
Edit track	the tape page's edit track (Track / `tape.edit.track`)
Tape 1 … 8	one track

Source	Material
Tape Mix	all eight tracks summed (raw track audio, before the mixer)
Input	records 1, 2, 4 or 8 bars of the audio input at the current tempo (at most 60 s), then bounces

From the tape, a bounce takes the **recorded part of the track** (from the start to the last sound) – or, while the tape **loop** is on, exactly the **loop range** (Loop In … Loop Out). That is the quickest way to an endless loop in time: set a loop of one or two bars, switch the loop on, press BOUNCE. At most 60 seconds are bounced.

The live input needs the audio input enabled (standalone: Audio Settings; plug-in: the host's input routing); with silence nothing is bounced and the display says so.

Endless loop or one-shot

Mode (`bounce.loop`)	Preparation	Grain	Sampler
Endless loop (default)	the last 50 ms (at most a quarter) are cross-faded into the first 50 ms with an equal-power curve and removed, so the last frame flows into the first. Normalised to -1 dBFS	grains wrap across the seam: position, scan and spray run round and round the loop without stopping at an end	forward loop over the whole bounce (Loop = Forward, Start 0): a held note sustains the loop for as long as the key
One-shot	silence at both ends trimmed, 3 ms fades, normalised to -1 dBFS (at most +24 dB)	grains stay inside the sample (as with any loaded file)	no loop: the bounce plays once

In the Sampler, **C4** plays the bounce at its original pitch and speed; other keys transpose it. In the Grain engine the note transposes the grains the same way around C4.

Where the bounces go

Every bounce is stored as a 24-bit WAV in the bounce library:

System	Folder
macOS	`~/Library/Application Support/groovex/Ohpe/Bounces`
Windows	`%APPDATA%\groovex\Ohpe\Bounces`
Linux	`~/.config/groovex/Ohpe/Bounces`

The environment variable `OHPE_BOUNCE_DIR` overrides the folder. File names are the source and the time (`Tape 2 2026-10-17 141502.wav`). The engines keep the file's path, so the session and project presets reload the bounce like any loaded sample. An endless loop carries a sample loop chunk: Ohpe recognises it when the file is loaded again (also via **Load sample**) and other samplers can loop it too. If the folder cannot be written, the bounce still plays – for this session only, which the message tells you.

Parameters

Appended after the character macros (`p4.char.wide`), 5 IDs, not part of presets. See PARAMETERS.md.

ID	Name	Values
`bounce.src`	Bounce Source	Edit Track, Tape 1 … 8, Tape Mix, Input
`bounce.to`	Bounce Target	Grain, Sampler
`bounce.loop`	Endless Loop	on / off (default on)
`bounce.len`	Input Length	1, 2, 4, 8 bars (default 4)
`bounce.go`	Bounce	trigger

The settings are on the tape page's fifth encoder page (**Bounce**) and in the pro view of the tape mode. Bouncing is a one-off action, so it has no modulation-matrix targets; the bounced material is modulated through the engines' own targets (grain position, scan, sampler start …).

Motion recorder.



The Motion view

Every part has a **motion recorder**: turn its knobs while the transport runs and Ohpe records the movements as **looping lanes** – a filter sweep, a resonance swell, a stepped detune – and plays them back in time with the music. The loop has its **own length** (1 – 64 sixteenths), so a 5-step lane against a 16-step pattern drifts around the bar and comes back together after five bars: polymetric movement without any programming.

Recording

1. LFO mode, fourth tab **Motion** (the edit part's four lanes).
2. Press the soft key **Rec** (or turn encoder 2) – the lane rows wait, the playhead turns red once the transport runs.
3. Press **PLAY** and turn any knob of the part: the first **continuous** parameter you touch takes lane 1, the next one lane 2 … up to four lanes. While you hold or move a knob, its lane follows you (the row lights red) and you hear the knob; let go and the lane plays what you recorded.
4. Touch the same knob again later to **overdub**: only the part of the loop you touch is replaced.
5. Press **Rec** again to stop recording – **Play** stays on (it switches on with the first recording you play back).

Gaps between two screen updates are bridged smoothly. Choice, switch and whole-number parameters (engine, filter type, voices …), the motion settings themselves and the part's MIDI setup are never recorded. When many parameters change at once (loading a preset, a breeding step) nothing is recorded.

Playing

Control	What it does
Play (`pN.motion.on`)	loops the lanes with the transport (also on while recording)
Rec (`pN.motion.rec`)	records touches into the lanes

Control	What it does
Length (`pN.motion.len`)	loop length in sixteenths, 1 – 64 (eight points per sixteenth); all lanes of the part share it. A shorter loop plays the start of the recording, a longer one plays the knob where nothing was recorded – nothing is lost
Depth (`pN.motion.amt`)	0 = the knob's own value, 1 = the recorded movement; in between the movement is scaled towards the knob
Clear (soft key, or `pN.motion.clear`)	clears the part's four lanes

All five are host parameters: automate them, or map them to a MIDI controller (right-click → MIDI learn) – e.g. a footswitch on Rec.

A lane lays its value over the parameter **while the transport runs**; it sits below the step sequencer's locks (a locked step wins) and above the knob, and the modulation matrix still moves the parameter around it. The knob on screen and the host automation keep their own value. With the transport stopped, Play off or nothing recorded, the part sounds exactly as without the recorder.

Modulation sources

The four lanes are also sources of the modulation matrix: **Motion 1 … Motion 4** (0 … 1, the recorded value at the play position, 0 where nothing is recorded or the transport stops). Record one gesture and let it move other targets too – the pan, an FX mix, the morph position.

Storage

The lanes belong to the **project**: they are saved with the plug-in state and with project presets. Part presets neither carry nor reset the motion settings – load another sound into a part and its lanes keep moving the new sound's parameters. Morph corners and preset breeding ignore the motion block.

The Motion view

Four rows, one per lane: the lane number in its encoder colour, the parameter's name (or Empty), and the lane with a sixteenth grid over the loop (beats brighter), the recorded movement as a filled curve, and the playhead – green while playing, red while recording. A lane being touched has a breathing red rim. The strip below says what the recorder does right now. Soft keys: **Rec**, **Play**, **Clear**; encoders: Play, Rec, Length, Depth.

Performance pads.



The Pads view

Four **performance pads** throw effects on the whole main mix – the moves you reach for in a live set, a build-up or a break: a **stutter**, a **tape stop**, a **filter sweep** and a **reverb freeze.** Hold a pad and it plays; let go and the mix comes back. The pads lock to the beat: a pad you press waits for the next grid line of **Quantise** while the transport runs, so a stutter always catches a clean slice.

The pads

Pad	What it does	Settings
Stutter	from the grid line, catches one slice of the mix and repeats it while held	Stutter Length: 1/4, 1/8, 1/16, 1/32, 1/8T, 1/16T
Tape Stop	the mix slows to a standstill – pitch and speed fall together like a stopping tape; silent once stopped	Tape Stop Time: 1/8, 1/4, 1/2, 1 bar, 2 bars
Sweep	a resonant low-pass closes (to ~150 Hz) or a high-pass opens (to ~8 kHz) over the sweep time; on release it sweeps back in a quarter of that time	Sweep Filter (low-pass / high-pass), Sweep Time (1/8 … 4 bars), Sweep Resonance
Freeze	throws the next quarter second of the mix into a reverb and holds it endlessly while held, under the dry mix that plays on; on release the wash decays in about three seconds	Freeze Level

The pads stack: stutter → tape stop → sweep → freeze. A frozen wash under a closing low-pass, a stutter that then tape-stops – all at once is fine.

Playing them

1. **Effects** mode, sixth tab **Pads.**
2. **Press and hold** a pad on the display: it plays while you hold it. A violet blinking rim means it waits for the grid; green means it plays.

3. **Shift-click** a pad, or press its **soft key**, to latch it; again to release.
4. **Right-click** a pad for MIDI learn: map it to a button on your controller (button pressed = pad on).

Quantise (Off, 1/16, 1/8, 1/4, 1/2, 1 bar; default 1/16) sets the grid. With the transport stopped, or Quantise Off, a pad engages at once. Letting go is always immediate, with a short fade. Times follow the tempo.

Where they sit

The pads work on the **main mix** in the master chain: after the master FX rack, before the master compressor, tape saturation and limiter – the limiter keeps a resonant sweep or a dense freeze in check. Channels routed to the aux outputs are not affected. With no pad held and nothing ringing, the mix passes untouched.

Automation, MIDI, modulation

Every pad and setting is a host parameter (`pads.*`, see PARAMETERS.md): automate the throws in your DAW or map them by MIDI learn. The modulation matrix gains two sources – **Pad Sweep** (the sweep position, 0 … 1) and **Pads Held** (1 while any pad plays) – so a part can, say, open its own filter against the master sweep or duck its reverb send while a pad is held.

The four pads themselves are performance: project presets and the saved host session keep the settings but not whether a pad was held or latched – reopening a song always starts with all pads off.

PART 4

Sound design.

Modulation matrix.

../PARAMETERS.md*

Every part has a modulation matrix with **16 routings**, two extra LFOs, a third envelope, four macros and a sequencer lane. Held notes keep moving: route an LFO to the wavetable position, an envelope to the cutoff, pressure to the level – each routing is one line of the matrix.

```
source ──curve──×──(via)──× amount ──▶ target
  LFO 1-3, envelopes, velocity, key, wheel, bend, pressure, slide, random, macros, seq lane
```

A routing

Field	Values	Notes
Source (`pN.mod.sK.src`)	see the source table	Off = the routing is silent
Via (`pN.mod.sK.via`)	any source	scales the routing (e.g. LFO via mod wheel: the wheel opens the vibrato); Off = 1
Target (`pN.mod.sK.dst`)	see the target table	
Amount (`pN.mod.sK.amt`)	−100 … +100 %	bipolar; negative amounts invert
Curve (`pN.mod.sK.curve`)	Linear, Exponential, Logarithmic, S-Curve, Steps	shapes the source (mirrored for bipolar sources); Steps = 8 steps

All fields are ordinary parameters: host-automatable, MIDI-learnable, stored in presets and projects. The routing value is `curve(source) × via × amount`. Several routings to the same target add up.

Sources

#	Source	Range	Per voice	Notes
1	LFO 1	±1	yes	the part's LFO (LFO page), before its depth – usable even with LFO depth 0
2, 3	LFO 2, LFO 3	±1	yes	see below
4	Amp Env	0 … 1	yes	the amplitude envelope of the voice
5	Mod Env	0 … 1	yes	the mod envelope of the voice
6	Env 3	0 … 1	yes	delay – attack – hold – decay – sustain – release
7	Velocity	0 … 1	yes	
8	Key Track	±1	yes	C4 = 0, ±5 octaves = ±1
9	Mod Wheel	0 … 1	–	CC 1
10	Pitch Bend	±1	–	
11	Channel Pressure	0 … 1	–	channel aftertouch of the part
12	Note Pressure	0 … 1	yes	poly aftertouch or MPE pressure of the note (channel pressure acts on all notes)
13	Slide (Timbre)	±1	yes	CC 74 / MPE slide, centre = 0
14	Note Bend	±1	yes	per-note pitch bend of an MPE member channel
15	Random	±1	yes	a new value with every note
16 – 19	Macro 1 – 4	0 … 1	–	the parameters `pN.macro1` … `pN.macro4`: knobs to automate or map

#	Source	Range	Per voice	Notes
20	Seq Lane	0 … 1	–	the parameter `pN.seqlane`, made for step locks: each step sets the lane
21, 22	Morph X, Morph Y	0 … 1	–	p16: the glided position of the part's morph pad
23 – 26	Motion 1 … Motion 4	0 … 1	–	p22: lane 1 … 4 of the part's motion recorder at the play position (0 where nothing is recorded or the transport stops)
27	Pad Sweep	0 … 1	–	p25: the position of the filter sweep pad (0 = open, 1 = fully swept; the same for every part)
28	Pads Held	0 / 1	–	p25: 1 while any performance pad is engaged
29 – 31	Chord Root, Chord Change, Tension	0 … 1	–	p24: the harmony brain (the same in every part): the remembered chord's root (C = 0 … B = 1), 1 at every new chord fading in about 0.3 s, the share of held notes outside the key
32	Vibrato (5.5 Hz)	±1	yes	p33: a 5.5 Hz sine, starting with each note – the Vibrato function of the Expression page (Note Pressure via Vibrato → Pitch); e.g. as via for a vibrato that only the wheel or pressure brings in

Per voice sources have their own value in every voice: with a chord, every note gets its own envelope, velocity and random value.

p68 appends the **Voice Control** sources (Part scope, the same in every part; 0 while Voice Control is off or there is no input – see Voice Control):

#	Source	Range	Bipolar	What
34	Voice Level	0 … 1	–	the input's level above the gate (1 at about -6 dBFS)
35	Vowel X	±1	yes	back vowels (U, O) -1 … front vowels (I, E) +1 (second formant)
36	Vowel Y	±1	yes	closed vowels (I, U) -1 … open (A) +1 (first formant)
37	Voice Brightness	0 … 1	–	share of the voice's energy above about 2 kHz
38	Voice Noisiness	0 … 1	–	sibilants and fricatives (s, sh, f, t) high, sung vowels low
39	Voice Pitch	0 … 1	–	the sung note, C2 = 0 … C6 = 1 (only with pitch tracking on)
40	Voice Onset	0 … 1	–	1 at each new syllable or attack, fading in about 80 ms

LFO 2 and LFO 3 (`pN.lfo2.*`, `pN.lfo3.*`)

Parameter	Values
Rate	0.01 … 40 Hz (when Sync = Free)
Sync	Free, 4/1, 2/1, 1/1, 1/2, 1/4, 1/8, 1/16, 1/32, dotted 1/2 … 1/16, triplet 1/2 … 1/16
Shape	Sine, Triangle, Saw Up, Saw Down, Square, S&H, Smooth Random
Phase	start phase 0 … 1
Fade In	0 … 10 s: the LFO grows in after each note
Mode	Free-Running (one phase for the part; synced to the song position while the transport runs), Retrigger (restarts with each note, own phase per voice), One-Shot (one cycle, then holds: an extra envelope)
Depth	0 … 1

Envelope 3 (`pN.env3.*`)

Delay 0 … 4 s, Attack 1 ms … 10 s (linear), Hold 0 … 4 s, Decay 1 ms … 10 s, Sustain 0 … 1, Release 1 ms … 10 s (exponential).

Targets

#	Target	Per voice	Range at 100 %
1	Pitch	yes	±12 semitones
2	Level	yes	gain × (1 + value), at least 0 – ramped sample by sample
3	Pan	yes	full left … right
4	Timbre	yes	the engine's own brightness (Analog / Cluster / Grain / Sampler / Wavetable: ±3 octaves of the filter, FM: index, Pluck: brightness, Vocal: vowel)
5 – 28	Engine 1 – 24	–	parameter 1 … 24 of the active engine (its own order, the first four are the macros: e.g. Wavetable position, FM index, Grain position, Vocal formant), full range
29	Volume	–	part volume
30 – 35	LFO 1 – 3 Rate / Depth	–	
36	Mod Env Amount	–	
37 – 42	FX 1 – 6 Mix	–	mix of the FX rack slot
43 – 66	FX 1 – 6 Param 1 – 4	–	four parameters of the effect in that slot (follows the slot's type: the first four of a 0.2 type, the Mod 1–4 of a rack type)
67 – 71	Azimuth, Elevation, Distance, Motion Path Rate / Phase	–	1.0: the part's 3D stage `pN.space.*` (SPACE.md), full range (azimuth ±180°, elevation −60 … 90°, distance 0.2 … 40 m logarithmic, rate ×−4 … 4, phase 0 … 360°); e.g. Slide → Elevation, Note Pressure → Distance for MPE
72	Life	–	p15: the part's life amount `pN.life` (USER-INTERFACE.md), 0 … 100 %; e.g. Mod Wheel → Life to bring held notes alive, Note Pressure → Life for MPE
73, 74	Morph X, Morph Y	–	p16: the position of the part's morph pad (`pN.morph.x` / `.y`), full width; e.g. LFO 2 → Morph X, Note Pressure → Morph Y
75 – 80	Body Tune, Body Size, Body Material, Body Decay, Body Position, Body Mix	–	p19: the part's resonator body (`pN.reso.tune` … `.mix`), full range (tune ±24 semitones); e.g. LFO 2 → Body Position for a moving pluck point, Env 3 → Body Mix
85	Sub Level	–	p31: the part's SUB layer level `pN.sub.level`, 0 … 100 %; e.g. Velocity → Sub Level for more low end on hard notes, Mod Env → Sub Level for a sub that only thumps at the start (81 – 84: the character macros)
86 – 89	Glide Time, Amp Attack, Amp Decay, Amp Release	–	p33: the part's glide time `pN.glide` (added to the parameter or a CC 5 portamento time) and amp envelope `pN.amp.a` / `.d` / `.r`, full range; e.g. Note Pressure → Glide Time (the factory glide sounds), Velocity → Amp Attack (negative: hard notes start faster)
90 – 94	Key Gravity, Gravity Glide, Gravity Curve, Gravity Formant, Gravity Mix	–	p66: the part's Key Gravity `pN.grav.amt` / `.glide` / `.curve` / `.formant` / `.mix`; e.g. Mod Wheel → Key Gravity to play a drum loop into the held chord
95	3D Amount	–	p77: the strength of the part's 3D stage `pN.space.amount`; e.g. Mod Wheel → 3D Amount to bring the 3D effect in

The Expression page adds four more routings per part – velocity and aftertouch to two functions each – that work exactly like the 16 below.

Targets 5 and up move a **parameter** by `amount × source` of its full range (normalised, so a cutoff moves in octaves): one value for the part, computed from part-wide sources – the free-running LFOs, the wheel and macros, and the envelopes / velocity / key / random / expression of the newest note. Targets 1 – 4 are computed in every voice. Use Timbre for brightness that follows each note of a chord on its own.

Smoothness and real time

- Per-voice routings are evaluated every 32 samples; the level is ramped per sample (no zipper noise even on fast tremolo), pitch, pan and timbre glide over about 1 ms, so stepped sources (S&H, Steps) stay crisp without clicks.
- Parameter routings are applied every 32 samples and glide over about 1.5 ms; effects glide their own parameters on top. The parameter itself (host automation, UI value) is never changed – the UI shows the modulation as a ring around the value.
- Everything runs on the audio thread without allocation; a routing with amount 0, source Off or a via at 0 leaves the sound bit-identical.

Movement presets

Factory presets set routings like any other parameter (`"mod.s1.src": 2` …). Useful starting points: LFO 2 (Smooth Random, 0.2 Hz) → Engine 1 (position) 15 %; Env 3 (slow attack) → Timbre 20 %; Random → Pan 30 %; LFO 3 (synced 1/1, Free-Running) → FX 1 Mix 20 %; Note Pressure → Level or Timbre for MPE.

Expressive playing – velocity and aftertouch.



The Expression page: aftertouch to vibrato and cutoff

Every part has a small **Expression** page: how hard you play (**velocity**) and how hard you press a held key (**aftertouch**) each move up to **two musical functions** – with an amount and a curve. It is the quick way to make a sound respond to your hands, without building routings in the modulation matrix.

Where it is

- **Envelope** mode, the last three pages: **Velocity** (target 1, amount 1, target 2, amount 2), **Aftertouch** (the same) and **Expression Curves** (velocity curve 1 / 2, aftertouch curve 1 / 2). The display shows the response of each slot as a line (left = soft / no pressure, right = hard / full pressure) and a dot that follows what you play.
- **Pro view** (Synth): the group **Expression** with all twelve controls.
- Everything is a host parameter and MIDI-mappable: `pN.expr.vel.dst1` … `pN.expr.at.curve2` (PARAMETERS.md).

Sources

Source	What it reads
Velocity	the note's velocity, per voice (every note of a chord its own)
Aftertouch	channel pressure, poly aftertouch and MPE pressure alike – per voice where the controller sends it per note. Held channel pressure also applies to new notes.

Functions

Function	What it moves
Cutoff	the engine's filter / tone (Analog: cutoff; Wavetable, Grain, Sampler, Cluster: tone; Pluck: brightness; Vocal: formant; FM: the BRIGHT character)
Resonance	the filter resonance (Analog, Wavetable, Cluster; Pluck: body)
Level	the level of each voice
Timbre	the engine's own brightness per voice (see Timbre)
Amp Attack / Amp Decay / Amp Release	the amplitude envelope (e.g. softer notes swell in slower)
Mod Env Amount	how strongly the mod envelope works
LFO Rate / LFO Depth	the part's LFO
Vibrato	a 5.5 Hz vibrato on the pitch – ±1 semitone at 100 % (independent of the LFO settings)
Glide Time	the portamento time (on top of the Glide parameter or a CC 5 time)
FM Amount	FM index (FM), FM amount (Wavetable)
Position	wavetable / grain position, vowel (Vocal), sample start, pick position
Grain Spray	grain spray (Grain), jitter (Vocal), drift (Cluster), else the ALIVE character
Morph X / Morph Y	the morph pad
Drive	Analog drive, FM feedback, Pluck excitation, Vocal tension, else the HARD character
FX Mix	the mix of the part's first FX rack slot
Sub Level	the sub oscillator (the part's SUB layer once it exists, else the Analog / Wavetable sub)
Pan	the pan of each voice
Pitch	the pitch of each voice (±12 semitones at 100 %)
Life	the part's life amount

A function the active engine does not have stays silent there (e.g. Resonance on FM).

Amount and curve

- **Amount** −100 … +100 %: how far the function moves at full velocity / pressure, in its full range (a cutoff moves in octaves). Negative amounts move it the other way (harder = darker, more pressure = shorter attack …). **0 % (the default) does nothing** – the sound is exactly as without the page.
- **Curve:** Linear, Exponential (only strong playing really moves it), Logarithmic (already a light touch moves it), S-Curve, Steps.

Defaults and factory sounds

The slots start on musical choices so that turning up an amount is all it takes: velocity 1 → Cutoff, velocity 2 → Amp Attack, aftertouch 1 → Vibrato, aftertouch 2 → Cutoff – all at 0 %.

Factory sounds with glide (portamento) use **aftertouch → Glide Time** with a positive amount: **more pressure = a longer, slower glide.** Leaning into a key while you play legato turns a tight slide into a slow, vocal swoop, and a light touch keeps the line tight. (A longer glide under pressure felt more natural than the other way round: the pressure is usually already there when the next note starts, so the glide stretches exactly when you are pushing into the phrase.)

Expression and Glide Pressure

There are two ways to let aftertouch shape the glide, and they add up if both are used:

- **Glide Pressure** (`pN.glide.press`, pro view **Glide** group, see MIDI) scales the glide time: up to 4× longer at full pressure (positive) or down to no glide (negative). It needs no slot and keeps the ratio to the Glide parameter or CC 5 time.
- **Expression → Glide Time** (an aftertouch slot here) adds time through the modulation matrix, with its own amount and curve, and can sit alongside a second aftertouch function.

When both are set, the Expression offset is added first and Glide Pressure then scales the result, so the glide is stretched twice. Each gliding factory sound uses exactly one of them (currently the Expression slot; a test checks this); for your own sounds pick one, or use both on purpose.

How it works

Each of the four slots is one more routing of the part's modulation matrix – same sources, curves, scaling and smoothing as the 16 routings of the Mod view (source Velocity or Note Pressure, the Vibrato source for Vibrato). The matrix gained the source **Vibrato (5.5 Hz)** (32) and the targets **Glide Time**, **Amp Attack**, **Amp Decay** and **Amp Release** (86 – 89), usable in the Mod view as well.

SUB layer.



SUB layer in the pro view

Every part has a **SUB layer:** one sub oscillator under every voice of the part's engine, one or two octaves below the played note, to dial in extra low end without changing the sound above it. It sits on the Synth mode's own encoder page **Sub** (after Glide and Pitch Envelope) and in the pro view as the card **Sub.**

Control	Parameter	What it does
Level	`pN.sub.level`	0 = off (the part sounds exactly as without the SUB layer) … 100 %
Octave	`pN.sub.oct`	−1 Oct or −2 Oct below the voice
Shape	`pN.sub.shape`	**Sine** (pure, deepest), **Round** (a rounded triangle: a little warmer, still clean), **Square** (a soft square: hollow, more push)
Direct	`pN.sub.direct`	Off: the sub goes through the engine's filter (Analog, Wavetable). On: it is added after the filter and stays full under a closed filter

How it behaves

- **Locked to the voice.** The sub follows exactly the pitch the voice plays – the note, glide, pitch bend, the pitch envelope, matrix pitch, MPE bends and the slow drift of Life. Its phase is derived from the voice's own cycles and starts with every note, so it never beats against the voice.
- **Clean.** All three shapes are band-limited: Sine and Round have no edges; Square is a soft square whose edges soften as the pitch rises, so even the top keys alias no audible products.
- **Mono.** The sub is the same signal on both channels – never panned, no stereo width, safe for club systems and vinyl. (The part's Bass Mono stays available for everything else.)
- **Level-safe.** It rides on the voice's amp envelope and velocity; changing the level ramps sample by sample (no zipper steps). At 100 % a sine sub is about as loud as one voice of the engine.
- **Main engine only.** The part's layer engine has no sub of its own (one SUB layer per part).

Engines

Engine	SUB layer	Through the filter (Direct off)
Analog	yes	yes – into the ladder / state-variable filter (drive included)
Wavetable	yes	yes – into the tone filter
FM	yes	always direct (no voice filter)
Pluck	yes	always direct (after the string)
Cluster	yes	always direct (after the stack)
Grain	yes	always direct, at the played key
Vocal	yes	always direct (after the voice)
Sampler	yes	always direct, at the played key (tuned for samples at their root key)
Drum kit	no – pads have no keyboard pitch	–

Analog and Wavetable also have their own engine sub (a sine one octave down, a macro of the engine); the SUB layer is independent of it and adds the octave choice, the shapes and the direct path.

Picture (pro view)

The card shows the voice as a thin grey wave and the sub under it in yellow – its shape, its octave (one sub cycle for every two or four voice cycles) and its level; the small fader on the right is the level, a violet strip shows what the modulation matrix moves. **Drag** up / down = level, a quick **swipe** = shape, a **click** on the octave caption = the other octave, a click on INTO FILTER / DIRECT = the route, **wheel** = level, double-click = off.

Modulation and MIDI

The modulation matrix has the target **Sub Level** (85, key `sublevel`): an envelope, velocity or a macro can bring the sub in. Every SUB parameter is MIDI-learnable (right-click) and part of part presets and projects.

Key Gravity.

Every part has a **Key Gravity** stage right after its engines (any of the eight): it pulls every partial of the sound toward the current key and the chord that is held. A slightly detuned pad locks into the key, a melody leans onto the chord tones, and at high amounts noise and drum loops turn into pitched texture that plays the chord. It sits on the Effects mode's last two encoder pages **Key Gravity** and in the pro view as the card **Key Gravity**.

Quickest way in: the small ring on the display's header, right next to the key, shows the edit part's Gravity (empty at 0 %). Click it to open the Key Gravity page directly; drag or turn the mouse wheel on it to set the amount; right-click for MIDI learn. See User interface.

Control	Parameter	What it does
Gravity	`pN.grav.amt`	0 = off (the part sounds exactly as without the stage: bit-identical, no latency, no CPU) … 100 %
Target	`pN.grav.target`	**Scale** (the key's notes), **Chord** (only the held chord's notes) or **Chord+Scale** (the chord tones pull harder, the scale fills between them)
Glide	`pN.grav.glide`	5 ms … 2 s: how fast the partials flow to the new notes when the chord or key changes
Curve	`pN.grav.curve`	−100 % (gentle: partials only lean toward the notes) … +100 % (magnetic: even a little gravity snaps them on)
Formant	`pN.grav.formant`	formant keep: the tone colour (spectral envelope) stays where it was while the partials move
Mix	`pN.grav.mix`	blend of the pulled sound with the original (in time with each other)
Quality	`pN.grav.quality`	**Low Latency** (≈ 11 ms), **Normal** (≈ 21 ms), **High** (≈ 43 ms, resolves low notes best)

Where the notes come from

- **Key / scale:** the key and scale of the instrument (KEY on the plate), or the host's key when the host sends one.
- **Chord:** the notes held on all four parts right now, named like the chord display (a C with E and G is C major, stray notes do not count). With nothing held, the last chord the harmony memory heard; before any chord, the key's tonic triad.
- Every octave counts: a partial goes to the nearest target pitch above or below it.

How it works

A short-time spectrum (Hann window, four times overlap) finds the peaks of the sound; the true frequency of each comes from its phase from frame to frame. Each peak carries the bins around it (phase-locked, so the partial stays one partial) and moves to its target pitch – partly, by Gravity and Curve. Partial tracks glide to a new target when the chord changes. Attacks pass with their own phases, so drums keep their hit. Above 50 % the energy between the target pitches fades and what lands on a target note takes that note's running phase – noise becomes a chord.

- **Latency:** one analysis frame (above). The host is told, and every part – and the drum kit – waits for the slowest running stage, so all stay in time. At Gravity 0 for one second the stage stops and the latency goes away again.
- **Real time:** no memory is taken while playing; any block size works (one sample too).
- **CPU:** about 1.5 % of one core per part (48 kHz, any quality); four parts about 6 % of one core.
- **Sample rates:** 44.1 – 192 kHz (the frame grows with the rate, so the pitch resolution stays the same).

Modulation and MIDI

All seven controls are host parameters and MIDI-learnable. Matrix targets (appended): 90 **Key Gravity**, 91 **Gravity Glide**, 92 **Gravity Curve**, 93 **Gravity Formant**, 94 **Gravity Mix** – e.g. the mod wheel → Key Gravity to play a drum loop into the chord. See Modulation.

Factory sounds

Gravity Noise Choir (pure noise pulled into the held chord), Gravity Hat Pad (a hat cluster melted into a pad in key), Gravity Bell Lock and Gravity Drum Wash show the stage; in the library you find them via the "Gravity …" name prefix (search "Gravity" – tags are categories only). The mod wheel lowers the gravity on all four.

Harmony brain.



The Harmony view with the grid keyboard

Ohpe listens to what you play and keeps a **scale and chord memory shared by all four parts** – the harmony brain. Play chords on one part (keys, MIDI, a sequencer, the arpeggiator) and the other parts can **follow** them: a bass that only hits chord tones, a lead that cannot leave the key, an arpeggio that moves with your changes – without setting a key by hand. An **isomorphic grid keyboard** with MPE plays it from the screen.

What the brain hears and remembers

- **Who is heard:** every note that reaches a part whose Follow is **Off** – the leading parts. Parts that follow are not heard (no feedback). The drum kit is not heard.
- **Chord:** as soon as two or more different notes are held, the brain names the chord – major, minor, diminished, augmented, sus2, sus4, maj7, m7, 7 or a bare fifth; the lowest note decides between two readings (A–C–E over A is Am). A chord counts once it stood for 40 ms (the notes of a played chord arrive one by one). It **stays in memory** after you let go, until the next chord. Letting go of one note of a triad keeps the chord.
- **Key:** a pitch-class memory gathers every note (its start and how long it is held) and fades with a half-life of **Memory** bars (1 – 32, default 8). The brain picks the major or minor key that fits best; another key must win clearly before it switches. Until anything was heard, the key is the global **Key / Scale** (`key.root`, `key.scale`).
- **History:** the last eight chords, newest first.

Follow

Each part has **Follow** (`harmony.pN.follow`):

Follow	The part's notes …
Off	… play as before (and are heard by the brain) – neutral, the default
Scale	… snap to the nearest note of the remembered key
Chord	… snap to the nearest tone of the remembered chord (before any chord: the key's tonic triad)
Root	… snap to the chord's root (any octave)

Snapping uses the nearest note (a tie goes down) – live keys, host MIDI and all sequencers alike. The part's one-finger **Chord** still works and builds its chord in the remembered key. Notes of a part with **MPE** on keep their pitch (per-note expression); play the grid with In Key on there.

Controls

Control	ID	What it does
Follow (per part)	`harmony.pN.follow`	Off / Scale / Chord / Root
Memory	`harmony.memory`	half-life of the key memory in bars (1 – 32)
Hold	`harmony.hold`	freezes the memory: new notes change neither chord nor key
Take Key	`harmony.take`	trigger: writes the remembered key into the global Key / Scale
Forget	`harmony.clear`	trigger: forgets key, chord, history
Grid Layout	`harmony.kb.layout`	Fourths / Thirds / Whole Tones / Sequential
Grid In Key	`harmony.kb.inkey`	off = Chromatic (every note; notes outside the key dark but playable), on = In Key (only key notes)
Grid Fixed C	`harmony.kb.fixed`	off (default) = the bottom-left cell is the key's root, on = it is C

All are host parameters: automatable and MIDI-learnable (right-click). The memory (key, chord, history, pitch-class weights) is stored in projects and the plug-in state; part presets neither carry nor change Follow.

The Harmony view

LFO mode, fifth tab **Harmony**: the remembered **key** and **chord** (the chord glows when it changes), twelve bars for the pitch-class memory starting at the key's root (key notes light, other notes dim, **chord tones violet**, **held notes green**, the root marked **yellow**) and the **history** chips. The strip says what the brain does and whether the edit part follows. Soft keys: **Take Key**, **Hold**, **Forget**. Encoders: page Harmony = Follow (edit part), Memory, Hold, Grid Layout; page Grid = Grid Layout, In Key / Chromatic, Fixed C.



Encoder page Grid: layout, In Key / Chromatic, Fixed C

Chord names and the key

While notes sound, the view's **Chord** shows the chord they make – every part together, with the notes following parts and one-finger chords add – with the full naming of the display's voice strip: sixths, sevenths (also without the fifth), m7b5, dim7, sus, add9, ninths, and **slash chords** for inversions (D/F#) or a foreign bass (C/D); the legend adds the position (1st inversion). When nothing sounds it shows the remembered chord as before. The status line adds Playing F#m7. Note and chord names are spelled in the global key: flats in flat keys (Bb, Eb), sharps in sharp keys (F#, C#), German H / B. **Scale Lock** (on the display header, see Key, chord and Scale Lock) uses the same global key; a following part (Follow Scale / Chord / Root) follows the brain as before and ignores Scale Lock's snapping (White Keys still maps its keys first).

The grid keyboard



The four layouts of the grid keyboard, lowest 12 columns

The **GRID** key (between OCTAVE + and MIDI, yellow bar while on) swaps the piano keys for a grid of **4 rows × 24 cells**. It is an **isomorphic** keyboard. This section explains it from scratch – no music theory needed beyond counting semitones.

The idea: one shape, every key

On a piano, C major and D major look different: different white and black keys, different hand shapes. On an isomorphic grid every cell is a fixed number of semitones away from its neighbours – one step **right** is always the same interval, one step **up** is always the same interval. So an **interval, a chord or a scale is a shape**, and the shape does not change when you move it. Learn the shape of a major chord once and you can play every major chord by putting the same shape somewhere else. Moving a shape is **transposing**.

Two words used below: a **semitone** is the smallest step (C → C#, the next piano key); an **interval** is the distance between two notes in semitones (a major third = 4, a fifth = 7, an octave = 12).

Switching it on

- **GRID** shows the grid in place of the piano keys; press it again for the piano. Ohpe remembers which one was shown.
- It plays the **edit part** (the part selected with the part keys) and follows when you choose another part.
- **OCTAVE – / +** shift it like the piano: the bottom-left cell is the **root of the key** in the piano's lowest octave (C3 by default; C0 … C7) – or the C of that octave with **Fixed C** on. The readout shows `GRID` and the grid's lowest and highest note.
- Three parameters set it up, all global (the same for all parts), automatable and MIDI-mappable: **Grid Layout** (`harmony.kb.layout`), **Grid In Key** (`harmony.kb.inkey`) and **Grid Fixed C** (`harmony.kb.fixed`). On screen: LFO mode, view **Harmony**, encoder page Grid (the layout next to the two switches).
- Hiding the grid, switching part or octave releases every held cell (no hanging notes).

Reading the cells

Cell	Means
light	a note of the **key** the harmony brain remembers (before it has heard anything: the global Key / Scale)
graphite	a note outside the key
bold name with octave (e.g. **C3**)	the key's **root**
violet bar	a note of the **chord** the brain remembers
sunk, red bar	a cell you are holding

Note names follow the language: C# / B in English, Cis / H in German. **Where** you strike a cell sets the velocity: the lower edge plays softly (velocity 56), the upper edge at full strength (127).

The four layouts

The steps, exactly as the code computes them (row 1 = bottom row, column 1 = left). **Chromatic** counts semitones, **In Key** counts steps of the key's scale (see below):

Layout	Chromatic: right / up	In Key: right / up	Range (Chromatic, default octave, key C)
Fourths (default)	+1 / +5 semitones (a fourth)	+1 / +3 scale steps	C3 – D6
Thirds	+1 / +4 semitones (a major third)	+1 / +2 scale steps (a third)	C3 – B5
Whole Tones	+2 / +5 semitones	+1 / +3 scale steps (every scale step is about a whole tone, so In Key plays like Fourths)	C3 – C#8
Sequential	+1 / +8 semitones: continuous rows of eight	+1 / +8 scale steps	C3 – B6

Sequential – the eight pads of a row continue in the row above, like reading a page: the last cell of a row + 1 is the first cell of the next row. Easy to find any note, but chord shapes change from row to row.

In Key or Chromatic (every layout)

Grid In Key (`harmony.kb.inkey`) works with every layout:

- **Chromatic** (off, the default): every note. Notes outside the key are **dark but playable**; with **Scale Lock** on they are also silent on the pads (on the screen they snap like every note).
- **In Key** (on): only the notes of the key, so you cannot play a "wrong" note. One cell right is the next note of the scale, one row up is the same number of **scale steps** everywhere (Fourths 3, Thirds 2, Sequential 8). In a seven-note key three steps are a fourth except from the fourth degree (a tritone, in C major F → B).
- A chromatic key scale plays In Key like Chromatic. When the brain moves to a new key the grid relabels at once.
- The old layout In Key of earlier versions is now **Fourths + In Key**; projects and presets that used it open that way.

Fixed C or the key's root

Grid Fixed C (`harmony.kb.fixed`):

- **Off (the default):** the bottom-left cell is the **root of the key set on the display header** (Key, or the host's key), and every row is relative to it. It stays put while you play, even when the harmony brain's remembered key moves (that key still decides which notes are In Key and how the cells are lit). A chord shape therefore plays the same chord degree in **every key** – learn it once in C, play it in F# without moving your hand.
- **On:** the bottom-left cell is **C** (In Key: the first key note at or above C) – the behaviour of earlier versions.
- **OCTAVE − / +** shift the grid by an octave in both cases.

Fourths + In Key in C major (the bottom-left cell is C with Fixed C on or off):

```
4 | E4    F4    G4    A4    B4    C5    D5    E5
3 | B3    C4    D4    E4    F4    G4    A4    B4
2 | F3    G3    A3    B3    C4    D4    E4    F4
1 | C3    D3    E3    F3    G3    A3    B3    C4
  +-----------------------------------------------
    1     2     3     4     5     6     7     8
```

Thirds + In Key, and Sequential (Chromatic), C major:

```
4 | B3    C4    D4    E4    F4    G4    A4    B4
3 | G3    A3    B3    C4    D4    E4    F4    G4
2 | E3    F3    G3    A3    B3    C4    D4    E4
1 | C3    D3    E3    F3    G3    A3    B3    C4
  +-----------------------------------------------
    1     2     3     4     5     6     7     8
```

```
4 | C5    C#5   D5    D#5   E5    F5    F#5   G5
3 | E4    F4    F#4   G4    G#4   A4    A#4   B4
2 | G#3   A3    A#3   B3    C4    C#4   D4    D#4
1 | C3    C#3   D3    D#3   E3    F3    F#3   G3
  +-----------------------------------------------
    1     2     3     4     5     6     7     8
```

The Chromatic diagrams below show the layouts with the key C (the root is C, so Fixed C makes no difference there).

Fourths – like the lower strings of a bass or guitar: each row starts a fourth above the one below.

```
4 |  D#4   E4    F4    F#4   G4    G#4   A4    A#4
3 |  A#3   B3    C4    C#4   D4    D#4   E4    F4
2 |  F3    F#3   G3    G#3   A3    A#3   B3    C4
1 |  C3    C#3   D3    D#3   E3    F3    F#3   G3
  +-----------------------------------------------
     1     2     3     4     5     6     7     8
```

Thirds – rows a major third apart: compact chords, small hand spans.

```
4 |  C4    C#4   D4    D#4   E4    F4    F#4   G4
3 |  G#3   A3    A#3   B3    C4    C#4   D4    D#4
2 |  E3    F3    F#3   G3    G#3   A3    A#3   B3
1 |  C3    C#3   D3    D#3   E3    F3    F#3   G3
  +-----------------------------------------------
     1     2     3     4     5     6     7     8
```

Whole Tones – two semitones per cell; each row alone is a whole-tone scale, the next row (a fourth up) holds the other six notes. Wide range in little space.

```
4 |  D#4    F4     G4     A4     B4     C#5    D#5    F5
3 |  A#3    C4     D4     E4     F#4    G#4    A#4    C5
2 |  F3     G3     A3     B3     C#4    D#4    F4     G4
1 |  C3     D3     E3     F#3    G#3    A#3    C4     D4
  +-------------------------------------------------------
      1      2      3      4      5      6      7      8
```

Worked examples

The diagrams show the bottom-left 4 × 8 cells at the default octave (key C major); cells in `[ ]` are the ones to press. Every shape works anywhere on the grid – if it runs off an edge, use the same shape somewhere else or shift the octave.

Major triad (C – E – G). Fourths: root, four cells right (the third), and between them one row up (the fifth).

```
4 |  D#4    E4     F4     F#4    G4     G#4    A4     A#4
3 |  A#3    B3     C4     C#4    D4     D#4    E4     F4
2 |  F3     F#3   [G3]    G#3    A3     A#3    B3     C4
1 | [C3]    C#3    D3     D#3   [E3]    F3     F#3    G3
  +-------------------------------------------------------
      1      2      3      4      5      6      7      8
```

The same chord has a second shape: third one row up and one cell left, fifth one row up and two right (here F – A – C):

```
4 |  D#4    E4     F4     F#4    G4     G#4    A4     A#4
3 |  A#3    B3     C4     C#4    D4     D#4    E4     F4
2 |  F3     F#3    G3     G#3   [A3]    A#3    B3    [C4]
1 |  C3     C#3    D3     D#3    E3    [F3]    F#3    G3
  +-------------------------------------------------------
      1      2      3      4      5      6      7      8
```

Minor triad (C – Eb – G). Only the third moves: one cell left of the major third (three cells right of the root).

```
4 |  D#4    E4     F4     F#4    G4     G#4    A4     A#4
3 |  A#3    B3     C4     C#4    D4     D#4    E4     F4
2 |  F3     F#3   [G3]    G#3    A3     A#3    B3     C4
1 | [C3]    C#3    D3    [D#3]   E3     F3     F#3    G3
  +-------------------------------------------------------
      1      2      3      4      5      6      7      8
```

Power chord / fifth (C – G – C). Root, one row up and two right (the fifth), the cell above that (the octave).

```
4 |  D#4    E4     F4     F#4    G4     G#4    A4     A#4
3 |  A#3    B3    [C4]    C#4    D4     D#4    E4     F4
2 |  F3     F#3   [G3]    G#3    A3     A#3    B3     C4
1 | [C3]    C#3    D3     D#3    E3     F3     F#3    G3
  +-------------------------------------------------------
      1      2      3      4      5      6      7      8
```

Octave (C3 → C4). In Fourths always **two rows up, two cells right:**

```
4 |  D#4    E4     F4     F#4    G4     G#4    A4     A#4
3 |  A#3    B3    [C4]    C#4    D4     D#4    E4     F4
2 |  F3     F#3    G3     G#3    A3     A#3    B3     C4
1 | [C3]    C#3    D3     D#3    E3     F3     F#3    G3
  +-------------------------------------------------------
      1      2      3      4      5      6      7      8
```

Major scale (C D E F G A B C). Three notes per row, a pattern that repeats:

```
4 |  D#4    E4     F4     F#4    G4     G#4    A4     A#4
3 |  A#3   [B3]   [C4]   C#4    D4     D#4    E4     F4
2 | [F3]    F#3   [G3]   G#3   [A3]   A#3    B3     C4
1 | [C3]    C#3   [D3]   D#3   [E3]   F3     F#3    G3
  +-------------------------------------------------------
      1      2      3      4      5      6      7      8
```

Transposing. Move the whole shape: two cells right = two semitones up. C major becomes D major (D – F# – A) with the same fingers:

```
4 |  D#4    E4     F4     F#4    G4     G#4    A4     A#4
3 |  A#3    B3     C4     C#4    D4     D#4    E4     F4
2 |  F3     F#3    G3     G#3   [A3]   A#3    B3     C4
1 |  C3     C#3   [D3]   D#3    E3     F3    [F#3]   G3
  +-------------------------------------------------------
      1      2      3      4      5      6      7      8
```

One row up transposes by the row's interval (a fourth in Fourths: C major → F major).

The same chords in the other layouts (offsets as cells right, rows up from the root; negative = left):

Shape	Fourths	Thirds	Whole Tones	Fourths + In Key
major triad	3rd (4, 0) or (−1, 1), 5th (2, 1)	3rd (0, 1), 5th (3, 1)	3rd (2, 0), 5th (1, 1)	3rd (2, 0), 5th (1, 1) – major or minor as the key decides
minor triad	3rd (3, 0) or (−2, 1), 5th (2, 1)	3rd (3, 0) or (−1, 1), 5th (3, 1)	3rd (−1, 1), 5th (1, 1)	same shape as above
fifth	(2, 1)	(3, 1)	(1, 1)	(1, 1) – diminished from the 7th degree
octave	(2, 2)	(0, 3) or (4, 2)	(1, 2)	(1, 2)

Thirds, major triad (C – E – G) and minor triad (C – Eb – G):

```
4 |  C4     C#4    D4     D#4    E4     F4     F#4    G4
3 |  G#3    A3     A#3    B3     C4     C#4    D4     D#4
2 | [E3]    F3     F#3   [G3]    G#3    A3     A#3    B3
1 | [C3]    C#3    D3     D#3    E3     F3     F#3    G3
  +-------------------------------------------------------
      1      2      3      4      5      6      7      8
```

```
4 |  C4     C#4    D4     D#4    E4     F4     F#4    G4
3 |  G#3    A3     A#3    B3     C4     C#4    D4     D#4
2 |  E3     F3     F#3   [G3]    G#3    A3     A#3    B3
1 | [C3]    C#3    D3    [D#3]   E3     F3     F#3    G3
  +-------------------------------------------------------
      1      2      3      4      5      6      7      8
```

Whole Tones, major triad (C – E – G) and minor triad (D – F – A):

```
4 |  D#4    F4     G4     A4     B4     C#5    D#5    F5
3 |  A#3    C4     D4     E4     F#4    G#4    A#4    C5
2 |  F3    [G3]    A3     B3     C#4    D#4    F4     G4
1 | [C3]    D3    [E3]    F#3    G#3    A#3    C4     D4
  +-------------------------------------------------------
      1      2      3      4      5      6      7      8
```

```
4 |  D#4    F4     G4     A4     B4     C#5    D#5    F5
3 |  A#3    C4     D4     E4     F#4    G#4    A#4    C5
2 | [F3]    G3    [A3]    B3     C#4    D#4    F4     G4
1 |  C3    [D3]    E3     F#3    G#3    A#3    C4     D4
  +-------------------------------------------------------
     1      2      3      4      5      6      7      8
```

Fourths + In Key: the scale is simply a row from left to right, and **one** triad shape plays every chord of the key: from C it is C major, moved one cell right it is D minor (D – F – A) – the key decides major or minor.

```
4 |  E4     F4     G4     A4     B4     C5     D5     E5
3 |  B3     C4     D4     E4     F4     G4     A4     B4
2 |  F3    [G3]    A3     B3     C4     D4     E4     F4
1 | [C3]    D3    [E3]    F3     G3     A3     B3     C4
  +-------------------------------------------------------
     1      2      3      4      5      6      7      8
```

```
4 |  E4     F4     G4     A4     B4     C5     D5     E5
3 |  B3     C4     D4     E4     F4     G4     A4     B4
2 |  F3     G3    [A3]    B3     C4     D4     E4     F4
1 |  C3    [D3]    E3    [F3]    G3     A3     B3     C4
  +-------------------------------------------------------
     1      2      3      4      5      6      7      8
```

Chord guide: playing chords on the grid

This guide uses the easiest way to play: **Fourths + In Key, Fixed C off** (switch In Key on; Fourths and Fixed C off are the defaults). The bottom-left cell is always the key's root, so every shape below plays the same chord degree in every key. Offsets are cells right, rows up from the chord's root cell; the diagrams show C major (and A minor) at the default octave, cells in `[ ]` are pressed. The same shapes work on the screen grid and on the Grid-Controller pads (its four lower rows are the screen's rows).

Triad – root, 2 right, 1 right + 1 up. The key decides major or minor: on the root of a major key it is a major triad, on the root of a minor key a minor triad.

```
4 | E4     F4     G4     A4     B4     C5     D5     E5
3 | B3     C4     D4     E4     F4     G4     A4     B4
2 | F3    [G3]    A3     B3     C4     D4     E4     F4
1 |[C3]    D3    [E3]    F3     G3     A3     B3     C4
  +-------------------------------------------------------
    1      2      3      4      5      6      7      8
```

```
4 | C5     D5     E5     F5     G5     A5     B5     C6
3 | G4     A4     B4     C5     D5     E5     F5     G5
2 | D4    [E4]    F4     G4     A4     B4     C5     D5
1 |[A3]    B3    [C4]    D4     E4     F4     G4     A4
  +-------------------------------------------------------
    1      2      3      4      5      6      7      8
```

Every chord of the key with one shape. Move the shape one cell right and it plays the next chord of the key – in C major: C, Dm, Em, F, G, Am, Bdim. Here Dm:

```
4 | E4     F4     G4     A4     B4     C5     D5     E5
3 | B3     C4     D4     E4     F4     G4     A4     B4
2 | F3     G3    [A3]    B3     C4     D4     E4     F4
1 | C3    [D3]    E3    [F3]    G3     A3     B3     C4
  +-------------------------------------------------------
    1      2      3      4      5      6      7      8
```

Seventh chord – the triad + 3 right + 1 up. Major 7 on the 1st and 4th degree, dominant 7 on the 5th, minor 7 on the 2nd, 3rd and 6th, m7b5 on the 7th – again the key decides.

```
4 | E4     F4     G4     A4     B4     C5     D5     E5
3 | B3     C4     D4     E4     F4     G4     A4     B4
2 | F3    [G3]    A3    [B3]    C4     D4     E4     F4
1 |[C3]    D3    [E3]    F3     G3     A3     B3     C4
  +-------------------------------------------------------
    1      2      3      4      5      6      7      8
```

Inversions – another note in the bass, the same chord. First inversion (third in the bass): 2 right, 1 right + 1 up, 4 right + 1 up. Second inversion (fifth in the bass): 1 right + 1 up, 4 right + 1 up, 3 right + 2 up.

```
4 | E4     F4     G4     A4     B4     C5     D5     E5
3 | B3     C4     D4     E4     F4     G4     A4     B4
2 | F3    [G3]    A3     B3    [C4]    D4     E4     F4
1 | C3     D3    [E3]    F3     G3     A3     B3     C4
  +-------------------------------------------------------
    1      2      3      4      5      6      7      8
```

```
4 | E4     F4     G4     A4     B4     C5     D5     E5
3 | B3     C4     D4    [E4]    F4     G4     A4     B4
2 | F3    [G3]    A3     B3    [C4]    D4     E4     F4
1 | C3     D3     E3     F3     G3     A3     B3     C4
  +-------------------------------------------------------
    1      2      3      4      5      6      7      8
```

Power chord – root, 1 right + 1 up (fifth), the cell above that (octave). In Key the fifth from the 7th degree is diminished (B – F in C major); play it Chromatic for a pure fifth everywhere.

```
4 | E4     F4     G4     A4     B4     C5     D5     E5
3 | B3    [C4]    D4     E4     F4     G4     A4     B4
2 | F3    [G3]    A3     B3     C4     D4     E4     F4
1 |[C3]    D3     E3     F3     G3     A3     B3     C4
  +-------------------------------------------------------
    1      2      3      4      5      6      7      8
```

Octave – 1 right + 2 up (also 4 right + 1 up, or 7 right).

```
4 | E4     F4     G4     A4     B4     C5     D5     E5
3 | B3    [C4]    D4     E4     F4     G4     A4     B4
2 | F3     G3     A3     B3     C4     D4     E4     F4
1 |[C3]    D3     E3     F3     G3     A3     B3     C4
  +-------------------------------------------------------
    1      2      3      4      5      6      7      8
```

Chromatic shapes (Fourths, In Key off). Every chord type has its own fixed shape; the key does not change it:

Chord	Offsets from the root
major	(4, 0), (2, 1)
minor	(3, 0), (2, 1)
dominant 7	(4, 0), (2, 1), (0, 2)
major 7	(4, 0), (2, 1), (1, 2)

```
4 | D#4    E4     F4     F#4    G4     G#4    A4     A#4
3 | A#3    B3     C4     C#4    D4     D#4    E4     F4
2 | F3     F#3   [G3]    G#3    A3     A#3    B3     C4
1 |[C3]    C#3    D3     D#3   [E3]    F3     F#3    G3
  +-------------------------------------------------------
    1      2      3      4      5      6      7      8
```

```
4 | D#4    E4     F4     F#4    G4     G#4    A4     A#4
3 | A#3    B3     C4     C#4    D4     D#4    E4     F4
2 | F3     F#3   [G3]    G#3    A3     A#3    B3     C4
1 |[C3]    C#3    D3    [D#3]   E3     F3     F#3    G3
  +-------------------------------------------------------
    1      2      3      4      5      6      7      8
```

```
4 | D#4    E4     F4     F#4    G4     G#4    A4     A#4
3 |[A#3]   B3     C4     C#4    D4     D#4    E4     F4
2 | F3     F#3   [G3]    G#3    A3     A#3    B3     C4
1 |[C3]    C#3    D3     D#3   [E3]    F3     F#3    G3
  +-------------------------------------------------------
    1      2      3      4      5      6      7      8
```

```
4 | D#4    E4     F4     F#4    G4     G#4    A4     A#4
3 | A#3   [B3]    C4     C#4    D4     D#4    E4     F4
2 | F3     F#3   [G3]    G#3    A3     A#3    B3     C4
1 |[C3]    C#3    D3     D#3   [E3]    F3     F#3    G3
  +-------------------------------------------------------
    1      2      3      4      5      6      7      8
```

Hearing what you play. While notes sound the Harmony view and the display's voice strip name the chord live (sevenths, inversions as slash chords, see Chord names and the key); the brain's **chord memory** marks the remembered chord's tones with a violet bar on the screen and dim green on the pads, so the next chord's notes are easy to find. The diagrams are generated from the layout code (`tools/docs/grid_diagrams.py`).

Playing with MPE

What the grid sends depends on whether the edit part has **MPE** on (`pN.mpe`, see MIDI).

MPE on – every touch (mouse or finger, up to 10 at once) is its own note with its own expression:

- Each touch gets a free **member channel** (2 – 16, in turn). It starts with the bend centred and slide at 64.
- **Slide sideways = pitch bend** of that note only. The bend follows the cells: one cell's width bends by the layout's step to the right – 1 semitone in Fourths and Thirds, 2 in Whole Tones, about 1.7 (12 / 7) with In Key. It is sent as 14-bit pitch bend scaled to the part's **MPE bend range** (`pN.mpe.bend`, default 48 semitones) and stops at the end of that range. The note stays the note you struck; letting go ends it.
- **Up / down = slide** (CC 74, centre 64): one and a half cell heights up reach 127, down reach 0. The part sends it to its timbre target (`pN.mpe.timbre`).

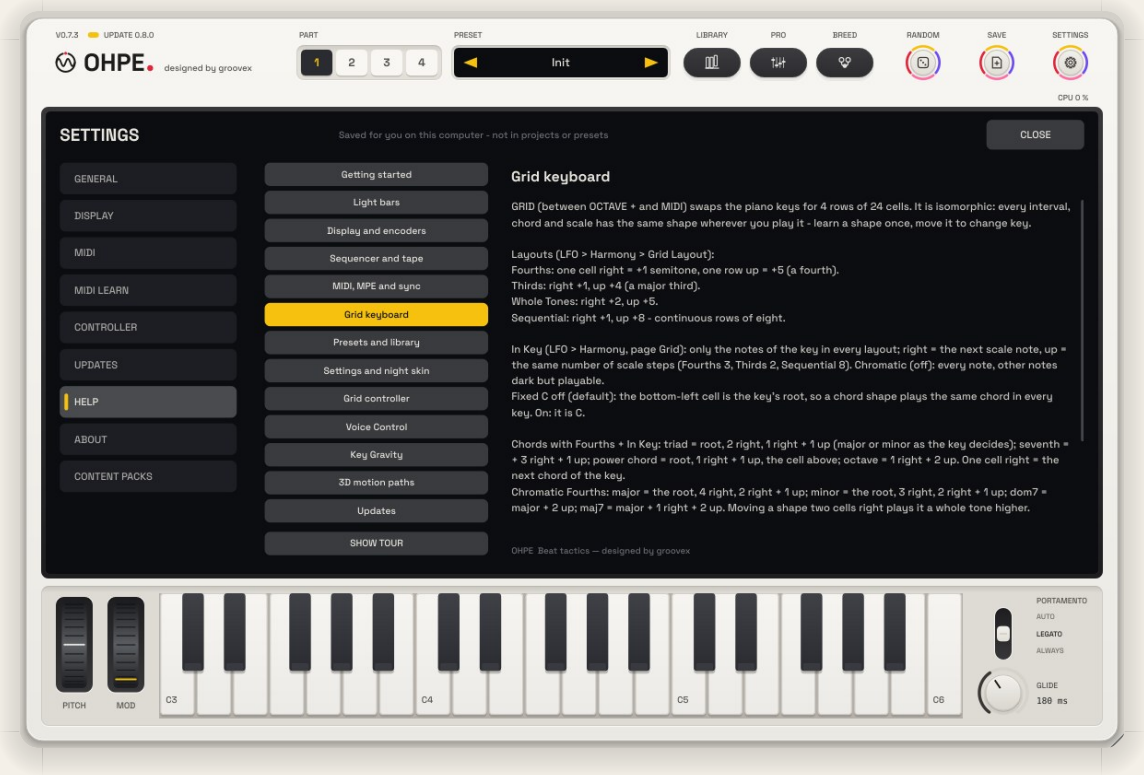
- **Pressure** (from a pressure-sensitive touch screen or pen; a mouse has none) is sent as channel pressure on the note's channel and goes to the part's pressure target (`pN.mpe.press`).
- MPE notes keep their pitch: they are not snapped by Scale, Chord or Follow.

MPE off – the notes play on the part's MIDI channel and go through the part like the piano keys (scale, one-finger chord, Follow, the arpeggiator when it plays this part). Sliding does nothing; there is no per-note expression.

In both cases the harmony brain hears the grid's notes (they shape the remembered key and chord). Like the piano keys, the grid is not sent to the MIDI output.

Help in the app

Settings > Help > Grid keyboard sums this page up (layouts, In Key / Chromatic, Fixed C, chord shapes, cells, MPE). With the **help overlay** on (Settings > General, or **F1**) and the grid shown, a line over the keyboard names the steps of every layout.



Settings – Help – Grid keyboard



The help overlay with the grid keyboard

On a grid controller

Grid controller profile 1 (standalone app and plug-in, Settings > Controller) puts this grid on an external 8 × 8 pad controller: the **same layouts** (a tap on Scales steps through them; holding Scales shows the layout, In Key / Chromatic, Fixed C, root and scale on the display, the upper buttons choose), the **same note** for the same column and row from the same bottom-left cell (the pads' four lower rows are the screen grid's four rows), the **same key** with **Scale Lock** (other notes dark and silent), the root lit **amber**, key notes **yellow**, chord tones **green (dim)**, held pads **green**, and **Octave − / +** moving pads and screen grid together. Velocity and pressure play the edit part (MPE: a member channel per pad). Details: Grid controller.

Modulation

Three matrix sources (the same value in every part):

#	Source	Range	
29	**Chord Root**	0 … 1	the chord's root, C = 0 … B = 1
30	**Chord Change**	0 … 1	1 at every new chord, fades in about 0.3 s – a filter accent on each change
31	**Tension**	0 … 1	share of held notes outside the key (smoothed)

Resonator body.

Every part can play through a **physically modelled body:** a string, a plate or a tube that the sound of the engine excites. A pluck strikes a tuned metal plate, noise bows a bore, a pad wakes a set of sympathetic strings tuned to the chord you hold. The body follows your keyboard – the note, the chord, pitch bend and MPE expression – so it is an instrument, not just an effect.



The Body view

Quick start

1. Press **Effects**, then the **Body** tab of the display.
2. Turn **Resonator** on (encoder 1, first page). The part now plays through a string body tuned to each note you play.
3. Choose the **Body** (encoder 2): String, Plate, Open Tube or Closed Tube.
4. Set **Body Mix** (encoder 4): 0 = the dry part (an exact bypass), 1 = only the body.

Or load one of the resonator factory presets: Plucked Plate, Gamelan Plate, Bowed Tube, Breath Bore, Sympathetic Pad, Drone Strings, Glass Body Keys.

Where it sits

The body is the first stage after the engine and the layer: **engine → resonator → volume → FX rack → bass mono → 3D space → mixer.** So the rack's effects colour the body like a microphone and a room would. The same body is also an FX rack type, **Resonator** (DE Resonanzkörper), for any slot of a part rack or the master rack – put it after a distortion, a reverb or another body. In a part rack it follows that part's keyboard; on the master rack it stays at its fixed tuning.

The models

Body	What it is	Sound
String	two coupled strings (a slightly detuned pair that exchanges energy at the bridge), stiffness for stretched partials, a frequency-dependent loss	plucked and struck strings, a beating, two-stage decay like a piano unison
Plate	48 modes of a rectangular plate, excited at one point and picked up at two (natural stereo)	bells, gongs, metal and glass plates, gamelan
Open Tube	a bore open at both ends (all harmonics)	flutes, pipes, breathy organ tones
Closed Tube	a bore closed at one end (odd harmonics, a quarter-wave resonance)	clarinet-like, hollow, woody

Every body can add up to six **sympathetic strings** that the body's vibration drives: they ring on at the octaves and fifths of the note, or at the notes of the chord you hold – and keep ringing after you let go (in Chord tuning they keep the last chord).

Parameters

Page	Encoder 1	Encoder 2	Encoder 3	Encoder 4
Resonator	Resonator (on / off)	Body (model)	Tuning (Fixed, Note, Chord)	Body Mix
Body Tuning	Body Tune (± 24 semitones)	Size	Material	Decay (0.05 – 20 s)
Body Character	Position	Inharmonicity	Sympathetic Strings	Body Width
Drive & Expression	Drive	Pressure to Body	Slide to Position	–

- **Tuning** – Note: the body follows the newest note (plus pitch bend, per-note bend and the part's tuning); Chord: as Note, and the sympathetic strings tune to the held keys; Fixed: C3 plus **Body Tune**, whatever you play.
- **Size** – String: how many sympathetic strings (2 – 6) and how far the string pair is detuned; Plate: its proportions (square and sparse … long and dense); Tubes: the bore width (wider = darker inside, more radiated highs).
- **Material** – how the highs decay: low = soft, dark materials (gut, wood), high = steel, bronze, glass.
- **Decay** – how long the fundamental rings (T60). Higher partials fade faster unless Material is high.
- **Position** – where the body is excited: near the end of a string (thin, bright) or in the middle (round, hollow); on the plate it changes which modes speak; on a tube where along the bore.
- **Inharmonicity** – String: stiffness (upper partials sharp, piano- and bell-like); Plate: the mode pattern from almost harmonic (low) over the ideal plate (middle) to stretched (high); Tubes: bore deviations.
- **Drive** – a smooth saturation of the excitation and, for strings and tubes, inside the loop (it never adds energy); on the plate the modes bend upward when it rings loud (gong-like pitch glide).
- **Pressure to Body** – MPE / aftertouch pressure of the newest note: positive = it rings longer and is excited harder, negative = pressure damps the body (like a palm mute).
- **Slide to Position** – MPE slide (CC 74) moves the excitation point.

All parameters are automatable and MIDI-mappable. The modulation matrix has the targets **Body Tune**, **Body Size**, **Body Material**, **Body Decay**, **Body Position** and **Body Mix** (75 – 80). For the rack type, FX Param 1 – 4 are Body Tune, Material, Decay and Position.

The display

The Body view draws the body on the dark display: the string pair between its bridges with the sympathetic strings below, the plate with its moving mode pattern, or the bore with its standing wave (closed end as a solid cap, open ends radiating). The yellow dot is the excitation point; the motion grows with the level. The line under the rows shows the model, the tuning and the pitch the body is tuned to. A Resonator rack slot shows the same picture in the Rack view.

Sound and safety

- Stable at every setting: every loop loses energy, coupling is lossless, saturation never amplifies, states are flushed (no denormals) and guarded; the body's output passes a peak limiter (it never goes above full scale).
- **Body Mix 0** or **Resonator off** is an exact bypass: the part sounds bit-identical to no body at all. Switching on and off crossfades over 20 ms.
- One body per part (as a real instrument has one body for all its strings), so the cost does not grow with the number of voices.

3D Space – binaural motion.

Every part can leave the stereo line and move **around your head:** in front, to the sides, behind you, above you, near and far – with motion paths that run in time with the song. On headphones Ohpe renders the part **binaurally:** your ears receive the time and level differences, the head shadow and the ear's direction cues of a real source at that place. On speakers the same scene is rendered without binaural colouration and stays fully mono-compatible.



The 3D Space view

Quick start

1. Press **Effects**, then the **3D Space** tab of the display (or click the **3D** badge in the display header).
2. Set the **3D Space** switch (encoder 1, first page) to Headphones or Speakers – turn encoder 1 or click a segment of the switch on the display. The part now sits in front of you, 1.5 m away.
3. Choose a **Motion Path** (encoder 2) and its **Motion Length** (encoder 3) – e.g. Circle over 2 Bars. Press Play: the dot of the part flies around the head in the display.
4. Put on headphones for Headphones (binaural); choose Speakers when you listen on speakers.

Or load one of the presets with the **3D** tag in the library – they carry a headphones pictogram. The factory bank has 75 of them (72 part presets and three projects; 49 binaural showcase presets were added in p71, listed in FACTORY-SOUNDS.md).

The 3D badge and the headphones pictogram

Where	Lit (ultraviolet, pulsing)	Outlined yellow, speakers	Dim
Display header, after the part badge: headphones pictogram + **3D**	the edit part plays binaural 3D (Space on or a Space 3D rack slot, monitoring on Headphones)	3D is on, monitoring is on Speakers	the part does not use 3D (the dim pictogram shows the monitoring a switch-on would use: headphones or speakers)
Part selector 1–4 (top)	the same per part, a small headphones pictogram in the corner	a small speakers pictogram	nothing
Library list	presets tagged **3D** show the headphones pictogram after their name; the category chip **3D** lists them		

Hover the badge for the tooltip (Binaural 3D – best with headphones); click it to open the 3D Space view of the edit part.



The 3D badge in the display header

The display: head-centred 3D view

You look over the listener's shoulder: the head in the middle (wearing headphones while you monitor binaurally, two speakers in front of it on speaker monitoring), the floor rings at 1 m, 5 m and 40 m, the ring 45° up, the front / back arc. Every part with 3D on is a dot in its colour (part 1 red, 2 yellow, 3 ultraviolet, 4 rose) with a fading **trail** and a **drop line** to the floor that shows the height. Nearer dots are larger; the glow follows the level. Top right: the edit part's azimuth, elevation and distance. The thin bar at the bottom is the **master correlation** (red = out of phase, see Mono compatibility). The view drifts slowly and follows the mouse a little (parallax; off with Animated 3D visuals).

The same view can be the display's left picture in any mode: click the picture to cycle Character → Stereo scope → Spectrum terrain → 3D space, or right-click → 3D space. A Space 3D rack slot shows its own source in the strip below the rack.

Parameters – the part's stage `pN.space.*`

The stage sits at the very end of the part: engine → FX rack → bass mono → **3D** → mixer. Space off is an exact bypass (every older sound plays bit-identically); switching crossfades over 20 ms.

Page	Parameter	Range	What it does
3D Motion	3D Space (`pN.space.mode`, p41)	Off, Headphones, Speakers	the 3D switch: Off = the stage off; Headphones = on, binaural; Speakers = on, speaker rendering. Drawn as a segmented switch; encoder 1 steps through the three states
	Motion Path	Off, Circle, Figure-8, Fly-By, Random Orbit, Pendulum, Spiral Up, Ping-Pong F/B, Swarm, Whisper, Distance Walk, Over the Head, Weave, Ping-Pong L/R, Four Corners, Breathe, Helix, Fly-Over, Bounce, Wander, Loop Orbit	see below
	Motion Length	1/16 … 32 Bars, 1/4.	one cycle of the path, in time with the song
	Motion Rate	×−4 … ×4 (default ×1)	multiplies the speed; negative = backwards, 0 = stopped. At exactly ×1 the motion locks to the song position (it repeats identically on every playback; with Retrigger Free only). Fastest: Motion Length 1/16 at ×4 = one cycle per 1/64 note – for short notes use 1/2 … 1/16 with Retrigger Note
3D Position	Azimuth	−180 … 180°	direction (0 = front, +90 = right, ±180 = behind); rotates the whole path
	Elevation	−60 … 90°	height; tilts the whole path
	Distance	0.2 … 40 m	base distance (the path scales from it)
	Source Spread	0 … 180°	how far apart the left and right channel of the part are placed (0 = a point source)
Motion Shape	Motion Size	0 … 100 %	how big the path is (per path, see below)
	Motion Phase	0 … 360°	offset of the path – give two parts the same path half a cycle apart
	Motion Glide	5 ms … 2 s	smoothing of every position change (automation, modulation, jumps)
	Doppler	0 … 100 %	pitch shift of moving sources (the distance becomes a delay; 0 = no delay at all)
Room & Distance	Room	0 … 100 %	early reflections of a room that move with the source, plus a short diffuse tail – they put the sound outside your head
	Room Size	0 … 100 %	from a small room (4 m) to a hall (30 m)
	Air Absorption	0 … 100 %	far sources lose their highs
	Distance Rolloff	0 … 100 %	how much quieter far sources get (100 % = inverse distance law; 1.5 m plays at the input level)
Monitoring	Mono Below	40 … 300 Hz	everything below stays **mono and unspatialised** (Linkwitz-Riley crossover, default 120 Hz) – the sub never moves
	3D Monitoring (`space.out`, global)	Headphones, Speakers	binaural rendering or speaker rendering for every 3D part and rack slot
	Mono Check (`space.monocheck`, global)	off / on	folds the main output to mono – listen for what speakers in mono would play

Page	Parameter	Range	What it does
Amount & Retrigger	Amount (`pN.space.amount`, p77)	0 … 100 % (default 100 %)	strength of the 3D effect: 100 % = the full position, motion and binaural colouring; lower values shrink the motion towards the front, the direction, distance, Doppler, air and room, and fade to the plain stereo sound at 0 %. The level stays about the same (the plain sound plays at the base distance level). Smoothed, click-free; matrix target
	Motion Retrigger (`pN.space.retrig`, p77)	Free, First, Note (default Free)	Free = the motion runs on with the song (as before); First = it restarts from its start position on a note played after all keys were up; Note = on every note (in mono, legato notes – a key still down – keep it running). The position glides to the start in about 8 ms, so a restart never clicks. With First / Note the motion runs from the note, not from the song position

3D Monitoring and Mono Check belong to the listener: project presets neither save nor change them.

The 3D switch

The switch on encoder 1 is one control for two older parameters, which stay what presets and sessions store: the part's stage on / off (`pN.space.on`) and the global **3D Monitoring** (`space.out`). Off switches the part's stage off; Headphones and Speakers switch it on and set the monitoring. Monitoring is how **you** listen, so it is the same for every part: choosing Speakers on part 2 also moves the switch of every other part that has 3D on to Speakers (the header badge and the head in the 3D view change with it – headphones on the head, or two speakers in front of it). The three stay in step both ways: automating or MIDI-mapping the switch (`pN.space.mode`) works, and so do the older `pN.space.on` / `space.out` – the switches follow them. Presets and sessions made before the switch existed show the right state when they load.



The 3D switch on Speakers: the badge and the caption show the speaker pictogram



3D off: the hint has the first line of the strip to itself

- **Headphones** – binaural: interaural time difference, spherical-head shadow, pinna notches, shoulder reflection, near-field level (see How it works).
- **Speakers** – the speaker path of the same stage: constant-power amplitude panning by direction, **no** interaural delay and **no** head / pinna colouration (speakers and the room do that for real), depth by level, air and a darker back, the moving room reflections; mono-compatible (see below).

Motion paths

Path	Movement (one cycle)	Motion Size
Circle	a full orbit around the head	tilts the orbit's plane (0 = level, 100 % = 60°: it rises on one side and dips on the other)
Figure-8	a lying eight: wide left / right sweeps crossing in front, rising and falling	width of the sweep (30 … 180°)
Fly-By	a straight line passing in front, left → right and back, slowest at the far ends	length of the line (2 … 12 × the distance) – with Doppler the classic fly-by
Random Orbit	a smooth random path, four new points per cycle (every part its own)	how far it wanders (direction, height, distance)
Pendulum	swings from in front over your head to behind you and back	how far it swings (100 % = front to back)
Spiral Up	climbs from below ear height to above your head in 1–4 turns, then drops back	number of turns
Ping-Pong F/B	jumps from in front to behind you and back, in time: each half cycle opens with a quick move on the beat, then the sound holds. Front → back passes up on the right, back → front up on the left	how snappy (0 = a smooth swing over the whole half, 100 % = a short hop in the first 30 %)
Swarm	a flutter around the front: twelve small random jumps per cycle on top of a slow random drift – like a bee or a cloud of sparks (every part its own, never repeats exactly)	how wide it flutters and drifts (direction, height, distance)
Whisper	very close behind your head (0.4 m at the default distance), lingering and nearest at one ear, then the other – the near-field level difference makes it intimate. Lower Distance for an even closer whisper	how far round to the front it reaches at the ears

Path	Movement (one cycle)	Motion Size
Distance Walk	walks off straight ahead into the distance and comes back, swaying a little left and right – level, air absorption and Doppler change on the way	how far (2 … 16 × the distance; nearest 0.35 ×)
Over the Head	a vertical circle through both ears: above your head → right ear → underneath → left ear	tilts the circle's top forward (0 = straight through the ears, 100 % = 45°)
Weave	a full orbit that weaves up and down three times per turn and comes a little closer at its low points	height of the waves (10 … 40°) and their depth
Ping-Pong L/R	jumps from your left ear to your right ear and back, in time: each half cycle opens with a quick move on the beat, then the sound holds. Left → right passes in front of you, right → left behind you	how snappy (0 = a smooth swing over the whole half, 100 % = a quick move in the first 45 %)
Four Corners	steps through four corners around you – front left → front right → back right → back left – one per quarter cycle, rising a little on each move, then holding	how snappy (0 = smooth, 100 % = a quick move in the first 40 % of each quarter)
Breathe	breathes in and out in front of you: from far away to close (0.45 × the distance) and back, log-even, a little higher when close	how far out it breathes (1.2 … 3 × the distance)
Helix	orbits two to four times while rising evenly from below ear height to above your head (first half) and sinking back down (second half), a little closer at the top	number of turns per cycle (2 … 4)
Fly-Over	a straight line over your head (one distance up): from far in front, right overhead, to far behind and back, slowest at the far ends – with Doppler a plane passing over	length of the line (2 … 12 × the distance)
Bounce	hops like a ball across the front, left → right → left, six hops per cycle	hop height (up to 60° above the floor point)
Wander	a smooth random loop – direction, height and distance – that repeats exactly every cycle, so it stays in time with the song (every part its own shape)	how far it wanders
Loop Orbit	a full orbit with four small loops: on the near side of each loop the sound curls back against the turn, on the far side it lifts a little	size of the loops (and how near it comes: 0.75 … 0.33 × the distance)

The six paths from Ping-Pong F/B on and the eight from Ping-Pong L/R on were added later; they sit at the end of the list, so sounds and automation saved with the older paths keep their path and sound the same. All paths are continuous (also where a cycle starts again) and stay in time with the song.

In the 3D view the edit part's path is drawn ahead of its dot as a faint dotted curve – one whole cycle (for Swarm and Random Orbit: the next cycle from now) – so you see the shape of a path, its size and its orientation before a note plays. The path's name sits bottom left.



Weave, Swarm and Over the Head in the 3D view – part 1's path drawn ahead of its dot

Azimuth and Elevation rotate every path: a Pendulum at azimuth 90° swings left over right.

Modulation and MPE

The modulation matrix (LFO mode → Mod) moves **Azimuth, Elevation, Distance, Motion Path Rate** and **Motion Path Phase** of the part's stage (targets 67 – 71) and its **3D Amount** (target 95, p77; MODULATION.md). With an MPE controller or poly aftertouch:

- **Slide (Timbre) → Elevation:** sliding up lifts the sound above your head;
- **Note Pressure → Distance** with a negative amount: pressing harder pulls the sound closer;
- an LFO on Motion Path Rate makes an orbit accelerate and slow down; a macro on Azimuth turns the whole scene.

The modulation uses the newest note (one position per part). Every 3D factory preset has Slide → Elevation and Pressure → Distance routed.

Space 3D as a rack effect

The FX rack has the type **Space 3D** (DE 3D-Raum, type 22) in every slot of the part racks and the master rack, with the same parameters (`pN.fxK.space3d.*`, `mfxK.space3d.*`). Use it to place a single effect return, or the whole master, in the room. Its FX Param 1 – 4 targets are Azimuth, Elevation, Distance and Motion Rate. Keep the slot's mix at 100 % (the dry signal is not delayed like the 3D one).

Mono compatibility and the low end

- The band below Mono Below is summed to mono and never spatialised – basses keep a centred, solid sub while their mid range moves (the 3D bass presets use 150 – 220 Hz).
- **Speakers:** amplitude panning without any time differences – the mono fold loses at most 3 dB anywhere on the circle (tested).
- **Headphones:** binaural signals contain small time differences between the ears. They fold to mono without hollow combing (straight ahead the ears are identical; to the side the far ear is much quieter), but check important mixes with **Mono Check** and the correlation bar.

How it works

Ohpe uses its **own parametric model** of the head and ears – no measured head-related data set is included or needed:

- **Spherical head:** interaural time difference (Woodworth's formula, 0.65 ms at 90°) through fractional, cubic-interpolated delay lines, and a first-order head-shadow filter per ear whose coefficients glide sample by sample;
- **Ear cues:** two pinna notches that rise from about 6 kHz to 11 kHz with elevation, directional bands for front (presence), behind (pinna shadow, 1 kHz band) and above (8 kHz band), a shoulder reflection, and a near-field level difference below 1 m;
- **Distance:** level rolloff, air absorption, a distance delay that makes the Doppler shift (its speed is limited, so the pitch never bends more than about ±12 %);
- **Room:** six image-source reflections of a shoebox room that follow the source, rendered with their own directions, into a small four-line diffuse tail;
- **Stereo sources:** the part's mid signal is a point source, its side signal a pair of sources Source Spread apart, so mono material never comb-filters.

Everything is interpolated – positions glide, delays ramp per sample, filters update every 16 samples – so moving sources stay click-free (tests sweep, jump and switch while measuring the second difference of the output). CPU: about 1.3 % of one core per stage at 48 kHz on the build machine; four part stages plus eight rack instances stay well within the real-time budget (measured by the core tests).

Tips for bass music

- Neurofunk / dubstep basses: a short path (Circle or Figure-8, 1 – 2 bars) with a small distance (1.2 – 1.5 m), little room and Mono Below 150 – 180 Hz: the growl moves, the sub punches mono.
- Atmospheres: Random Orbit over 8 – 16 bars, distance 3 – 5 m, Room 35 – 50 %, Spread 70 – 90°.
- Risers: Spiral Up over the riser's length; fly-bys: Fly-By with Doppler 100 %.
- Two parts with the same path and Motion Phase 0° / 180° chase each other around the head.

Morph space.



The Morph view

Every part has a **morph pad:** an XY field whose four corners each hold a **complete sound** – engine, envelopes, filter, modulation matrix, FX rack, 3D stage, everything a part preset stores. Moving the position across the pad blends the four sounds, smoothly, at even loudness, also between **different engines.** Play it with the mouse, the host's automation, a MIDI controller, the modulation matrix, the mod wheel or the pressure and slide of an MPE keyboard.

Corners

Corner	Position
A	top left (x = 0, y = 1)
B	top right (x = 1, y = 1)
C	bottom left (x = 0, y = 0)
D	bottom right (x = 1, y = 0)

A corner keeps every **sound parameter** of the part (`pN.*`), except the morph block itself and the part's MIDI routing (MIDI channel, scale / chord, the MPE switches), which stay the part's own. An **empty corner** stands for the part's **own sound** – the parameters as you edit them with the encoders. So one assigned corner already makes a morph between your sound and another one; four assigned corners make a four-way morph.

Assign a corner in the **Morph** view (LFO mode, third tab): click a corner chip on the pad or one of the soft keys **A … D:**

Menu	Result
Current sound	the edit part's sound as it is now (the corner is named after the loaded preset)
Preset ▸	any part preset of the library (by category) – loaded into the corner only, the part keeps its sound

Menu	Result
Clear corner	the corner stands for the part's own sound again
Clear all corners	all four

Assigning a corner switches the morph on. The chips show the corner's name and, as a thin bar in the corner's colour, its share of the sound right now.

How the sounds are blended

The position gives each corner a weight (bilinear: A = (1 − x) · y, B = x · y, C = (1 − x)(1 − y), D = x (1 − y); together always 100 %).

- **Same structure – parameters are interpolated.** Corners whose discrete settings agree (same engine, same layer engine, the same effect in each rack slot, the same filter type, LFO shapes, routings …) are one sound: every continuous parameter moves between the corner values. Ohpe interpolates in normalised units, so a cutoff or a time moves evenly in octaves / ratios, not in Hz. One sound costs no extra CPU.
- **Different structure – the sounds play in parallel.** Corners that differ in a discrete setting (another engine, another effect, another filter type …) play as parallel sounds ("lanes") that are cross-faded with **equal-power** gains (√weight): the loudness stays even across the pad, nothing switches abruptly, and a changed engine never clicks. Each lane is a full part (engine, rack, 3D stage), so a pad with four different engines in the middle costs up to four parts of CPU; a lane whose weight has been zero for 0.1 s stops rendering, and when the position comes back it starts again with the keys you hold.
- On the pad, **solid edges** join corners that morph their parameters, **dotted edges** corners that cross-fade as parallel sounds.
- **On a corner the sound is exactly that corner** – every parameter has the corner's value, bit for bit.

Moving the position

Control	Parameter / source
The pad (drag)	`pN.morph.x`, `pN.morph.y` – host-automatable, **MIDI-learnable** (right-click the pad or an encoder)
Encoders (page Morph)	Morph on, X, Y, Glide
X Control / Y Control (page Control)	Off, Pressure (channel or MPE pressure), Slide (MPE / CC 74, centre = 0, ±half the pad), Mod Wheel – added to X / Y
Modulation matrix	targets **Morph X**, **Morph Y** (73, 74): any source moves the pad, e.g. LFO 2 → Morph X for an evolving sound
Glide	`pN.morph.glide` 0 … 2 s – the position follows every change smoothly (also stepped MIDI CCs)

The glided position is also a **matrix source: Morph X, Morph Y** (21, 22, 0 … 1), so the pad can drive anything else as well (an effect mix, the 3D distance …). Routings of the part's own sound that move the pad itself stay active whatever the corners hold.

A typical MPE setup: Y Control = Pressure, X Control = Slide – press harder to move up, slide forward to move right, every note of the part follows.

Presets and projects

- A **part preset** stores the part's corners in an optional `"morph"` block; loading it into any part brings the corners along, a preset without the block clears them (PRESETS.md).
- **Projects** and the plug-in state keep the corners of all four parts.
- The **factory bank** has seven morph presets (tag Morph; four of them are described below). They start in corner C (the part's own sound); the mod wheel moves right, pressure moves up:

Preset	Corners	Lanes
Morph Strings Filter	one analog string ensemble in four shapes (open, edged, home, slow and dark)	1 – pure parameter morph
Morph Four Worlds	Glass Shimmer (wavetable), Breath Choir Pad (vocal), Warm Strings (analog), Crystal FM Pad (FM)	4
Morph Keys and Bells	FM Vibes, Glass Keys, Digital EP, Dream Tine	several (different engines)
Morph Bass Shapes	Growl FM Bass, Classic Reese, Round Sub, Acid Line	several (different engines)

Notes

- With the morph on, the part plays the morph: the encoders still edit the part's own sound, which you hear in the empty corners. To change a corner, play it, edit, and assign **Current sound** to it again.
- User samples loaded into a Sampler / Grain part play in the part's own lane; parallel lanes use the factory instruments their corner selects.
- Parameter locks of the step sequencer win over the morph on their step.
- Parameters: PARAMETERS.md.

Voice Control.

Sing, hum, talk or whisper into an audio input and your voice moves the sound. Voice Control is **not a vocoder:** the voice is never heard through Ohpe, it only modulates. It is analysed in real time into seven modulation sources that go anywhere in the modulation matrix.



Voice view

Connect an input

Where	How
Standalone app	**Settings > Audio, Input device** (none by default). macOS asks once for the microphone; the text explains why. Choose No input to close it again.
Plug-in (VST3 / AU)	Ohpe's optional stereo input bus – the one the tape and bounce already use – is the voice input. Enable the instrument's input or sidechain in the host (e.g. route a microphone track or a vocal bus to Ohpe's sidechain input). It stays off until the host enables it, so existing projects and layouts are unchanged.

No input connected = every source rests at 0, so a preset or routing that uses them plays exactly as programmed.

The sources

#	Source	Range	What it follows
34	**Voice Level**	0 … 1	how loud you are above the gate (1 at about −6 dBFS)
35	**Vowel X**	−1 … +1	back (U, O) to front (I, E) – the second formant
36	**Vowel Y**	−1 … +1	closed (I, U) to open (A) – the first formant
37	**Voice Brightness**	0 … 1	how much of the voice lies above about 2 kHz
38	**Voice Noisiness**	0 … 1	consonants: s, sh, f, t are high, sung vowels low

#	Source	Range	What it follows
39	**Voice Pitch**	0 … 1	the sung note, C2 … C6 (only with Pitch Tracking on)
40	**Voice Onset**	0 … 1	a short trigger at every new syllable or attack (fades in about 80 ms)

The vowel is found from the voice's formants (linear prediction on a 16 ms window, updated every 2.7 ms): the analysis lags the voice by about **8 ms** plus the smoothing you set. The vowel map is tuned to adult average formants; higher voices still land on the right side of the map. After a pause of about 0.4 s the vowel, brightness, noisiness and pitch glide home to 0.

The Voice view (LFO mode)

LFO > Voice shows the vowel map – A E I O U at their places, your vowel as a yellow dot (grey while you are silent) – and the input meter with the gate as a white line, then small bars for level, brightness, noisiness, pitch and onset. The line under it says what Voice Control hears (vowel and level, Listening below the gate, or No input).

Encoder	Parameter	Range	Default
Voice 1	**Voice Control** (`voice.on`)	Off / On	On
Voice 2	**Input Gain** (`voice.gain`)	-24 … +24 dB	0 dB
Voice 3	**Gate** (`voice.gate`)	-80 … -20 dBFS	-50 dBFS
Voice 4	**Smoothing** (`voice.smooth`)	1 … 250 ms	20 ms
Options 1	**Pitch Tracking** (`voice.pitch`)	Off / On	Off
Options 2	**Monitor** (`voice.monitor`)	Off / On	**Off** – hear the input on the main output; keep it off with speakers (feedback)

Soft keys: **On** switches Voice Control, **Vowel → Filter + Morph** is the quick mapping for the edit part (below). All parameters are host-automatable and MIDI-learnable; they belong to your setup and are not stored in presets or projects.

Quick mapping: Vowel → Filter + Morph

One press (or the MIDI-learnable trigger `pN.voice.map`) writes three routings into the edit part's matrix: **Vowel Y → Timbre** (+40 %: open vowels open the filter), **Vowel X → Morph X** and **Vowel Y → Morph Y** (+50 % each: your vowel walks the morph pad – switch the part's morph space on to hear it). Existing routings are kept; pressing again only resets the three amounts; a full matrix is left alone (the display says so).

Factory presets

Three part presets start with **Voice** in their name (search "Voice" in the library). Without an input they are complete sounds:

Preset	Without input	With your voice
Voice Vowel Lead	a clean vowel lead with vibrato	your vowel moves its vowel, brightness opens it, syllables lean in
Voice Filter Pad	a warm pulse pad	open vowels open the filter, level brings it alive, the vowel pans it
Voice Wah Bass	a round talking bass	vowels sweep the formant table, consonants bite

Tips

- Start with the gate just above the room noise: the dot should turn yellow only when you sing.
- A longer smoothing (60 – 120 ms) makes vowel sweeps glide; 5 – 10 ms follows fast talking.
- Voice Onset as via of an envelope or as a Level routing makes rhythmic syllables play the part.
- Use headphones or keep **Monitor** off – Ohpe never needs to hear itself.

Preset breeding.



The breeding view

Breeding crosses two sounds and lets you pick the best of their children: choose two parents (your current sound, any part preset of the library, a child of an earlier brood, or a roll of the die), set how much the children may **mutate**, and press **BREED**. Six children appear; click one to hear it, give the good ones a **heart** and press **AGAIN** – the next generation descends from the kept ones. Every generation stays in a history with undo / redo, and **SAVE** stores a child in the library as a favourite.

Opening the breeding view

How	Result
BREED in the header (next to RANDOM)	opens / closes the breeding view; parent A = the current sound; the key's light bar is lit while the view is open
BREED in the library's details (part presets)	closes the library (your sound comes back), opens the breeding view with A = the current sound and B = that preset and breeds at once
MIDI: `lib.breed`	opens / closes the view (MIDI learn in the MIDI menu, Remote controls)
MIDI: `lib.breed.go` / `lib.breed.next`	breed again / hear the next child

The view lies on the plate like the library. **Keep** (Enter) closes it with the child you listened to last; **Cancel** (Esc) brings back the sound from before the view opened. While the view is open, a short press of **RANDOM** rolls a new parent B and breeds, and **SAVE** saves the selected child.

Parents, mutation, generations

Control	Function
A / **B** chips	menu: Current sound, Random preset, Selected child, Preset ▸ (part presets by category; favourites marked with a heart)
Die	a random part preset of the library becomes parent B
MUTATION	drag: 0 % = the children are pure crosses of the parents (every value lies between the parents' values), 100 % = wild
BREED	a new brood of six from A and B
AGAIN	the next generation: two or more kept children are crossed with each other (every pair in turn), one kept child is mutated (at least 10 %), no kept child = the parents again
< / >	undo / redo a generation (a new brood after undo drops the redo branch); the last 24 generations are kept
♪ NOTE	each click on a child plays a short C (on / off)
INHERIT chips	per group (Engine, Filter, Envelope, FX, Mod): Auto (a random choice per child), A, B or Mix

The children

Each tile shows the child's name (the parents' first words and generation.child), a **waveform thumbnail** coloured by its **brightness** (violet = dark, rose = middle, yellow = bright), its level and brightness, its engine (with the layer engine of a hybrid) and an **inheritance strip:** one segment per group, white = from A, violet = from B, in between = blended. The selected child has a yellow frame and light bar (green while its note plays). Right-click a tile: Listen, Keep, Make parent A / B, Save to the library.

A saved child is an ordinary part preset (tag "Bred", a comment naming its parents) and is marked as a favourite, so the library's Favourites filter finds it at once.

How a child is made

A sound is treated as a **genome:** every parameter of the part, sorted into groups.

Group	Parameters
Engine	engine choice, layer, the engines' own sound parameters, mono bass
Filter	cutoff, resonance, filter type, tone / brightness, drive, key tracking of the engines
Envelope	amp, mod and third envelope, pitch envelope, glide, velocity
FX	the six rack slots (types and parameters), the 3D stage
Mod	LFOs, the 16 matrix routings, macros, MPE amounts, life

Group	Parameters
Level	volume, part gain, layer level: blended, never mutated, then level-matched
Fixed	tuning, voices, pitch-bend range, MIDI channel, scale / chord, MPE switches, morph block: from parent A

- **Crossover:** per group the child takes A, B or a blend (one blend position per group, in the parameters' own curve, so a frequency blends musically). Choices and switches are never interpolated – they come from one parent; whole numbers are rounded.
- **Coherent across engines:** an engine's parameters come from the parent that plays that engine; an effect's parameters come from the parent whose slot holds that effect; a matrix routing (source, via, target, curve, amount) travels as a unit unless both parents wire it alike. Two parents on **different engines:** the child plays the engine of one; when the Engine group blends, it may keep the other engine as its **layer** (a hybrid, at a moderate layer level, no cross-modulation).
- **Mutation:** small Gaussian steps on continuous parameters (inside 2 … 98 % of their range), rare steps on choices and whole numbers, very rarely another effect type (above 50 %). Never mutated: the engine and sample choice, the routing wiring, levels. Resonance, feedback, drive and gain-like parameters cannot rise more than a small step above the louder parent.
- **Level match:** every child is rendered offline (a private engine core on a worker thread, C4 + G4 for 0.7 s); its part gain is set so it plays as loud as its parents (at most 6 dB louder than its blended gain, any amount quieter), with the peak below -1.5 dBFS. The same render gives the thumbnail and the brightness.
- **Deterministic:** every random choice comes from Ohpe's own number generator; a seed gives the same brood on every platform.

The breeding session – parents, generations, kept children, the group choices – is saved with the project (plug-in state, section `"breed"`). The breeding core (`src/core/breed`) is engine-agnostic: it reads only the parameter descriptions, so new engines and effects breed without changes.

Learning breed

The breeding learns locally what you like. It watches, per sound (a library preset or a bred child): how long you play it with notes, whether you save it, load it again, favourite it or keep it (heart) – and whether you skip it quickly (a note or two, replaced within a few seconds: counts against it). From this it keeps a small **taste profile per engine:** for every sound parameter the region you liked and the one you skipped.



Learning breed

BREED and AGAIN then lean towards the liked region of each child's engine: twice as many candidates are bred and the best-scoring ones are shown, and mutation steps are pulled towards what you liked (and away from what you skipped). The die picks its random parent B with the same taste. Levels are never touched – every child is still level-matched to its parents, and the safety caps on resonance, feedback and drive stay.

The foot row of the breeding view:

Control	What it does
LEARN: ON / OFF	Learning on or off. Off = nothing is recorded and the brood is not biased.
CURIOSITY n %	How much keeps exploring (click +10 %, right-click −10 %). 100 % = the brood ignores the profile, 0 % = it follows it fully. Default 35 %.
WHY	Shows the strongest learned traits, e.g. "Analog · Cutoff: higher (82 %)".
RESET	Press twice: forgets the whole profile (the settings stay).

The profile is a local file (`<user data>/groovex/Ohpe/Taste/taste.json`, see FORMATS); nothing leaves the computer. A damaged file starts an empty profile. With a fixed seed the brood is reproducible.

Living display.

The fifth display view, **Living object**, shows the sound as a 3D object in the left half of the display. It listens to what shapes the sound right now and lets you see it at a glance:



The living display

What moves	What it means
Size, glow	the amp envelope of the newest note and the level – the object swells on the attack, holds on the sustain and shrinks on the release
Squash / stretch upright	LFO 1
A twisting waist	LFO 2
Tilt	LFO 3
Ripples on the surface	the mod envelope and how much the modulation matrix moves right now (more depth = faster ripples, faster turning)
Width	the stereo width of the signal (the side share): mono is round, wide sounds stretch to the left and right
Two lobes	the channels drift out of phase (correlation below about 0.2) – the object pinches into a left and a right half: a warning for mono playback

An LFO only moves the object as much as it is used: the part LFO by its depth, LFO 2 / 3 by the routings of the matrix that take them as source or via. A free-running LFO that drives nothing leaves the object still.

The equator ring wears the part's colour (P1 red, P2 yellow, P3 violet, P4 pink). Three small bars under the object read out **Envelope**, **Mod** and **Width**. When the sound has faded the object settles into a still sphere (Play to wake it).

Choosing it

Click the left half of the display to cycle the views (Character → Stereo scope → Spectrum terrain → 3D space → Living object) or right-click it and choose **Living object**. The view is saved with the project.

- **Synth** pages and the other part pages: the edit part (its envelopes, LFOs, matrix and its own signal after its FX rack), labelled P1–P4.
- **Mixer**: the master signal for the stereo picture (labelled Master), the edit part for the modulation.
- The Mid/Side scope still takes over by itself on Mid/Side slot pages; the tape page keeps its transport.

Animated 3D visuals off (view menu): the object stops turning and follows the sound without smoothing.

Light on the CPU

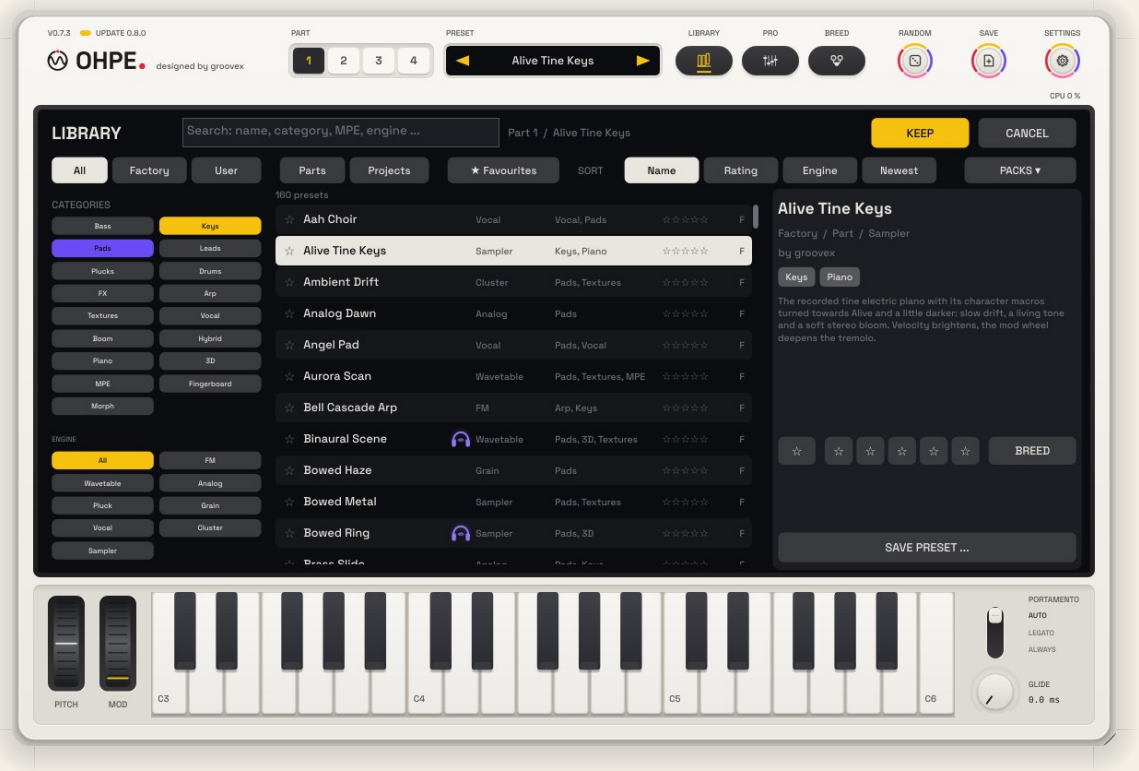
The object is a small mesh (9 rings x 28 points) drawn as three strokes. The display repaints only the object's own rectangle (and its readouts), and only while it moves: a resting object costs nothing – the rest of the display is not repainted for it. The values come from the core's lock-free UI feedback and the scope tap; nothing waits or allocates on the audio thread.

The object has no host parameters of its own – it is a view; everything it shows (envelopes, LFOs, matrix, width) is automatable and MIDI-learnable where it is set.

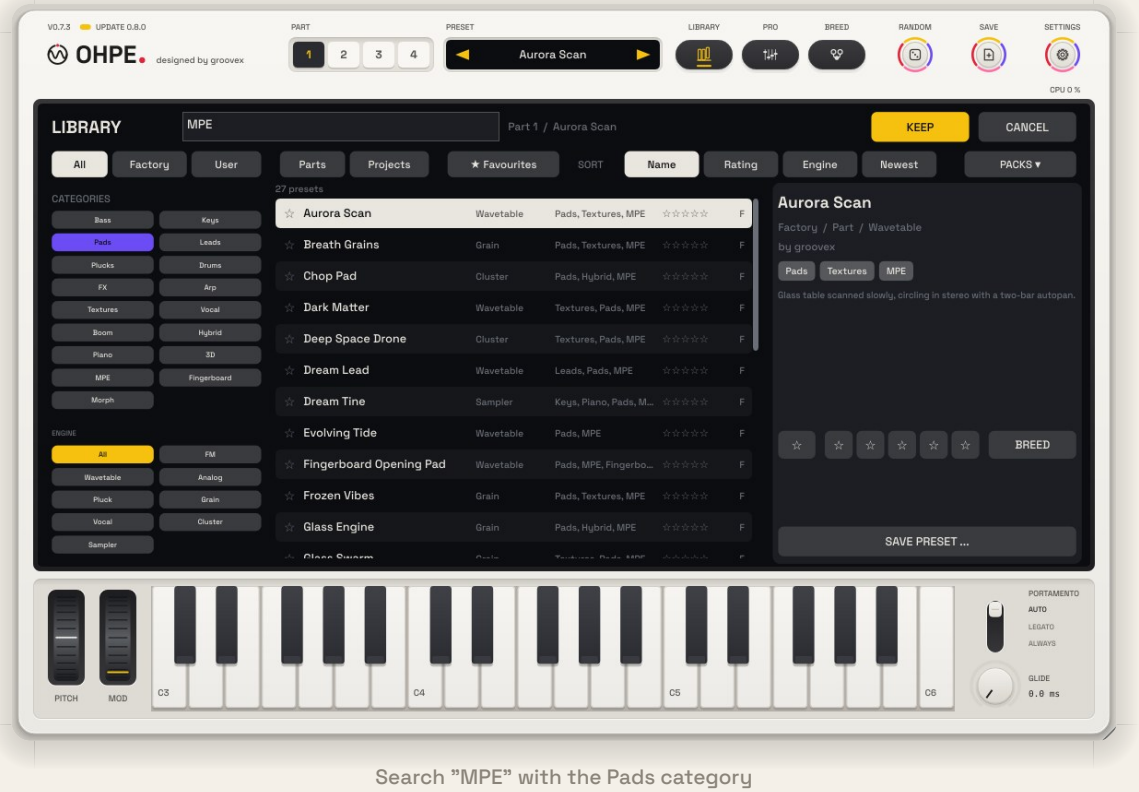
PART 5

Presets and sounds.

Preset library.



Preset library



Search "MPE" with the Pads category

Using the library

Press **LIBRARY** (top of the instrument) to open the library. It covers the keys and the display; the keyboard stays playable, so you can audition sounds while you browse.

Area	What it does
Categories (left)	Bass, Keys, Pads, Leads, Plucks, Drums, FX, Arp, Textures, Vocal, Boom (0.6: deep tuned kicks and sub basses), Hybrid (0.8: two engines in one sound), Piano (0.8: acoustic and electric pianos). Select one or several: the list shows presets that carry **any** of the selected tags.
Engine (left)	Only presets for one engine (or All).
Quick chips (left, under the categories)	**MPE**, **Fingerboard**, **Morph**: presets that carry **all** of the selected chips. Combined with the categories, Pads + MPE shows the MPE pads. Only chips that some preset uses are shown.
Search (top)	p54: every word must match (AND) one of: name, author, comment, category and tags (English and German, also singular and other spellings: Pad, Fläche, Flaeche, Flache), the tags **MPE**, **Fingerboard** (also MPE-HD) and **Morph**, engine (id, English and German name), content pack (id and name, or pack), bank (factory / werk, user / eigene), kind (part, project / projekt), favourite / favorit and the rating (`***` = at least three stars, `5star`). Case and umlauts do not matter. Examples: `MPE`, `bass mpe`, `fläche fm`, `favorit ***`. The search works together with all chips.
All / Factory / User	Which bank: the built-in factory presets, your own presets, or both.
Parts / Projects	Only part presets (one sound) or only project presets (the whole instrument).
★ Favourites	Only presets you marked as favourite.
Sort	By name, rating (best first), engine or newest.
List	Selecting a preset **loads it at once** (preview) into the edit part – part presets into the part chosen with the part selector, project presets replace everything. Click the star on the left to mark a favourite, click the stars on the right to rate (click the same star again to clear). Arrow keys move through the list.
Details (right)	Name, bank, kind, engine, author, tags, comment, favourite and rating. User presets can be deleted (moved to the trash).
Keep / Cancel	Keep (or Enter, or a double-click on a preset) closes the library with the preset loaded. Cancel (or Esc) restores the sound you had before the library was opened.

The **preset strip** next to the part selector shows the preset loaded last into the **edit part**: every part remembers its own preset, so switching parts (part selector, MIDI, grid controller part buttons) switches the name too – on the strip, in the library and on the controller display. A project preset names all four parts; a part with nothing loaded shows "Init". Its arrows load the previous / next preset of the library's current list (with its filters and sort order) into the edit part, starting from that part's preset; a click on the name opens the library.

First start

A new Ohpe – the first start of the app, or a new plug-in instance in a song – starts with the **first factory preset of the library** (the built-in bank in the library's default order, by name), not with a blank "Init" sound: a part preset is loaded into all four parts (each keeps its MIDI channel), a project preset would replace everything. The preset strip shows its name; tape and sequencers are empty. A saved session – the host's song, the standalone app's last session or a loaded state file – restores itself instead, together with the names of the presets that were loaded (one per part; a session from version 0.6.1 or older gives its one name to every part). In the standalone app, Audio Settings … > Reset to default state returns to this start.

Saving

Save preset … opens the save panel: name, author (remembered), tags (categories plus your own tags, comma separated) and what to save – **This part** (the edit part's sound) or **Whole project** (every part, drum kit, mixer, effects, sequencer patterns and settings). The panel starts with the name and tags of the loaded preset, so an edited sound keeps its categories.

If one of your presets already has that name, Ohpe asks: **Replace** overwrites it (favourite and rating stay), **Keep both** saves the new one as "Name 2" (or the next free number), **Cancel** goes back to the panel. Factory presets are never overwritten.

After saving, the new preset is selected in the list. If a filter would hide it (a category, the engine, the search text, favourites, the Factory bank), the filters are cleared and the list shows **User**. The line above the list confirms the save – or says which folder could not be written.

User presets are files in the user folder, so you can copy, share and back them up:

System	User folder
macOS	`~/Library/Application Support/groovex/Ohpe/Presets`
Windows	`%APPDATA%\groovex\Ohpe\Presets`
Linux	`~/.config/groovex/Ohpe/Presets`

Part presets go to `Parts/`, project presets to `Projects/`; sub-folders are scanned too. The environment variable `OHPE_PRESET_DIR` points Ohpe at another folder. Favourites, ratings and the author name are kept in `library.json` in the same folder. You can also drop a `.ohpepreset` file onto the instrument to load it.

MIDI

Three remote-control parameters make the library playable from a controller (right-click the LIBRARY key, or MIDI → Remote controls, then MIDI Learn):

Parameter	ID	Action
Library	`lib.open`	opens / closes the library
Previous Preset	`lib.prev`	loads the previous preset of the current list
Next Preset	`lib.next`	loads the next preset of the current list
Edit Part	`ui.part`	selects the edit part 1–4 (where part presets load)

A trigger fires when its control goes to "on" (CC value 64 or more, or a note); pressing and releasing a button sends one step.

File format (version 2)

A preset is one UTF-8 JSON file with the extension `.ohpepreset`:

```
{
  "format": "ohpe-preset",
  "version": 2,
  "kind": "part",
  "name": "Glass Keys",
  "author": "groovex",
  "tags": ["Keys", "Plucks", "bright"],
  "engine": "fm",
  "comment": "",
  "params": { "engine": 0, "fm.algo": 2, "fm.index": 0.45, "amp.d": 1.2 },
  "arp": { "on": 1, "mode": 2, "oct": 2, "gate": 0.5, "rate": 2, "latch": 1 },
  "sections": { }
}
```

Field	Meaning
`format`	always `"ohpe-preset"`
`version`	`2`. Version-1 files (written by `state::savePreset`, without `kind` / `tags` / `engine`, with a `prefix` field) load as part presets.

Field	Meaning
`kind`	`"part"` – one part's sound; `"project"` – the whole instrument. Missing = part.
`name`, `author`, `comment`	text (UTF-8)
`tags`	list of strings. The **categories** `Bass`, `Keys`, `Pads`, `Leads`, `Plucks`, `Drums`, `FX`, `Arp`, `Textures`, `Vocal`, `Boom` (0.6), `Hybrid`, `Piano` (0.8) are fixed English keys (append-only; the UI translates them). Any other string is a free tag. A preset may carry several.
`engine`	engine ID of a part preset (`fm`, `wavetable`, `analog`, `pluck`, `grain`, `vocal`, `cluster`, `sampler`); for a project the engine of part 1. Used by the engine filter. Derived from `params.engine` when missing.
`params`	parameter values as **plain values** (the unit shown on the display: Hz, seconds, dB …). Part presets use IDs **without** the part prefix (`"analog.cutoff"`, not `"p1.analog.cutoff"`), so they load into any part. Project presets use full IDs.
`arp`	part presets only, optional (0.7): arpeggiator settings that travel with the sound – see Arp block.
`sections`	project presets only: `"sequencer"` (step patterns, chain, parameter locks), `"orbit"` (Orbit steps) and optionally `"samples"` (paths of loaded sample files).

Rules when loading:

- Parameters of the part (or, for a project, of the instrument) that the preset does **not** list return to their defaults – presets may be sparse.
- Unknown IDs are ignored, so presets from newer versions load in older ones and vice versa.
- A part preset keeps the part's MIDI channel (`pN.midich`, a setup value). Since 0.8 every part has an FX slot, so a part preset's effect loads into any part (before 0.8 only part 1 had one).
- Project presets neither save nor load the remote controls (`lib.*`, `ui.*`, `tape.edit.*`) and the sync setup (`sync.*`). The MIDI map and the tape audio are never part of a preset.
- Loading a preset never adds, removes or reorders parameters (tested).

Arp block

The arpeggiator is global (`seq.arp.*`, one arp that plays one part). A **part** preset can carry the arp settings that belong to its sound in an optional `"arp"` object:

Key	Parameter	Values
`on`	`seq.arp.on`	`1` (default when the block is there) or `0`; `true` / `false` work too
`mode`	`seq.arp.mode`	`0` Up, `1` Down, `2` Up/Down, `3` Random, `4` As Played, `5` Chord
`oct`	`seq.arp.oct`	octaves `1`–`4`
`gate`	`seq.arp.gate`	`0.05`–`1.0` (note length as part of a step)
`rate`	`seq.arp.rate`	`0` 1/4, `1` 1/8, `2` 1/16, `3` 1/32, `4` 1/4T, `5` 1/8T, `6` 1/16T, `7` 1/32T, `8` 1/4., `9` 1/8., `10` 1/16.
`latch`	`seq.arp.latch`	read for compatibility but **not applied** (since 0.9 fix batch p28): Latch is the player's setting, off by default; Save preset no longer writes it

What happens when a part preset loads into part N:

- **With an arp block:** the arpeggiator switches on and plays part N (`seq.arp.on` = 1, `seq.arp.part` = N) with the listed settings; keys the block does not list return to their defaults. There is no `part` key – the target is always the part the preset loads into. Like every other parameter change of a preset load, this is one change the host sees (automation / undo in hosts that record it).
- **Without an arp block:** the arpeggiator is left alone – with one exception: if a preset's arp block switched the arp on for part N and nobody has changed the arp since, loading a preset without arp block into part N switches the arp off again (its settings stay). An arp you set up by hand, an arp you changed after loading, or an arp playing another part is never touched. Ohpe remembers which preset switched the arp on only until the instrument is closed; in a reopened project the arp counts as set by hand.
- Project presets carry the arp in their `params` (`seq.arp.*`); an `"arp"` block there is ignored.
- Save preset → This part writes the arp block when the arp is on and plays that part, so a saved arp sound loads with its arp.

The factory Arp presets play while you hold keys and stop when you let go. To keep an arpeggio running hands-free, switch Latch on on the sequencer page (`seq.arp.latch`, MIDI-mappable): then the next key or chord pressed with no key held starts a new one.

Morph block

A **part** preset can carry the corners of the part's morph pad in an optional `"morph"` object (p16):

```
"morph": { "corners": [ { "name": "Glass Shimmer", "params": { "engine": 1, "wavetable.pos": 0.4 } },
                        { "name": "Breath Choir Pad", "params": { "engine": 5 } },
                        null,
                        { "name": "Crystal FM Pad", "params": { "engine": 0 } } ] }
```

- Four entries, corners A (top left), B (top right), C (bottom left), D (bottom right); `null` = the corner stands for the part's own sound.
- `params` like the preset's own `params`: local IDs without the part prefix, plain values, sparse (missing = default). The morph block (`morph.*`) and the MIDI routing (`midich`, `quant`, `chord`, `mpe`, `mpe.bend`, `mpe.press`, `mpe.timbre`) are not part of a corner.
- Loading a part preset into part N sets part N's corners from the block; a part preset **without** the block clears them. Save preset → This part writes the block when a corner is assigned.
- Projects and the plug-in state keep the corners of all parts in the section `"morph"` (`{"version": 1, "parts": [ {"corners": [...]}, ... ]}`).

Factory bank

The factory presets are the `*.ohpepreset` files in `resources/presets/factory/`. They are embedded into the plug-in and the app as binary data (CMake target `ohpe_presets`, namespace `OhpePresets`) and are read-only. To add or replace factory sounds, put files there and re-run CMake – nothing else needs to change. The test "library: the factory bank is embedded and every factory preset is valid" checks that every factory preset parses, uses only categories as tags and only existing parameter IDs.

The factory bank holds 412 part presets and 18 project presets (the free tag **MPE** marks presets with MPE / aftertouch mappings, see MIDI.md) – see FACTORY-SOUNDS.md. They are generated by `tools/presets/make_factory.py`; the plug-in test "factory sounds: every factory preset renders cleanly …" renders each of them and checks level, peaks and spectrum.

Factory sounds.

Ohpe ships with **412 part presets** for all eight engines, **18 project presets** (demo patterns that use all four parts, the sequencers and the drum kit), **six drum kits, ten recorded instruments** and (0.8) **four keyboard instruments** (grand and upright piano, tine and reed electric piano) for the Sampler and Grain engines. Open them with **LIBRARY**: pick a category on the left (Bass, Keys, Pads, Leads, Plucks, Drums, FX, Arp, Textures, Vocal, Boom, Hybrid, Piano), or an engine, and click through the list – every preset loads at once, so you hear it while you browse.

Playing the presets

- A **part preset** loads into the edit part (part selector 1–4). Its **Part Gain** (`pN.gain`, dB) matches its loudness to the other factory sounds: all of them sit around −18 … −12 dBFS RMS when played, so switching presets does not jump in level.
- **Chord Headroom** (`pN.chordroom`, 0–24 dB, p74) keeps chords clean: the part's sum is lowered smoothly by the number of notes sounding – 1 note 0 dB, 2 notes a third, 4 notes two thirds and 8 notes the full amount (a released note fades out of the count with its envelope). The gain moves down within a few milliseconds when notes start and back over about a quarter of a second when they end, so single notes are never touched and nothing pumps. It works right after the engines, so the body, every rack effect and the 3D stage get the room. Every factory part preset carries its own value: a held 8-note chord at full velocity stays at or below −1 dBTP without the master limiter, while single notes keep their level. Older sessions and new parts use 4.5 dB. MIDI-mappable like every parameter.
- The **FX rack** (up to six effects per part, see TAPE-MIXER-FX.md) is part of the preset and exists on **every part**, so a preset sounds the same in any part. (The projects made before 0.8 keep their parts 2–4 dry, as they were designed.)
- The macros on the four encoders are the engine's first four parameters; most presets are built so that turning them gives musical results (cutoff, tone, position, grain size …). The modulation envelope and the LFO are part of every preset.
- Presets with a recorded instrument (Sampler, Grain) need the factory sample bank, which is built into the plug-in and the app.

Project presets (demo patterns)

A **project preset** replaces the whole instrument: four parts, drum kit, mixer, effects, tempo and the patterns. Press **Play** (Sequencer Run) to hear them. They show how the sequencers combine: the step sequencer, Euclid and Orbit can each play a part – or, since 0.4, the **drum kit** (`seq.step.drums`, `seq.euclid.drums`, `seq.orbit.drums`). Orbit's four lanes make a natural drum machine (one lane per drum, each with its own length); Euclid's four rings play polyrhythmic drums. Parts without a sequencer are left for you to play along.

Sound revision 3

120 new part presets and four projects, made for modern productions and for playing:

- **Bass music:** reese basses (classic detuned, phased, flanged, hollow comb, distorted neuro, vowel and moving reeses) built on the Wavetable engine's new unison stack (up to eight voices) with a mono-safe low end (Bass Mono below 150 Hz after the FX rack); drum & bass subs and rollers; wobble and growl basses with the tempo-synced Wobble (it restarts with every note), the vowel Formant filter, FM on the wavetable, wavefolding (Distortion) and Multiband squash – chains of up to six effects per preset.
- **Leads** with stereo stacks, delays and halls; **stereo effects** (risers, impacts, zaps, whooshes).
- **Pianos** (category Piano): grand and upright piano recordings, modelled tine and reed electric pianos (suitcase autopan, tremolo, dream chorus) and an FM electric piano with velocity bark.

- **Thicker granular sounds and fuller arps:** eight grain voices with up to 95 grains per second, instrument and piano sources, echoes and halls; arps with stacks, ping-pong delays and halls.
- **Evolving atmospheres:** slow wavetable scans, two-bar autopans and filter tides, drifting combs and endless halls.
- **Hybrid** (category Hybrid, 18 presets): two engines in one part with the new **layer engine** and cross-modulation (engines/layer.md) – a pluck dissolving into a grain cloud, a reese with a singing vocal layer, ring-modulated FM grains, a piano with a halo of its own frozen recording, a pad chopped by a grain envelope.
- **Movement:** every new preset moves when a note is held (slow LFOs on the wavetable position, filter or pan, synced effects).
- **MPE:** 93 of the new presets carry the free tag **MPE**: pressure, timbre and per-note bend slides are mapped to suit each sound (see MIDI.md); pressure also works from any aftertouch keyboard.

New projects: **Neuro Engine** (neurofunk, 174 BPM), **Halftime Pressure** (dubstep, 140 BPM), **Liquid Roll** (liquid drum & bass, 174 BPM), **Evolving Space** (ambient, 66 BPM).

Boom

Boom is a category of its own (library chip Boom, German Boom): deep, tuned kicks and sub basses – the synthesised **Boom** kit (project Boom Kit), Sub Kick, the sub basses (Long Sub, Dirty Sub, Punch Sub, Glide Sine Bass) and five new ones built on a pure sine sub with a pitch dive (the 0.6 pitch envelope): Boom Long Sub (long, tuned, with a dive), Boom Distorted (driven hard), Boom Kick Bass (short: kick and bass in one), Boom Slide (overlap notes and it glides – legato portamento, for trap-style slides) and Boom Click (a bright click so it cuts on small speakers). Play them in the key of the song; Tune and the amp Decay shape the length. The project Sub & Glide is tagged Boom too.

Lo-fi

The project **Lo-Fi Beat** is classic lo-fi hip-hop: 78 BPM with a lazy, heavy swing, muffled kick, soft snare and quiet shuffled hats on the Lo-Fi kit, rootless ninth voicings (Fm9 – Bb13 – Ebmaj9 – C7#9) on **Dusty Keys** (strong tape wow, a bit-reduced edge), a warm round bass (**Mellow Lofi Bass**) on the roots and **Vinyl Crackle** under it all; the master is darkened by the 0.6 master low-pass (4.2 kHz) and tape saturation. **Lo-Fi Kit** is the same kit set-up as a starting point for your own beats. Vinyl Crackle is the Analog engine with its oscillators off (0.6 `analog.osclevel`) and the new crackle generator (`analog.crackle`) – hold any note.

Life

Eight part presets use **LIFE** so their held notes keep evolving: Analog Dawn, Warm Strings, Angel Pad, Evolving Tide, Crystal FM Pad, Bowed Haze, Warm Horizon (15 – 40 %) and Dusty EP (a hint, 15 %). Turn LIFE to 0 for the static version (USER-INTERFACE.md); the list lives in `tools/presets/factory_parts_life.py`.

Mod wheel

Every factory preset answers the **mod wheel** (the on-screen MOD wheel, CC 1 from a controller): a routing of the part's modulation matrix with the source (or via) Mod Wheel, chosen by the sound's role. With the wheel down every preset sounds exactly as before.

Sounds	Wheel up
Leads, solo voices, organs, brass	vibrato (about ±0.35 semitones; LFO 1 where it already runs at a vibrato rate, otherwise LFO 2 at 5.3 Hz with a short fade-in)
Tines, reeds, vibes with an auto-pan	deeper auto-pan / tremolo (LFO 1 depth)

Sounds	Wheel up
Pads, basses, plucks, keys, FX	the filter / tone opens (Timbre: cutoff, FM index, pluck brightness); where it is already fully open (recorded pianos and instruments at full tone) it closes instead – a softer, darker tone; struck bars and plates get a harder strike
Grain pads, textures and FX	grain spray
Vocal-engine choirs	the vowel moves
Subs, booms, kicks	harder, driven character (Character Hard)
Arps with a delay / echo	echo throw (longer delay / echo feedback, at most 0.75)
Morph presets	Morph X (already, through the morph pad's wheel control)
Project presets	every part, by its engine (spray, vowel, tone)

The routing sits in the first free matrix slot, so it can be changed or removed in the Mod view. Amounts were checked with internal renders: with the wheel fully up no preset gets more than 3 dB louder or clips. The script is `tools/presets/add_modwheel.py`; the core tests check that every factory preset has a wheel routing and that the wheel fully up changes the sound cleanly.

Showcase presets

48 part presets and 4 projects built around the newer features: Life, the morph pad, user wavetables analysed from the factory recordings, the resonator body (stage and rack effect), the character macros, the 3D motion paths (best with headphones), the SUB layer, the Expression page (velocity / aftertouch), pressure glide and tempo-synced movement (LFO 2 / 3 and motion paths measured in bars follow the tempo, whether the metronome clicks or not). The projects show what only a project carries: the harmony brain's Follow, recorded motion lanes, the performance pads' settings and four 3D parts at once. Every one is level-matched to the factory median, has a musical mod wheel and answers velocity; only the glide presets glide. They are listed in the library with the other factory presets (search for a feature, e.g. 3D or Morph).

Name	Category	Engine	Features	Description
Living Reese	Bass	Wavetable	Life, Character macros, SUB layer	A wide reese whose unison voices never settle: Life lets every voice drift on its own, Alive and Wide push the stack apart while a mono sine sub holds the floor. The mod wheel opens the wavetable tone.
Pressure Glide Bass	Bass	Analog	Glide pressure, Expression	A legato mono bass that slides between overlapping notes. Pressure stretches the slide (pressure glide): a light touch keeps it tight, leaning in turns it into a slow swoop. The mod wheel opens the filter.
Round Square Sub	Bass, Boom	FM	SUB layer, Expression	A deep FM sine bass whose weight comes from the SUB layer's soft square one octave down: hollow push in the low mids, clean on big systems. The mod wheel adds FM grit; velocity sets the punch.
Sub Pressure Bass	Bass	Analog	SUB layer, Expression, Character macros	A tight saw bass over a clean sine sub two octaves down that stays full under the closed filter. Velocity bites into the filter, pressure opens it; the mod wheel hardens the attack and drive.
Tempo Wobble Sub	Bass	Analog	Tempo-synced, SUB layer	A growling filter wobble locked to the tempo (LFO 2 on eighth notes, retriggered with each note) over a steady mono sub that never wobbles. The mod wheel deepens the wobble.
Wood Body Bass	Bass	Pluck	Resonator, Expression	A plucked string bass ringing a short wooden body tuned to each note: round, woody thumps. Velocity strikes harder and brighter; the mod wheel changes the body from wood to metal.
Alive Tine Keys	Keys, Piano	Sampler	Character macros, Life, Expression	The recorded tine electric piano with its character macros turned towards Alive and a little darker: slow drift, a living tone and a soft stereo bloom. Velocity brightens, the mod wheel deepens the tremolo.

Name	Category	Engine	Features	Description
Breathing Organ	Keys	Analog	Life, Character macros	A drawbar-like organ that is never quite still: Life drifts every voice a few cents and breathes in its tone, like an old instrument warming up. The mod wheel speeds up the rotating chorus.
Morph Living Keys	Keys, Morph	Analog	Morph pad	Four showcase keyboards on four engines: the breathing organ at home, the harp with its sympathetic strings to the right, the FM piano above, the struck plates in the far corner. Mod wheel = right, pressure = up.
Piano Scan Table	Keys	Wavetable	User wavetable, Expression	A wavetable analysed from the recorded grand piano, eight frames from the hammer to the fading tail. The mod envelope scans through it with every note, so each key starts bright and settles like the piano did; velocity opens the tone. The mod wheel holds it on the bright frames.
Pressure Reed Keys	Keys, Piano	Sampler	Expression	The recorded reed electric piano made for aftertouch: press into a chord and it growls (drive) and wavers (vibrato); velocity sets the bark. The mod wheel brings in a pulsing auto-pan.
Struck Plate Keys	Keys	Cluster	Resonator, Expression	Soft mallet keys striking small tuned metal plates: the attack blooms into bell partials that follow every note. Velocity strikes the plate harder; the mod wheel lengthens its ring.
Sympathetic Harp Keys	Keys, Plucks	Pluck	Resonator, Expression	A bright harp-like pluck that wakes a bank of sympathetic strings tuned to the chord you hold: chords keep shimmering after you let go. Velocity plucks harder; the mod wheel brings the sympathetic strings forward.
Velocity Glass EP	Keys	FM	Expression	An FM electric piano that answers the hands: soft notes are round and bloom in slowly, hard ones bark with a glassy tine; pressure adds a gentle vibrato. The mod wheel brings in a stereo tremolo.
Bright Wide Pad	Pads	Wavetable	Character macros, Expression	A pad played with its character macros: pressure turns up Bright (tone, warp and FM open together) and the mod wheel turns up Wide (unison and layer spread). Soft and narrow at rest, huge when you push.
Halo Spiral Pad	Pads, 3D	Wavetable	3D motion path, Tempo-synced	A glassy wavetable pad that spirals upwards around your head over sixteen bars and starts again below (motion path Spiral Up, measured in bars: it follows the tempo). Best with headphones; the mod wheel speeds up the spiral.
Living Strings	Pads	Analog	Life, Character macros	An ensemble string pad that keeps moving while you hold it: Life gives every voice its own slow drift and breathing tone, Alive adds detune motion. The mod wheel opens the filter.
Morph Pressure Pads	Pads, Morph	Analog	Morph pad	Living strings at home that turn into a dark pad with a sub floor (mod wheel), a tempo-synced wavetable tide (pressure) or a choir over sympathetic strings (both). Mod wheel = right, pressure = up.
Overhead Swell	Pads, 3D	Vocal	3D motion path, Expression	Breathy voices that rise from in front of you, pass over your head and land behind you (motion path Over the Head, four bars, in tempo). Pressure swells them; the mod wheel adds a choir vibrato. Best with headphones.
Recorder Breath Table	Pads	Wavetable	User wavetable, Life, Tempo-synced	A wavetable analysed from the recorded recorder, eight frames from the breathy onset to the singing body, slowly scanned in two-bar cycles of the tempo and drifting with Life: a soft, woody pad. The mod wheel opens the tone.
Sub Floor Pad	Pads	Analog	SUB layer, Expression	A dark, warm pad with a round SUB layer one octave down that gives chords a floor without a bass part. Velocity brings in more of the sub; the mod wheel opens the pad above it.
Sympathy Choir	Pads, Vocal	Vocal	Resonator, Life	A soft 'ah' choir that sets a ring of sympathetic strings humming in the chord you hold, with Life drifting each voice. The strings keep ringing when the choir stops; the mod wheel adds a choir vibrato.
Tempo Tide Pad	Pads	Wavetable	Tempo-synced, Expression	A warm wavetable pad whose tone rises and falls over two bars and drifts through the stereo field over one bar - both locked to the tempo, they run with any song speed. The mod wheel deepens the tide.
Weave Cluster 3D	Pads, 3D	Cluster	3D motion path, Tempo-synced	A wide cluster pad weaving left-right and front-back around the listener over eight bars (motion path Weave, in bars: it follows the tempo). Pressure lifts it overhead; the mod wheel widens the stack.

Name	Category	Engine	Features	Description
Bowed Body Lead	Leads	Analog	Resonator, Expression	A soft saw bowing a long open tube tuned to each note: a woody, cello-like lead. Pressure presses the bow (the tube's material darkens to wood); the mod wheel slides the bowing position.
Hard Edge FM Lead	Leads	FM	Character macros, Expression	A cutting FM lead voiced with the Hard macro (shorter attack, more feedback, more edge). Velocity adds bite, pressure vibrato; the mod wheel pushes Hard further into a snarl.
Living Flute Lead	Leads	Vocal	Life, Expression	A breathy flute-like voice whose pitch and breath drift a little with Life, like a real player. Pressure brings in vibrato and more breath, the mod wheel a wider vibrato.
Morph Lead Characters	Leads, Morph	Analog	Morph pad	One melody, four players: a bowed wooden tube at home, a breathy flute to the right, a hard FM lead above and a biting digital lead in the far corner. Mod wheel = right, pressure = up.
Pendulum Lead	Leads, 3D	Analog	3D motion path, Tempo-synced	A bright pulse lead swinging like a pendulum from one side of your head to the other, once per bar from the first note of each phrase (motion path Pendulum, in tempo). Pressure adds vibrato; the mod wheel slows the swing. Best with headphones.
Pressure Glide Lead	Leads	Analog	Glide pressure, Expression	A singing mono saw lead built for legato: pressure stretches the glide (pressure glide) and the Expression page adds vibrato as you lean in. The mod wheel opens the filter.
Velocity Bite Lead	Leads	Wavetable	Expression, Character macros	A digital wavetable lead that bites harder the harder you play: velocity opens the tone and drives it (Hard), pressure adds vibrato. The mod wheel adds wavetable FM.
Glass Plate Pluck	Plucks	FM	Resonator, Expression	A short FM pluck striking a thin glass plate tuned to the note: crisp attack, then bell partials that bloom. Velocity strikes harder; the mod wheel lets the plate ring longer.
Kalimba Body	Plucks	Sampler	Resonator, Expression	The recorded kalimba over a small wooden box body that rings with every note: warmer, rounder, a little hollow. Velocity brightens; the mod wheel opens the box (more body).
Living Nylon	Plucks	Pluck	Life, Expression	A warm nylon string with a little Life: every note drifts and breathes a touch differently, like a hand-played guitar. Velocity plucks brighter; the mod wheel moves the pluck towards the bridge.
Ping-Pong Space Pluck	Plucks, 3D	Pluck	3D motion path, Tempo-synced	A bright pluck that bounces front to back every half bar, starting again with every note (motion path Ping-Pong front/back, in tempo) inside a small room. Velocity plucks harder; the mod wheel speeds up the bounce. Best with headphones.
Living Bell Arp	Arp	FM	Life, Character macros	Soft FM bells arpeggiated in sixteenths, with Life and Alive keeping every repeat a little different - drift, tone and pan move on their own. The mod wheel throws the echoes.
Orbit Space Arp	Arp, 3D	FM	3D motion path, Tempo-synced	Bell arpeggios circling the listener once per bar (motion path Circle, one bar, in tempo) at arm's length, in a soft room. Pressure pulls them closer; the mod wheel moves the orbit further out. Best with headphones.
Resonant Comb Arp	Arp	Pluck	Resonator, Tempo-synced	A plucked-string arpeggio through the Resonator as a rack effect: a string body tuned to the chord you hold rings under every step. The mod wheel lets the body ring longer, the echo is a synced dotted eighth.
Tempo Gate Arp	Arp	Wavetable	Tempo-synced, Expression	A glassy arpeggio chopped by a sixteenth-note gate that is locked to the tempo (LFO 2 on the level, retriggered with each note). Holds a chord, plays in time at any tempo; the mod wheel lifts the echoes.
Bowed Metal Body	Textures	Grain	Resonator, Expression	The recorded bowed metal, stretched into a slow grain bed and resonating in a closed tube tuned to the chord: dark, singing, endless. Pressure moves through the recording; the mod wheel changes the tube's material.
Slow Weather	Textures	Cluster	Life, Character macros, Tempo-synced	A cluster drone that keeps changing like weather: full Life, Alive turned up, and a tone swell locked to four-bar cycles of the tempo. Hold a note and listen; the mod wheel brightens the sky.
Swarm Field 3D	Textures, 3D	Grain	3D motion path, Life	A cloud of glassy grains swarming around the listener (motion path Swarm, eight bars, in tempo) while Life scatters them further. Pressure brings the swarm closer; the mod wheel adds grain spray.

Name	Category	Engine	Features	Description
Whisper Room	Textures, 3D	Vocal	3D motion path, Expression	Breath and whispered vowels close to your ears, moving like someone walking around you (motion path Whisper, four bars, in tempo). Pressure adds breath; the mod wheel changes the vowel. Best with headphones.
Distance Walk Drone	FX, 3D	Cluster	3D motion path, Life	A low drone that walks towards you and away again over eight bars (motion path Distance Walk, in tempo): it swells, darkens in the distance, fills the room up close. The mod wheel lifts it above you.
Resonant Impact	FX	Analog	Resonator, Expression	A noise burst slamming into a large metal plate tuned to the note: a cinematic hit with a long, metallic ring. Velocity hits harder; the mod wheel shrinks the plate to a smaller, brighter ring.
Rising Halo Sweep	FX, 3D	Wavetable	3D motion path, Tempo-synced	A riser that spirals up and over your head in four bars (motion path Spiral Up, in tempo) while its tone opens. Hold a note for the build; the mod wheel speeds it up. Best with headphones.
Living Choir	Vocal, Pads	Vocal	Life, Character macros	A wide 'ah' choir where every singer drifts on their own (Life full up, Alive and Wide on): it sounds like a group, not a copy. The mod wheel adds a wider choir vibrato.
Pressure Vowel	Vocal, Leads	Vocal	Expression, Glide pressure	A solo voice that opens its mouth as you press: aftertouch moves the vowel from 'oo' to 'ah' and velocity sets the volume. Overlapping notes slide, longer with pressure. The mod wheel adds vibrato.
Harmony Follow Jam	Keys, Bass, Arp	Project	Harmony follow, Expression	Play chords and the band follows: Orbit plays Am-F-C-G on the glass piano (part 1, heard by the harmony brain); the bass plays one fixed riff and follows the chord root, the bells play fixed notes and snap to the chord tones, the pad holds one note that moves to the nearest chord tone. Replace the chords with your own on part 1. Press Play.
Motion Lanes Groove	Bass, Drums, Pads	Project	Motion lanes, Tempo-synced, Character macros	A groove whose sounds move by themselves: the motion recorder plays a filter and resonance lane on the bass that loops every five sixteenths, a twelve-sixteenth Bright lane on the pad and a seven-sixteenth decay lane on the plates - against the four-four beat on the Tight kit they drift apart and meet again. Record your own: LFO mode, Motion. Press Play.
Pads Performance Set	Drums, Bass, FX	Project	Performance pads, Tempo-synced	A house loop set up for the performance pads (Effects mode, Pads): stutter on sixteenths, a one-bar tape stop, a low-pass sweep over two bars and a reverb freeze, all on the beat. While the sweep closes, the pad opens its own filter against it; while any pad is held the plucks step back. Press Play, then hold the pads.
Binaural Scene	Pads, 3D, Textures	Project	3D motion path, Tempo-synced	Four sounds moving around your head at once: a pad spiralling upwards over sixteen bars, bells circling once per bar, plucks bouncing front to back on the half bar and a glassy grain swarm. Every path is measured in bars, so the whole scene follows the tempo. Headphones on, press Play.

Binaural presets

49 part presets made for headphones, each built around the part's binaural 3D stage: pads circling the head, near whispers and far-off voices, rain-like grain fields, plucks and bells placed around you, percussion whose hits land at different places (pitch or chance picks the direction), basses whose low end stays centred and mono while the overtones move, and arps that travel - some combined with Key Gravity or Voice Control. They carry the **3D** category: the library's 3D chip or a search for 3D / binaural lists them, and the headphones pictogram marks them in the list. Every one is level-matched to the bank target and has an audible mod wheel.

Name	Category	Engine	Motion	Description
Bowed Ring	Pads, 3D	Sampler	Circle	Bowed metal turning slowly against the clock around the listener (circle, four bars, reversed), with a long ringing room. The mod wheel darkens the bow. Best with headphones.
Cathedral Ring	Pads, 3D	Cluster	Over the Head	Major-seventh stacks hanging in a high vault: the pad passes over your head and back every eight bars while a slow drift keeps the stack shimmering. The mod wheel opens the tone. Best with headphones.

Name	Category	Engine	Motion	Description
Choir Around You	Pads, 3D	Vocal	Random Orbit	A soft 'aah' choir drifting on a random orbit around the listener over sixteen bars, sometimes above, sometimes behind. The mod wheel turns the vowel towards 'ooh'. Best with headphones.
Glass Horizon	Pads, 3D	Grain	Fly-Over	Long glass grains spread into a wide horizon that drifts over your head from far in front to far behind and back over eight bars. The mod wheel scatters the grains into a sparkle. Best with headphones.
Gravity Halo	Pads, 3D, Textures	Analog	Spiral Up	A breathy noise-and-saw pad pulled into the key by Key Gravity, spiralling up around you over sixteen bars. The mod wheel releases the gravity, so the pad drifts free of the key. Best with headphones.
Halo Carousel	Pads, 3D	Wavetable	Circle	A glassy unison pad that circles your head once every eight bars at a few steps' distance, its wide image turning with it. Pressure draws it closer; the mod wheel speeds the carousel up. Best with headphones.
Morning Mist	Pads, 3D	FM	Figure-8	Soft two-operator FM tones tracing a slow figure-eight through the space around your head, close in front, further behind. The mod wheel adds bell-like brightness. Best with headphones.
Slow Planet	Pads, 3D	Analog	Weave	A warm detuned saw pad weaving around the head on a wave-shaped orbit that rises and dips over sixteen bars. Velocity softens the attack; the mod wheel brings the planet in from far away. Best with headphones.
Close Whispers	Textures, 3D, Vocal	Vocal	Whisper	Breath and whispered vowels right at your ears, slipping from one side to the other like someone leaning in (motion path Whisper, four bars). Pressure adds breath; the mod wheel changes the vowel. Best with headphones.
Distant Voices	Textures, 3D, Vocal	Vocal	Breathe	Hummed voices far off in a large space, breathing nearer and further over sixteen bars. The mod wheel brings them into the room with you. Best with headphones.
Ember Field	Textures, 3D	Grain	Wander	Short crackling marimba grains like embers, popping up around you on a wandering loop that returns every four bars. The mod wheel stretches the crackle into glowing tones. Best with headphones.
Night Insects	Textures, 3D	Grain	Swarm	Tiny bright chirps of plucked strings buzzing around your head on a fast swarm path, near and far at once. The mod wheel scatters them wider. Best with headphones.
Rain Canopy	Textures, 3D	Grain	Swarm	Dense short grains falling like rain on leaves above and around you: a swarm of drops spread over the whole sphere, slightly overhead. The mod wheel makes the drops finer. Best with headphones.
Voice Halo	Textures, 3D, Vocal	Vocal	Circle	A sung halo whose place follows your own voice: sing into the input and your vowel turns it around your head, your level draws it nearer. Without input a gently orbiting choir. The mod wheel speeds the orbit. Best with headphones.
Wind Spiral	Textures, 3D, FX	Analog	Spiral Up	Filtered wind that spirals up and over your head every four bars and starts again at your feet, its resonance whistling softly. The mod wheel opens the gusts. Best with headphones.
Ceiling Harpsichord	Keys, 3D	Sampler	Over the Head	A harpsichord that rises over your head and settles behind you every two bars from the first note of each phrase, the room answering from all sides. The mod wheel darkens the quill. Best with headphones.
Glass Pendulum	Keys, 3D	Sampler	Pendulum	Glass harmonica tones swinging from one ear to the other on a pendulum that starts again with each phrase. The mod wheel softens the glass. Best with headphones.
Orbiting Tines	Keys, 3D	FM	Circle	A bell-tine electric piano drifting in a small circle in front of you once per bar, starting again with every note, like the instrument sways as you play. Velocity adds bite; the mod wheel adds tine brightness. Best with headphones.
Rotor Organ	Keys, 3D	Cluster	Circle	A drawbar-style organ whose sound really spins around your head (circle, one bar). The mod wheel throws the rotor into the fast speed. Best with headphones.
Vibes in the Round	Keys, 3D	Sampler	Ping-Pong F/B	Vibraphone bouncing between the front and the back of your head every half bar, the motor-like shimmer becoming real movement. The mod wheel moves the vibes further away. Best with headphones.

Name	Category	Engine	Motion	Description
Glass Droplets	Plucks, 3D	Pluck	fixed position	Bright glassy plucks falling like droplets at random points around your head, a little above. Velocity sets the splash; the mod wheel lets them ring longer. Best with headphones.
Harp Constellation	Plucks, 3D	Sampler	fixed position	Every harp note appears as a star at its own random place around and above you, then the next one lights up elsewhere. The mod wheel lifts the constellation higher. Best with headphones.
Kalimba Circle	Plucks, 3D	Sampler	Circle	Kalimba notes placed around a circle by pitch: low notes on your left, high notes behind you to the right, the whole ring turning once per bar. The mod wheel brightens the tines. Best with headphones.
Marimba Ricochet	Plucks, 3D, Drums	Sampler	Four Corners	Marimba hits ricocheting through the four corners around you, one corner per quarter note, crisp and close. The mod wheel throws them further out into the room. Best with headphones.
Zither Drizzle	Plucks, 3D	Sampler	Swarm	Zither strings scattered by a swarm path, every pluck coming from a slightly different place like a light drizzle. The mod wheel darkens the strings. Best with headphones.
Bell Ricochet	Arp, 3D	Sampler	Bounce	Glockenspiel notes in random order bouncing like a ball from left to right and back in front of you, six hops every two bars. The mod wheel speeds the bounce. Best with headphones.
Helix Sequence	Arp, 3D	Wavetable	Helix	A glassy wavetable arpeggio on a helix around your head: it circles up above you over two bars and spirals back down over the next two. The mod wheel sweeps the wavetable. Best with headphones.
Pulse Orbit	Arp, 3D	Analog	Figure-8	A resonant square arpeggio tracing a figure-eight through your head space every two bars. The mod wheel opens the filter. Best with headphones.
Satellite Arp	Arp, 3D	FM	Loop Orbit	FM bell arpeggios orbiting your head every two bars like a satellite, curling through four small loops on the way. The mod wheel pushes the orbit further out. Best with headphones.
Swarm Sequence	Arp, 3D	Cluster	Swarm	Detuned cluster stabs arpeggiated in eighths, each note flying off with a swarm that never lands twice in the same place. The mod wheel widens the stack. Best with headphones.
Gravity Bass Halo	Bass, 3D	Analog	Figure-8	A vowel-like analog bass whose partials Key Gravity tunes into the key; the halo of overtones circles in a figure-eight over the centred sub. The mod wheel deepens the gravity. Best with headphones.
Halo Reese	Bass, 3D	Wavetable	Weave	A wide detuned reese bass with a mono sub core; the moving top weaves around you on a four-bar orbit. The mod wheel sweeps the wavetable for growl. Best with headphones.
Orbit Sub	Bass, 3D	Analog	Circle	A round saw bass whose low end stays centred and mono while its upper harmonics circle your head every two bars. The mod wheel opens the filter. Best with headphones.
Pendulum Growl	Bass, 3D	FM	Pendulum	A growling FM bass whose bright edge swings from ear to ear every half bar while the fundamental stays planted in the middle. The mod wheel raises the FM index. Best with headphones.
Spinning Pluck Bass	Bass, 3D	Pluck	Circle	A plucked string bass: the woody attack spins once around your head every half bar, the low note stays centred. The mod wheel lets the string sustain. Best with headphones.
Comet Lead	Leads, 3D	Wavetable	Fly-By	A bright wavetable lead that streaks past you with every note it plays, rising in pitch as it approaches and falling as it leaves (Doppler). The mod wheel sweeps the tone. Best with headphones.
Ear to Ear Lead	Leads, 3D	Analog	Ping-Pong L/R	A singing analog lead hopping from one ear to the other every half bar, starting again with every new note – in front of you on the way right, behind you on the way back. Velocity opens the filter; the mod wheel opens it further. Best with headphones.
Overhead Flute	Leads, 3D	Sampler	Over the Head	A breathy recorder melody floating over your head and down behind you every two bars from the first note of each phrase, as if played from a gallery. The mod wheel brightens the breath. Best with headphones.
Vowel Orbit Lead	Leads, 3D, Vocal	Vocal	Circle	A singing vowel lead circling you every two bars from the first note of each phrase like a voice moving through the room. Pressure adds vibrato; the mod wheel walks the vowel from 'ah' to 'ee'. Best with headphones.

Name	Category	Engine	Motion	Description
Circling Hats	Drums, 3D	Cluster	Circle	Metallic cluster hats travelling around your head once every beat, so a straight hat pattern becomes a ring of ticks. The mod wheel speeds the ring up. Best with headphones.
Hand Drum Ring	Drums, 3D	FM	Circle	Hand-drum tones sitting in a ring around you: low strokes behind, open tones in front, each note at its own angle, and the ring slowly turning. The mod wheel speeds the turn. Best with headphones.
Round Toms	Drums, 3D	FM	fixed position	Tuned FM toms placed around you by pitch, low toms on the left and behind, high toms to the right and in front, so a fill rolls around your head. The mod wheel moves the kit further away. Best with headphones.
Drone Flyover	FX, 3D	Wavetable	Fly-By	A gritty engine drone that approaches from far in front, roars past overhead and fades out behind you (Fly-By with strong Doppler). Hold a note; the mod wheel speeds up the passes. Best with headphones.
Grain Vortex	FX, 3D, Textures	Grain	Swarm	Bursts of reversed vibraphone grains whirling around the listener on a swarm path with Doppler swoops. The mod wheel sprays the grains wider. Best with headphones.
Thunder Walk	FX, 3D, Textures	Analog	Distance Walk	A low, rolling rumble wandering near and far over eight bars like a storm crossing the valley, the sub kept mono and steady. The mod wheel brings the storm closer. Best with headphones.
Vortex Riser	FX, 3D	Analog	Spiral Up	A noise riser caught in a vortex: it spirals upwards faster and faster around your head over four bars. Hold a note for the build; the mod wheel tightens the vortex. Best with headphones.
Angel Overhead	Vocal, 3D	Vocal	Over the Head	A pure, high 'ooh' voice floating over your head and down behind you every eight bars, like a singer in the gallery. The mod wheel opens the vowel. Best with headphones.
Breath Circle	Vocal, 3D, Textures	Vocal	Circle	Close, rhythmic breaths circling your head every bar from the first breath of each phrase at whispering distance: intimate and slightly uncanny. The mod wheel adds a sung tone to the breath. Best with headphones.
Choir Weave	Vocal, 3D, Pads	Vocal	Weave	A dark male-formant choir weaving a wavy orbit around you over eight bars, rising and dipping as it turns. Pressure pulls it closer; the mod wheel lifts the formant. Best with headphones.

Drum kits

`drum.kit` chooses the recordings the 24 pads play. All kits have the same pad layout, so every pattern plays on every kit. Your own pad samples always win over the kit.

Kit	Sound	Made from
Studio	the original kit: close, room-style and overhead microphones mixed per piece; since 0.5 with punchier kicks (shorter ring, scooped box, beater click, a short pitched sub layer)	Virtuosity Drums (CC0), Ohpe's own processing
Tight	dry and short: close microphones, muted toms, short cymbals, a recorded hand clap; punchy kicks (0.5)	Virtuosity Drums, VCSL claps (CC0)
Room	big and live: room and overhead microphones, long tails, a layered clap with a short room tail	Virtuosity Drums, VCSL claps (CC0)
Lo-Fi	dry close microphones (0.5; before: the source's roomy vintage microphone) with a weighty kick, then band-limited, saturated and quantised to 12 bits – dusty but punchy	Virtuosity Drums, VCSL claps (CC0)
Hybrid	electronic kick, toms, hats, cowbell, claves and congas synthesised by Ohpe, layered with acoustic transients, real cymbals and claps	Ohpe's own synthesis, Virtuosity Drums, VCSL (CC0)
Boom (0.5)	every pad synthesised: a deep sub kick with a pitch envelope on pad 1 (C2), a long, driven sub kick on pad 2 (C#2) – tune and shorten them with the pad's Tune and Decay – a punchy snare, a crisp layered clap, closed / open / pedal hats, cowbell, rim, toms and percussion	Ohpe's own synthesis (no recordings)

Recorded instruments

`pN.sampler.inst` (Sampler) and `pN.grain.inst` (Grain) choose a factory instrument. The Sampler plays it when no sample of your own is loaded; Grain granulates one of its notes. All recordings are from the Versatile Community Sample Library (VCSL, CC0), multisampled (4–8 notes, marimba with two velocity layers), level-matched and mono.

Instrument	Notes	Loop
Kalimba	7	–
Marimba	7 × 2 velocity layers	–
Vibraphone	6	–
Glockenspiel	6	–
Harp	8	–
Harpsichord	7	–
Recorder	7	yes – set Loop to Forward
Singing Glass	4	yes
Zither	7	–
Bowed Metal (bowed vibraphone)	6	yes

Sustained instruments carry loop points; the presets that use them switch the Sampler's Loop to Forward. Licences and sources: THIRD-PARTY.md.

Bank curation

For 1.0 the whole factory bank was rendered offline and curated (`tools/presets/bank_curate.py`, record in `tools/presets/bank_curation.json`):

- **Near-duplicates** – presets of one engine and category were compared by their parameters and by their sound (spectrum and envelope of the standard render). Three that sounded the same as another preset were retired; the better one stays: Sine Slide (→ Glide Sine Bass), Phase Reese (→ Glide Reese, which keeps its phaser) and Sparkle Bell (→ FM Arp Bell). Variants that play differently (glide, pitch rise, morph, arp, Fingerboard, Expression) were kept on purpose.
- **Sessions stay safe** – a session or project that used a retired preset keeps its sound (the sound is stored in the session); the library simply shows no preset name for that part. Presets that stay keep their identity, so favourites, ratings and sessions still find them, even where the display name changed.
- **Names and tags** – every name is unique; Funk Clav is now Funk Snap Keys, the lo-fi presets are spelt Lo-Fi (Lo-Fi Bass, Mellow Lo-Fi Bass, Lo-Fi Kit); the FM electric pianos and Dream Tine Arp are found under Piano, Glide Sine Bass under Boom, Digital Pluck Arp under Plucks.
- **Loudness** – every part preset plays the standard phrase of its category (120 BPM, 8 s), every project its pattern with the master limiter as heard. Target: the loudest 3-second window (short-term loudness, EBU R128 K-weighting) at **−11 LUFS ±1.5 LU, true peak at or below −1 dBTP**. −11 LUFS is the bank's median, so switching between factory presets, RANDOM and BREED sounds (matched to the same median) does not jump in level. Only the level parameter was changed (Part Gain for part presets, Master Volume for projects); where the peak or the plug-in's level window (−21 … −10 dBFS active RMS) does not allow the full step, the preset stays as close as it can. Bright leads came down, soft subs and pianos went up; presets that clipped were brought below the ceiling.

How the sounds were checked

Every factory preset is rendered offline playing a typical phrase of its category (a bass line, chords, a lead line, an arpeggio, a groove …) and measured: loudness (target window −18 … −12 dBFS RMS), peak (no clipping, never more than −1.2 dBFS for part presets), DC, denormals, whether tails end, and energy above 16 kHz (no fizz unless intended, e.g. hi-hats). Spectrograms of all renders were reviewed and weak presets reworked. The plug-in tests repeat these checks for every embedded preset ("factory sounds: every factory preset renders cleanly …").

Tools, for contributors:

- `tools/presets/factory_parts.py`, `factory_projects.py` – the preset definitions; `make_factory.py` writes `resources/presets/factory/*.ohpepreset`, `make_docs.py` the lists below.
- `ohpe_render` (built with the core tests) renders presets to WAV and prints the metrics; `tools/presets/analyse.py` runs it on every factory preset, flags problems, draws spectrograms and writes the level calibration (`levels.json`).
- `tools/samples/build_samples.py` builds the drum kits and instruments (`resources/samples`) from the CC0 sources at pinned revisions.

Preset list

Bass (66)

Name	Engine	Also in	Description
Acid Line	Analog	Leads	Squelchy resonant saw with a sub for weight, accent-friendly velocity and a short glide.
Analog Reese	Analog	MPE	Two analog saws detuned wide through the ladder filter, phaser and squash: warm and fat.
Bounce Bass	Analog	–	Bright, punchy saw bass for octave-bouncing eighths.
Classic Reese	Wavetable	MPE	Two detuned saws beating slowly against each other, a clean sub underneath; mono below 120 Hz.
Click Sub	Analog	–	Sine sub with a sharp click on top, so the bass line reads on small speakers.
Dark Reese	Wavetable	MPE	Low, smoky reese: dark filter, slow chorus, light squash - for liquid and deep tracks.
Deep Pluck Sub	Analog	–	Plucked sub with a bright attack that fades into a clean sine; tight for fast lines.
Digital Monster	Wavetable	Hybrid, MPE	Digital FM table gated by a glass-grain layer an octave down, then wobbled: unstable and huge.
Dirty Sub	Analog	Boom	Sub driven into the ladder filter: growling harmonics that stay audible on small speakers.
Distorted Reese	Wavetable	MPE	Tube-driven reese with a grinding midrange; the squash keeps it dense and upfront.
Fat Saw Bass	Analog	–	Two detuned saws and a sub through the ladder filter; big but not boomy.
Flanged Reese	Wavetable	MPE	Saw stack with a slow jet flanger and chorus: swirling stereo top, solid mono bottom.
Fold Bass	Wavetable	Hybrid, MPE	Sine sub and an analog saw folded into each other: the fold amount makes it roar.
Glide Reese	Wavetable	MPE	Saw-stack reese through a slow, deep phaser, with constant-rate portamento: overlap notes and it slides like a fretless rubber band.
Glide Sine Bass	FM	Boom	Pure sine bass with legato glide for sliding bass lines; detached notes stay clean.
Grind Bass	Wavetable	MPE	Grinding, folded Grit table squashed hard: a constant, dense wall of mid-range.
Grit Bass	Wavetable	–	Growly Grit table with an envelope sweep; aggressive but controlled.
Growl FM Bass	FM	–	Feedback-edged FM bass with a slow timbre LFO for movement.

Name	Engine	Also in	Description
Growl Machine	Wavetable	MPE	Formant table with FM and an LFO on the position, a synced vowel filter and heavy squash: a talking growl.
Half-Step Wobble	Wavetable	MPE	Dotted-quarter wobble that runs across the bar line - heavy and lopsided.
Halftime Sub	Wavetable	–	Long, heavy sine sub with always-on glide: slow slides for half-time sections.
Hollow Reese	Wavetable	MPE	Reese through a negative-feedback comb that drifts slowly: hollow, metallic and dark.
House Organ Bass	Analog	–	Hollow square with an octave-down second oscillator; short and punchy.
Laser Wobble	Wavetable	MPE	Rising saw LFO on a band-pass wobble at sixteenth notes: zapping, laser-like movement.
Liquid Sub	FM	–	Pure, warm sine sub with a soft second harmonic; slides between overlapping notes.
Lo-Fi Bass	Analog	–	Round, warm bass: a soft square body over a strong sine sub, gently driven.
Log Drum Bass	Sampler	–	Low marimba bars played as a bass: woody, round and short.
Long Sub	Analog	Boom	Deep, long-decaying sub that slides between overlapping notes (glide); a soft click on the attack.
Mellow Lo-Fi Bass	Analog	–	Warm, round bass: a soft, low-passed body over a sine sub with gentle saturation and a hint of glide.
Metal Growl	Wavetable	MPE	Clanging metallic growl: FM table, a resonant comb on top and hard squash.
Moving Reese	Wavetable	MPE	Reese whose table position and filter breathe with a slow LFO - movement on held notes without automation.
Neuro Hybrid	FM	Hybrid, MPE	FM growl and a wavetable reese ring-modulating each other, folded and squashed: a neuro monster.
Neuro Reese	Wavetable	MPE	Distorted six-voice reese: driven filter, folded harmonics and a hard mono low end - made to be automated.
Neuro Swirl	Wavetable	3D, MPE	Talking formant growl swirling in a figure eight once a bar - the vowel moves, the space moves, the sub does not.
Orbit Reese	Wavetable	3D, MPE	Neurofunk reese whose growling mid range circles your head every two bars while the sub below 160 Hz stays dead centre and mono - club-safe movement.
Pendulum Wobble	Wavetable	3D, MPE	Half-time dubstep wobble that swings over your head and back with the bar: the wobble and the space breathe together.
Punch FM Bass	FM	–	Solid digital bass with a bright attack and a clean sine body.
Punch Sub	Analog	Boom	Short, punchy sub with a snappy filter click; locks in with the kick.
Reese Drift	Analog	–	Wide detuned saws slowly beating against each other; dark and moving.
Riddim Saw	Wavetable	MPE	Hard-clipped saw with a quarter-note band-pass wobble: sparse, heavy and minimal.
Rise Bass	Analog	–	Driven bass that slides up a fifth into every note; great for syncopated lines.
Roller Bass	Analog	–	Punchy square-ish roller: a short filter snap on every note, solid sub, dense with squash.
Rolling Bass	Analog	–	Tight, driven techno bass for rolling sixteenths: short filter snap, solid sub.
Rolling Sub	Wavetable	–	Round sub with a tiny filter blip and gentle saturation for rolling bass lines; mono and steady.
Round Sub	Analog	–	Warm square and sub oscillator with a soft filter blip; sits under anything.
Rubber Bass	Analog	Plucks	Bouncy plucked bass: fast filter decay, round sub.
Screech Bass	Wavetable	MPE	Screaming high-ratio FM with a folded top: for the high, aggressive bass answers.
Slap FM Bass	FM	–	Percussive FM bass: the index snaps shut like a slapped string.
Square Wobble	Wavetable	MPE	Hard sixteenth on / off wobble with a square LFO - choppy and rhythmic.
Step Sub	Wavetable	–	Sub that drops a fifth into every note: a little pitch kick for steppy, half-time lines.
Sub Reese	Wavetable	MPE	Gentle reese on top of a big sine sub - the sub stays mono and steady, the top moves.

Name	Engine	Also in	Description
Table Sub	Wavetable	–	Near-sine wavetable bass with a sub; pure low end with a touch of edge.
Talker Bass	Wavetable	MPE	Formant table walked by a fast LFO and an envelope: each note says something different.
Talking Bass	Wavetable	–	Formant table swept by the envelope: a vowel 'wow' on every note.
Tear Reese	Wavetable	MPE	Aggressive neuro reese: FM-edged digital table, heavy multiband squash with lifted highs.
Tech Stab Bass	FM	–	Hard, short tech bass: FM bite, hard clipper and squash for clinical, percussive lines.
Triplet Wobble	Wavetable	MPE	Eighth-triplet wobble on a driven saw stack: the bouncing half-time classic.
Unison Bass	Cluster	–	Octave-stacked saws with gentle detune; thick and wide for drops.
Upright Pluck	Pluck	Plucks	Plucked string with a big wooden body: an acoustic-style bass.
Vowel Growl	Wavetable	MPE	Quarter-note vowel sweep (a - e - i - o - u) over a folded saw: the classic 'talking' bass.
Vowel Reese	Wavetable	MPE	Reese that speaks: formant table under an LFO plus a synced vowel filter walking a - o - i.
Vox Reese	Wavetable	Hybrid, MPE	A reese with a singing vocal layer on top: the formants ride the bass, a synced vowel filter makes it talk.
Warm Roller	Analog	–	Ladder-filtered saw with sub for warm, round rolling basslines in liquid tracks.
Wide Reese	Wavetable	MPE	Eight-voice reese spread across the stereo field, with a Haas spread on top and a mono low end.
Wobble Bass	Analog	FX	Tempo-ish filter wobble on a driven saw stack; turn the LFO rate to taste.
Wub Machine	Wavetable	MPE	Saw stack through a tempo-synced eighth-note wobble filter that restarts with every note.

Keys (38)

Name	Engine	Also in	Description
Bright Grand	Sampler	Piano, MPE	Grand piano opened up for pop: brighter, lightly squashed, short room.
Bright Poly Wide	Analog	MPE	Classic bright poly synth, chorused and widened - chords that fill the room.
Classic Tine	Sampler	Piano, MPE	Modelled tine electric piano: bell-like when soft, barking when hit hard.
Cluster Organ	Cluster	–	Octave stack of soft squares: a cheesy combo organ.
Concert Grand	Sampler	Piano, MPE	Grand piano recordings, two velocity layers, a touch of hall.
Crystal Organ	FM	Hybrid, MPE	Additive FM organ with a glass wavetable layer, circling in stereo with a slow autopan.
Digital EP	FM	Piano	Brighter electric piano with more bark when played hard.
Drawbar Organ	FM	–	Additive organ: six sine drawbars, gentle vibrato and a rotary-style chorus.
Dream Tine	Sampler	Piano, Pads, MPE	Tine piano drenched in chorus, dotted delay and a long hall.
Dub Stab	Analog	Pads	Dark detuned chord stab whose filter drifts slowly open and shut; give it delay and reverb.
Dusty EP	FM	Piano	Warm electric piano with tape wow and flutter (slow pitch LFO) and chorus; made for lo-fi chords.
Dusty Keys	FM	Piano	Mellow, dark electric piano for lo-fi chords: strong tape wow on the pitch, a bit-reduced edge, soft attack.
FM Bark Piano	FM	Piano, MPE	Two-pair FM electric piano: the tine pair rings, the bark pair bites with velocity.
FM Vibes	FM	Plucks	Mellow metallic bars with tremolo.
Felt Grand	Sampler	Piano, MPE	Dark, soft grand: the top rolled off, a slow release and a warm hall - intimate.
Funk Snap Keys	FM	Plucks	Snappy, nasal FM keys with a quick bite; play short and syncopated.
Glass Body Keys	Cluster	MPE	Soft keys through a glass plate tuned to each note: a glassy halo over the chord.
Glass Keys	Wavetable	–	Bell-like Glass table that darkens as the note decays.

Name	Engine	Also in	Description
Glide Poly Synth	Analog	Leads	Poly synth whose chords always slide from the chord before (poly glide).
Halo Grand	Sampler	Hybrid, Piano, MPE	Grand piano with a halo: its own recording frozen into a grain cloud that blooms behind every chord.
Harmonic Organ	Wavetable	–	Additive Harmonic table: a soft, slightly breathy organ.
Harpsichord	Sampler	Plucks	Recorded harpsichord (seven notes, CC0); baroque lines and bright comping.
Mallet Keys	Pluck	Plucks	Physically modelled wooden bars; soft mallet.
Marimba	Sampler	Plucks	Recorded marimba with two velocity layers.
Minor Stab	Cluster	–	One finger plays a minor-seventh chord stab; detuned and punchy.
Music Box	Sampler	Plucks	Glockenspiel softened into a music box; lullabies and toy melodies.
Pop Brass	Analog	Pads	Lush, detuned poly brass with a quick swell and chorus: classic synth-pop chords.
Reed Keys	Sampler	Piano, MPE	Modelled reed electric piano: nasal and woody, with a gritty bark when driven.
Reed Tremolo	Sampler	Piano, MPE	Reed piano through its classic amplitude tremolo - soul and gospel.
Scoop Keys	FM	Piano	Electric piano where every note scoops up a whole tone: soulful, a little lazy.
Soft Brass	Analog	Pads	Polyphonic brass: a slow filter swell on detuned saws.
Suitcase Tine	Sampler	Piano, MPE	Tine piano with the stereo autopan of a suitcase amp and a small room.
Tape Ghost Keys	Sampler	Hybrid, Piano, MPE	Tine piano with a ghost of itself: a detuned grain copy, crushed and flanged like an old resampled tape.
Tine Piano	FM	Piano	Electric piano: soft body plus a bell-like tine that fades with the key. Chorus in the FX slot.
Upright Keys	Sampler	Piano, MPE	Upright piano recordings with a little room: honest and close.
Velvet Stab	Analog	–	Short filtered chord stab for house and disco.
Vibraphone	Sampler	–	Recorded vibraphone with a gentle motor tremolo.
Wide Organ	Wavetable	–	Drawbar-like organ with a rotating stereo pan and a short hall.

Pads (48)

Name	Engine	Also in	Description
Analog Dawn	Analog	–	Pulse pad whose filter opens slowly, with a slow phaser.
Angel Pad	Vocal	Vocal	High 'oo' voices morphing slowly towards 'ah'.
Aurora Scan	Wavetable	Textures, MPE	Glass table scanned slowly, circling in stereo with a two-bar autopan.
Bowed Haze	Grain	–	Bowed vibraphone grains: a soft metallic halo.
Bowed Metal	Sampler	Textures	Bowed vibraphone recordings (CC0), looped: glassy and eerie.
Brass Slide	Analog	Keys	Swelling poly brass with quick portamento between chords.
Breath Choir Pad	Vocal	Vocal	Airy 'ah' choir with plenty of breath.
Breath Grains	Grain	Textures, MPE	Recorder breath sliced into a soft, wide grain pad.
Chop Pad	Cluster	Hybrid, MPE	A wide chord pad chopped and stuttered by the envelope of a kalimba grain layer.
Crystal FM Pad	FM	–	Five slightly detuned sine carriers sharing one modulator.
Crystal Slide	FM	–	Shimmering FM pad: detuned sine carriers sliding between chords.
Drift Stack	Cluster	Textures	Detuned saw stacks drifting slowly from chord to chord.
Evolving Tide	Wavetable	MPE	Harmonic table drifting through its frames on a very slow LFO, eight-voice stack, flanger and hall.
Fall Haze	Wavetable	FX	Hazy pad whose notes sink a fifth from above into the chord.
Fifth Cloud	Cluster	–	Open fifths with slow drift: cinematic and neutral.

Name	Engine	Also in	Description
Frozen Cloud	Grain	Textures	Granular freeze of the built-in harmonic texture.
Frozen Vibes	Grain	Textures, MPE	Vibraphone grains held still, with a slow flanger: a frozen, shimmering chord.
Ghost Organ	Wavetable	Keys	Hollow harmonic organ with a slow autopan and a cavernous hall.
Glass Engine	Grain	Hybrid, MPE	Singing-glass grain cloud fused with a harmonic wavetable layer that scans slowly; a touch of ring modulation.
Glass Glide	Wavetable	Textures	Wide glass pad with slow, dreamy slides between chords.
Glass Grains	Grain	Textures	Grains of a singing wine glass (CC0 recording), slowly scanned.
Glass Shimmer	Wavetable	Textures	Glass table with wide unison: airy, glittering pad.
Glide Strings	Analog	–	Ensemble strings whose chords slide into each other (poly glide); legato chord changes.
Halo Drift	Cluster	3D, MPE	Detuned cluster choir wandering around and above you on a slow random orbit - never the same place twice.
Harmonic Swell	Wavetable	–	Harmonics bloom in as the pad swells.
Infinite Hall	Cluster	MPE	Major-seventh stack in an enormous hall that never seems to end.
Metal Strings	Pluck	Hybrid, MPE	Metallic plucked strings folded into a slow cluster pad - an orchestra made of wire.
Night Drive	Analog	Textures	Dark resonant pad breathing with a slow filter LFO.
Orbit Atmos	Wavetable	3D, MPE	Wide wavetable atmosphere circling your head once every eight bars on a gently tilted orbit, in a soft room. Best with headphones: slide lifts it, pressure pulls it closer.
Orbit Pad	Grain	Hybrid, MPE	Harp grains orbiting an FM layer, ring-modulated softly and panning over two bars.
Overhead Choir	Vocal	Vocal, 3D, MPE	Breathy 'ooh' choir swinging from in front of you, over your head to behind you and back - a pendulum through the room.
Piano Grains	Grain	Textures, Piano, MPE	Grand piano recordings frozen into a slow-moving grain cloud: a piano that breathes.
Rise Bloom	FM	–	Shimmering FM pad blooming up from an octave below; long and bright.
Rise Choir	Vocal	Vocal	Airy 'ah' choir that scoops up a fourth into every chord.
Rise Glass	Wavetable	Textures	Wide glass pad: each note glides up a fifth into place, dreamy and slow.
Rise Strings	Analog	–	Ensemble strings whose notes rise an octave up into the chord on every new note.
Rise Warmth	Cluster	–	Soft detuned saw stack whose notes lean up a whole tone into the chord.
Slide Choir	Vocal	Vocal	Airy 'ah' choir; each voice slides to the next chord.
Slow Motion Choir	Vocal	Vocal, MPE	Ooh choir that swells in slowly, chorused and wide, in a long cathedral hall.
Spectral Morph	Wavetable	Hybrid, MPE	Formant table morphing under a breathy vocal layer, both through a drifting comb: spectra that melt into each other.
Super Stack	Cluster	–	Seven detuned saws per key, wide and lush.
Sus Horizon	Cluster	–	Suspended-second stacks that never resolve; ambient beds.
Sympathetic Pad	Analog	MPE	A warm pad that wakes a set of sympathetic strings tuned to the chord you hold - they keep ringing after release.
Tidal Pad	Analog	MPE	Warm saw pad whose filter rolls in and out over two bars like a tide.
Tine Bloom	Grain	Textures, Piano, MPE	Tine-piano grains blooming slowly, chorused, with a dotted echo.
Vowel Pad	Wavetable	–	Formant table drifting between vowels; slow attack.
Warm Horizon	Cluster	MPE	Soft, wide add-nine pad with gentle drift: warm and open, sits behind everything.
Warm Strings	Analog	–	Ensemble strings: detuned saws, an octave up, gentle chorus.

Leads (34)

Name	Engine	Also in	Description
Binaural Lead	Wavetable	3D, MPE	Expressive lead placed in front of you: slide moves it up over your head, pressure brings it right up to your ear, a slow drift keeps it alive. Made for MPE controllers and headphones.
Bowed Tube	Analog	MPE	Breathy noise bowing a closed bore: a hollow, clarinet-like tone that swells with pressure.
Breath Lead	Vocal	Hybrid, MPE	Breathy sung tone fused with a soft saw stack: a lead between a voice and a synth.
Chip Square Lead	Wavetable	MPE	Narrow pulse lead with a crunchy bit crusher and a quick vibrato - retro, but wide.
Classic Mono Lead	Analog	–	Saw lead with a little vibrato, glide and a dotted delay.
Digital Lead	Wavetable	–	Warped Digital table with unison: modern and cutting.
Dream Lead	Wavetable	Pads, MPE	Soft triangle lead with a slow flanger and a long, airy reverb.
FM Brass Lead	FM	–	Brassy FM lead: the index rises with a short swell.
Fall Lead	Analog	FX	Square lead that drops a fifth from above into every note.
Formant Lead	Wavetable	Vocal	Vowel-like lead that opens up with the envelope.
Growl Lead	Wavetable	MPE	Biting FM-formant lead that growls with an LFO: a bass-music lead for the drop's answer phrase.
Hot Wire	Analog	–	Driven, resonant saw for rock-ish solos.
Hyper Saw Lead	Cluster	MPE	Huge detuned saw stack squashed upfront, with a stereo delay: festival main-room energy.
Legato Glide Lead	Analog	–	Mono saw lead with legato portamento: overlapping notes glide, detached notes do not ret-rigger a slide.
Mono Glide Lead	Wavetable	MPE	Expressive mono lead with legato glide, light drive and a dotted ping-pong delay.
Octave Lead	Cluster	–	Octave-stacked pulses with a quick filter blip.
Pluck Lead Wide	Wavetable	Plucks, MPE	Snappy plucked lead: a quick filter snap, stereo stack and a dotted delay.
Pop Lead	Analog	–	Bright saw and an octave square with vibrato: a clear, singing synth-pop lead.
Portamento Square	Analog	–	Hollow square that always glides, at a constant rate: big jumps slide longer than small steps.
Recorder	Sampler	–	Recorded alto recorder, looped, with light vibrato.
Ring Machine	Analog	Hybrid, MPE	Saw lead ring-modulated by an FM sine a fifth up: metallic, vocal and aggressive, with delay.
Rise Lead	Analog	–	Saw lead that swoops up an octave into every note; a dotted delay behind it.
Saw Unison Lead	Wavetable	–	Classic table at the saw position with wide unison.
Screamer Lead	Analog	MPE	Resonant, overdriven saw lead that screams on high notes; ping-pong delay keeps it wide.
Shimmer Lead	FM	Keys, MPE	Glassy FM lead with a bell edge, chorus and a long bright hall.
Sine Whistle	FM	–	Almost pure sine with a breath of modulation; delicate melodies.
Soft Flute Lead	Analog	–	Breathy, resonant low-pass on a square: a soft whistle tone.
Square Hero	Analog	–	Hollow square plus an octave: a bright, singing lead.
Stereo Saw Lead	Analog	MPE	Classic two-saw lead made wide: chorus, Haas spread and a short room.
Super Lead	Cluster	–	Seven-saw unison lead: festival size.
Super Wide Lead	Wavetable	MPE	Eight-voice saw stack spread across the stereo field, chorus, ping-pong delay and a big hall.
Twist Lead	Wavetable	MPE	Twisted, sync-like lead: the envelope bends the warp, the stack spreads it wide.

Name	Engine	Also in	Description
Vox Lead	Vocal	Vocal	Solo synthetic voice with expressive vibrato and glide.
Vox Synth Lead	Vocal	Vocal, MPE	Sung synth lead: an 'ah' with vibrato, widened by a Haas spread, with delay.

Plucks (23)

Name	Engine	Also in	Description
Concert Harp	Sampler	–	Recorded concert harp (eight notes, CC0).
Drone Strings	Pluck	MPE	Plucked steel over sympathetic strings tuned to the chord: buzzing, droning, endlessly resonant.
Echo Harp	Pluck	MPE	Plucked nylon-like string with dotted ping-pong echoes and a wide room.
FM Koto	FM	–	Plucked FM: a bright attack that quickly mellows.
Future Chord	Cluster	Keys, MPE	Wide minor-seventh chord stab with squash and a short shimmer - one key, one chord.
Gamelan Plate	Pluck	MPE	Short mallet hits on a long bronze plate, stretched partials and a shimmering tail.
Glass Pluck	Wavetable	–	Glass table pluck with a bell-like sparkle.
Harp Pluck	Pluck	–	Long-ringing harp-like string with a soft attack.
Kalimba	Sampler	Keys	Recorded thumb piano with its buzzy, woody tines (CC0).
Koto Pluck	Pluck	–	Bright, twangy string with a resonant body.
Muted Pick	Analog	–	Short palm-muted square pluck for fast lines.
Nylon Guitar	Pluck	–	Warm nylon string plucked near the soundhole.
Pluck Nebula	Pluck	Hybrid, MPE	A plucked string that dissolves into a kalimba grain cloud: the string fades, the cloud swells and lingers.
Plucked Plate	Pluck	MPE	A bright pluck striking a tuned metal plate: bell-like partials that bloom after the attack.
Poly Pluck	Analog	–	Classic synth pluck: snappy filter on a detuned saw pair.
Resonator Hit	Pluck	Hybrid, MPE	A struck metal bar feeding a short grain resonance and a tuned comb: percussive, ringing, strange.
Steel Drum FM	FM	Keys	Tropical pan tones from inharmonic FM.
Steel String	Pluck	–	Bright steel string plucked near the bridge.
Stereo Pluck	Wavetable	MPE	Bright six-voice pluck spread wide with a dotted ping-pong delay and hall.
Table Pluck	Wavetable	–	Wavetable pluck: bright saw frame closing fast.
Wide Glass Pluck	FM	Keys, MPE	FM glass pluck with a Haas spread and a bright, airy reverb.
Zip Pluck	Wavetable	–	Bright pluck with a fast upward zip into the note.
Zither	Sampler	–	Recorded zither: shimmering plucked strings (CC0).

Drums (11)

Name	Engine	Also in	Description
Cluster Hat	Cluster	–	Stacked narrow pulses detuned into a metallic hat.
FM Snare	FM	–	Feedback noise plus a body tone: a crisp digital snare.
FM Tom	FM	–	Pitched FM drum with a woody knock.
Filter Kick	Analog	–	Kick from a self-oscillating ladder swept down by the envelope. Play around C1-C2.
Metal Hat FM	FM	–	Inharmonic FM noise: closed and open hats depending on the key length.
Noise Snare	Analog	–	Band-passed noise and tone: an electronic snare.
Sub Kick	Analog	Bass, Boom	Tunable deep kick: a self-oscillating filter tuned to the key, swept down from above; long boom, driven.

Name	Engine	Also in	Description
Tick Hat	Analog	–	High-passed noise tick for hi-hat patterns.
Tuned Tom	Pluck	–	Modelled drum-like bar with a big body; play it in octaves.
Wood Block	Pluck	–	Hard mallet on a short wooden bar.
Zap Perc	Wavetable	FX	Digital table snapped shut: zaps, clicks and lasers.

FX (19)

Name	Engine	Also in	Description
Alien Signal	FM	–	Random-stepping FM timbre: radio chatter from far away.
Detune Swarm	Cluster	Textures	Seven wildly drifting oscillators: a swarm.
Doppler Fly-By	Wavetable	3D	A siren tone racing past from left to right and back with every note - the pitch bends with the Doppler effect as it passes.
Downlifter	Analog	–	Falling noise and saw sweep over three seconds with a jet flanger - the end of the build.
Glitch Spray	Grain	Textures	Tiny, fast grains crushed and scattered into a stereo delay: digital spray.
Gong Hit	FM	Textures	Inharmonic FM gong with a long shimmer.
Granular Rise	Grain	–	Grains scan forward and climb while the key is held.
Grit Sweep	Wavetable	–	Noisy Grit table opening over two seconds.
Impact Boom	FM	Boom	Sub impact with a metallic crack, squashed and thrown into a huge hall.
Laser Drop	Analog	Drums	Resonant falling sweep; short hits become lasers.
Noise Riser	Analog	–	Hold a key: filtered noise climbs for four seconds.
Reverse Spray	Grain	Textures	Mostly reversed harp grains sprayed across the stereo field.
Siren Pan	FM	–	Wailing FM siren whose pitch swings and pans across the speakers.
Spiral Riser	Analog	3D	Hold a key: noise and saws climb for four seconds while spiralling from below your ears to high above your head.
Stereo Zap	Wavetable	–	Laser zap diving two octaves, thrown left and right by a ping-pong delay.
Stutter Glitch	Grain	–	Tiny, sparse grains: glitchy stutters from the marimba.
Table Scanner	Wavetable	Textures	Digital table swept by a slow LFO: robotic movement.
Whoosh	Analog	–	Filtered noise swept by a one-bar band-pass wobble, spread wide: a passing whoosh.
Wide Riser	Analog	–	Noise and saws rising over four seconds, flanged and wide, into a long hall.

Arp (22)

Presets marked Arp: switch the arpeggiator on for the part they load into; it plays while you hold keys and stops when you let go (switch Latch on on the sequencer page to keep it running). Loading a preset without arp into that part switches it off again. See PRESETS.md.

Name	Engine	Also in	Description
Analog Arp	Analog	Plucks	Resonant square blips with a ping-pong delay. Arp: Up/Down · 1/16 · 2 oct
Bell Cascade Arp	FM	Keys	FM bells cascading down three octaves in triplets, with echoes and a bright hall. Arp: Down · 1/16T · 3 oct
Bubble Arp	Analog	–	Narrow pulse through a resonant filter: bubbly and round. Arp: Up · 1/16 · 2 oct
Chord Stab Arp	Cluster	Keys	Rhythmic chord stabs (chord mode, eighths) from a wide saw stack - instant house groove. Arp: Chord · 1/8 · 1 oct
Digital Pluck Arp	FM	Plucks	Clicky digital pluck that loves fast sixteenths. Arp: Up · 1/16 · 2 oct

Name	Engine	Also in	Description
Dream Tine Arp	Sampler	Keys, Piano	Tine piano arpeggio in eighths with chorus, delay and hall: dreamy and full. Arp: Up/Down · 1/8 · 2 oct
FM Arp Bell	FM	Keys	Short glassy FM bells for sequences. Arp: Up/Down · 1/16 · 2 oct
Figure Eight Arp	Wavetable	3D	Digital arp tracing a lying eight around you - left front, across, right behind - over two bars. Arp: Up/Down · 1/16 · 2 oct
Formant Arp	Wavetable	Vocal	Vowel blips for talking sequences. Arp: Up · 1/16 · 2 oct
Glock Arp	Sampler	Keys	Recorded glockenspiel for sparkling arpeggios (CC0). Arp: Up · 1/16 · 2 oct
Mallet Echo Arp	Sampler	Plucks	Glockenspiel arpeggio with long dotted echoes - sparkling and wide. Arp: Up · 1/8 · 3 oct
Neuro Arp	Wavetable	Bass	Growling FM sixteenths in one octave: a rolling neuro sequence from a single held note. Arp: As Played · 1/16 · 1 oct
Orbit Arp	Pluck	3D	Sixteenth-note pluck arpeggio flying round your head once a bar - close, bright and fast. Arp: Up · 1/16 · 2 oct
Pizzicato Arp	Pluck	Plucks	Short plucked strings for ostinatos. Arp: Up/Down · 1/16 · 2 oct
Random Spark Arp	FM	FX	Random thirty-second FM blips over four octaves through a ping-pong delay: sparks. Arp: Random · 1/32 · 4 oct
Ripple Arp	Analog	–	Analog saw arp in sixteenth triplets with a sweeping phaser and stereo echo. Arp: Up · 1/16T · 2 oct
Spiral Arp	FM	3D	Bell arpeggio spiralling up from ear height to above your head over four bars, then dropping back for the next round. Arp: Up · 1/16 · 3 oct
Table Arp	Wavetable	–	Wavetable sequence sound: the position snaps with each note. Arp: Up/Down · 1/16 · 2 oct
Trance Gate Arp	Cluster	Leads	Supersaw stack on a sixteenth arp over two octaves, ping-pong delay and hall. Arp: Up · 1/16 · 2 oct
Wide Pluck Arp	Wavetable	Plucks	Six-voice stereo pluck, up / down over three octaves, dotted ping-pong delay. Arp: Up/Down · 1/16 · 3 oct

Textures (26)

Name	Engine	Also in	Description
Backwards Kalimba	Sampler	FX	The kalimba played in reverse: swells that end in a pluck.
Breath Bore	Vocal	MPE	A whispering voice blown through an open pipe: airy flute tones with a soft edge.
Breath Texture	Vocal	Vocal	Nearly unvoiced breath through slowly moving vowels.
Breathing Machine	Vocal	Vocal, MPE	Breathy vocal drone whose vowel opens and closes over two bars.
Dark Matter	Wavetable	Pads, MPE	Low Grit table with a crawling phaser and a slowly moving position - ominous.
Deep Space Drone	Cluster	Pads, MPE	Fourths stacked low, a drifting comb on top and an endless hall.
Dust Orchestra	Grain	MPE	Zither grains thrown backwards and forwards, a stereo delay trail - an orchestra of dust.
FM Grains	Grain	Hybrid, MPE	Bowed-metal grains ring-modulated by an FM bell layer: grains that clang in tune.
Fourths Drift	Cluster	Pads	Stacked fourths with heavy drift: an unresolved haze.
Glacier	Grain	MPE	Glass grains crawling at a glacial scan rate, very wide, endless reverb.
Glass Swarm	Grain	Pads, MPE	Dense cloud of singing-glass grains, eight voices deep, chorused into a huge hall.
Grit Drone	Wavetable	–	Slowly churning Grit table; hold one low note.
Harp Cloud	Grain	Pads, MPE	Harp recordings scattered into a dense cloud with ping-pong echoes.
Harp Mist	Grain	Pads	Long, soft harp grains blurred into a mist.
Kalimba Dust	Grain	–	Tiny kalimba grains scattered like dust.
Kalimba Storm	Grain	MPE	Kalimba grains at full density with reversed fragments - a rattling, tuned storm.

Name	Engine	Also in	Description
Marimba Rain	Grain	–	Sparse marimba grains falling like rain.
Metal Choir	Grain	Pads, MPE	Bowed-metal grains stacked into a choir; slow scan, shimmering stereo.
Night Rain	Grain	MPE	Random kalimba droplets falling into a ping-pong delay and a dark hall.
Night Swarm	FM	3D, MPE	Glassy FM bells scattered on a fast random orbit at different distances: a swarm of lights around your head.
Recorder Breath	Grain	Vocal	Breathy recorder grains: airy, vocal-like tones.
Rust Texture	Grain	FX, MPE	Bowed metal grains played backwards through a slow comb: rusty, creaking, alive.
Singing Glass	Sampler	Pads	Rubbed wine-glass recordings (CC0), looped: pure and haunting.
Starfield	FM	Keys, MPE	Random high FM bells drifting across the stereo field into delay and hall.
Vinyl Crackle	Analog	FX	Record crackle and soft hiss (Analog engine: oscillators off, crackle and noise on); hold a note under a beat.
Zither Cloud	Grain	–	Shimmering zither grains, octave sprinkles.

Vocal (9)

Name	Engine	Also in	Description
Aah Choir	Vocal	Pads	Mixed choir singing 'ah'.
Choir Grains	Vocal	Hybrid, MPE	Ah choir doubled by a recorder grain cloud an octave up - human and synthetic at once.
Formant Voices	Wavetable	Pads	Formant wavetable with unison: synthetic choir.
Gospel Hum	Vocal	–	Closed-mouth hum: warm, low and intimate.
Male Ensemble	Vocal	Pads	Low voices on 'oh': a men's choir.
Ooh Choir	Vocal	Pads	Soft 'oo' choir, round and dark.
Solo Soprano	Vocal	Leads	High solo voice: bright formants, generous vibrato.
Talking Synth	Vocal	Leads	The LFO walks through the vowels: a talking lead.
Vowel Morph	Vocal	–	Each note sings 'ee' and opens to 'ah' with the envelope.

Boom (11)

Name	Engine	Also in	Description
Boom Click	Analog	Bass	Sine sub with a bright click on the attack, so the boom cuts through on small speakers.
Boom Distorted	Analog	Bass	Sine sub driven hard into the ladder filter and the FX slot: a growling, distorted boom.
Boom Kick Bass	Analog	Drums, Bass	Short boom: a two-octave pitch dive into a tuned sine - kick and bass in one.
Boom Long Sub	Analog	Bass	Long, tuned sine sub with a quick pitch dive on the attack; rings out for seconds.
Boom Slide	Analog	Bass	Long sine sub made for trap-style slides: overlap notes and it glides (legato, constant time).
Dirty Sub	Analog	Bass	Sub driven into the ladder filter: growling harmonics that stay audible on small speakers.
Glide Sine Bass	FM	Bass	Pure sine bass with legato glide for sliding bass lines; detached notes stay clean.
Impact Boom	FM	FX	Sub impact with a metallic crack, squashed and thrown into a huge hall.
Long Sub	Analog	Bass	Deep, long-decaying sub that slides between overlapping notes (glide); a soft click on the attack.
Punch Sub	Analog	Bass	Short, punchy sub with a snappy filter click; locks in with the kick.
Sub Kick	Analog	Drums, Bass	Tunable deep kick: a self-oscillating filter tuned to the key, swept down from above; long boom, driven.

Hybrid (18)

Name	Engine	Also in	Description
Breath Lead	Vocal	Leads, MPE	Breathy sung tone fused with a soft saw stack: a lead between a voice and a synth.
Choir Grains	Vocal	Vocal, MPE	Ah choir doubled by a recorder grain cloud an octave up - human and synthetic at once.
Chop Pad	Cluster	Pads, MPE	A wide chord pad chopped and stuttered by the envelope of a kalimba grain layer.
Crystal Organ	FM	Keys, MPE	Additive FM organ with a glass wavetable layer, circling in stereo with a slow autopan.
Digital Monster	Wavetable	Bass, MPE	Digital FM table gated by a glass-grain layer an octave down, then wobbled: unstable and huge.
FM Grains	Grain	Textures, MPE	Bowed-metal grains ring-modulated by an FM bell layer: grains that clang in tune.
Fold Bass	Wavetable	Bass, MPE	Sine sub and an analog saw folded into each other: the fold amount makes it roar.
Glass Engine	Grain	Pads, MPE	Singing-glass grain cloud fused with a harmonic wavetable layer that scans slowly; a touch of ring modulation.
Halo Grand	Sampler	Keys, Piano, MPE	Grand piano with a halo: its own recording frozen into a grain cloud that blooms behind every chord.
Metal Strings	Pluck	Pads, MPE	Metallic plucked strings folded into a slow cluster pad - an orchestra made of wire.
Neuro Hybrid	FM	Bass, MPE	FM growl and a wavetable reese ring-modulating each other, folded and squashed: a neuro monster.
Orbit Pad	Grain	Pads, MPE	Harp grains orbiting an FM layer, ring-modulated softly and panning over two bars.
Pluck Nebula	Pluck	Plucks, MPE	A plucked string that dissolves into a kalimba grain cloud: the string fades, the cloud swells and lingers.
Resonator Hit	Pluck	Plucks, MPE	A struck metal bar feeding a short grain resonance and a tuned comb: percussive, ringing, strange.
Ring Machine	Analog	Leads, MPE	Saw lead ring-modulated by an FM sine a fifth up: metallic, vocal and aggressive, with delay.
Spectral Morph	Wavetable	Pads, MPE	Formant table morphing under a breathy vocal layer, both through a drifting comb: spectra that melt into each other.
Tape Ghost Keys	Sampler	Keys, Piano, MPE	Tine piano with a ghost of itself: a detuned grain copy, crushed and flanged like an old resampled tape.
Vox Reese	Wavetable	Bass, MPE	A reese with a singing vocal layer on top: the formants ride the bass, a synced vowel filter makes it talk.

Piano (20)

Name	Engine	Also in	Description
Bright Grand	Sampler	Keys, MPE	Grand piano opened up for pop: brighter, lightly squashed, short room.
Classic Tine	Sampler	Keys, MPE	Modelled tine electric piano: bell-like when soft, barking when hit hard.
Concert Grand	Sampler	Keys, MPE	Grand piano recordings, two velocity layers, a touch of hall.
Digital EP	FM	Keys	Brighter electric piano with more bark when played hard.
Dream Tine	Sampler	Keys, Pads, MPE	Tine piano drenched in chorus, dotted delay and a long hall.
Dream Tine Arp	Sampler	Arp, Keys	Tine piano arpeggio in eighths with chorus, delay and hall: dreamy and full. Arp: Up/Down · 1/8 · 2 oct
Dusty EP	FM	Keys	Warm electric piano with tape wow and flutter (slow pitch LFO) and chorus; made for lo-fi chords.
Dusty Keys	FM	Keys	Mellow, dark electric piano for lo-fi chords: strong tape wow on the pitch, a bit-reduced edge, soft attack.
Felt Grand	Sampler	Keys, MPE	Dark, soft grand: the top rolled off, a slow release and a warm hall - intimate.
FM Bark Piano	FM	Keys, MPE	Two-pair FM electric piano: the tine pair rings, the bark pair bites with velocity.
Halo Grand	Sampler	Keys, Hybrid, MPE	Grand piano with a halo: its own recording frozen into a grain cloud that blooms behind every chord.

Name	Engine	Also in	Description
Piano Grains	Grain	Pads, Textures, MPE	Grand piano recordings frozen into a slow-moving grain cloud: a piano that breathes.
Reed Keys	Sampler	Keys, MPE	Modelled reed electric piano: nasal and woody, with a gritty bark when driven.
Reed Tremolo	Sampler	Keys, MPE	Reed piano through its classic amplitude tremolo - soul and gospel.
Scoop Keys	FM	Keys	Electric piano where every note scoops up a whole tone: soulful, a little lazy.
Suitcase Tine	Sampler	Keys, MPE	Tine piano with the stereo autopan of a suitcase amp and a small room.
Tape Ghost Keys	Sampler	Keys, Hybrid, MPE	Tine piano with a ghost of itself: a detuned grain copy, crushed and flanged like an old resampled tape.
Tine Bloom	Grain	Pads, Textures, MPE	Tine-piano grains blooming slowly, chorused, with a dotted echo.
Tine Piano	FM	Keys	Electric piano: soft body plus a bell-like tine that fades with the key. Chorus in the FX slot.
Upright Keys	Sampler	Keys, MPE	Upright piano recordings with a little room: honest and close.

3D (13)

Name	Engine	Also in	Description
Binaural Lead	Wavetable	Leads, MPE	Expressive lead placed in front of you: slide moves it up over your head, pressure brings it right up to your ear, a slow drift keeps it alive. Made for MPE controllers and headphones.
Doppler Fly-By	Wavetable	FX	A siren tone racing past from left to right and back with every note - the pitch bends with the Doppler effect as it passes.
Figure Eight Arp	Wavetable	Arp	Digital arp tracing a lying eight around you - left front, across, right behind - over two bars. Arp: Up/Down · 1/16 · 2 oct
Halo Drift	Cluster	Pads, MPE	Detuned cluster choir wandering around and above you on a slow random orbit - never the same place twice.
Neuro Swirl	Wavetable	Bass, MPE	Talking formant growl swirling in a figure eight once a bar - the vowel moves, the space moves, the sub does not.
Night Swarm	FM	Textures, MPE	Glassy FM bells scattered on a fast random orbit at different distances: a swarm of lights around your head.
Orbit Arp	Pluck	Arp	Sixteenth-note pluck arpeggio flying round your head once a bar - close, bright and fast. Arp: Up · 1/16 · 2 oct
Orbit Atmos	Wavetable	Pads, MPE	Wide wavetable atmosphere circling your head once every eight bars on a gently tilted orbit, in a soft room. Best with headphones: slide lifts it, pressure pulls it closer.
Orbit Reese	Wavetable	Bass, MPE	Neurofunk reese whose growling mid range circles your head every two bars while the sub below 160 Hz stays dead centre and mono - club-safe movement.
Overhead Choir	Vocal	Pads, Vocal, MPE	Breathy 'ooh' choir swinging from in front of you, over your head to behind you and back - a pendulum through the room.
Pendulum Wobble	Wavetable	Bass, MPE	Half-time dubstep wobble that swings over your head and back with the bar: the wobble and the space breathe together.
Spiral Arp	FM	Arp	Bell arpeggio spiralling up from ear height to above your head over four bars, then dropping back for the next round. Arp: Up · 1/16 · 3 oct
Spiral Riser	Analog	FX	Hold a key: noise and saws climb for four seconds while spiralling from below your ears to high above your head.

Projects / demo patterns (14)

Name	Categories	Description
Lo-Fi Beat	Drums, Keys, Bass	Classic lo-fi hip-hop at 78 BPM with a lazy, heavy swing: muffled kick, soft snare and quiet shuffled hats on the Lo-Fi kit (Euclid), rootless ninth voicings on Dusty Keys with strong tape wow (Orbit lanes 1-3), a warm round bass on the roots (step sequencer), vinyl crackle (Orbit lane 4). Fm9 - Bb13 - Ebmaj9 - C7#9; dark master (low-pass, tape saturation). Press Play.
Techno Groove	Drums, Bass	Driving four-on-the-floor on the Hybrid kit from Orbit (claps on 2 and 4, rims, accented 16th hats, off-beat open hat), a rolling bass from the step sequencer and dark seventh stabs from Euclid with a slowly drifting filter and a synced dotted-eighth delay.
Ambient Drift	Pads, Textures	A slow fifths pad, a generative kalimba from four Orbit lanes of 5, 7, 11 and 16 steps with a tempo-synced dotted-eighth delay, sparse singing-glass tones from Euclid. No drums.
Synth-Pop	Drums, Bass, Leads	Bright four-on-the-floor on the Tight kit from Euclid (snare and clap on 2 and 4, eighth hats), an octave-bouncing bass, lush poly brass and a sparkle arpeggio from three Orbit lanes over Am-F-C-G, and a lead melody from the step sequencer.
Euclid Polyrhythm	Drums, Arp	Four Euclid rings of 8, 12, 16 and 9 steps on the Room kit, a marimba from Orbit lanes of 5, 6, 7 and 8 steps and a pedal bass.
Arp Sequence	Arp, Drums	A 64-step arpeggio over Am, Fmaj7, G and Em on an A pedal, Studio-kit drums from Orbit. The arpeggiator plays part 3: hold chords.
Deep House	Drums, Bass, Keys	Shuffled hats and a deep kick on the Tight kit (Orbit), a syncopated organ bass (step sequencer) and Euclid chord stabs.
Sub & Glide	Pads, Bass, Drums, Boom	Seventh chords that slide into each other (Glide Strings, poly glide), a long sliding sub on Orbit lane 1 and a half-time beat on the Boom kit from Orbit lanes 2-4, sparkle bells from Euclid. Cm7 - Abmaj7 - Ebmaj7 - Bb7.
Lofi Kit	Drums	The Lo-Fi kit set up for lo-fi hip-hop: muffled, round kick, soft snare with ghost notes and a rim, dull shuffled hats with velocity variation (Orbit, lazy swing), vinyl crackle on part 4 (Orbit lane 4) and a dark master. Play your own keys on part 1.
Boom Kit	Drums, Bass, Boom	The synthesised Boom kit in a half-time beat with rolling hats (Orbit), Boom Slide on the step sequencer - overlapping notes glide (legato portamento) - and a dark pad that rises into its chord (Rise Glass, Euclid). C minor, 70 BPM.
Neuro Engine	Bass, Drums, Hybrid, 3D	Neurofunk at 174 BPM in F minor: a two-step beat on the Hybrid kit (Orbit), Neuro Reese riffing on the step sequencer, Growl Machine answering from Euclid and a dark seventh-chord pad underneath. Mono low end, squashed and loud. Press Play. In 3D: the growl answers swirl around your head, the pad drifts above you (best with headphones).
Halftime Pressure	Bass, Drums	Modern dubstep at 140 BPM in E minor: half-time drums on the Studio kit (Orbit), Wub Machine on the step sequencer - its eighth-note wobble restarts with every note - Vowel Growl answering from Euclid and a cold glass texture. Press Play.
Liquid Roll	Bass, Drums, Keys	Liquid drum & bass at 174 BPM in D minor: a rolling beat on the Tight kit (Orbit), dreamy tine-piano seventh chords and a warm pad (Orbit lanes 3 and 4), a smooth sliding sub (step sequencer) and glassy plucks from Euclid. Press Play.
Evolving Space	Pads, Textures, Hybrid, 3D	Ambient at 66 BPM in A minor: Evolving Tide chords drifting through their wavetable (step sequencer), a generative grand piano with a grain halo from three Orbit lanes of 5, 7 and 11 steps, a glacial grain drone and random starfield bells (Euclid). Let it run. In 3D: the chords circle you slowly, the starfield bells scatter all around and the drone sways overhead (best with headphones).

Engines: FM 42, Wavetable 79, Analog 68, Pluck 17, Grain 27, Vocal 20, Cluster 23, Sampler 26

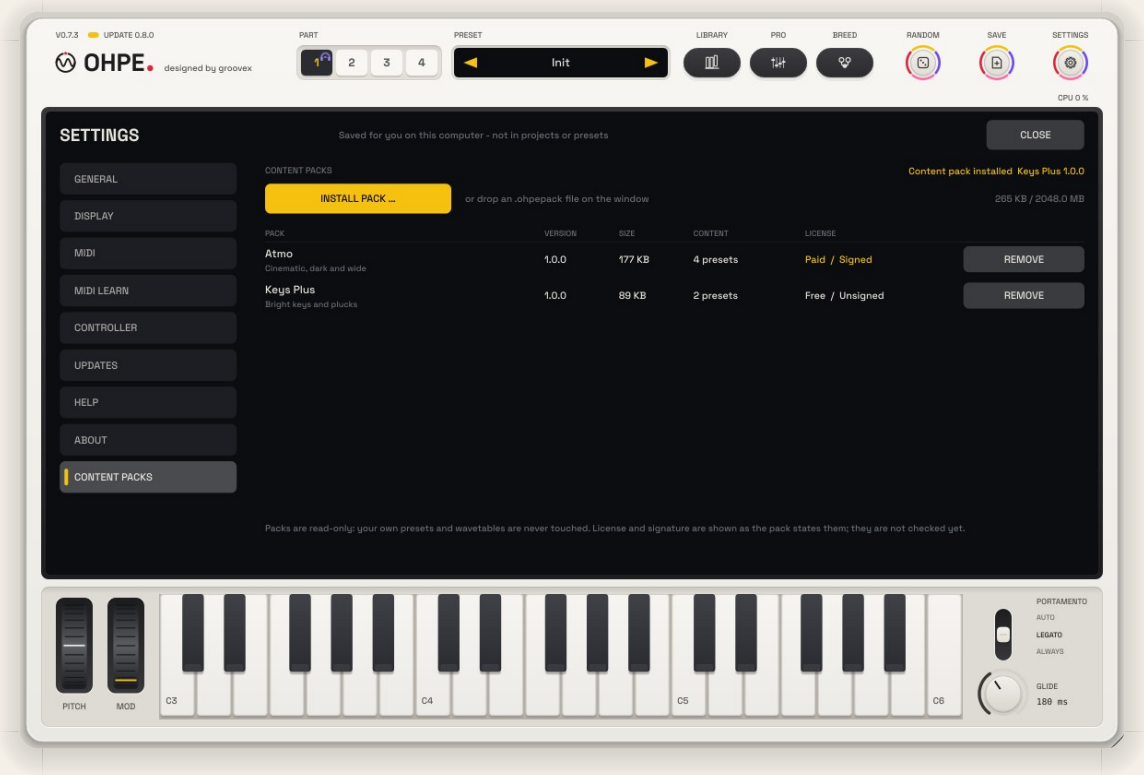
Fingerboard

Five sounds tuned for continuous-surface fingerboard controllers in Fingerboard mode (tags **Fingerboard** and **MPE**): polyphonic, no portamento (the slide is the finger's), pressure and forward / back on musical targets. They do not switch MPE on – the Fingerboard mode button on the MIDI settings page does, and a part in Fingerboard mode keeps that setup when a sound loads. With an aftertouch keyboard they play as normal sounds.

Name	Category	Engine	Description
Fingerboard Bowed String	Leads	Analog	A bowed string for continuous fingerboards: pressure is the bow (level and brightness), sliding forward and back moves the bowing position, the slide is free. Vibrato comes from your finger. The mod wheel slides the bowing position too.
Fingerboard Breath Lead	Leads	Analog	A breathy reed lead for continuous fingerboards: pressure is the breath (level, air and a brighter bore), forward / back opens the tone. Glide-free slides, quick response.
Fingerboard Cello	Bass	Analog	A dark, woody cello for continuous fingerboards in the low register: pressure bows harder (level, rosin and body), forward / back darkens or opens the tone. Slide into notes and lean in for vibrato.
Fingerboard Expressive Bass	Bass	Analog	A warm analog bass for continuous fingerboards: pressure opens the filter as it raises the level, forward / back adds drive. Slides glide exactly with your finger over the whole range.
Fingerboard Opening Pad	Pads	Wavetable	A wide wavetable pad for continuous fingerboards that opens as you slide your finger forward (Y opens the filter and the table position); pressure swells it. Hold chords and move single fingers.

Content packs.

A content pack adds sounds to Ohpe without a new version of the program: preset collections (kind `presets`) and wavetable banks (kind `wavetables`: single-cycle sets for the Wavetable engine, listed read-only in its table menu under **Content packs** – see Wavetable). A pack is one file ending in `.ohpepack`. Packs are read-only – your own presets, wavetables, bounces and settings are never touched.



Content packs in the settings

Installing a pack

- **Settings > Content packs > INSTALL PACK ...** (also on **Settings > Updates** and in the library's **PACKS** menu) opens a file dialog for `.ohpepack` files, or
- **drop** an `.ohpepack` file anywhere on the Ohpe window.

The display says when the pack is installed or why it was not. Installing a newer version of an installed pack keeps the old one as a backup (one step of rollback, as with packs from the updater).

Before anything changes, Ohpe checks the pack:

Check	When it fails
every path stays inside the pack (no `..`, no absolute paths, no links)	the pack is refused as a whole
`pack.json` is there and valid (id, version, kind, names)	refused
every file has its SHA-256 in `pack.json`, and every listed file is there	refused – "A file of the content pack is damaged (checksum)"
the pack fits the size limits of its kind and the 2 GB room for all packs	refused
the pack's `minApp` is not newer than this Ohpe	refused – "This pack needs a newer Ohpe"

The pack is unpacked into a staging folder first and only replaces the installed version when every file passed – a failed install leaves everything as it was.

Managing packs

Settings > Content packs lists every installed pack: name, version, size on disk, what it holds, license and signature, and **REMOVE** (click twice: the chip asks REMOVE? first). The line above the list shows how much of the room for packs is used.

In the **library**, the **PACKS** chip (right end of the filter row) opens a menu: All presets, one entry per installed pack (shows only that pack's presets), Install pack … and Manage packs …. Pack presets carry a **P** in the list and the pack's name in the details.



The library filtered by a pack

License and signature

`pack.json` may name a license (`free` or `paid`, with a product id) and carry a signature (Ed25519 over the pack's canonical `pack.json`). Ohpe shows both in the pack list (Free / Paid, Signed / Unsigned) but **does not check or enforce them yet** – there is no shop and no online activation. The interface for a later unlock is in place (`update::PackGate`); today every pack is allowed.

Presets that use pack content

A preset can point at content of a pack by a reference `"<pack>/<item>"` – for example the wavetable `atmo/glass-choir` is `wavetables/glass-choir.wav` in the pack `atmo`. Such references are never copied into the preset. When the pack is not installed, Ohpe says **"Content pack missing – install it in Settings > Updates"** with the pack's id; the part plays its built-in material until the pack is installed and the preset is loaded again.

Where packs live

`<user data>/groovex/Ohpe/Content/<pack id>/` (macOS: `~/Library/Application Support/groovex/Ohpe/Content`, Windows: `%APPDATA%\groovex\Ohpe\Content`, Linux: `~/.config/groovex/Ohpe/Content`), the installed versions in `Content/installed.json`, the backups in `Content/.backup/`. `OHPE_DATA_DIR` moves the whole data folder.

PART 6

MIDI and controllers.

MIDI guide.

This guide covers how Ohpe receives and sends MIDI: channels per part, MPE, MIDI clock, host sync and Link. Mapping controllers to parameters (MIDI learn) is described in the main manual.

Channels and parts

Every part has its own **MIDI channel** (`pN.midich`): part 1 listens to channel 1, part 2 to channel 2 and so on. `0` means **omni** (all channels). A controller on channel 3 therefore plays part 3; set several parts to the same channel to layer them. The on-screen keyboard plays on channel 1.

Channel 10 plays the drum kit. Omni parts ignore it; a part set to channel 10 (or in MPE mode) hears it as well, layered with the kit.

The sequencers address their part directly (their target part), whatever its channel.

Message	Effect
Note on / off	plays the part (through scale / chord mode and the arpeggiator, if active)
Pitch bend	bends the part (range `pN.bend`)
CC 5	portamento time (0.6): overrides the part's Glide time (0 … 2 s) until `pN.glide` itself changes
CC 64	sustain
CC 65	portamento off (< 64) / on (≥ 64, the default) (0.6)
CC 74	timbre (target `pN.mpe.timbre`) for all notes of the part
Channel pressure	pressure (target `pN.mpe.press`, amount `pN.mpe.pressamt`) for all notes of the part
Polyphonic aftertouch	pressure (same target and amount) for that one note (0.8: documented; it always worked)
CC 120 / 123	all sound off / all notes off
Mapped CCs and notes	change their parameter (MIDI learn) and do not play

MPE (MIDI Polyphonic Expression)

Switch on **MPE** (`pN.mpe`) for a part to play it from an MPE controller (lower zone: master channel 1, member channels 2–16). The part then listens to all channels, and every note keeps its own expression:

- **Per-note pitch bend** on the member channel, range `pN.mpe.bend` (default 48 semitones, the MPE standard). Pitch bend on the master channel bends the whole part.
- **Pressure** (channel pressure on the member channel, or polyphonic aftertouch) goes to the **pressure target** (`pN.mpe.press`, default Level).
- **Timbre** (CC 74 on the member channel, centre 64) goes to the **timbre target** (`pN.mpe.timbre`, default Timbre).

Targets: Off, Timbre (the engine's brightness: filter cutoff in Analog, Wavetable, Cluster, Grain and Sampler, brightness in Pluck, modulation index in FM), Level (up to +6 dB), LFO Depth (adds vibrato / tremolo per note), Pan, and since 0.8:

Target	What it moves
Cutoff	filter / tone of the engine, ±3 octaves (Pluck: brightness)
Position	Wavetable position, Grain position (±half the range)
FM Index	FM engine index, Wavetable FM amount
Grain Density	grains per second, ±2 octaves

Target	What it moves
Drive	Analog drive, a per-note saturation in Wavetable, a longer ring in Pluck
Vibrato	a per-note vibrato (5.5 Hz, up to ±½ semitone)
Formant	Vocal formant shift (±12 semitones), Wavetable warp
Layer Blend	the balance between the main engine and the layer engine (part-wide: the latest value)

Amounts (0.8). `pN.mpe.pressamt` and `pN.mpe.timbreamt` (−1 … +1, default +1) scale what pressure and timbre do; a negative amount turns it round (pressure closes the filter, timbre below the centre opens it).

Smoothing and slides (0.8). Per-note pitch bends glide to their new value over `pN.mpe.slide` (0 … 1 s) – at least 2 ms, so a controller sending coarse steps never zippers. A note started with a bend starts at that pitch (it does not slide in). Pressure and timbre glide over `pN.mpe.smooth` (0 … 0.5 s, 0 = at once, as before 0.8).

Portamento and poly glide work as without MPE: a new note glides from the previous one (part glide settings), its own per-note bend is added on top.

Pressure glide. `pN.glide.press` (Part N Glide Pressure, −1 … +1, default 0) lets the pressure change the glide time. The pressure is the larger of the channel pressure and the newest note's own pressure (polyphonic aftertouch or MPE). A positive amount makes the glide slower the harder you press (time × (1 + 3 × amount × pressure), up to four times as long at +1 and full pressure); a negative amount makes it faster (time × (1 + amount × pressure); at −1 and full pressure the pitch jumps). It scales the Glide time and the CC 5 portamento time alike and acts on new glides and on glides that are running; CC 65 off still switches glide off. At 0, or without pressure, the glide is exactly as before. It is in the pro view's **Glide** group and part of part presets (key `glide.press`). For how it relates to an Expression aftertouch slot on Glide Time (scaling vs. adding; both together stretch twice) see Expressive playing.

Without MPE the same targets, amounts and smoothing apply to channel pressure, polyphonic aftertouch and CC 74 on the part's channel, so an aftertouch keyboard drives them too.

Set your controller's pitch-bend range to the same value as `pN.mpe.bend`.

Factory presets tagged **MPE** (0.8, search "MPE" in the library) come with pressure / timbre mappings, amounts, slide and smoothing that suit the sound (e.g. pressure opens the filter or adds drive on basses, timbre scans the wavetable on pads, pressure thickens the grain cloud). They do not switch MPE on – do that for the part (`pN.mpe`) when you play from an MPE controller; with an aftertouch keyboard the pressure mapping works right away.

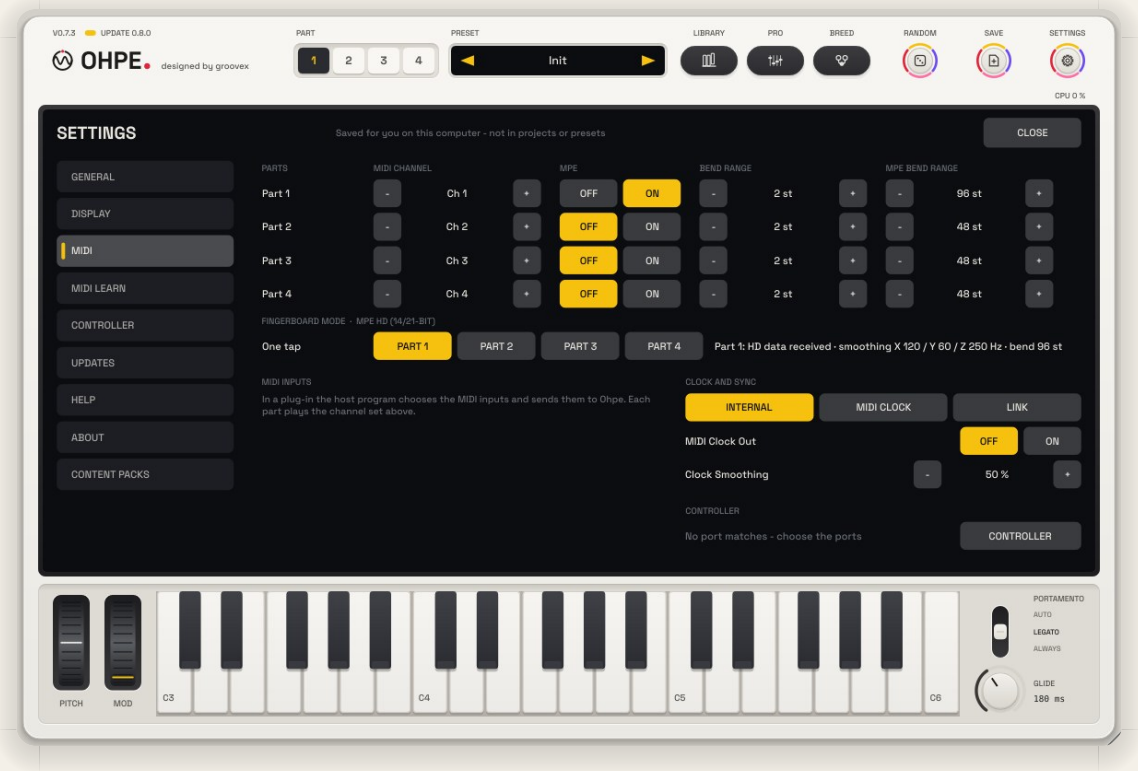
Fingerboard mode (MPE HD 14/21-bit)

For **continuous-surface fingerboard controllers** – a playing surface without keys where the finger slides freely along it (X = pitch), moves forward and back (Y) and presses (Z), usually with pedal inputs. These controllers send MPE with extra resolution (**MPE HD**: 14-bit pressure and Y, 21-bit pitch bend) and always send note velocity 127. Ohpe receives all of it; plain MPE keeps working exactly as before.

Setup

1. On the controller, choose its **MPE HD** output (the high-resolution MPE mode; plain MPE works too, with 7-bit pressure / Y and 14-bit bend).
2. In Ohpe open **Settings → MIDI**. Under **Fingerboard mode · MPE HD (14/21-bit)** tap the part you play (Part 1 … Part 4). One tap sets: MPE on, MPE bend range 96 semitones, note onset from pressure, MPE HD Auto. Tap again to go back to velocity onset and 48 semitones.

3. Play. The status next to the buttons shows HD data received as soon as the controller sends high-resolution values, and the smoothing cutoffs (X / Y / Z in Hz) and bend range it sent.
4. Pick a sound – the presets tagged **Fingerboard** (search "Fingerboard" in the library) are tuned for it. A part in Fingerboard mode keeps its controller setup (MPE, bend range, MPE HD, onset) when you load another sound.



Fingerboard mode on the MIDI settings page: part 1 active, HD data received, the controller's smoothing cutoffs and bend range

What the controller does in Ohpe

Controller	Ohpe
X – slide left / right	per-note pitch, continuous over the whole range (21-bit bend, 96 semitones); a note starts exactly at the pitch where the finger lands
Y – forward / back	the **timbre target** (`pN.mpe.timbre`) and the matrix source Slide (Timbre), 14-bit
Z – pressure	the note's **level** (Fingerboard mode: no pressure = silent, the finger shapes the attack and the swell), the **pressure target** (`pN.mpe.press`) and the matrix source Note Pressure, 14-bit
Pedals on the master channel	sustain (CC 64) and sostenuto (CC 66) hold every note of the part; other pedals / CCs (e.g. CC 4, CC 11) work through MIDI Learn and act on the whole part

The matrix source **Key + Bend** follows the continuous position of the finger (key track including the note's own bend), e.g. to darken a sound further up the surface.

What is received

- **CC 87 (LSB prefix).** A CC 87 on a member channel adds 7 low bits to the next channel pressure, CC 74 or pitch bend on that channel: pressure and Y run 0x0000 … 0x3F80 (0x3F80 = full), the bend has 21 bits. Messages without the prefix are read as plain 7-bit / 14-bit values – the controller may mix both at any time.
- **NRPN 100 / 101 / 102** (CC 99 = 0, CC 98 = 100 / 101 / 102, CC 6 = value): the controller's smoothing cutoff for X / Y / Z in 2 Hz steps (e.g. 60 = 120 Hz). Ohpe smooths each dimension of every note with it, instead of Expression Smoothing / Per-Note Slide. Before anything arrives (MPE HD On, or Auto once HD data came in) Ohpe uses X 120 Hz, Y 60 Hz, Z 250 Hz.
- **RPN 0** (bend range) on a member channel sets the bend range of all member channels (these controllers send 96). **RPN 6** (MPE configuration message) on channel 1 or 16 selects the lower or upper zone.

Settings (part parameters, MIDI-mappable; on the MPE page of the pro view):

Parameter	What it does
MPE HD (`pN.mpe.hd`)	Auto (default): fine values and smoothing are used as soon as they arrive – plain MPE sounds exactly as before. On: the smoothing also runs before the controller sent its cutoffs. Off: CC 87 and the (N)RPNs are ignored.
Note Onset (`pN.mpe.onset`)	Velocity (default, as before), Pressure (velocity ignored; the level follows the pressure, 0 = silent), Velocity x Pressure (both).
Pitch Rounding (`pN.mpe.round`)	Off (default) = free slides. Initial: a note starts on the nearest semitone (the slide after it is free). Continuous: the pitch is also pulled to the nearest semitone while the finger holds still.
Rounding Time (`pN.mpe.round.time`)	how fast continuous rounding pulls (0.01 … 1 s, default 0.15 s).
Pressure Curve (`pN.mpe.pcurve`)	−1 … +1, 0 = linear. Right: a light touch stays soft; left: it speaks early.

Velocity onset with a fingerboard controller plays every note at full velocity – choose Pressure (the Fingerboard mode button does). Portamento and the part glide add to the slide and never fight it; for free slides keep Glide at 0 (the Fingerboard presets do).

Key and Scale Lock

The key (`key.root`, `key.scale`) and Scale Lock (`key.lock`, `key.lock.mode`) are host parameters: MIDI learn them from the key readout on the display header (click > MIDI Learn, or right-click on the padlock), e.g. a footswitch for Scale Lock or an encoder (relative) for the root. With Scale Lock on, incoming notes from any channel that reaches a part snap to the key (or, in White Keys, the white keys play the key; black keys are silent). MPE parts keep their pitch. Hosts pass no key to a plug-in: the key is set in Ohpe (or automated). See Key, chord and Scale Lock.

Tempo and transport

Situation	Transport source
Plug-in in a host	always the host: tempo and position (ppq), sample-accurate
Standalone, `sync.src` = Internal	Ohpe's own clock (`tempo`, start with `seq.run`)
Standalone, `sync.src` = MIDI Clock	an external MIDI clock on any enabled MIDI input
Standalone, `sync.src` = Link	the Link session on the local network

In every case the sequencers start and stop with **Sequencer Run** (`seq.run`) – in a host they run while the host plays and `seq.run` is on.

Transport keys from a controller. Map a button to Transport Play/Stop (`tape.tx.play`), Transport Record (`tape.tx.rec`), Transport Stop (`tape.tx.stop`), Transport Loop (`tape.tx.loop`), Tape Play/Stop (`tape.tx.tplay`), Arm Edit Track (`tape.tx.arm`) or Tape Return to Zero (`tape.edit.rtz`): right-click the key on Ohpe, or MIDI → Remote controls. They fire on a press (CC ≥ 64), like the keys on the body; see tape transport. p38: Tape Clear Track (`tape.edit.clear`) empties the edit track at once (undoable); Tape Clear All (`tape.edit.clearall`) opens the confirmation on the screen and clears only when it is confirmed. p42: Tape Set Loop In (`tape.edit.loopin`) / Tape Set Loop Out (`tape.edit.loopout`) set the loop at the playhead (on the grid of Tape Loop Snap, `tape.loopsnap`; loop out switches the loop on), Tape Loop Half / Double (`tape.edit.loophalf`, `tape.edit.loopdouble`), Tape Loop Undo / Redo (`tape.edit.loopundo`, `tape.edit.loopredo`) – map two buttons to set in / out and play loops by hand; see Setting the loop.

On-screen wheels. The PITCH and MOD wheels left of the keyboard send pitch bend and the mod wheel (CC 1) on the edit part's channel, like the on-screen keyboard. Hardware pitch bend and CC 1 reach the parts directly as before. The wheels are also host parameters (`perf.bend`, `perf.mod`): right-click a wheel to map a fader or a controller wheel to it.

Host sync (plug-in)

Ohpe reads the host's tempo and song position every block and places every step at the exact sample where the host's timeline reaches it, for any tempo and buffer size. Loops and jumps in the host land every sequencer on the right step (and in song mode on the right pattern).

MIDI clock in

With `sync.src` = **MIDI Clock** Ohpe follows 24 clock ticks per quarter note, Start, Stop, Continue and Song Position Pointer. The tempo is averaged over recent ticks and the position is pulled gently towards each tick; **Clock Smoothing** (`sync.smooth`) sets how much: low values react at once, high values ignore jittery clocks (USB, wireless). A single missing or doubled tick is ignored; a real tempo change is picked up within a few ticks.

MIDI out (standalone)

Choose a **MIDI output** under Sync & MIDI … in the standalone menu.

- **Send MIDI clock** (`sync.clockout`): 24 ticks per quarter note while the transport runs, Start (or Song Position + Continue when starting mid-song), Stop.
- **Send sequencer notes** (`sync.notesout`): the notes of all sequencers, each on its part's MIDI channel – to play external instruments from Ohpe's sequencers.

Link (standalone)

Ohpe's standalone app can join a **Link** session (Ableton Link, a third-party library, see THIRD-PARTY.md) with other apps on the same network. With `sync.src` = **Link** the transport takes tempo and beat phase from the session, so Ohpe's bars line up with everybody else's. Changing Ohpe's tempo changes the session tempo. Start and stop remain local. Link is a build option (`-DOHPE_WITH_LINK=ON`, default on); the plug-in always follows its host.

Grid controller (standalone and plug-in)

An 8 × 8 pad controller with a four-line display can play the grid keyboard and edit the shown page – it is on by default and connects by itself when plugged in (Settings > Controller: profile, ports, status, virtual controller). Its ports are taken out of the plain MIDI inputs while Ohpe holds them. In a plug-in Ohpe opens the controller's user port itself (one instance at a time) – switch that port off for tracks and remote in the DAW. See Grid controller.

Settings at a glance

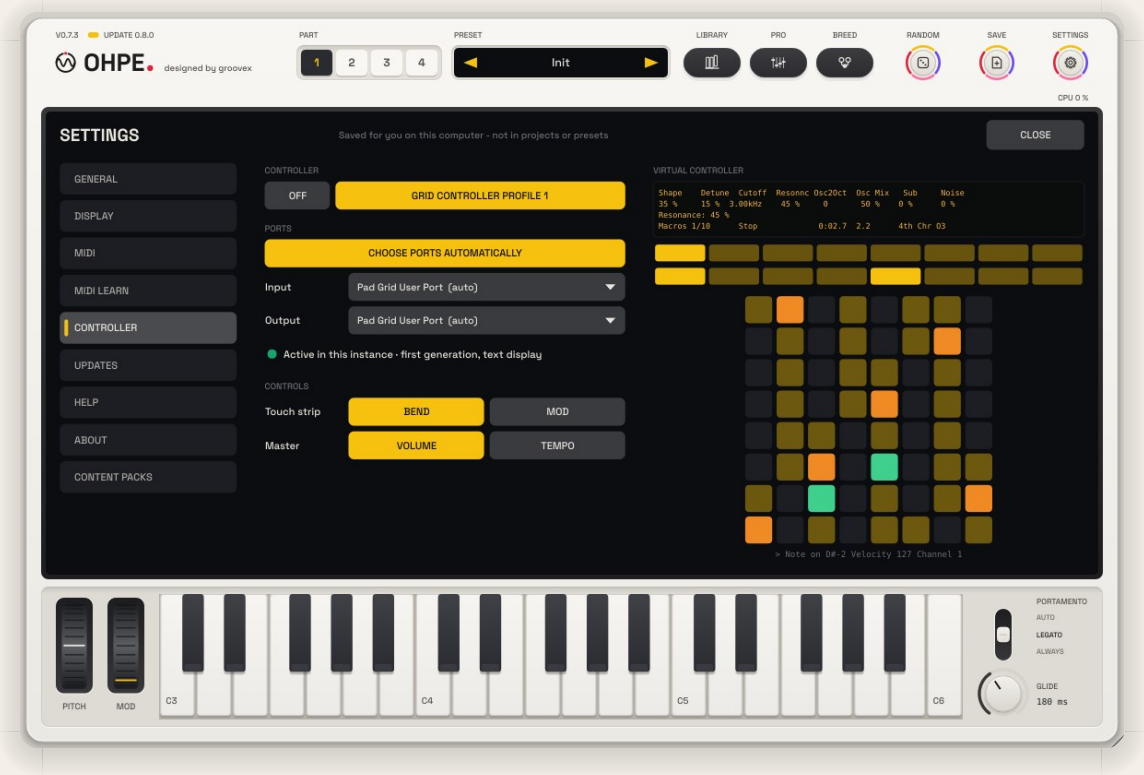
ID	Range	Default
`pN.midich`	0 (omni) … 16	N
`pN.mpe`	off / on	off
`pN.mpe.bend`	0 … 96 semitones	48
`pN.mpe.press`, `pN.mpe.timbre`	Off, Timbre, Level, LFO Depth, Pan, Cutoff, Position, FM Index, Grain Density, Drive, Vibrato, Formant, Layer Blend	Level, Timbre
`pN.mpe.pressamt`, `pN.mpe.timbreamt` (0.8)	−1 … +1	+1
`pN.mpe.smooth` (0.8)	0 … 0.5 s	0

ID	Range	Default
`pN.mpe.slide` (0.8)	0 … 1 s (at least 2 ms)	0
`sync.src`	Internal, MIDI Clock, Link	Internal
`sync.smooth`	0 … 1	0.5
`sync.clockout`, `sync.notesout`	off / on	off

Grid controller (profile 1).

Ohpe (the standalone app and, since p49, the plug-in) can be played and edited from an external **8 × 8 pad grid controller with a four-line character display:** 64 pads with velocity and pressure, 8 touch-sensitive encoders and a master encoder, two rows of 8 buttons under the display, transport and navigation buttons and a touch strip. In Ohpe this is **Grid-Controller (Profil 1) / Grid controller profile 1.**

The pads play exactly like the on-screen grid keyboard: same layouts, same notes, same colours for root and key notes – what you learn on one works on the other.



Settings – Controller: ports, connection, controls and the virtual controller

Setting it up

Nothing to do in most cases. Since 0.6.3 the profile is **on by default with automatic ports:** plug the controller in and start Ohpe (standalone app, or open the plug-in's window). Ohpe finds the controller's user port (input **and** output of the same device), switches the controller into its **User mode** by itself (no need to press its User button), shows **OHPE · connecting …** and then **connected** on its display for a moment (German UI: verbindet … / verbunden), and then the normal Ohpe display. Plugged in later? Ohpe connects within a second. When Ohpe quits, the profile is switched off or the instance goes, it clears the display and lights and hands the controller back to its own mode.

Settings > Controller (its own page):

- **Off / Grid controller profile 1.**
- **Ports: Choose ports automatically** (on by default), then a menu for **Input** and **Output** with the full port names. Picking a port in a menu switches automatic off (the other side keeps the port in use); choosing automatic in the menu switches it back on.
- The **status line: Connected · first generation, text display** (or later generation, no text display), **Checking the device …, No controller found, Port in use by another program** (standalone) / **Port in use - free it in the host** (plug-in), **Waiting for the controller.**

- **Controls: Touch strip** Bend / Mod; **Master** encoder Volume / Tempo.
- The **virtual controller**, large on the right: the four display lines, the two button rows under the display and the 64 pads in the controller's colours, live – pressed pads light green, the mode / part buttons light as on the device. It works with the mouse too: click a pad to play it, click a button, turn the mouse wheel over an eighth of the display to turn that encoder. It also follows the pads when no output is connected.

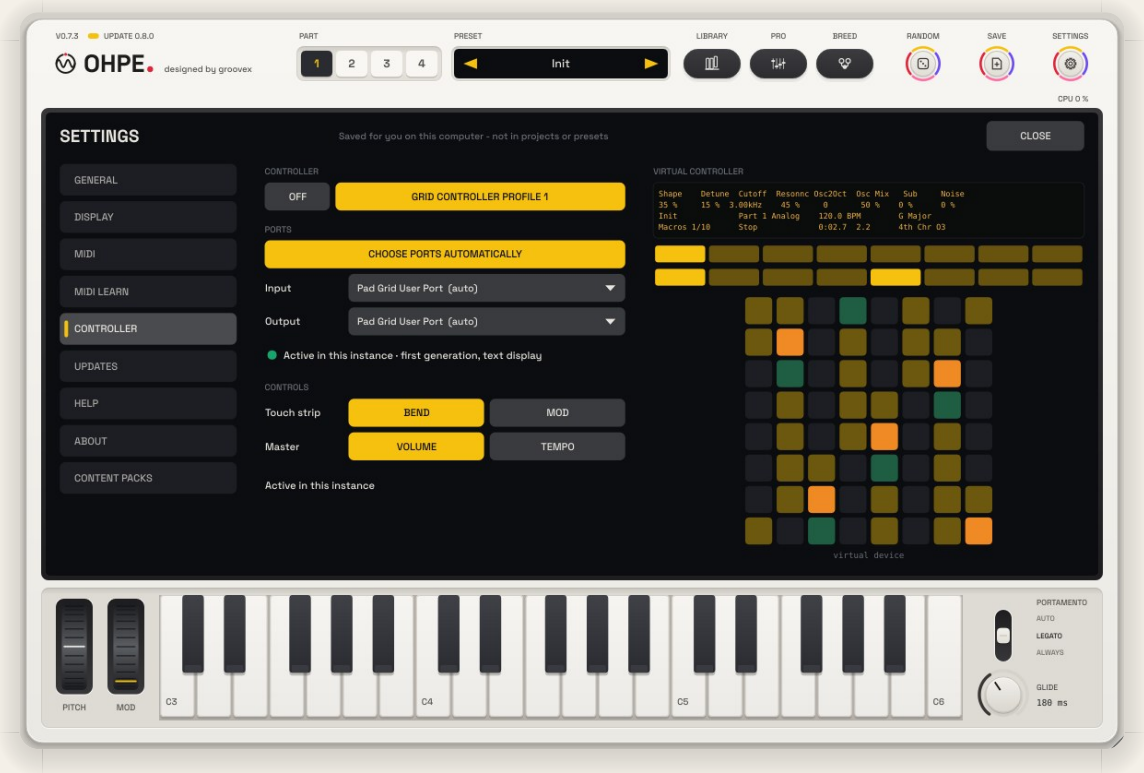
How the controller is found. Port names that match the pattern `User; MIDIIN2; MIDIOUT2` (any part, case does not matter; editable as `"pattern"` under `"controller"` in `settings.json`) are candidates; the input and output of the same device are paired (on macOS / Linux its … User Port, on Windows MIDIIN2 (…) / MIDIOUT2 (…)). Ohpe sends a standard identity request: the answer tells the controller's **generation.** A device that also has its live port next to the user port connects at once; another matching device must answer as the controller within two seconds, otherwise Ohpe leaves it alone and tries the next candidate.

Generations. The first generation has the four-line character display written over MIDI – Ohpe uses all of it. A later generation (pixel display over USB) gets pads, buttons, encoders, touch and lights, but no text on its display (its pad colours may differ).

The settings are saved per user in `settings.json` (never in projects or presets). Ohpe switches the controller's ports (user and live port) off in the plain MIDI input list while it holds them, so pads don't play twice and the encoder / strip touch notes never reach the sound as notes, and switches them back on when it lets go.

In a plug-in

Since p49 the controller also works inside the **plug-in** (VST3 / AU in a DAW). The controller has two modes: in its usual mode it talks to the DAW; in its **User mode** (the controller's User button) it hands its pads, display and lights to whatever program opens its user port. Ohpe opens that port itself – past the host – with the full display, pad colours, pressure, encoders (with touch), touch strip and buttons, exactly as in the standalone app.



Settings – Controller in a plug-in

1. **In the DAW:** in its MIDI preferences, switch the controller's user port (often called … User Port, on Windows MIDIIN2 (…) / MIDIOUT2 (…)) **off for tracks and remote** (input and output), so the DAW releases it and the pads don't play twice. Leave the controller's main port as it is.
2. **In Ohpe:** nothing (**Settings > Controller:** the profile and automatic ports are on by default; pick the ports in the menus to choose them yourself).
3. Ohpe switches the controller to its **User mode** itself. The page shows **Active in this instance** once both ports are open.

One instance at a time. Only one Ohpe of the DAW can hold the controller: the one whose window you **opened or clicked last.** The others show **Active in another instance** and a **Take over here** button; when the holding instance is removed, the next one takes over. The holding instance keeps the controller while its window is **closed** (the encoders keep the last page, the part, mode and transport buttons work). After loading a project, open the window of the instance you want to play once – until then no instance holds the controller.

Port in use. On Windows a MIDI port can be open in only one program. If the DAW still holds the user port, the legend shows **Port in use - free it in the host;** free the port in the DAW's MIDI preferences – Ohpe tries again every second, and also reconnects when the controller is unplugged and plugged in again. MIDI from the host keeps working in parallel. Closing the instance or switching the profile off clears the display and the lights.

Display

Line	Shows
1	names of the parameters on the 8 encoders (two over each display segment)
2	their values

Line	Shows
3	preset · part and engine · tempo · key (and **Lock** with Scale Lock) – while an encoder is touched: its full name and value
4	page (e.g. Filter 2/5; Volume, Pan & Send, Browse) · transport (Play / Stop / REC / Count-in, Click with the metronome) · tape position · Keys / Steps: grid layout, In Key / Chromatic, Fixed C and octave (e.g. `4th Key Fix 03`: Fourths, In Key, Fixed C, octave 3; `Seq Chr 03`), other pad modes: their name

Umlauts are written as ae / oe / ue (the display has no umlauts).

Encoders

- **Encoders 1 – 8:** the parameters of the shown page and the next one – encoders 1 – 4 are the four screen encoders, 5 – 8 the page after it (in the order of the pages on screen). Hold **Select** (or Shift) for fine steps.
- **Master encoder:** master volume; **Master** button switches it to the tempo.
- **Tempo encoder:** tempo (1 BPM per step, Shift / Select 0.1).

Pads

The second button row, keys 5 – 8 (or **Note** / **Session**), chooses the pad mode:

Mode	Pads
Keys	the on-screen grid keyboard on 8 × 8 pads
Pads	the four performance pads (stutter, tape stop, sweep, freeze) – one 4 × 4 quarter each, held = on
Drum	the 24 drum pads on the three bottom rows
Steps	steps 1 – 16 of the step sequencer on the two top rows (top row 1 – 8); the six rows below play like Keys

Keys – the grid keyboard on the pads

- **Same layout** as Grid Layout (`harmony.kb.layout`): **Fourths** (right +1, up +5), **Thirds** (+1 / +4), **Whole Tones** (+2 / +5) or **Sequential** (+1 / +8, continuous rows), each **In Key** (scale steps, only key notes) or **Chromatic** (`harmony.kb.inkey`), with the bottom-left pad on the key's root or on C (**Fixed C**, `harmony.kb.fixed`) – see Harmony. A **tap** on **Scales** steps through the layouts; **holding Scales** shows the grid options on the display and the upper button row chooses them (below). Since p53 the pads follow the on-screen keyboard: while the **grid keyboard** is shown (GRID key) they take its layout, root, scale and colours; with the **piano keys** shown they use the default layout (Fourths). Scales with the piano keys shown first switches the screen to the grid keyboard, then steps its layout – screen and pads always show the same. Held notes show both ways: a pad held sinks its cell on the screen, a cell clicked on the screen lights its pads green.
- **Same mapping:** pad (column, row) plays the note of the screen grid's cell at the same column and row, from the same lowest C. The four lower pad rows are exactly the screen grid's four rows; the upper four continue it.
- **Same key:** the key the grid shows (the remembered key – the set key until anything was heard). With **Scale Lock** (Shift + Scales) the other notes stay dark and silent in Chromatic; In Key has only key notes anyway.
- **Colours:** the key's root **amber**, key notes **yellow (dim)**, tones of the remembered chord **green (dim)**, held pads **green**, other notes dark (graphite on screen).
- **Octave − / +** moves pads, screen keyboard and screen grid together (the bottom-left pad is the key's root in that octave, or its C with Fixed C).
- **Velocity and pressure** play the **edit part.** With MPE on for the part each pad gets its own member channel and pressure is that note's pressure; without MPE the notes go through scale, chord and follow and pressure is sent as poly aftertouch of the part.

Scales held – the grid options

While **Scales** is held the display shows the options over the upper button row, and the eight upper buttons choose:

Upper button	Chooses	Light
1 – 4	layout Fourths / Thirds / Whole Tones / Sequential	the chosen one yellow, the others yellow dim
5	In Key ↔ Chromatic	green = In Key
6	Fixed C ↔ the key's root bottom left	green = Fixed C
7	key root + 1 semitone (Shift: − 1)	amber
8	next key scale (Shift: previous)	amber

```
  Fourths  Thirds  Whole T  Sequent   Notes   Bottom    Root    Scale
    [x]                              In Key    Root      C      Major
Grid: Fourths - In Key - bottom left root
Hold Scales + upper buttons to choose (Shift: root / scale back)
```

Letting go of Scales without choosing anything steps to the next layout (a tap).

Buttons

Button	Action
first row under the display	the eight mode keys in screen order, Synth to Drum (lit like the screen keys)
second row 1 – 4	parts 1 – 4
second row 5 – 8	pad mode Keys / Pads / Drum / Steps
◀ / ▶	page − / +
▲ / ▼	previous / next preset (in browse mode: previous / next in the list, loaded at once)
Octave − / +	octave of the keyboard, the grid and the pads
Play	play / stop (Shift: tape play / stop)
Record	record (Shift: arm the edit track)
Stop	stop (Shift: to the start)
Repeat	tape loop on / off
Metronome	metronome on / off (Shift: count-in Off / 1 bar / 2 bars)
Tap Tempo	tap the tempo
Undo	tape undo (Shift: redo)
Scales	tap: next grid layout; hold: grid options on the upper row (above); Shift: Scale Lock – lit with Scale Lock, blinks while held
Master	master encoder: volume ↔ tempo
Shift / Select	modifiers (Select: fine encoder steps)
Mute	alone: mute the edit part's mixer channel; held + part button 1 – 4: mute that part – lit when a part is muted, the part buttons show mute red while held
Solo	the same for solo (part buttons amber while held)
Volume	encoders 1 – 4 = volume of parts 1 – 4, 5 – 8 = their pan (again: back to the page) – lit while on
Pan & Send	encoders 1 – 4 = send A of parts 1 – 4, 5 – 8 = send B – lit while on

Button	Action
Device	opens / closes the FX editor (switches to the Effects racks first; in browse mode: leaves it) – lit while open
Browse	browse mode on / off – the preset library on the display (below) – lit while on
Track	mixer mode – lit there
Clip	tape mode – lit there
Accent	fixed full velocity (127) for every pad on / off – lit while on
Quantize	scale quantise of the edit part on / off (`pN.quant`) – lit while on
Duplicate	copies the step pattern into the next pattern, which is then edited and plays
Double	doubles the step pattern's length, the steps copied into the new half (up to 64)
Delete	held + a step pad (Steps mode) = clear that step – lit while held
New	loads the start sound (the first preset of the library, into every part)
Fixed Length	step length 4 → 8 → 16 → 32 → 64 → 4 – lit when not 16
Automation	motion record of the edit part on / off (`pN.motion.rec`) – lit while armed, blinks while the tape rolls
Add Effect	the selected FX slot (part rack if the Effects page shows no rack) gets the next effect type
Add Track	CAPTURE: the tape memory becomes an instrument in the target part (`tapemem.go`, see TAPE-MEMORY.md) – dim while the memory holds sound, blinks while the capture is analysed
User	opens / closes ZUCHT (breeding) – lit while open
1/4 … 1/32t (right)	held while playing pads = note repeat: the arpeggiator on the edit part at that rate (1/4, 1/4T, 1/8, 1/8T, 1/16, 1/16T, 1/32, 1/32T), as before when released; Shift = set the arp rate and keep it – green while held, yellow = the arp's rate

Not mapped (no matching Ohpe function): Add Track, In, Out.

Browse – the library on the display

Browse turns the controller into a preset browser you can use without looking at the screen; the screen's library opens and follows every move (filters, cursor, loading). The display:

- **Line 1** – the values of the eight columns, one over each encoder: **1** position in the list, **2** category, **3** engine, **4** source (All / Factory / User = own presets / a content pack), **5** favourites only, **6** tag (MPE / Fingerboard / Morph – the library's quick filters), **7** minimum rating, **8** sort. A touched encoder shows its column's name and value in full.
- **Lines 2 – 4** – the preset before, the one under the cursor (marked `>`), the next one: name · category · engine · author or pack, `+` = favourite, `4*` = rating.

```
 12/240    Pads     All      All      Off      MPE      All      Name
 Warm Glass Pad    Pads             Analog             groovex
>Velvet Layers     Pads             Wavetable          groovex  + 4*
 Wide Air          Pads             Grain              Factory
```

Control	In browse mode
encoder 1	moves the cursor through the list (nothing is loaded)
encoders 2 – 8	change that column's filter (the list and the screen follow)
upper button 1	loads the preset under the cursor (lit green)
upper buttons 2 – 8	reset that column's filter – lit yellow while the filter is set
▲ / ▼	previous / next preset in the list, **loaded at once** (audition)
◀ / ▶	cursor 4 up / down

Control	In browse mode
Select	loads the preset under the cursor into the edit part
part buttons 1 – 4	load it into that part (the edit part lit green)
Shift (pressed alone)	favourite on / off for the preset under the cursor
Browse or Device	leave browse mode (the loaded preset stays)

Transport, octave and the other buttons keep working. Outside browse mode ▲ / ▼ step the presets as before.

Touch strip

Bend: pitch bend of the edit part (the screen's pitch wheel follows), back to the middle when you let go.
Mod: the mod wheel.

Foot pedal

A sustain pedal in the controller's pedal jack works like one on a MIDI keyboard: **sustain** (CC 64) holds the notes of the part the pads play after you let go of the pads, **sostenuto** (CC 66) holds the notes that were down when it was pressed; both let go when the pedal comes up. In an MPE / Fingerboard part the pedal holds the notes of all member channels. The pedal goes up on the part it went down on, also after a part change, and when the controller is unplugged. With the part's arpeggiator on, the pedal holds the arpeggio's keys too – see Sequencers.

Notes

- No host parameters of its own: the controller only changes existing parameters (host notified, automatable as before).
- Own implementation of the controller's publicly documented MIDI / display protocol; Ohpe contains no third-party code.

PART 7

Installation and settings.

Installing Ohpe.

Ohpe runs on macOS and Windows; Windows is described in section 5. Ohpe for macOS comes as one disk image, `Ohpe-<version>-macOS-Installer.dmg`. It runs on macOS 11 or newer, on Apple Silicon and Intel Macs (all binaries are universal).

File on the disk image	What it does
`Ohpe-Installer.pkg`	Installer with three parts you can pick individually
`Ohpe Gatekeeper Fix.command`	Makes an installed Ohpe openable if macOS still blocks it
`Uninstall Ohpe.command`	Removes Ohpe again; keeps your own presets unless you say otherwise
`Read me first.txt` / `Bitte zuerst lesen.txt`	The short version of this page
`LICENSE.txt`	GNU Affero General Public License v3.0 or later

Where to get it

Download the newest version from the releases page: `Ohpe-<version>-macOS-Installer.dmg` for the Mac, `Ohpe-<version>-Windows-app.zip` and `Ohpe-<version>-Windows-VST3.zip` for Windows, plus `SHA256SUMS.txt` to check the files. Once Ohpe is installed it finds and loads new versions itself (UPDATES.md).

Builds between releases: every push to main, every manual run of the **CI** workflow and every pull request labelled `installer` builds the installer on the macOS runner and attaches it to the run as the artifact **Ohpe-macOS-Installer**: open the repository's Actions tab → CI → a green run → Artifacts. GitHub delivers artifacts as a zip; unzip it to get the `.dmg` (and `SHA256SUMS.txt`).

1. Open the installer

The installer is not signed with an Apple Developer ID, so on the first double-click macOS says it cannot be opened or comes from an unidentified developer. Use one of these ways – each only affects this one file:

1. **Open Anyway** (every macOS version; on macOS 15 and newer the only way from the Finder):
    1. Double-click `Ohpe-Installer.pkg`. macOS blocks it – click **Done** (or **OK**).
    2. Open **System Settings → Privacy & Security** and scroll down to **Security**. There is a line saying that `Ohpe-Installer.pkg` was blocked.
    3. Click **Open Anyway**, confirm with your password or Touch ID, then click **Open Anyway** (or **Open**) once more in the dialog. The installer starts.
2. **Right-click → Open** (macOS 14 and older): in Finder, right-click (or Control-click) `Ohpe-Installer.pkg`, choose **Open**, then **Open** again in the dialog.
3. **Terminal:** copy the pkg out of the disk image (for example into Downloads) and remove the quarantine flag:

    ```
    xattr -d com.apple.quarantine ~/Downloads/Ohpe-Installer.pkg
    ```

    Then double-click it as usual.

2. Choose what to install

Click **Customize** in the installer to switch parts on or off:

Part	Installed to
Standalone app	`/Applications/Ohpe.app`
VST3 plug-in	`/Library/Audio/Plug-Ins/VST3/Ohpe.vst3`
Audio Unit	`/Library/Audio/Plug-Ins/Components/Ohpe.component`

The welcome, licence and read-me pages appear in English or German depending on your system language. Quit your music software before installing and restart it afterwards so it rescans its plug-ins.

3. How the Gatekeeper fix works

macOS blocks software that is downloaded and not notarised by Apple, and Apple Silicon Macs refuse to run code that carries no signature at all. Every part of the installer therefore runs a small `postinstall` script right after copying its bundle:

1. `xattr -dr com.apple.quarantine <bundle>` – removes the "downloaded from the internet" flag.
2. `codesign --force --deep -s - <bundle>` – an ad-hoc signature made on your Mac. It proves nothing about the author, but it satisfies Apple Silicon's "code must be signed" rule. If the bundle already carries a valid Developer ID signature (a signed build), this step is skipped so the real signature is kept.
3. For the Audio Unit: `killall -9 AudioComponentRegistrar` – restarts the Audio Unit registry so hosts see the new version without logging out (errors are ignored).

The script never stops the installation; problems are written to the installer log (`/var/log/install.log`).

If macOS still blocks Ohpe

Double-click **`Ohpe Gatekeeper Fix.command`** on the disk image. It asks for your administrator password (sudo) and repeats steps 1–3 for every Ohpe copy it finds – and only for these exact paths, and only bundles whose identifier starts with `com.groovex.ohpe`:

- `/Applications/Ohpe.app`, `/Library/Audio/Plug-Ins/VST3/Ohpe.vst3`, `/Library/Audio/Plug-Ins/Components/Ohpe.component`
- the same three below your home folder (`~/Applications`, `~/Library/Audio/Plug-Ins/…`)

Symbolic links are never followed. Running it again does no harm. Messages are in English or German depending on your system language. If macOS also blocks the `.command` file itself, right-click it → **Open**, or run it in Terminal:

```
bash "/Volumes/Ohpe/Ohpe Gatekeeper Fix.command"
```

4. Uninstall

Double-click **`Uninstall Ohpe.command`**. After confirming and entering your administrator password it removes the app, the VST3, the Audio Unit (same paths and checks as the Gatekeeper fix), forgets the installer receipts (`pkgutil --forget com.groovex.ohpe.app`, `….vst3`, `….au`) and restarts the Audio Unit registry.

Your own presets (`~/Library/Application Support/groovex/Ohpe`) and the app settings (`~/Library/Application Support/Ohpe.settings`) are **kept**; the script asks whether you want to delete them as well (default: no). In Terminal:

```
sudo bash "/Volumes/Ohpe/Uninstall Ohpe.command" --yes          # no questions, keeps presets
sudo bash "/Volumes/Ohpe/Uninstall Ohpe.command" --yes --purge  # also deletes presets/settings
```

5. Windows

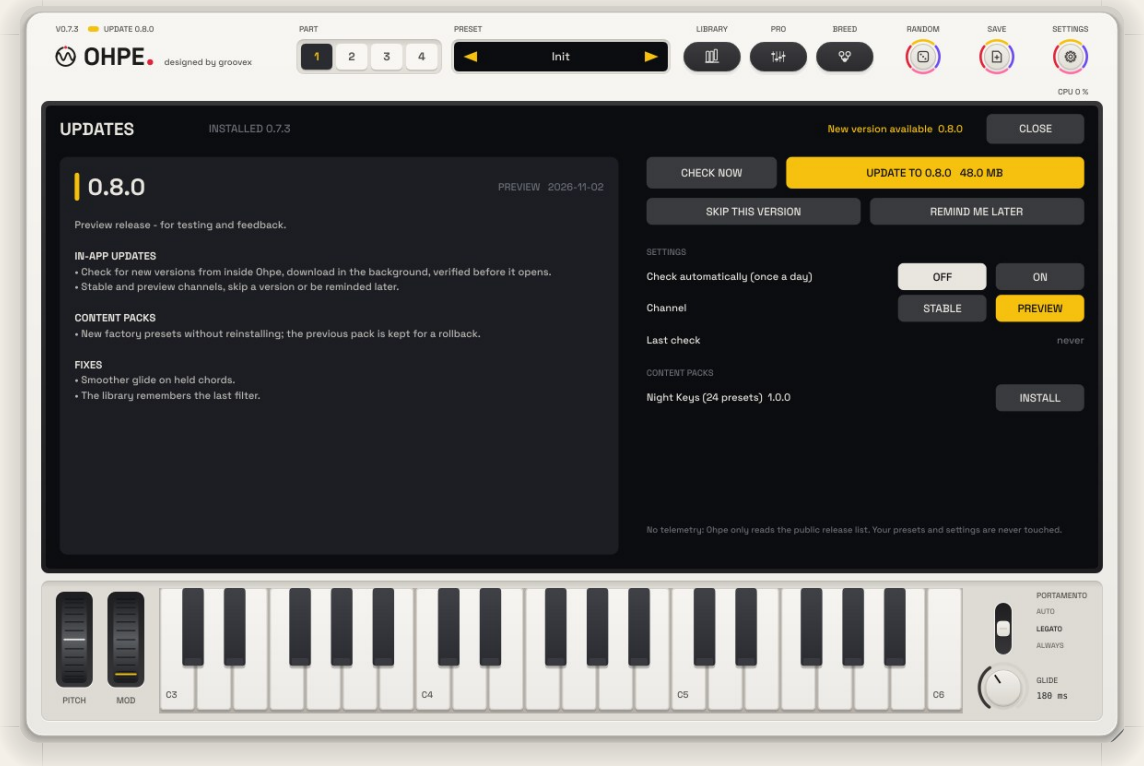
Ohpe for Windows (Windows 10 or 11, 64-bit) comes as two zip files: `Ohpe-<version>-Windows-app.zip` holds the standalone app `Ohpe.exe`, `Ohpe-<version>-Windows-VST3.zip` the plug-in folder `Ohpe.vst3`. The builds are not signed with a code-signing certificate, so Windows SmartScreen warns once.

1. **Unblock the download:** right-click the zip → **Properties** → tick **Unblock** at the bottom → **OK**. Then right-click → **Extract All…**
2. **App:** move `Ohpe.exe` wherever you like (for example `C:\Program Files\Ohpe\`) and double-click it.
3. **SmartScreen:** if Windows says Windows protected your PC, click **More info**, check that the app is `Ohpe.exe`, then click **Run anyway**. Windows remembers this; the warning does not come back for this file.
4. **VST3 plug-in:** copy the whole `Ohpe.vst3` folder to `C:\Program Files\Common Files\VST3\` (Windows asks for administrator rights), then let your music software rescan its plug-ins.

Uninstall: delete `Ohpe.exe` and the `Ohpe.vst3` folder. Your own presets, tables and settings stay in `%APPDATA%\groovex\Ohpe`; delete that folder too if you want to remove them.

Updates from inside Ohpe.

Ohpe can find, download and open a new version itself – in the standalone app and in the plug-in. Factory content (preset and sample packs) can be added the same way without reinstalling.



The update view

Opening the update view

Click the **version legend** above the wordmark (e.g. `V0.9.0`) or the wordmark itself. When a newer version is on offer, the legend shows a small **yellow light bar** and `UPDATE 0.8.0`. **Esc** or **CLOSE** closes the view; opening the library, the pro view or the breeding view closes it too.

What it shows

Area	Content
Top row	installed version, the last result (yellow = something to act on, red = an error)
Left	the version on offer: number, channel, date and its **release notes in the UI language** (scroll with the wheel)
CHECK NOW	looks for a new version now
UPDATE TO x.y.z	downloads the installer for your system in the background; a **violet** bar shows the progress, **CANCEL** pauses it
INSTALL	after the download: opens the verified installer (macOS installer image; Windows archives are shown in Explorer)
SKIP THIS VERSION	this version is not offered again – a newer one is
REMIND ME LATER	the automatic check keeps quiet about this version for three days
Settings	**Check automatically (once a day)** – off until you switch it on; **Channel** Stable / Preview; last check

Area	Content
Content packs	factory preset / sample packs of the release: **INSTALL**, **UPDATE**, **ROLL BACK**
PREVIOUS VERSION	downloads the newest release older than yours (rollback) – installed the same way

How it works

- **Where it looks.** Ohpe reads the public release list of the project on GitHub. No account, no telemetry: nothing about you, your system or your music is sent. Without a connection the view says so; nothing else happens. A server that answers with an error (rate limit, maintenance) or with something that is not a release list is reported as such – never as "up to date". Redirects are followed (at most 8); a request that gets no answer gives up after 15 s.
- **When.** Only when you press CHECK NOW – or, if you switch it on, automatically **at most once a day** while an Ohpe window is open, and **never while the transport or the tape is running.**
- **Channels.** Stable offers releases only; Preview also offers pre-releases. Builds before 1.0 start on Preview.
- **Download.** The download runs on its own thread. A broken connection is retried with growing pauses and the download **continues where it stopped** (also after a restart of Ohpe). A connection that stops delivering data (no byte for twice the 15 s timeout) is dropped and continued in a new one, so a download never hangs; CANCEL pauses it at once. If the server keeps breaking off without getting further, the download stops with an error.
- **Verification.** Every file is checked against the **SHA-256 checksum** published with the release (and its size). A file that fails the check is **deleted and never opened.** The check is repeated right before INSTALL opens it. Code signature / notarisation checks follow once the installers are signed. **Size limits:** a download larger than 2 GB (declared or while it streams) and a release list or manifest larger than 4 MB are refused.
- **Installing.** The standalone offers to quit so the installer can replace the program. In a plug-in host Ohpe asks you to save the project and quit the host before the installer continues.

Content packs

A release can offer content packs (factory presets, samples). INSTALL downloads the pack, verifies it and unpacks it into its own folder; its presets appear in the library next to the factory bank (read-only). Installing a newer version keeps the previous one as a backup – **ROLL BACK** brings it back (one step). Every path inside a pack is checked to stay in the pack's folder; a pack that tries to write elsewhere is refused as a whole.

Your presets, user wavetables, bounces and settings are never touched – they live in other folders.

Packs you have as a file (`.ohpepack`) are installed with **INSTALL PACK ...** or by dropping them on the window, and managed (version, size, remove) on **Settings > Content packs** – see Content packs.

Files

What	Where (inside the user data folder `groovex/Ohpe`)
Update settings (per user, not per project)	`Settings/update.json`
Downloads (partial files end in `.part`)	`Updates/`
Content packs, their registry and backups	`Content/<pack>/`, `Content/installed.json`, `Content/.backup/`

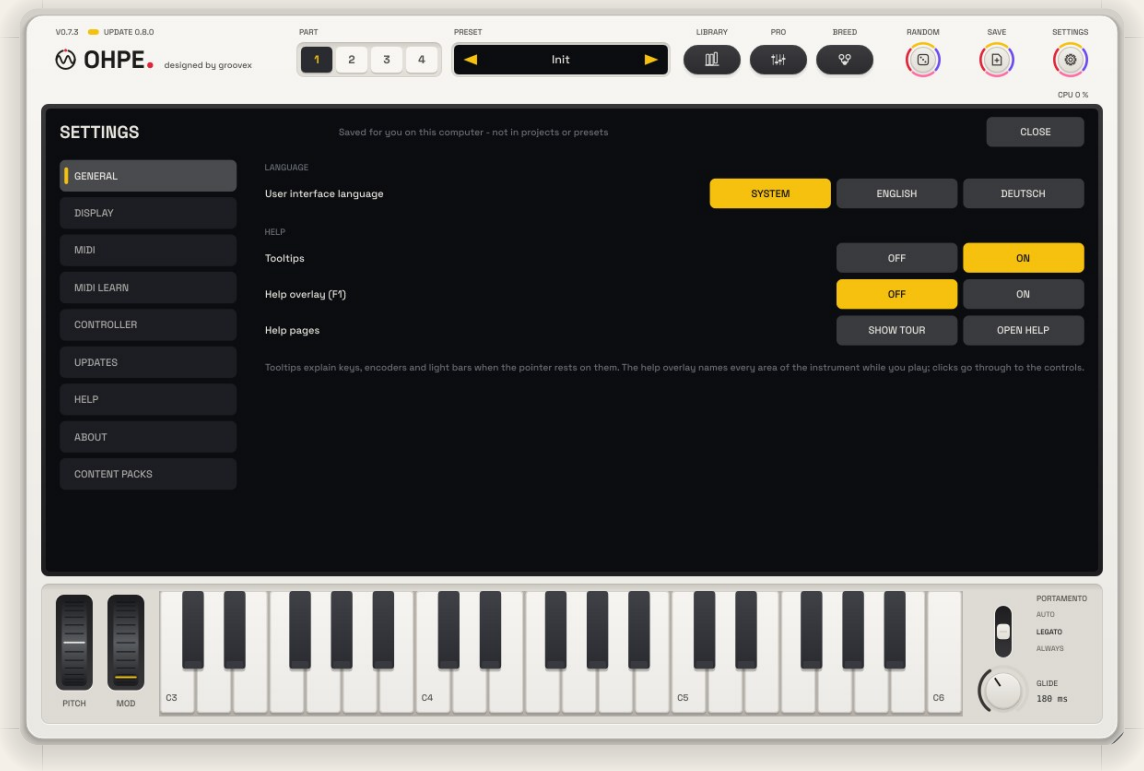
The user data folder is `~/Library/Application Support/groovex/Ohpe` on macOS, `%APPDATA%\groovex\Ohpe` on Windows and `~/.config/groovex/Ohpe` on Linux. The environment variable `OHPE_DATA_DIR` moves it (portable setups, tests).

The updater is not a sound control: it has no host parameters and no MIDI mapping.

Settings.

One settings view for the standalone app and the plug-in. Open it with the **gear key** at the right of the header, from the standalone window's menu (**Settings …**), or click the wordmark for the **About** page. **Esc** closes it, **↑ / ↓** switch the tabs.

Settings are saved **for you on this computer** – not in projects, sessions or presets – and apply to every Ohpe you open (all open plug-in windows follow at once). File: `Settings/settings.json` in the Ohpe user folder (macOS `~/Library/Application Support/groovex/Ohpe`, Windows `%APPDATA%\groovex\Ohpe`, Linux `~/.config/groovex/Ohpe`; `OHPE_DATA_DIR` overrides it). The updater keeps its own settings next to it in `update.json`. None of the settings is a host parameter: nothing here is automated or changes the sound of a project.



Settings – General

General

Setting	Choices
Language	**System** (follows the operating system), **English**, **Deutsch**. The language pills of earlier versions moved here.
Tooltips	On / Off – explanations on keys, encoders, light bars and the display
Help overlay	On / Off (also **F1**) – names and one-line explanations over every area of the plate; clicks go through to the controls
Help pages	**SHOW TOUR** starts the first-start tour again, **OPEN HELP** opens the **Help** tab



Help overlay

Display

Setting	Choices
Skin	**Light** (matte warm white), **Dark** (night skin), **Auto** (follows the system's light / dark appearance)
Display brightness	40 – 100 % – dims the display for dark rooms
Animations	**Full**, **Reduced** (slower breathing lights and 3D pulse), **Off** (running lights stand still, fully lit)
Animated 3D visuals	the moving visuals of the display (same switch as in the display's visual menu)
Window size	UI size 75 / 100 / 125 / 150 / 175 / 200 % – the window takes it at once and a new window opens with it (you can still drag the corner); also with a right-click on the housing or the settings key – see accessibility

The **night skin** covers the whole instrument – housing, keys, display, pro view, library, update view and the settings themselves. It uses warm dark greys, never pure white; the accents and light bars are dimmed. Every text colour keeps a contrast of at least 4.5 : 1 to its background (checked by the tests).

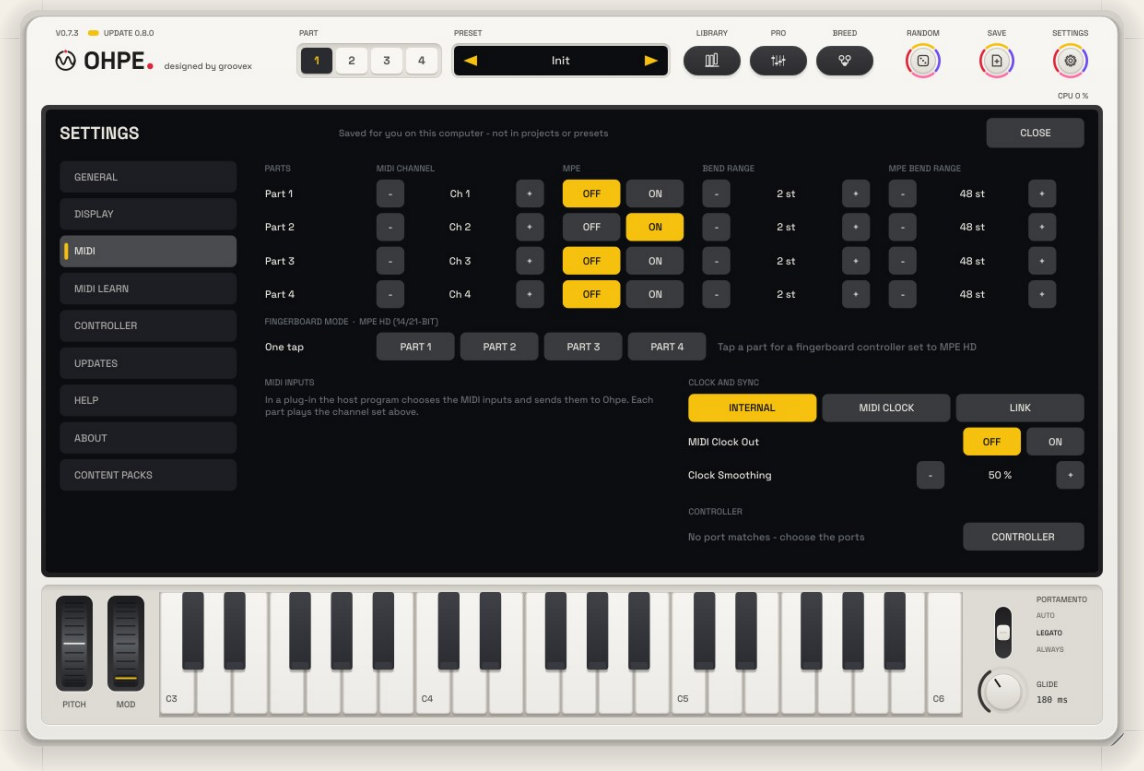


Night skin



Settings – Display in the night skin

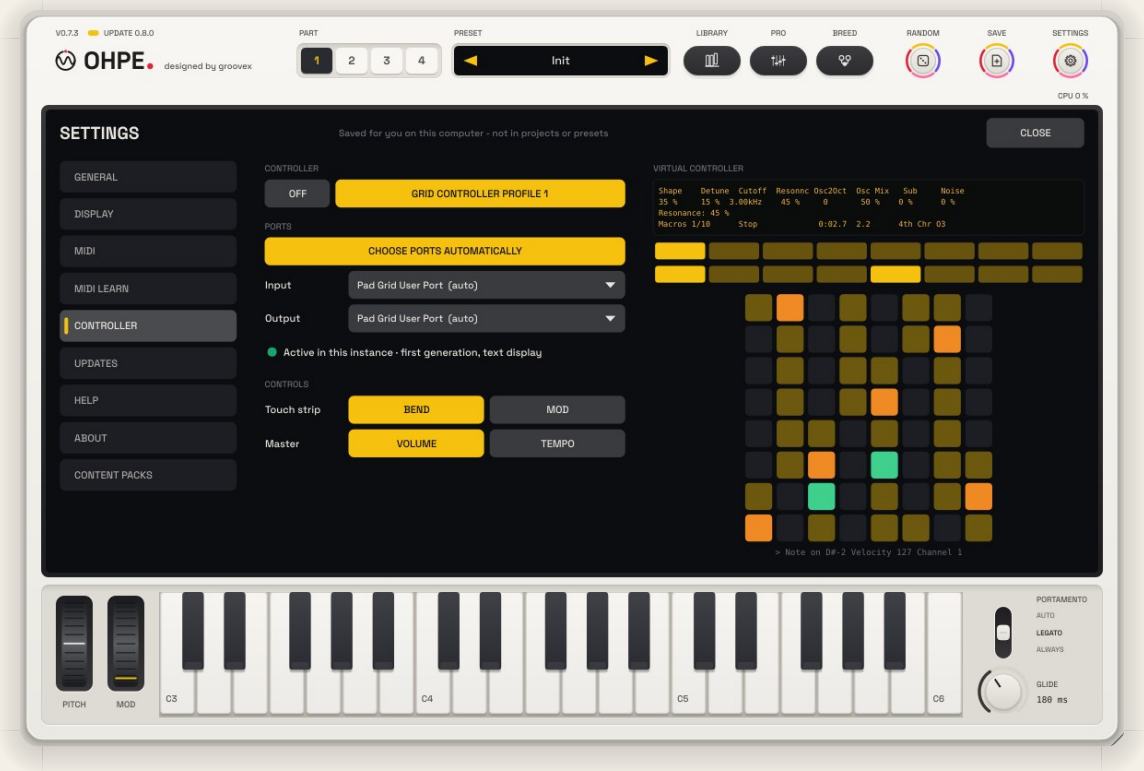
MIDI



Settings – MIDI

- **Parts:** MIDI channel (Omni or 1 – 16), MPE on / off, bend range and MPE bend range of each part.
- **MIDI inputs:** in the standalone app every MIDI input with On / Off and a button to the audio settings; in a plug-in the host chooses the inputs.
- **Clock and sync:** source Internal / MIDI clock / Link, MIDI clock out, clock smoothing.
- **Controller:** a status line and a key to the **Controller** page.

Controller

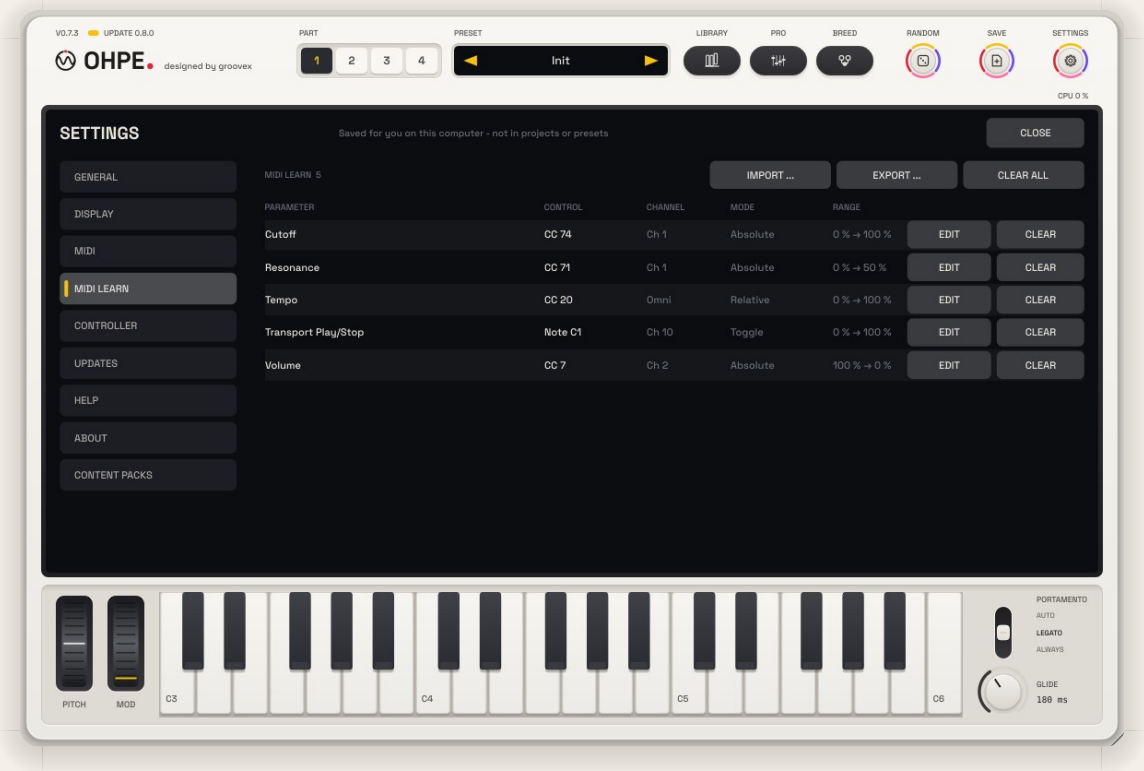


Settings – Controller

Profile Off / **Grid controller profile 1** (on by default), **Choose ports automatically** (on by default) and a menu each for the controller's **Input** and **Output** with the full port names, the connection status (Connected with the controller's generation / Checking the device / No controller found / Port in use / Waiting), touch strip Bend / Mod, master encoder Volume / Tempo; in a plug-in which instance holds the controller and **Take over here.** On the right the live virtual controller (display, button rows, pads; click pads and buttons, mouse wheel over the display turns an encoder). Saved in `settings.json` – see Grid controller.

These are the existing parameters (`pN.midich`, `pN.mpe`, `pN.bend`, `pN.mpe.bend`, `sync.*`) – changing them here is the same as anywhere else: the host is notified, they stay automatable and MIDI-mappable and are saved with the project.

MIDI learn

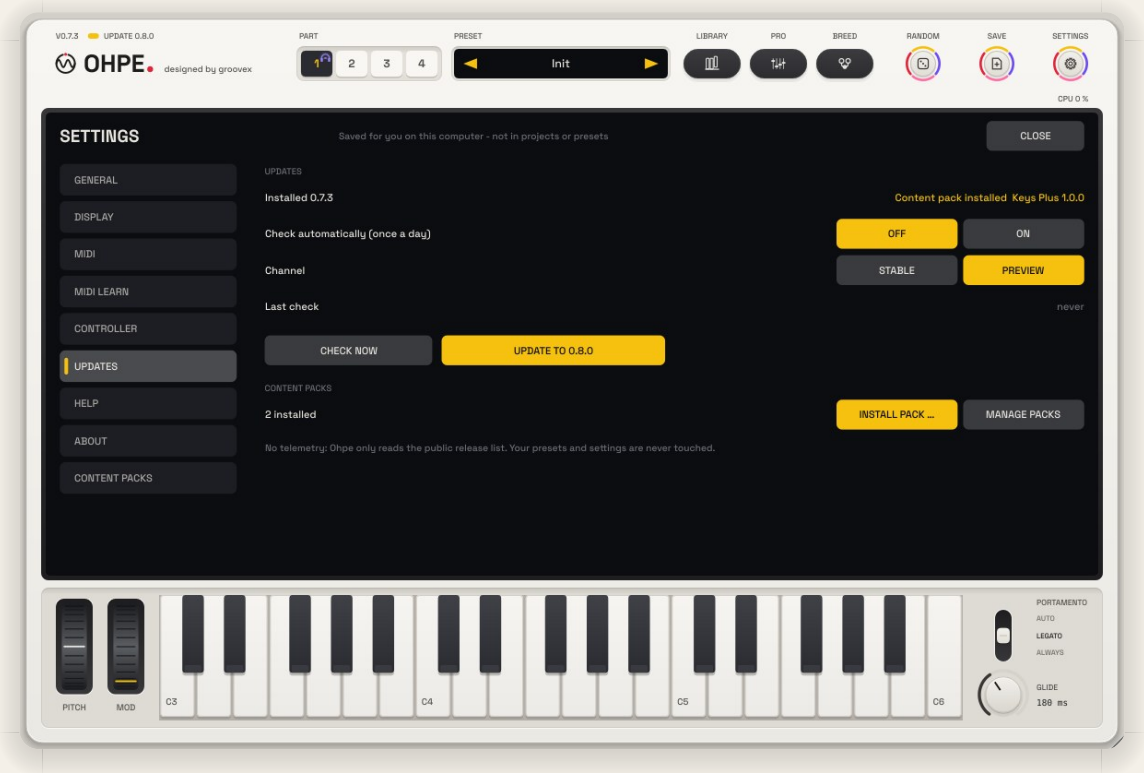


Settings – MIDI learn

Every mapping in one list: parameter, control (CC or note), channel, mode and range. **EDIT** opens a menu – channel (Omni, 1 – 16), mode (Absolute, Toggle, Relative), range (full, inverted, lower half, upper half), **Learn again** (the old mapping is removed and MIDI learn waits for a new control) and **Clear**. **CLEAR** removes one mapping, **CLEAR ALL** all of them. **IMPORT ... / EXPORT ...** read and write MIDI map files (`.ohpemidi`). New mappings come from MIDI learn: right-click an encoder, key or wheel. The map is saved with the project.

Updates

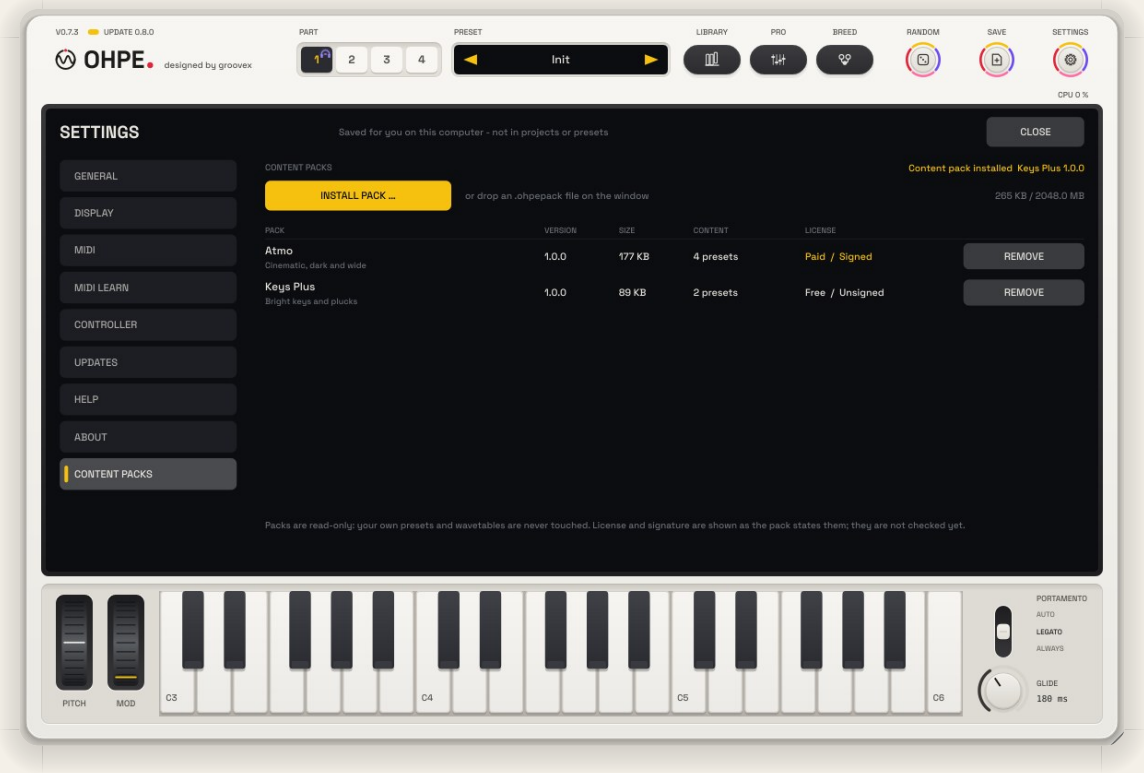
The settings of the updater: daily automatic check (off until switched on), channel Stable / Preview, last check, **CHECK NOW** and the update view. Below them: **Content packs** – how many are installed, **INSTALL PACK ...** (file dialog for `.ohpepack` files) and **MANAGE PACKS**.



Settings – Updates

Content packs

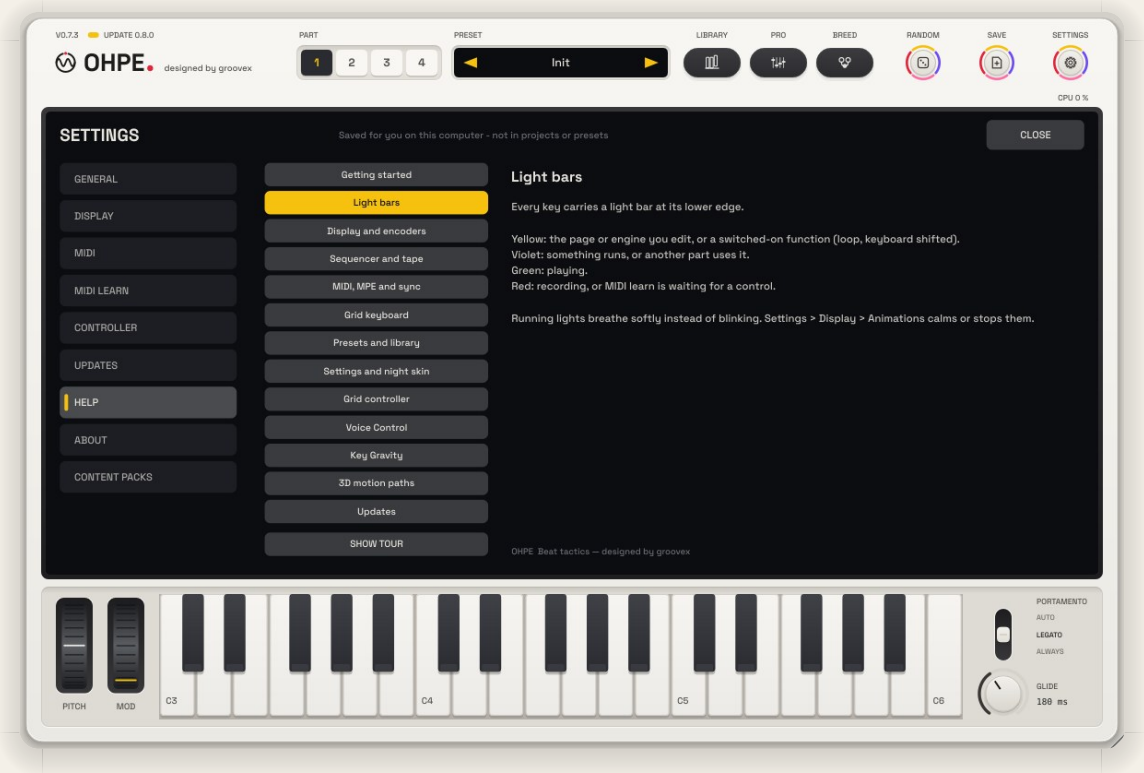
The pack manager (a tab after About): **INSTALL PACK ...**, the room the packs use, and every installed pack with version, size, content, license and signature (shown, not checked) and **REMOVE** (click twice). Dropping an `.ohpepack` file on the window installs it too. See Content packs.



Settings – Content packs

Help

Help pages in English and German: getting started, light bars, display and encoders, sequencer and tape, MIDI / MPE / sync, the grid keyboard (layouts, shapes, MPE – the full guide is in Harmony), presets, settings and the night skin, updates. With the grid keyboard shown, the help overlay adds a line with its steps over the keyboard.

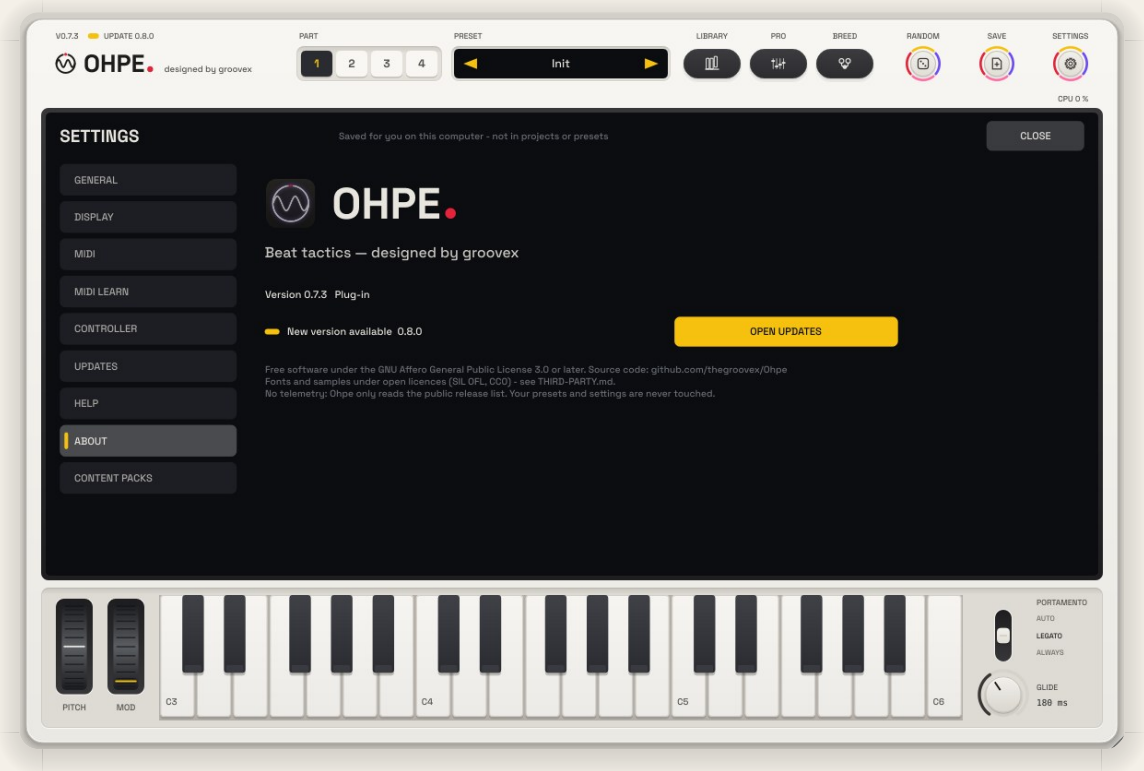


Settings – Help

SHOW TOUR under the topic list starts the first-start tour again.

About

The app icon and the wordmark at the top, then version and format (standalone app / plug-in), the update status with **CHECK NOW** (or **OPEN UPDATES** when a newer version is on offer), the project line, licence (AGPL-3.0-or-later) and source code.


V0.7.3
UPDATE 0.8.0
OHPE.
designed by groovex
PART
1
2
3
4
PRESET
Init
LIBRARY
PRO
BREED
RANDOM
SAVE
SETTINGS
CPU 0 %
SETTINGS
Saved for you on this computer - not in projects or presets
CLOSE
GENERAL
DISPLAY
MIDI
MIDI LEARN
CONTROLLER
UPDATES
HELP
ABOUT
CONTENT PACKS
OHPE.
Beat tactics — designed by groovex
Version 0.7.3 Plug-in
New version available 0.8.0
OPEN UPDATES
Free software under the GNU Affero General Public License 3.0 or later. Source code: github.com/thegroovex/Ohpe
Fonts and samples under open licences (SIL OFL, CC0) - see THIRD-PARTY.md.
No telemetry: Ohpe only reads the public release list. Your presets and settings are never touched.
PITCH
MOD
C3
C4
C5
C6
PORTAMENTO
AUTO
LEGATO
ALWAYS
GLIDE
180 ms

About

Accessibility and UI size.

Ohpe can be played and edited without a mouse, read by a screen reader, and shown at six sizes. None of this changes the sound or the project: the UI size lives in the settings of this computer (`settings.json`), never in a project, preset or host session, and no host parameter was added.

UI size (75 – 200 %)

Six sizes: **75 / 100 / 125 / 150 / 175 / 200 %** of the 1200 × 800 plate.

- **Settings > Display > Window size**, or
- **right-click on the housing** (any empty spot of the plate) or on the **settings key** – the menu UI size.

The window (standalone) or the plug-in editor takes the size at once and can still be dragged at the corner; it keeps the plate's aspect ratio (3 : 2) between 67 % and 200 %. Everything is drawn as vectors: legends, pictograms, light bars and the display stay sharp at every size, and on high-density (Retina / HiDPI) screens the housing is rendered at the screen's own pixel density. On a small screen choose a size that fits it – the window is never larger than you set.

Keyboard operation

A click never moves the keyboard focus: after using the mouse, the computer keyboard still plays notes. The focus moves only with these keys:

Key	Action
Tab / Shift+Tab	next / previous control, area by area: header (version, parts, preset, LIBRARY … settings), engines and LIFE, display (display, Key Gravity, TAP, tempo), the four encoders, the mode keys, transport and tempo, octave / MIDI / grid keys, wheels, portamento, glide and the keyboard
F6 / Shift+F6 (or Ctrl+Tab)	first control of the next / previous area
← → ↑ ↓	on a value control (encoder, wheel, LIFE, portamento, glide, Key Gravity, tempo, part): one step – a choice one position, a continuous value 1 %
Shift + arrow	finer (0.1 %)
Page Up / Page Down	ten steps (choices: three positions; tempo: 10 BPM)
Delete / Backspace	the default value (like a double-click)
Enter / Space	press the focused key; on the preset strip: open the library; on the tempo: type the tempo
Shift+F10	the focused key's menu (MIDI learn, as a right-click)
1 – 4	on the part selector: part 1 – 4
← → and 1 – 9	on the display: page back / forward and the soft keys (numbered from the left)
Esc	the focus back to the keyboard (notes on the computer keyboard again)

The focused control carries a **focus ring**: a dark outline with the yellow edit light inside, so it stands out on the light housing, the night housing and the dark display. It shows only while you move with Tab / F6; a click hides it.

Notes on the computer keyboard play while the on-screen keyboard has the focus (by default, after Esc or a click on the keys): A W S E D F T G Y H U J K O L P play C to D♯ an octave up. None of the navigation keys above is a note key, so Tab, F6, the arrows, Page, Enter, Space, Delete and Esc never play notes. While another control has the focus, the letters do not play notes – they go on to the shortcuts (standalone: R, L, I, O, T,

K). The letters T, K, L and O are both notes and standalone shortcuts: with the keyboard focused they play notes; press Tab once to use them as shortcuts. Space is PLAY / stop in the standalone app – unless a key has the focus, then Space presses that key.

Screen reader

Every control has a name and a value in the UI language (English or German, switched with the language): keys (with their state, e.g. a selected engine or mode is checked), encoders (the bound parameter's full name and its value as the plate shows it), wheels, LIFE, portamento, glide, part selector, preset strip, tempo, TAP, Key Gravity, the key readout, the metronome strip, the tape and keyboard readouts and the display (header title, page, status and the numbered soft keys). Value controls can be set by the screen reader (in percent of their range). Tooltips are read as the control's help text (every control has one, English and German); **Ctrl / Cmd + F** opens the help search, which finds topics, parameters and controls by typing (see Tooltips and help search in the user interface chapter). Ohpe uses the operating system's accessibility interface (VoiceOver on macOS, Narrator / NVDA on Windows; on Linux as far as the desktop supports it).

Contrast

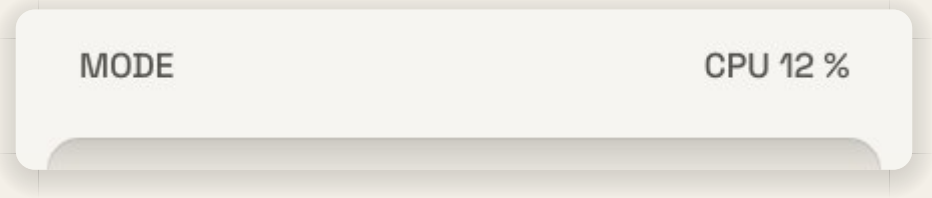
Both skins meet WCAG AA where the design allows it:

- legends on the housing and on the keys, display text and the dim display texts: at least 4.5 : 1 (light and night);
- every lit light bar has a fine edge in its own colour, deepened in the light skin and lifted in the night skin, to at least 3 : 1 on its key (non-text contrast) – the yellow bar on the warm-white housing is no longer only a hint;
- the focus ring: at least 3 : 1 on every surface (housing, panels, keys, display).

The coloured display accents (violet, red) keep their colour; they always come with text or a symbol, never as the only sign.

See also: user interface · settings.

Performance (p35, p79).



The CPU meter on the mode column's legend line

Ohpe computes four parts, their racks, the master and the display on your computer's processor. This page explains what it costs, how to read the CPU meter, what Ohpe does to stay light, and how to measure it yourself.

The CPU meter

`CPU 12 %` on the legend line of the mode column (right, next to MODE) shows how much of each audio block's time Ohpe needs to compute its sound, averaged over about half a second. 100 % would be the whole time: the audio would break up. The tooltip adds the slowest block of the last two seconds (held for two seconds, then falling). The legend turns red from 80 % on average or when a block took longer than its own time.

- It measures Ohpe alone (the plug-in's own processing), not the host or other plug-ins.
- It is a display only: no parameter, not automatable, not saved.
- Short peaks right after loading a preset or a project are normal (new sounds are prepared once).

What Ohpe does to stay light

Measure	Effect
Silent parts sleep	A part whose voices have not played for 4.5 s and whose output (after resonator body, FX rack and 3D stage) stayed below −120 dBFS all that time stops rendering its effect chain until a note starts again. The hold is longer than the longest delay line of any effect, so no echo or reverb tail is ever cut, and a note wakes the part within the same block.
Faster FM engine	The operators use a polynomial sine instead of the library call, only the links an algorithm actually has are summed, operators at level 0 skip their sine, and each operator is computed over a whole control block at once. The sound stays the same (tests compare it with the previous engine).
Lighter effects	The hall's line modulation and the phaser's sweep use the same polynomial sine. Mixer sends at 0 and switched-off outputs cost nothing.
Only changed parameters	Per audio block the core reads only the parameters that changed (instead of all of them), and the plug-in passes only changed host values on. Automation reaches the sound exactly as before.
Shared built-in tones	Built-in sample material (the Grain texture, the Sampler's tone) exists once per process, not once per part or instance: a second instance starts in a few milliseconds.
Limiter only when needed	The master limiter skips its true-peak interpolation in blocks where no peak can reach the ceiling. The output is bit-identical.
Sends and master rack sleep	Like the parts, the two send effects and the master rack stop computing after their input and output stayed below −120 dBFS for longer than their longest tail; a note wakes them in the same block.
No denormals	The core flushes tiny floating-point values to zero while it computes, so fading tails never slow the processor down.

Measure	Effect
Cheaper voices	Pluck recomputes its string filter only when a value changes (bit-identical); Grain's window and Vocal's glottal pulse use the polynomial sine; the 3D stage and Key Gravity compute once per block or per change what they computed per sample or per frame.
Quieter display	When nothing sounds, runs or changes for two seconds and nothing on the display moves with the clock (3D scene, character, meters, scopes, blinking keys), the display repaints at 10 instead of 30 frames per second. Any change brings it back at once.

Numbers (p79, path to 1.0)

Measured with `ohpe_bench --seconds 1.5 --hash` at 48 kHz, block 256, best of 3 runs, on a 2-core 64-bit Linux build machine. "Before" is main before this pass (0.7.3), "after" is this pass; both ran one after the other on the same machine. Percent of one CPU core – lower is better. "Output" compares a hash of the rendered audio: bit-identical means the sound did not change at all.

Scenario	Before avg %	After avg %	Change	Before peak %	After peak %	Output
`idle`	1.78	0.84	-53 %	2.50	1.07	bit-identical
`typical.fm`	7.59	7.25	-4 %	8.64	12.10	bit-identical
`typical.wavetable`	7.95	7.20	-9 %	20.38	10.14	bit-identical
`typical.analog`	6.98	6.06	-13 %	10.93	9.07	bit-identical
`typical.pluck`	6.66	3.93	-41 %	19.30	5.02	bit-identical
`typical.grain`	8.85	7.73	-13 %	23.15	11.58	changed: polynomial sine (window)
`typical.vocal`	7.51	7.07	-6 %	8.37	11.39	changed: polynomial sine (glottal pulse)
`typical.cluster`	8.39	7.73	-8 %	9.05	9.66	bit-identical
`typical.sampler`	3.83	2.97	-22 %	4.55	4.48	bit-identical
`worst.fm`	21.13	21.09	0 %	21.92	22.45	bit-identical
`worst.wavetable`	20.29	20.14	-1 %	20.96	21.65	bit-identical
`worst.analog`	18.83	15.91	-16 %	19.61	17.80	bit-identical
`worst.pluck`	21.00	18.14	-14 %	21.79	21.01	changed: grain layer
`worst.grain`	22.12	21.94	-1 %	23.02	24.87	changed: polynomial sine
`worst.vocal`	21.76	21.89	+1 %	22.51	24.89	changed: polynomial sine
`worst.cluster`	17.71	17.59	-1 %	18.65	20.82	bit-identical
`worst.sampler`	16.69	16.56	-1 %	17.55	19.41	bit-identical
`one-part.fm`	8.69	7.88	-9 %	11.20	11.20	bit-identical
`silent-racks`	1.68	0.67	-60 %	2.01	1.02	bit-identical
`morph.4x4`	25.68	23.41	-9 %	26.66	26.68	bit-identical
`chord16.fm`	3.23	2.64	-18 %	3.63	3.33	bit-identical
`chord16.wavetable`	3.32	2.40	-28 %	3.66	2.92	bit-identical
`chord16.analog`	3.01	2.10	-30 %	3.41	2.59	bit-identical
`chord16.pluck`	2.93	1.50	-49 %	3.31	1.92	bit-identical
`chord16.grain`	3.58	2.61	-27 %	4.13	3.26	changed: polynomial sine
`chord16.vocal`	3.18	2.34	-26 %	3.43	2.93	changed: polynomial sine
`chord16.cluster`	3.46	2.56	-26 %	3.75	3.13	bit-identical
`chord16.sampler`	2.17	1.28	-41 %	2.67	1.91	bit-identical
`drums`	2.13	1.44	-32 %	2.66	2.17	bit-identical

Scenario	Before avg %	After avg %	Change	Before peak %	After peak %	Output
fx.full	9.73	8.83	-9 %	12.16	13.78	bit-identical
space3d	11.46	11.19	-2 %	11.99	13.36	bit-identical
gravity	10.91	8.72	-20 %	11.59	9.95	changed: Key Gravity weights cached (RMS equal to 6 digits)
everything	26.55	25.27	-5 %	27.45	28.46	bit-identical
silent.after-play	3.06	0.75	-75 %	3.46	2.44	changed: Key Gravity in the setup
silent.master-fx	2.96	0.69	-77 %	3.42	0.79	bit-identical
silent.long-after (new)	–	0.72	–	–	0.83	–
sum	**346.8**	**311.0**	**-10 %**			

Startup: about 378 ms for the first instance in a process and 21 ms for each further one (369 / 19 ms before; within run-to-run noise). Peaks vary more from run to run than averages; the averages are the comparison that counts.

The quick runs on the CI runners of all three systems (`--quick`, 0.5 s × 2; before = main 0.7.3, after = this pass, different runs on shared machines):

Scenario	Linux before → after	macOS before → after	Windows before → after
idle	1.09 → 1.12	1.42 → 0.74	1.82 → 1.17
typical.fm	5.34 → 8.52	8.53 → 6.41	9.36 → 9.24
typical.pluck	5.68 → 4.66	8.24 → 3.93	8.10 → 4.93
typical.grain	6.24 → 10.09	8.87 → 7.81	14.01 → 11.88
worst.analog	15.31 → 19.92	18.72 → 16.27	25.26 → 21.84
silent-racks	0.98 → 0.76	2.54 → 0.60	1.72 → 0.88
morph.4x4	19.48 → 29.23	36.63 → 26.58	37.24 → 34.78
chord16.pluck	2.19 → 1.86	3.26 → 1.40	3.29 → 1.89
drums	1.54 → 1.72	3.96 → 1.10	2.60 → 1.88
gravity	7.59 → 10.40	17.58 → 11.36	16.58 → 13.71
everything	19.14 → 31.48	31.62 → 17.34	36.85 → 34.26
silent.master-fx	1.95 → 0.77	2.45 → 0.52	3.57 → 0.88
sum, all 35	**260 → 401** (+54 %)	**372 → 313** (-16 %)	**477 → 438** (-8 %)

The Linux runner of the ”after” run was a slower machine: bit-identical scenarios such as `typical.fm` rose just as much, so read the Linux column as noise; the silent scenarios still fell. The macOS and Windows columns agree with the local table.

New scenarios: chord16.* = one part, 16-note chord on 16 voices, no effects; drums = the drum kit, sixteenths at 120 BPM; fx.full = 6 effects on every part, 4 master effects, both sends; space3d = 3D stage moving with room on every part; gravity = Key Gravity 80 % on every part; everything = all of it at once; silent.after-play = the full setup played for 1 s and released, measured after 9 s; silent.long-after = the same after 19 s (everything asleep); silent.master-fx = no notes, master effects and sends.

Numbers

Measured with `ohpe_bench` (below) at 48 kHz, block 256, 2 s per scenario, best of 3 runs, on a 2-core 64-bit Linux build machine. ”Before” is the state before this pass (0.4.0 plus the features merged since), ”after” is this pass; both were run one after the other on the same machine. Percent of one CPU core – lower is better. Peaks vary more from run to run than averages; the averages are the comparison that counts.

Scenario	Before avg %	After avg %	Change	Before peak %	After peak %
idle	1.89	1.62	-14 %	2.05	2.08
typical.fm	12.76	7.46	-42 %	13.29	9.96
typical.wavetable	7.89	7.73	-2 %	8.47	9.22
typical.analog	6.94	6.79	-2 %	7.56	8.73
typical.pluck	6.59	6.44	-2 %	7.13	7.66
typical.grain	8.76	8.60	-2 %	9.48	10.26
typical.vocal	7.59	7.36	-3 %	8.30	8.34
typical.cluster	8.47	8.27	-2 %	9.06	10.05
typical.sampler	3.70	3.47	-6 %	4.52	4.06
worst.fm	27.15	20.98	-23 %	27.78	24.86
worst.wavetable	20.37	20.15	-1 %	21.02	21.49
worst.analog	18.86	18.67	-1 %	19.49	20.19
worst.pluck	21.04	20.90	-1 %	21.84	24.84
worst.grain	22.15	21.90	-1 %	22.90	23.54
worst.vocal	21.85	21.62	-1 %	22.61	22.56
worst.cluster	17.56	17.25	-2 %	18.45	18.03
worst.sampler	22.57	16.27	-28 %	23.54	17.16
one-part.fm	12.15	4.95	-59 %	12.80	10.27
silent-racks	9.11	1.61	-82 %	9.66	1.75
morph.4x4	30.09	25.02	-17 %	30.98	26.68
sum	**287.5**	**247.1**	**-14 %**		

Startup (build and prepare the core): about 370 ms for the first instance in a process and 18 ms for each further one (after this pass; the earlier bench did not measure startup).

Scenarios: idle = default sound, no notes; typical = 4 parts × 8 voices of one engine, 2 effects per part; worst = 4 parts × 16 voices plus a layer, 6 effects per part; one-part.fm = one FM part playing, three silent parts with full racks; silent-racks = no notes, 6 effects on every part, tails long gone; morph.4x4 = morph on with four sounds in every part.

What this means in practice: a typical project with four playing parts takes well under a tenth of one core on that machine; parts you are not using cost close to nothing once their tails have died away.

Measuring it yourself

`ohpe_bench` is built with the core tests (`cmake --build build --target ohpe_bench`, it lands in `build/tests/core/`). It renders the scenarios with the core alone – no plug-in, no UI – and prints average and slowest block per scenario.

Option	Meaning
(none)	all scenarios, 48 kHz, block 256, 2 s each, best of 3 runs
`--quick`	shorter runs (as in CI)
`--block N` / `--rate HZ`	another block size / sample rate
`--seconds S` / `--repeats N`	longer runs / more runs
`--filter TEXT`	only scenarios whose name contains TEXT
`--csv` / `--list`	machine-readable output / scenario names only

Option	Meaning
`--hash`	adds a hash and the RMS of each scenario's output (to prove an optimisation does not change the sound)
`--baseline FILE` / `--write FILE`	compare the averages with an earlier `--csv` run / write this run as a new one

Compare numbers from the same machine only. CI runs a quick bench on Linux, macOS and Windows, compares it with `tools/bench/baseline-<os>.csv`, prints it in the build summary and uploads it as the artifact `bench-<os>`, for information; nothing gates on timing. The tool exits with code 2 when a scenario renders a non-finite or runaway output (a correctness check).

Tips

- Larger buffer sizes in your host lower the load per block and make peaks less likely.
- Voices cost the most: the voice count per part and layers raise the load roughly in proportion.
- Long reverbs and delays keep a part awake until their tails fall below −120 dBFS.

PART 8

Appendix.

Parameters.

English · Deutsch: German names in the tables. The complete, ordered table of every parameter is printed by `ohpe_param_dump --markdown`; `tests/core/golden/param_ids_v1.txt` lists every ID in its permanent order. IDs are append-only (see ARCHITECTURE.md); sections are added at the end of this file. Note names: C4 = MIDI 60.

Grain, Vocal, Sampler engines and drum kit

Appended after the Wavetable, Pluck and Cluster parameters (`p1.cluster.drift`). The first four parameters of each engine are its macros. Guides: English · Deutsch.

Grain (`p1.grain.*`)

ID	English	Deutsch	Range	Default
p1.grain.pos	Position	Position	0 … 1	0.3
p1.grain.size	Grain Size	Korngröße	5 … 500	120
p1.grain.density	Density	Dichte	1 … 100	24
p1.grain.spray	Spray	Streuung	0 … 1	0.15
p1.grain.pitch	Grain Pitch	Korn-Tonhöhe	-24 … 24	0
p1.grain.reverse	Reverse Chance	Rückwärts-Anteil	0 … 1	0
p1.grain.spread	Stereo Spread	Stereobreite	0 … 1	0.6
p1.grain.scan	Scan Speed	Scan-Tempo	-2 … 2	0.1
p1.grain.source	Source	Quelle	Sample, Tape	0
p1.grain.tone	Tone	Klang	0 … 1	1

Vocal (`p1.vocal.*`)

ID	English	Deutsch	Range	Default
p1.vocal.vowel	Vowel	Vokal	0 … 4	0
p1.vocal.formant	Formant Shift	Formant-Verschiebung	-12 … 12	0
p1.vocal.breath	Breath	Atem	0 … 1	0.15
p1.vocal.tension	Tension	Spannung	0 … 1	0.5
p1.vocal.vibdepth	Vibrato Depth	Vibrato-Tiefe	0 … 1	0.2
p1.vocal.vibrate	Vibrato Rate	Vibrato-Tempo	0.5 … 10	5.5
p1.vocal.jitter	Jitter	Jitter	0 … 1	0.1

Sampler (`p1.sampler.*`)

ID	English	Deutsch	Range	Default
p1.sampler.start	Start	Start	0 … 1	0
p1.sampler.tone	Tone	Klang	0 … 1	1
p1.sampler.loopstart	Loop Start	Loop-Start	0 … 1	0
p1.sampler.loopend	Loop End	Loop-Ende	0 … 1	1
p1.sampler.root	Root Key	Grundton	0 … 127	60
p1.sampler.loop	Loop	Loop	Off, Forward, Ping-Pong	0
p1.sampler.reverse	Reverse	Rückwärts	0 … 1	0

ID	English	Deutsch	Range	Default
p1.sampler.mode	Mode	Modus	Zones, Slices	0
p1.sampler.slicesens	Slice Sensitivity	Slice-Empfindlichkeit	0 … 1	0.5
p1.sampler.slicebase	First Slice Key	Erste Slice-Taste	0 … 127	36

Drum kit (`drum.*`)

`drum.vol`, `drum.base`, then for every pad N = 1 … 24 the eight parameters `drum.padN.src`, `.type`, `.tune`, `.decay`, `.level`, `.pan`, `.choke`, `.tone` (pad 1 shown; pads 2 – 24 are identical apart from the defaults of the factory kit below).

ID	English	Deutsch	Range	Default
drum.vol	Drum Volume	Drum-Lautstärke	0 … 1	0.8
drum.base	Drum Base Note	Drum-Grundnote	0 … 104	36
drum.pad1.src	Pad 1 Source	Pad 1 Quelle	Sample, Synth	0
drum.pad1.type	Pad 1 Synth Model	Pad 1 Synth-Modell	Kick, Snare, Hat, Clap, Tom, Perc	0
drum.pad1.tune	Pad 1 Tune	Pad 1 Stimmung	-24 … 24	0
drum.pad1.decay	Pad 1 Decay	Pad 1 Abklingen	0 … 1	0.5
drum.pad1.level	Pad 1 Level	Pad 1 Pegel	0 … 1	0.95
drum.pad1.pan	Pad 1 Pan	Pad 1 Panorama	-1 … 1	0
drum.pad1.choke	Pad 1 Choke Group	Pad 1 Choke-Gruppe	0 … 8	0
drum.pad1.tone	Pad 1 Tone	Pad 1 Klangfarbe	0 … 1	0.5

Factory kit (MIDI channel 10, `drum.base` = 36). "Recording" = the pad plays a recorded sample (CC0, see THIRD-PARTY.md); the synth model plays without a sample or with Source = Synth.

Pad	Note	Sound	Recording	Synth model	Choke	Level	Pan
1	36 (C2)	kick	yes	Kick	0	0.95	+0.00
2	37 (C#2)	kick, snares on	yes	Kick	0	0.90	+0.00
3	38 (D2)	snare	yes	Snare	0	0.85	+0.00
4	39 (D#2)	rimshot	yes	Snare	0	0.75	+0.00
5	40 (E2)	clap	no (model)	Clap	0	0.70	+0.00
6	41 (F2)	closed hat	yes	Hat	1	0.60	+0.00
7	42 (F#2)	open hat	yes	Hat	1	0.50	+0.00
8	43 (G2)	pedal hat	yes	Hat	1	0.50	+0.00
9	44 (G#2)	low tom	yes	Tom	0	0.80	+0.00
10	45 (A2)	high tom	yes	Tom	0	0.80	+0.00
11	46 (A#2)	floor tom	yes	Tom	0	0.80	+0.00
12	47 (B2)	cross-stick	yes	Perc	0	0.70	+0.00
13	48 (C3)	ride	yes	Hat	0	0.45	+0.00
14	49 (C#3)	ride bell	yes	Perc	0	0.45	+0.00
15	50 (D3)	crash	yes	Hat	0	0.45	+0.00
16	51 (D#3)	sizzle crash	yes	Hat	0	0.40	+0.00
17	52 (E3)	half-open hat	yes	Hat	1	0.50	+0.00
18	53 (F3)	ghost snare	yes	Snare	0	0.40	+0.00
19	54 (F#3)	cowbell	yes	Perc	0	0.45	+0.30
20	55 (G3)	tambourine	yes	Hat	0	0.40	-0.30

Pad	Note	Sound	Recording	Synth model	Choke	Level	Pan
21	56 (G#3)	shaker	yes	Hat	0	0.35	+0.35
22	57 (A3)	claves	yes	Perc	0	0.45	-0.25
23	58 (A#3)	conga	yes	Tom	0	0.65	-0.35
24	59 (B3)	bongo	yes	Tom	0	0.60	+0.35

Tape machine, mixer dynamics, FX slot

85 parameters, appended in this order after the drum kit parameters (`drum.pad24.tone`). Guides: English · Deutsch.

Mixer dynamics (`mix.*`)

ID	English	Deutsch	Range	Default
`mix.ch1.comp`	Ch 1 Compressor	Kanal 1 Kompressor	0 … 1	0
`mix.ch2.comp`	Ch 2 Compressor	Kanal 2 Kompressor	0 … 1	0
`mix.ch3.comp`	Ch 3 Compressor	Kanal 3 Kompressor	0 … 1	0
`mix.ch4.comp`	Ch 4 Compressor	Kanal 4 Kompressor	0 … 1	0
`mix.ch5.comp`	Ch 5 Compressor	Kanal 5 Kompressor	0 … 1	0
`mix.ch6.comp`	Ch 6 Compressor	Kanal 6 Kompressor	0 … 1	0
`mix.ch7.comp`	Ch 7 Compressor	Kanal 7 Kompressor	0 … 1	0
`mix.ch8.comp`	Ch 8 Compressor	Kanal 8 Kompressor	0 … 1	0
`mix.master.comp`	Master Compressor	Master-Kompressor	0 … 1	0
`mix.master.comp.thresh`	Comp Threshold	Kompressor Schwelle	-40 … 0	-12
`mix.master.comp.ratio`	Comp Ratio	Kompressor Ratio	1 … 20	3
`mix.master.comp.attack`	Comp Attack	Kompressor Attack	0.0001 … 0.1	0.01
`mix.master.comp.release`	Comp Release	Kompressor Release	0.01 … 2	0.15
`mix.master.comp.makeup`	Comp Makeup	Kompressor Aufholverstärkung	0 … 24	0
`mix.master.tapesat`	Master Tape Saturation	Master-Bandsättigung	0 … 1	0
`mix.master.ceiling`	Limiter Ceiling	Limiter-Obergrenze	-12 … 0	-0.3

Tape transport and character (`tape.*`)

ID	English	Deutsch	Range	Default
`tape.play`	Tape Play	Tape Wiedergabe	0 … 1	0
`tape.rec`	Tape Record	Tape Aufnahme	0 … 1	0
`tape.speed`	Tape Speed	Tape-Geschwindigkeit	0.5 … 2	1
`tape.reverse`	Tape Reverse	Tape rückwärts	0 … 1	0
`tape.loop`	Tape Loop	Tape-Schleife	0 … 1	0
`tape.loopin`	Loop In	Schleife Start	0 … 3600	0
`tape.loopout`	Loop Out	Schleife Ende	0 … 3600	8
`tape.overdub`	Overdub	Overdub	0 … 1	1
`tape.sat`	Tape Saturation	Tape-Sättigung	0 … 1	0
`tape.wow`	Wow	Wow	0 … 1	0
`tape.flutter`	Flutter	Flutter	0 … 1	0
`tape.hiss`	Tape Hiss	Bandrauschen	0 … 1	0

Tape tracks (`tape.tr1.*` … `tape.tr8.*`)

Shown for track 1; tracks 2–8 are identical (`tape.tr2.arm` …). Order: track 1 arm, source, level, direct out, then track 2 …

ID	English	Deutsch	Range	Default
`tape.tr1.arm`	Track 1 Arm	Spur 1 Aufnahmebereit	0 … 1	0
`tape.tr1.src`	Track 1 Source	Spur 1 Quelle	Input, Part 1, Part 2, Part 3, Part 4, Drum	1
`tape.tr1.level`	Track 1 Level	Spur 1 Pegel	-60 … 6	0
`tape.tr1.out`	Track 1 Direct Out	Spur 1 Direktausgang	Mixer only, Out 1, Out 2, Out 3, Out 4, Out 5, Out 6, Out 7, Out 8	0

FX slot of part 1 (`p1.fx.*`)

`p1.fx.type`: Off, Chorus, Phaser, Bitcrush, Filter Drive, Delay, Reverb (append-only list). Each effect keeps its own parameters, so switching the type and back restores the sound.

ID	English	Deutsch	Range	Default
`p1.fx.type`	FX Slot	FX-Slot	Off, Chorus, Phaser, Bitcrush, Filter Drive, Delay, Reverb	0
`p1.fx.chorus.rate`	Chorus Rate	Chorus-Tempo	0.05 … 5	0.6
`p1.fx.chorus.depth`	Chorus Depth	Chorus-Tiefe	0 … 1	0.5
`p1.fx.chorus.mix`	Chorus Mix	Chorus-Mischung	0 … 1	0.5
`p1.fx.phaser.rate`	Phaser Rate	Phaser-Tempo	0.05 … 8	0.4
`p1.fx.phaser.depth`	Phaser Depth	Phaser-Tiefe	0 … 1	0.7
`p1.fx.phaser.feedback`	Phaser Feedback	Phaser-Feedback	0 … 0.95	0.5
`p1.fx.phaser.mix`	Phaser Mix	Phaser-Mischung	0 … 1	0.5
`p1.fx.bitcrush.bits`	Bit Depth	Bittiefe	1 … 16	8
`p1.fx.bitcrush.downsample`	Downsample	Downsampling	1 … 64	1
`p1.fx.bitcrush.mix`	Bitcrush Mix	Bitcrusher-Mischung	0 … 1	1
`p1.fx.filterdrive.cutoff`	Drive Cutoff	Drive-Cutoff	20 … 20000	2000
`p1.fx.filterdrive.reso`	Drive Resonance	Drive-Resonanz	0 … 1	0.3
`p1.fx.filterdrive.drive`	Drive	Drive	0 … 1	0.3
`p1.fx.filterdrive.mode`	Drive Filter Type	Drive-Filtertyp	Low-pass, Band-pass, High-pass	0
`p1.fx.filterdrive.mix`	Drive Mix	Drive-Mischung	0 … 1	1
`p1.fx.delay.time`	Delay Time	Delay-Zeit	0.01 … 2	0.375
`p1.fx.delay.feedback`	Feedback	Feedback	0 … 0.95	0.4
`p1.fx.delay.tone`	Tone	Klang	0 … 1	0.6
`p1.fx.delay.pingpong`	Ping-Pong	Ping-Pong	0 … 1	1
`p1.fx.delay.mix`	Delay Mix	Delay-Mischung	0 … 1	0.35
`p1.fx.reverb.size`	Size	Größe	0 … 1	0.6
`p1.fx.reverb.damping`	Damping	Dämpfung	0 … 1	0.4
`p1.fx.reverb.predelay`	Pre-Delay	Pre-Delay	0 … 0.2	0.01
`p1.fx.reverb.mix`	Reverb Mix	Hall-Mischung	0 … 1	0.35

Sequencers, parts, MIDI and sync / Sequencer, Parts, MIDI und Sync

ID	EN	DE	Range / Bereich	Default
`seq.step.pattern`	Pattern	Pattern	1–16	1
`seq.step.song`	Song Mode	Song-Modus	off / on	off
`p2.*`, `p3.*`, `p4.*`	parts 2–4: same common and engine parameters as `p1.*`	Parts 2–4: dieselben Parameter wie `p1.*`		
`pN.midich`	Part N MIDI Channel	Part N MIDI-Kanal	0 (omni) – 16	N
`pN.quant`	Part N Scale Quantise	Part N Skala quantisieren	off / on	off
`pN.chord`	Part N Chord	Part N Akkord	Off, Triad, Seventh, Sus2, Sus4, Power, Octave, Sixth, Add9	Off
`pN.mpe`	Part N MPE	Part N MPE	off / on	off
`pN.mpe.bend`	Part N MPE Bend Range	Part N MPE-Bend-Bereich	0–96 st	48
`pN.mpe.press`	Part N Pressure Target	Part N Druck-Ziel	Off, Timbre, Level, LFO Depth, Pan	Level
`pN.mpe.timbre`	Part N Timbre Target	Part N Timbre-Ziel	Off, Timbre, Level, LFO Depth, Pan	Timbre
`key.root`	Key	Tonart	C – B	C
`key.scale`	Scale	Skala	Chromatic, Major, Minor, Dorian, Phrygian, Lydian, Mixolydian, Locrian, Harmonic Minor, Melodic Minor, Major / Minor Pentatonic, Blues, Whole Tone	Major
`seq.euclid.on`	Euclid On	Euklid an	off / on	off
`seq.euclid.part`	Euclid Target Part	Euklid Ziel-Part	1–4	2
`seq.euclid.rate`	Euclid Rate	Euklid Raster	1/4, 1/8, 1/16, 1/32, 1/4T, 1/8T, 1/16T, 1/32T, 1/4., 1/8., 1/16.	1/16
`seq.euclid.gate`	Euclid Gate	Euklid Gate	5–100 %	50 %
`seq.euclid.vel`	Euclid Velocity	Euklid Anschlag	1–127	100
`seq.euclid.rN.steps`	Ring N Steps	Ring N Schritte	1–32	16
`seq.euclid.rN.pulses`	Ring N Pulses	Ring N Impulse	0–32	4 / 0 / 0 / 0
`seq.euclid.rN.rot`	Ring N Rotation	Ring N Rotation	0–31	0
`seq.euclid.rN.note`	Ring N Note	Ring N Note	0–127	36 / 38 / 42 / 46
`seq.arp.on`	Arp On	Arp an	off / on	off
`seq.arp.part`	Arp Target Part	Arp Ziel-Part	1–4	3
`seq.arp.mode`	Arp Mode	Arp Modus	Up, Down, Up/Down, Random, As Played, Chord	Up

ID	EN	DE	Range / Bereich	Default
`seq.arp.oct`	Arp Octaves	Arp Oktaven	1–4	1
`seq.arp.gate`	Arp Gate	Arp Gate	5–100 %	50 %
`seq.arp.rate`	Arp Rate	Arp Raster	as Euclid	1/16
`seq.arp.latch`	Arp Latch	Arp Halten	off / on	off
`seq.orbit.on`	Orbit On	Orbit an	off / on	off
`seq.orbit.part`	Orbit Target Part	Orbit Ziel-Part	1–4	4
`seq.orbit.gate`	Orbit Gate	Orbit Gate	5–100 %	50 %
`seq.orbit.lN.len`	Lane N Length	Spur N Länge	1–16	16 / 12 / 10 / 7
`seq.orbit.lN.rate`	Lane N Rate	Spur N Raster	as Euclid	1/16
`seq.orbit.lN.transpose`	Lane N Transpose	Spur N Transponieren	±24 st	0
`sync.src`	Sync Source	Sync-Quelle	Internal, MIDI Clock, Link	Internal
`sync.smooth`	Clock Smoothing	Clock-Glättung	0–1	0.5
`sync.clockout`	MIDI Clock Out	MIDI-Clock senden	off / on	off
`sync.notesout`	Sequencer Notes Out	Sequencer-Noten senden	off / on	off

Engines of parts 2–4 / Engines der Parts 2–4

Appended last: `pN.wavetable.*`, `pN.pluck.*`, `pN.cluster.*`, `pN.grain.*`, `pN.vocal.*` and `pN.sampler.*` for N = 2, 3, 4 (in this order per part) – the same parameters, ranges and defaults as their `p1.*` counterparts (tables above; Wavetable, Pluck and Cluster in the engine guides, `docs/en/engines/`). / Zuletzt angehängt: dieselben Parameter, Bereiche und Standardwerte wie bei `p1.*` (Tabellen oben; Wavetable, Pluck und Cluster in den Engine-Anleitungen, `docs/de/engines/`), für die Parts 2, 3 und 4.

Remote controls

Appended after `p4.sampler.slicebase`. They make UI functions MIDI-mappable and are not part of presets. A trigger (`off / on`) fires when it rises to "on" (CC ≥ 64 or a note); see en/PRESETS.md and en/USER-INTERFACE.md.

ID	English	Deutsch	Range	Default
`lib.open`	Library	Bibliothek	trigger	off
`lib.prev`	Previous Preset	Vorheriges Preset	trigger	off
`lib.next`	Next Preset	Nächstes Preset	trigger	off
`ui.part`	Edit Part	Bearbeiteter Part	1–4	1
`tape.edit.track`	Tape Edit Track	Tape-Bearbeitungsspur	1–8	1
`tape.edit.lift`	Tape Lift	Tape Heben	trigger	off

ID	English	Deutsch	Range	Default
`tape.edit.cut`	Tape Cut	Tape Schneiden	trigger	off
`tape.edit.drop`	Tape Drop	Tape Einsetzen	trigger	off
`tape.edit.splice`	Tape Splice	Tape Einfügen	trigger	off
`tape.edit.undo`	Tape Undo	Tape Rückgängig	trigger	off
`tape.edit.redo`	Tape Redo	Tape Wiederholen	trigger	off
`tape.edit.rtz`	Tape Return to Zero	Tape zum Anfang	trigger	off

Factory sounds

Appended after `tape.edit.rtz`. Guides: English · Deutsch.

ID	English	Deutsch	Range	Default
`p1.sampler.inst`, `p1.grain.inst` … `p4.sampler.inst`, `p4.grain.inst`	Instrument	Instrument	Built-in, Kalimba, Marimba, Vibraphone, Glockenspiel, Harp, Harpsichord, Recorder, Singing Glass, Zither, Bowed Metal	Built-in
`drum.kit`	Drum Kit	Drum-Kit	Studio, Tight, Room, Lo-Fi, Hybrid, Boom (0.5: the list grows at the end)	Studio
`seq.step.drums`	Step Seq to Drums	Step-Sequenzer an Drums	off / on	off
`seq.euclid.drums`	Euclid to Drums	Euklid an Drums	off / on	off
`seq.orbit.drums`	Orbit to Drums	Orbit an Drums	off / on	off
`p1.gain` … `p4.gain`	Part N Gain	Part N Gain	−24 … +24 dB	0 dB

Order: `p1.sampler.inst`, `p1.grain.inst`, `p2.sampler.inst`, `p2.grain.inst`, … `p4.grain.inst`, `drum.kit`, `seq.step.drums`, `seq.euclid.drums`, `seq.orbit.drums`, `p1.gain` … `p4.gain`.

Sound revision 1: groove, poly glide, synced delay

Appended after `p4.gain` (index 900 onwards). Guides: English · Deutsch, delay sync: English · Deutsch.

#	ID	English	Deutsch	Range	Default
900	`seq.orbit.swing`	Orbit Swing	Orbit Swing	0–75 % (off-beat sixteenths late by swing ÷ 2 of a sixteenth)	0 %
901	`seq.orbit.human`	Orbit Humanise	Orbit Humanisieren	0–100 % (up to 10 ms late, velocity ±16)	0 %
902	`seq.euclid.swing`	Euclid Swing	Euklid Swing	0–75 %	0 %
903	`seq.euclid.human`	Euclid Humanise	Euklid Humanisieren	0–100 %	0 %
904–907	`p1.polyglide` … `p4.polyglide`	Part N Poly Glide	Part N Poly-Glide	off / on (chords glide voice by voice)	off
908–911	`seq.euclid.r1.vel` … `seq.euclid.r4.vel`	Ring N Velocity	Ring N Anschlag	0–127 (0 = `seq.euclid.vel`)	0
912–915	`seq.orbit.l1.part` … `seq.orbit.l4.part`	Lane N Target Part	Spur N Ziel-Part	0–4 (0 = the Orbit target / drum kit)	0
916	`fxa.delay.sync`	Delay Sync	Delay-Sync	Off, 1/16, 1/8T, 1/8, 1/4T, 1/8., 1/4, 1/4., 1/2	Off

Behaviour change in 0.5: a step-sequencer step at gate 100 % (and every Orbit step at `seq.orbit.gate` 100 %) ties into the next step when it plays the same note (not on the drum kit).

Sound revision 2: portamento, pitch envelope, crackle, master low-pass

Appended after `fxa.delay.sync` (index 917 onwards). Guides: portamento and pitch envelope English · Deutsch, master low-pass English · Deutsch, crackle / Boom / lo-fi English · Deutsch.

#	ID	English	Deutsch	Range	Default
917, 922, 927, 932	`p1.glide.mode` … `p4.glide.mode`	Part N Glide Mode	Part N Glide-Modus	Auto, Legato, Always (DE: Auto, Legato, Immer)	Auto
918, 923, 928, 933	`p1.glide.rate` … `p4.glide.rate`	Part N Glide Timing	Part N Glide-Timing	Constant Time, Constant Rate (DE: Feste Zeit, Feste Rate)	Constant Time
919, 924, 929, 934	`p1.pitchenv.amt` … `p4.pitchenv.amt`	Part N Pitch Env Amount	Part N Tonhöhen-Hüllkurve Menge	−24 … +24 semitones (negative = rise from below)	0
920, 925, 930, 935	`p1.pitchenv.time` … `p4.pitchenv.time`	Part N Pitch Env Time	Part N Tonhöhen-Hüllkurve Zeit	0.005 … 4 s	0.25 s
921, 926, 931, 936	`p1.pitchenv.curve` … `p4.pitchenv.curve`	Part N Pitch Env Curve	Part N Tonhöhen-Hüllkurve Kurve	0 (linear) … 1 (fast start)	0.5
937, 939, 941, 943	`p1.analog.osclevel` … `p4.analog.osclevel`	Osc Level	Osz.-Pegel	0 … 1 (0 = noise / sub / crackle only)	1
938, 940, 942, 944	`p1.analog.crackle` … `p4.analog.crackle`	Crackle	Knistern	0 … 1 (vinyl-style clicks into the filter)	0
945	`mix.master.lowpass`	Master Low-Pass	Master-Tiefpass	500 … 20000 Hz (20 kHz = off)	20000

Order: the five glide / pitch-envelope parameters of part 1, then parts 2, 3, 4; then `p1.analog.osclevel`, `p1.analog.crackle`, … `p4.analog.crackle`; then `mix.master.lowpass`.

Behaviour changes in 0.6: with one voice, releasing the newest key returns to the previous key still held (legato). A step-sequencer or Orbit step at gate 100 % ends one sample after the next step starts, so a different next note overlaps it (legato). MIDI CC 65 switches portamento off / on and CC 5 sets the glide time of the part on that channel.

FX rack and modulation matrix

Appended after `mix.master.lowpass`. Guides: FX rack English · Deutsch, modulation matrix English · Deutsch. Order of the block (registry indices; 1151 parameters, 946–2096):

Range	Block
946–1082	part 1 rack: slots 2–6, then mix / bypass of slots 1–6
1083–1244, 1245–1406, 1407–1568	part 2, 3, 4 rack: slot 1 (`pN.fx.*`), slots 2–6, mix / bypass
1569–1676	master rack `mfx1` … `mfx4`
1677–1781, 1782–1886, 1887–1991, 1992–2096	modulation matrix of part 1, 2, 3, 4 (105 each)

1. **Part racks** – for part 1, then 2, 3, 4: slot 1 (`pN.fx.type` + the 0.2 type blocks; parts 2–4 only, part 1 has it since 0.2), slots 2–6 (`pN.fx2.type` … `pN.fx6.*`, each the type + the 0.2 type blocks in the order of the 0.2 slot: chorus, phaser, bitcrush, filter drive, delay (+ `mix`), reverb (+ `mix`)), then `mix` and `bypass` of slots 1–6 (`pN.fx.mix`, `pN.fx.bypass`, `pN.fx2.mix` …).
2. **Master rack** – `mfx1.*` … `mfx4.*` (type + 0.2 blocks each), then `mfx1.mix`, `mfx1.bypass` … `mfx4.bypass`.
3. **Modulation matrix** – for part 1, then 2, 3, 4: routings 1–16 (`pN.mod.sK.src`, `.via`, `.dst`, `.amt`, `.curve`), LFO 2 and LFO 3 (`rate`, `sync`, `shape`, `phase`, `fade`, `mode`, `depth`), envelope 3 (`pN.env3.delay`, `.a`, `.hold`, `.d`, `.s`, `.r`), `pN.macro1` … `pN.macro4`, `pN.seqlane`.

ID	English	Deutsch	Range	Default
`pN.fxK.type` (K = 2…6), `mfxK.type`	PN FXK FX Slot / Master FX K FX Slot	PN FXK FX-Slot / Master-FX K FX-Slot	as `p1.fx.type`	Off
`pN.fxK.mix`, `mfxK.mix`	Part N FX K Mix / Master FX K Mix	… Mischung	0 … 100 % (dry / wet of the slot)	100 %
`pN.fxK.bypass`, `mfxK.bypass`	… Bypass	… Bypass	off / on	off
`pN.mod.sK.src`	Part N Mod K Source	Part N Mod K Quelle	Off, LFO 1, LFO 2, LFO 3, Amp Env, Mod Env, Env 3, Velocity, Key Track, Mod Wheel, Pitch Bend, Channel Pressure, Note Pressure, Slide (Timbre), Note Bend, Random, Macro 1–4, Seq Lane	Off
`pN.mod.sK.via`	… Via	… Via	as the source	Off
`pN.mod.sK.dst`	… Target	… Ziel	Off, Pitch, Level, Pan, Timbre, Engine 1–24, Volume, LFO 1–3 Rate / Depth, Mod Env Amount, FX 1–6 Mix, FX 1–6 Param 1–4, Azimuth, Elevation, Distance, Motion Path Rate, Motion Path Phase, Life	Off
`pN.mod.sK.amt`	… Amount	… Menge	−100 … +100 %	0
`pN.mod.sK.curve`	… Curve	… Kurve	Linear, Exponential, Logarithmic, S-Curve, Steps	Linear
`pN.lfo2.rate`	Part N LFO 2 Rate	Part N LFO 2 Tempo	0.01 … 40 Hz	1 Hz
`pN.lfo2.sync`	… LFO 2 Sync	… LFO 2 Sync	Free, 4/1, 2/1, 1/1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/2., 1/4., 1/8., 1/16., 1/2T, 1/4T, 1/8T, 1/16T	Free
`pN.lfo2.shape`	… LFO 2 Shape	… LFO 2 Form	Sine, Triangle, Saw Up, Saw Down, Square, S&H, Smooth Random	Sine
`pN.lfo2.phase`	… LFO 2 Phase	… LFO 2 Phase	0 … 1	0
`pN.lfo2.fade`	… LFO 2 Fade In	… LFO 2 Einblenden	0 … 10 s	0 s
`pN.lfo2.mode`	… LFO 2 Mode	… LFO 2 Modus	Free-Running, Retrigger, One-Shot	Retrig-ger
`pN.lfo2.depth`	… LFO 2 Depth	… LFO 2 Tiefe	0 … 1	1
`pN.lfo3.*`	LFO 3 – as LFO 2			
`pN.env3.delay` / `.hold`	Env 3 Delay / Hold	Hüllkurve 3 Verzögerung / Halten	0 … 4 s	0 s
`pN.env3.a`	Env 3 Attack	Hüllkurve 3 Attack	1 ms … 10 s	0.5 s
`pN.env3.d` / `.r`	Env 3 Decay / Release	Hüllkurve 3 Decay / Release	1 ms … 10 s	1 s
`pN.env3.s`	Env 3 Sustain	Hüllkurve 3 Sustain	0 … 1	0.5
`pN.macro1` … `pN.macro4`	Part N Macro 1–4	Part N Makro 1–4	0 … 1	0
`pN.seqlane`	Part N Seq Lane	Part N Seq-Spur	0 … 1	0

Behaviour changes in 0.7: the 0.2 FX slot is slot 1 of the part's rack (identical sound); parts 2–4 have FX slots (before 0.7 their slot parameters did not exist). The MPE / aftertouch target "Timbre" now also brightens the Cluster, Grain, Pluck, Sampler, Vocal and Wavetable engines (before: Analog and FM only). A change of the part volume glides over one block instead of jumping.

Rack effect types / Effekttypen des Racks

Appended after `p4.seqlane`. Every slot of every rack (slot 1 – 6 of each part, `mfx1` … `mfx4`) has its own block for each of the nine rack types, 75 parameters per slot, in this order: hyper, distortion, flanger, compressor, multiband, EQ, filter, echo, hall. The block of a slot is `<slot prefix>.<type>.<parameter>`, for example `p2.fx4.echo.time` or `mfx1.hall.decay`. Guides: English · Deutsch.

Range	Block
2097–2546	part 1: `p1.fx.*`, `p1.fx2.*` … `p1.fx6.*` (450 = 6 × 75)
2547–2996, 2997–3446, 3447–3896	part 2, 3, 4 – the same
3897–4196	master rack: `mfx1.*` … `mfx4.*` (300 = 4 × 75)

The table shows the block with the prefix `p1.fx2`; host names carry the slot ("P1 FX2 Echo Time", "Master FX 1 Hall Decay").

ID	English	Deutsch	Range	Default
`p1.fx2.hyper.rate`	Hyper Rate	Hyper-Tempo	0.05 … 5 Hz	0.6 Hz
`p1.fx2.hyper.detune`	Hyper Detune	Hyper-Verstimmung	0 … 1	0.35
`p1.fx2.hyper.voices`	Hyper Voices	Hyper-Stimmen	2 … 7	5
`p1.fx2.hyper.spread`	Hyper Spread	Hyper-Breite	0 … 1	0.8
`p1.fx2.hyper.dimension`	Dimension	Dimension	0 … 1	0.4
`p1.fx2.hyper.retrig`	Hyper Retrigger	Hyper-Neustart	off / on	off
`p1.fx2.hyper.mix`	Hyper Mix	Hyper-Mischung	0 … 1	0.5
`p1.fx2.distortion.mode`	Distortion Mode	Verzerrungs-Modus	Tube, Soft Clip, Hard Clip, Diode, Linear Fold, Sine Fold, Asymmetric, Downsample	Tube
`p1.fx2.distortion.drive`	Distortion Drive	Verzerrung Drive	0 … 1	0.4
`p1.fx2.distortion.bias`	Distortion Bias	Verzerrung Bias	−1 … 1	0
`p1.fx2.distortion.prefilter`	Distortion Pre-Filter	Verzerrung Vorfilter	Off, Low-pass, Band-pass, High-pass	Off
`p1.fx2.distortion.prefreq`	Distortion Pre-Filter Freq	Verzerrung Vorfilter-Frequenz	40 … 18000 Hz	800 Hz
`p1.fx2.distortion.tone`	Distortion Tone	Verzerrung Klang	0 … 1	0.8
`p1.fx2.distortion.output`	Distortion Output	Verzerrung Ausgang	−24 … +12 dB	0 dB
`p1.fx2.distortion.mix`	Distortion Mix	Verzerrung Mischung	0 … 1	1
`p1.fx2.flange.rate`	Flanger Rate	Flanger-Tempo	0.02 … 8 Hz	0.25 Hz
`p1.fx2.flange.sync`	Flanger Sync	Flanger-Sync	Free, 1/1 … 1/32, dotted, triplets	Free
`p1.fx2.flange.depth`	Flanger Depth	Flanger-Tiefe	0 … 1	0.7
`p1.fx2.flange.delay`	Flanger Delay	Flanger-Verzögerung	0.3 … 12 ms	2 ms

ID	English	Deutsch	Range	Default
p1.fx2.flange.feedback	Flanger Feedback	Flanger-Feedback	−0.95 … 0.95	0.5
p1.fx2.flange.stereo	Flanger Stereo	Flanger-Stereo	0 … 0.5 (LFO phase offset)	0.25
p1.fx2.flange.mix	Flanger Mix	Flanger-Mischung	0 … 1	0.5
p1.fx2.compressor.thresh	Compressor Threshold	Kompressor Schwelle	−60 … 0 dB	−18 dB
p1.fx2.compressor.ratio	Compressor Ratio	Kompressor Ratio	1 … 20	4
p1.fx2.compressor.attack	Compressor Attack	Kompressor Attack	0.1 … 200 ms	10 ms
p1.fx2.compressor.release	Compressor Release	Kompressor Release	5 … 2000 ms	120 ms
p1.fx2.compressor.knee	Compressor Knee	Kompressor Knie	0 … 24 dB	6 dB
p1.fx2.compressor.makeup	Compressor Makeup	Kompressor Aufholverstärkung	0 … 24 dB	0 dB
p1.fx2.compressor.mix	Compressor Mix	Kompressor Mischung	0 … 1	1
p1.fx2.multiband.depth	Multiband Depth	Multiband Tiefe	0 … 1	0.5
p1.fx2.multiband.upward / .downward	Multiband Upward / Downward	Multiband aufwärts / abwärts	0 … 1	1
p1.fx2.multiband.time	Multiband Time	Multiband Zeit	0 … 1	0.5
p1.fx2.multiband.low / .mid / .high	Multiband Low / Mid / High	Multiband Low / Mid / High	−12 … +12 dB	0 dB
p1.fx2.multiband.lowsplit	Multiband Low Split	Multiband Trennung tief	40 … 500 Hz	120 Hz
p1.fx2.multiband.highsplit	Multiband High Split	Multiband Trennung hoch	800 … 8000 Hz	2500 Hz
p1.fx2.multiband.output	Multiband Output	Multiband Ausgang	−12 … +12 dB	0 dB
p1.fx2.multiband.mix	Multiband Mix	Multiband Mischung	0 … 1	1
p1.fx2.eq.lowfreq / .lowgain	EQ Low Shelf Freq / Gain	EQ Low Shelf Frequenz / Gain	20 … 1000 Hz / ±18 dB	100 Hz / 0 dB
p1.fx2.eq.p1freq / .p1gain / .p1q	EQ Band 1 Freq / Gain / Q	EQ Band 1 Frequenz / Gain / Q	30 … 18000 Hz / ±18 dB / 0.3 … 12	300 Hz / 0 dB / 1
p1.fx2.eq.p2freq / .p2gain / .p2q	EQ Band 2 …	EQ Band 2 …	as band 1	1200 Hz / 0 dB / 1
p1.fx2.eq.p3freq / .p3gain / .p3q	EQ Band 3 …	EQ Band 3 …	as band 1	4500 Hz / 0 dB / 1
p1.fx2.eq.highfreq / .highgain	EQ High Shelf Freq / Gain	EQ High Shelf Frequenz / Gain	1000 … 20000 Hz / ±18 dB	8000 Hz / 0 dB
p1.fx2.eq.output	EQ Output	EQ Ausgang	−18 … +18 dB	0 dB
p1.fx2.filter.type	Filter Type	Filtertyp	Low-pass 12, Low-pass 24, High-pass 12, High-pass 24, Band-pass, Notch, Comb +, Comb −, Formant	Low-pass 24
p1.fx2.filter.cutoff	Filter Cutoff	Filter-Cutoff	20 … 20000 Hz	2000 Hz

ID	English	Deutsch	Range	Default
`p1.fx2.filter.reso`	Filter Resonance	Filter-Resonanz	0 … 1	0.3
`p1.fx2.filter.drive`	Filter Drive	Filter-Drive	0 … 1	0
`p1.fx2.filter.mix`	Filter Mix	Filter-Mischung	0 … 1	1
`p1.fx2.echo.mode`	Echo Mode	Echo-Modus	Stereo, Ping-Pong, Tap	Ping-Pong
`p1.fx2.echo.sync`	Echo Sync	Echo-Sync	Free, 1/1 … 1/32, dotted, triplets	1/4.
`p1.fx2.echo.time`	Echo Time	Echo-Zeit	1 … 2000 ms (Free)	375 ms
`p1.fx2.echo.ratio`	Echo Right Ratio	Echo Verhältnis rechts	0.25 … 1 (Tap)	0.75
`p1.fx2.echo.feedback`	Echo Feedback	Echo-Feedback	0 … 1.05	0.4
`p1.fx2.echo.lowcut / .highcut`	Echo Low Cut / High Cut	Echo Low Cut / High Cut	20 … 2000 Hz / 1000 … 20000 Hz	120 Hz / 8000 Hz
`p1.fx2.echo.width`	Echo Width	Echo-Breite	0 … 1	1
`p1.fx2.echo.mix`	Echo Mix	Echo-Mischung	0 … 1	0.3
`p1.fx2.hall.mode`	Hall Mode	Raum-Modus	Plate, Hall	Hall
`p1.fx2.hall.size`	Hall Size	Raumgröße	0 … 1	0.6
`p1.fx2.hall.decay`	Hall Decay	Raum Nachhall	0.2 … 20 s	2.5 s
`p1.fx2.hall.damping`	Hall Damping	Raum Dämpfung	0 … 1	0.4
`p1.fx2.hall.predelay`	Hall Pre-Delay	Raum Pre-Delay	0 … 250 ms	12 ms
`p1.fx2.hall.width`	Hall Width	Raum-Breite	0 … 1	1
`p1.fx2.hall.mix`	Hall Mix	Raum-Mischung	0 … 1	0.3

The modulation targets FX K Param 1–4 move the first four parameters of the slot's selected type in this table's order (for the echo: mode, sync, time, ratio).

Mid/Side stereo effect

Appended after the blocks of the nine 0.7 rack effect types (Hyper … Hall, last ID `mfx4.hall.mix`). Guide: English · Deutsch. The FX type list gains **Mid/Side** (DE: Mitte/Seite) at index 16 (append-only list, after Hall at 15) in every rack slot. Its block of 10 parameters is registered for every slot, in rack order: part 1 slots 1–6 (`p1.fx.midside.*`, `p1.fx2.midside.*` … `p1.fx6.midside.*`), then parts 2, 3, 4, then the master slots `mfx1.midside.*` … `mfx4.midside.*` – 28 slots × 10 = 280 parameters. Host names carry the slot prefix like the other rack blocks ("P2 FX3 Stereo Width", "Master FX 1 Stereo Width").

Range	Block
4197–4256	part 1: `p1.fx.midside.*`, `p1.fx2.midside.*` … `p1.fx6.midside.*` (60 = 6 × 10)
4257–4316, 4317–4376, 4377–4436	part 2, 3, 4 – the same
4437–4476	master rack: `mfx1.midside.*` … `mfx4.midside.*` (40 = 4 × 10)

ID (`<slot>.midside.*`)	English	Deutsch	Range	Default
`width`	Stereo Width	Stereobreite	0 … 200 % (0 = mono)	100 %
`mid`	Mid Level	Mitte-Pegel	−24 … +12 dB	0 dB
`side`	Side Level	Seite-Pegel	−24 … +12 dB	0 dB
`balance`	Balance	Balance	−100 … +100 %	0
`bassmono`	Bass Mono	Bass mono	Off, On	Off
`monofreq`	Mono Below	Mono unter	40 … 500 Hz	120 Hz
`haas`	Micro-Shift	Mikroversatz	0 … 100 %	0
`haastime`	Shift Time	Versatzzeit	1 … 30 ms	12 ms
`midtilt`	Mid Tilt	Mitte-Tilt	−6 … +6 dB	0 dB
`sidetilt`	Side Tilt	Seite-Tilt	−6 … +6 dB	0 dB

All defaults together are an exact (bit-identical) bypass. The modulation targets FX k Param 1–4 of a Mid/Side slot are `width`, `mid`, `side`, `balance`.

Sound revision 3: FX types, layer engine, keyboard instruments, MPE, bass mono

Appended after the 0.7 block (index 4477 onwards, 1056 parameters). Guides: FX types English · Deutsch, layer engine English · Deutsch, wavetable unison / FM English · Deutsch, keyboard instruments English · Deutsch, MPE English · Deutsch. Order of the block:

1. **FX types Comb, Formant, Wobble, Tremolo, Width** (`type` 16 – 20) – their parameter blocks for every slot of every rack: part 1 slots 1 – 6, parts 2, 3, 4, then the master slots (`FxRack::registerTypeBlocks`).
2. **Layer engine** of parts 1 – 4 (`pN.layer.*`).
3. **Late engine parameters**, per part: Wavetable (unison stack, FM, LFO > position, Env > FM), FM (velocity > index), Sampler and Grain (keyboard instrument).
4. **MPE / aftertouch** of parts 1 – 4: target amounts, expression smoothing, per-note slide.
5. **Bass mono** of parts 1 – 4 (`pN.monobass`).

`ohpe_param_dump --markdown` prints every row with its own index. N = part 1 – 4, K = slot (`fx.` = slot 1, `fx2.` … `fx6.`), master slots `mfx1.` … `mfx4.`.

ID	English	Deutsch	Range	Default
`pN.fxK.comb.freq`, `mfxK.comb.freq`	Comb Frequency	Kamm-Frequenz	20 … 5000	300
`pN.fxK.comb.feedback`, `mfxK.comb.feedback`	Comb Feedback	Kamm-Feedback	-0.95 … 0.95	0.6
`pN.fxK.comb.damp`, `mfxK.comb.damp`	Comb Damping	Kamm-Dämpfung	0 … 1	0.3
`pN.fxK.comb.rate`, `mfxK.comb.rate`	Comb Rate	Kamm-Tempo	0.01 … 10	0.2
`pN.fxK.comb.depth`, `mfxK.comb.depth`	Comb Depth	Kamm-Tiefe	0 … 1	0.3
`pN.fxK.comb.spread`, `mfxK.comb.spread`	Comb Spread	Kamm-Breite	0 … 1	0.3
`pN.fxK.comb.mix`, `mfxK.comb.mix`	Comb Mix	Kamm-Mischung	0 … 1	0.5
`pN.fxK.formant.vowel`, `mfxK.formant.vowel`	Vowel	Vokal	0 … 1	0

ID	English	Deutsch	Range	Default
pN.fxK.formant.shift, mfxK.formant.shift	Formant Shift	Formant-Verschiebung	-12 … 12	0
pN.fxK.formant.reso, mfxK.formant.reso	Formant Resonance	Formant-Resonanz	0 … 1	0.5
pN.fxK.formant.rate, mfxK.formant.rate	Formant Rate	Formant-Tempo	0.01 … 20	1
pN.fxK.formant.sync, mfxK.formant.sync	Formant Sync	Formant-Sync	Free, 1/1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/2., 1/4., 1/8., 1/16., 1/2T, 1/4T, 1/8T, 1/16T, 2/1, 4/1	0
pN.fxK.formant.depth, mfxK.formant.depth	Formant LFO Depth	Formant-LFO-Tiefe	0 … 1	0
pN.fxK.formant.mix, mfxK.formant.mix	Formant Mix	Formant-Mischung	0 … 1	1
pN.fxK.wobble.rate, mfxK.wobble.rate	Wobble Rate	Wobble-Tempo	0.05 … 20	2
pN.fxK.wobble.sync, mfxK.wobble.sync	Wobble Sync	Wobble-Sync	Free, 1/1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/2., 1/4., 1/8., 1/16., 1/2T, 1/4T, 1/8T, 1/16T, 2/1, 4/1	4
pN.fxK.wobble.shape, mfxK.wobble.shape	Wobble Shape	Wobble-Form	Sine, Triangle, Saw Down, Saw Up, Square, S&H	0
pN.fxK.wobble.low, mfxK.wobble.low	Wobble Low	Wobble tief	30 … 8000	150
pN.fxK.wobble.high, mfxK.wobble.high	Wobble High	Wobble hoch	100 … 18000	3000
pN.fxK.wobble.reso, mfxK.wobble.reso	Wobble Resonance	Wobble-Resonanz	0 … 1	0.5
pN.fxK.wobble.mode, mfxK.wobble.mode	Wobble Filter	Wobble-Filter	Low-pass, Band-pass	0
pN.fxK.wobble.drive, mfxK.wobble.drive	Wobble Drive	Wobble-Drive	0 … 1	0.2
pN.fxK.wobble.retrig, mfxK.wobble.retrig	Wobble Retrigger	Wobble-Neustart	0 … 1	1
pN.fxK.wobble.mix, mfxK.wobble.mix	Wobble Mix	Wobble-Mischung	0 … 1	1
pN.fxK.tremolo.rate, mfxK.tremolo.rate	Tremolo Rate	Tremolo-Tempo	0.05 … 20	4
pN.fxK.tremolo.sync, mfxK.tremolo.sync	Tremolo Sync	Tremolo-Sync	Free, 1/1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/2., 1/4., 1/8., 1/16., 1/2T, 1/4T, 1/8T, 1/16T, 2/1, 4/1	0
pN.fxK.tremolo.depth, mfxK.tremolo.depth	Tremolo Depth	Tremolo-Tiefe	0 … 1	0.5
pN.fxK.tremolo.mode, mfxK.tremolo.mode	Tremolo Mode	Tremolo-Modus	Tremolo, Autopan	0
pN.fxK.tremolo.shape, mfxK.tremolo.shape	Tremolo Shape	Tremolo-Form	Sine, Triangle, Square	0
pN.fxK.width.width, mfxK.width.width	Stereo Width	Stereobreite	0 … 2	1
pN.fxK.width.spread, mfxK.width.spread	Stereo Spread	Stereo-Spreizung	0 … 1	0
pN.fxK.width.haas, mfxK.width.haas	Spread Time	Spreizungszeit	1 … 25	12
pN.fxK.width.mono, mfxK.width.mono	Mono Below	Mono unter	20 … 500	20
pN.fxK.width.output, mfxK.width.output	Width Output	Breite-Ausgang	-24 … 12	0

ID	English	Deutsch	Range	Default
pN.layer.engine	Layer Engine	Layer-Engine	Off, FM, Wavetable, Analog, Pluck, Grain, Vocal, Cluster, Sampler	0
pN.layer.level	Layer Level	Layer-Pegel	0 … 1	0.7
pN.layer.tune	Layer Tune	Layer-Stimmung	-24 … 24	0
pN.layer.fine	Layer Fine	Layer-Feinstimmung	-50 … 50	0
pN.layer.xmode	Cross-Mod Mode	Kreuzmod.-Art	Ring, AM, Fold	0
pN.layer.xmod	Cross-Mod	Kreuzmodulation	0 … 1	0
pN.layer.width	Layer Width	Layer-Breite	0 … 1	0.5
pN.layer.ownenv	Layer Own Envelope	Layer eigene Hüllkurve	0 … 1	0
pN.layer.a	Layer Attack	Layer Attack	0.001 … 10	0.005
pN.layer.d	Layer Decay	Layer Decay	0.001 … 10	0.3
pN.layer.s	Layer Sustain	Layer Sustain	0 … 1	0.8
pN.layer.r	Layer Release	Layer Release	0.001 … 10	0.3
pN.wavetable.univoices	Unison Voices	Unisono-Stimmen	1 … 8	3
pN.wavetable.uniwidth	Unison Width	Unisono-Breite	0 … 1	0.7
pN.wavetable.fm	FM Amount	FM-Menge	0 … 1	0
pN.wavetable.fmratio	FM Ratio	FM-Verhältnis	0.25 … 8	1
pN.wavetable.lfopos	LFO > Position	LFO > Position	-1 … 1	0
pN.wavetable.envfm	Env > FM	Hüllk. > FM	-1 … 1	0
pN.fm.velidx	Velocity > Index	Anschlag > Index	0 … 1	0
pN.sampler.inst2	Keyboard Instrument	Tasteninstrument	Off, Grand Piano, Upright Piano, Tine Piano, Reed Piano	0
pN.grain.inst2	Keyboard Instrument	Tasteninstrument	Off, Grand Piano, Upright Piano, Tine Piano, Reed Piano	0
pN.mpe.pressamt	Pressure Amount	Druck-Menge	-1 … 1	1
pN.mpe.timbreamt	Timbre Amount	Timbre-Menge	-1 … 1	1
pN.mpe.smooth	Expression Smoothing	Ausdrucks-Glättung	0 … 0.5	0
pN.mpe.slide	Per-Note Slide	Pro-Noten-Slide	0 … 1	0
pN.monobass	Bass Mono Below	Bass mono unter	20 … 400	20

Behaviour changes in 0.8: `pN.fxK.type` / `mfxK.type` gain Comb, Formant, Wobble, Tremolo and Width after Hall (existing values keep their meaning). The MPE pressure / timbre targets (`pN.mpe.press`, `pN.mpe.timbre`) gain Cutoff, Position, FM Index, Grain Density, Drive, Vibrato, Formant and Layer Blend after Pan. Per-note pitch bend glides over `pN.mpe.slide` seconds (at least 2 ms); the other per-note expressions glide over `pN.mpe.smooth` seconds.

Tape transport

Appended after `p4.monobass`. MIDI-mappable transport keys (triggers: fire when the value rises to "on", CC ≥ 64) and the Play starts tape setting; none of them is stored in presets. Guides: English · Deutsch.

ID	English	Deutsch	Range	Default
tape.tx.play	Transport Play/Stop	Transport Start/Stopp	trigger	off
tape.tx.tplay	Tape Play/Stop	Tape Start/Stopp	trigger	off
tape.tx.stop	Transport Stop	Transport Stopp	trigger	off

ID	English	Deutsch	Range	Default
tape.tx.rec	Transport Record	Transport Aufnahme	trigger	off
tape.tx.loop	Transport Loop	Transport Schleife	trigger	off
tape.tx.arm	Arm Edit Track	Bearbeitungsspur aufnahmebereit	trigger	off
tape.tx.link	Play Starts Tape	Start startet Tape	on / off	on

3D spatialiser

Appended after `tape.tx.link`. Guide: English · Deutsch. 518 parameters:

1. **Per-part stage** `pN.space.*` for parts 1 – 4 (17 each, group `partN/space`; host names "Part N 3D …"): `pN.space.on` and the Space 3D block below.
2. **Monitoring** (group `master`; not stored in project presets):

ID	English	Deutsch	Range	Default
space.out	3D Monitoring	3D-Abhöre	Headphones, Speakers	0
space.monocheck	Mono Check	Mono-Prüfung	0 … 1	0

1. **Rack type Space 3D** (DE 3D-Raum, `pN.fxK.type` / `mfxK.type` value 22, appended after Width): its block `<slot>space3d.*` for every slot of every rack, in the order parts 1 – 4 slots 1 – 6, then master slots 1 – 4 (28 × 16). FX Param 1 – 4 (matrix) = Azimuth, Elevation, Distance, Motion Rate.

The Space 3D block (`pN.space.*`, `<slot>space3d.*`)

Local ID	English	Deutsch	Range	Default
on (stage only)	Space	Raum	0 … 1	0
azimuth	Azimuth	Azimut	-180 … 180 °	0
elevation	Elevation	Elevation	-60 … 90 °	0
distance	Distance	Entfernung	0.2 … 40 m (skew 0.3)	1.5
pathrate	Motion Rate	Bewegungstempo	-4 … 4	1
pathphase	Motion Phase	Bewegungsphase	0 … 360 °	0
path	Motion Path	Bewegungspfad	Off, Circle, Figure-8, Fly-By, Random Orbit, Pendulum, Spiral Up, Ping-Pong F/B, Swarm, Whisper, Distance Walk, Over the Head, Weave, Ping-Pong L/R, Four Corners, Breathe, Helix, Fly-Over, Bounce, Wander, Loop Orbit (choices 7–12 appended in p30, 13–20 in p73; older indices unchanged)	0
pathlen	Motion Length	Bewegungslänge	1/16, 1/8, 1/4, 1/2, 1 Bar, 2 Bars, 4 Bars, 8 Bars, 16 Bars, 32 Bars, 1/4.	5 (2 Bars)
pathsize	Motion Size	Bewegungsgröße	0 … 1	0.5
spread	Source Spread	Quellenbreite	0 … 180 °	40
room	Room	Raumanteil	0 … 1	0.25
roomsize	Room Size	Raumgröße	0 … 1	0.4
doppler	Doppler	Doppler	0 … 1	0.5
air	Air Absorption	Luftdämpfung	0 … 1	0.6

Local ID	English	Deutsch	Range	Default
rolloff	Distance Rolloff	Entfernungsabfall	0 … 1	0.7
lowkeep	Mono Below	Mono unter	40 … 300 Hz (skew 0.5)	120
glide	Motion Glide	Bewegungsglättung	0.005 … 2 s (skew 0.3)	0.06

Behaviour changes in 1.0: the matrix targets 67 – 71 (Azimuth, Elevation, Distance, Motion Path Rate / Phase), reserved in 0.7, now move `pN.space.azimuth` … `pN.space.pathphase`. The preset category list gains **3D**. New parameter units: degrees (°) and metres (m).

Sidechain ducking and chorus ensemble

Appended after the 3D block (`mfx4.space3d.glide`). Guides: English · Deutsch.

ID	English	Deutsch	Range	Default
mix.duck.src	Ducking Key	Ducking-Auslöser	Off, Ch 1 … Ch 8	Off
mix.duck.release	Ducking Release	Ducking-Release	0.03 … 1 s	0.16
mix.chN.duck (N = 1 … 8)	Ch N Ducking	Kanal N Ducking	0 … 1	0
pN.fxK.chorus.voices, mfxK.chorus.voices	Chorus Voices	Chorus-Stimmen	1 … 4	1
pN.fxK.chorus.drift, mfxK.chorus.drift	Chorus Drift	Chorus-Drift	0 … 1	0
pN.fxK.chorus.spread, mfxK.chorus.spread	Chorus Spread	Chorus-Breite	0 … 1	0.8
pN.fxK.chorus.lowcut, mfxK.chorus.lowcut	Chorus Low Cut	Chorus Low Cut	20 … 1000 Hz	20

The chorus IDs exist for every slot of every rack (`p1.fx.` … `p4.fx6.`, `mfx1.` … `mfx4.`), registered in that order after the ducking parameters. Their defaults give the 0.2 chorus.

Behaviour changes in the punch pass: the master limiter looks ahead 1.5 ms (latency 83 samples at 48 kHz - the look-ahead plus a 24-tap true-peak interpolator -, constant - also with `mix.master.limit` off and on the aux outputs - and reported to the host) and limits true peaks; the compressors (master, compressor-lite, rack Compressor) hold a detected peak for 12 ms; the Hyper effect restarts its voices on a new note only with Retrigger on.

Performance wheels

Appended after the 0.8 punch pass IDs (last ID `mfx4.chorus.lowcut`), as the last two IDs. The PITCH and MOD wheels left of the on-screen keyboard. A change is sent as pitch bend / mod wheel (CC 1) on the edit part's MIDI channel, the same way the on-screen keyboard plays; the range of the bend is each part's `pN.bend`. The pitch wheel springs back to the centre when released on the screen. Both are MIDI-learnable (right-click) and are not stored in presets.

ID	English	Deutsch	Range	Default
perf.bend	Pitch Wheel	Pitch-Rad	-1 … +1	0
perf.mod	Mod Wheel	Mod-Rad	0 … 1	0

Life per part

Appended after the performance wheels (last ID `perf.mod`), one per part in part order. Guides: English · Deutsch.

ID	English	Deutsch	Range	Default
pN.life (N = 1 … 4)	Part N Life	Part N Leben	0 … 1 (0 – 100 %)	0

pN.life makes held notes keep evolving: every voice of the part gets its own slow pitch drift (at most about ±12 cents), free chaos LFOs at incommensurate rates, filter / brightness breathing, a little colour motion of the engine (wavetable position, FM index, formant, grain density), extra grain scatter (Grain engine) and a hint of pan and level motion (about ±0.25 dB). 0 = static: the sound is exactly the same as without the parameter. Each voice is seeded from its part, its voice slot, its note and its start count, so a render is repeatable. Changes are followed over about 60 ms (no steps). The value is part of part presets and projects and MIDI-learnable.

The modulation matrix gains the target 72 **Life** (pN.mod.sK.dst, key life), which moves pN.life of the routing's part.

Morph space

Appended after the life block (p4.life), part by part (p1.morph.* … p4.morph.*, 24 IDs). Guides: English · Deutsch. The corners themselves are not parameters: they live in part presets ("morph" block) and in projects / the plug-in state (section "morph").

ID	English	Deutsch	Range	Default
pN.morph.on	Part N Morph	Part N Morph	Off / On	Off
pN.morph.x	Part N Morph X	Part N Morph X	0 … 1 (left … right)	0.5
pN.morph.y	Part N Morph Y	Part N Morph Y	0 … 1 (bottom … top)	0.5
pN.morph.glide	Part N Morph Glide	Part N Morph-Gleiten	0 … 2 s	0.05
pN.morph.xsrc	Part N Morph X Control	Part N Morph X Steuerung	Off, Pressure, Slide, Mod Wheel	Off
pN.morph.ysrc	Part N Morph Y Control	Part N Morph Y Steuerung	Off, Pressure, Slide, Mod Wheel	Off

The modulation matrix gains the sources **Morph X / Morph Y** (21, 22) and the targets **Morph X / Morph Y** (73, 74); the source and target choice lists grow at their end.

Audio to wavetable

Appended after the morph block (p4.morph.ysrc), part by part (p1.usertable.* … p4.usertable.*, 8 IDs). Guides: English · Deutsch. The tables themselves are not parameters: they live in the user wavetable library and are referenced by part presets (block "usertables") and by projects / the plug-in state (section "usertables"; path, library file name, embedded copy up to 64 frames).

ID	English	Deutsch	Range	Default
pN.usertable.wt	Part N User Wavetable	Part N Eigene Wavetable	Off / On	Off
pN.usertable.vocal	Part N User Formants	Part N Eigene Formanten	Off / On	Off

On without a loaded table = the built-in material (no change to the sound). No new modulation matrix sources or targets: the Wavetable Position and Vocal Vowel targets scan the user material.

Resonator body

Appended after the audio to wavetable block (`p4.usertable.vocal`). Guide: English · Deutsch. 452 parameters:

1. **Per-part stage** `pN.reso.*` for parts 1 – 4 (15 each, group `partN/reso`; host names "Part N …"): `pN.reso.on` (Resonator / Resonanzkörper, 0 … 1, default 0) and the Resonator block below. The stage sits right after the engines and the layer, before the part volume and the FX rack.
2. **Rack type Resonator** (DE Resonanzkörper, `pN.fxK.type` / `mfxK.type` value 23, appended after Space 3D): its block `<slot>resonator.*` for every slot of every rack, in the order parts 1 – 4 slots 1 – 6, then master slots 1 – 4 (28 × 14). FX Param 1 – 4 (matrix) = Body Tune, Material, Decay, Position.

The Resonator block (`pN.reso.*`, `<slot>resonator.*`)

Local ID	English	Deutsch	Range	Default
`model`	Body	Körper	String, Plate, Open Tube, Closed Tube	0 (String)
`follow`	Tuning	Stimmung	Fixed, Note, Chord	1 (Note)
`tune`	Body Tune	Körper-Stimmung	-24 … 24 semitones	0
`size`	Size	Größe	0 … 1	0.5
`material`	Material	Material	0 … 1	0.55
`decay`	Decay	Ausklang	0.05 … 20 s (skew 0.3)	2
`position`	Position	Position	0 … 1	0.3
`inharm`	Inharmonicity	Inharmonizität	0 … 1	0.1
`sympathetic`	Sympathetic Strings	Resonanzsaiten	0 … 1	0.35
`width`	Body Width	Körper-Breite	0 … 1	0.6
`drive`	Drive	Drive	0 … 1	0
`mix`	Body Mix	Körper-Mischung	0 … 1 (0 = exact bypass)	0.5
`pressamt`	Pressure to Body	Druck auf Körper	-1 … 1	0.5
`slideamt`	Slide to Position	Slide auf Position	-1 … 1	0.5

The modulation matrix gains the targets **Body Tune, Body Size, Body Material, Body Decay, Body Position, Body Mix** (75 – 80, `pN.reso.tune` … `pN.reso.mix`); the target choice list grows at its end. The FX slot type lists grow by **Resonator** (23).

Preset breeding

Appended after the resonator block (`mfx4.resonator.slideamt`), 3 IDs. Remote controls for the breeding view (triggers like `lib.open`, group "remote", not part of presets). Guides: English · Deutsch. Breeding itself adds no sound parameters: a child is an ordinary part sound; the breeding session (parents, generations, kept children) is stored in the plug-in state (section `"breed"`).

ID	English	Deutsch	Range	Default
`lib.breed`	Breeding	Zucht	trigger	off
`lib.breed.go`	Breed Again	Weiterzüchten	trigger	off
`lib.breed.next`	Next Child	Nächstes Kind	trigger	off

Character macros

Appended after the breeding triggers (`lib.breed.next`), part by part (`p1.char.*` … `p4.char.*`, 16 IDs). Guides: English · Deutsch.

ID	English	Deutsch	Range	Default
pN.char.bright	Part N Bright	Part N Hell	−1 … +1 (−100 … +100 %)	0
pN.char.hard	Part N Hard	Part N Hart	−1 … +1	0
pN.char.alive	Part N Alive	Part N Lebendig	−1 … +1	0
pN.char.wide	Part N Wide	Part N Weit	−1 … +1	0

Each macro moves several parameters of the part at once, inside the part after the modulation matrix (the stored parameter values, automation and the UI stay as they are; all four at 0 leave the sound bit-identical): **Bright** – cutoff / tone / brightness, FM index, warp, formant, a little resonance, less pluck damping; **Hard** – amp attack shorter (−30 % of its normalised range per unit), drive, FM feedback, pluck excitation, crackle, vocal tension, noise; **Alive** – `pN.life` (+0.6 per unit), cluster drift, detune, vocal jitter and vibrato depth, grain spray / scan; **Wide** – unison width, cluster / grain spread, cluster side, wavetable unison, the layer's width. The modulation matrix gains the targets **Bright / Hard / Alive / Wide** (81 … 84, after the resonator body targets 75 … 80; keys `charbright`, `charhard`, `charalive`, `charwide`); the target choice list grows at its end.

Bounce to engine

Appended after the character macros (`p4.char.wide`), 5 IDs, group "bounce" (`bounce.go`: "remote"), not part of presets. Guides: English · Deutsch. A bounce adds no sound parameters: it loads a sample into the part's Grain or Sampler engine (stored as a WAV in the bounce library, the path in the `"samples"` state section). No modulation-matrix changes.

ID	English	Deutsch	Range	Default
bounce.src	Bounce Source	Bounce-Quelle	Edit Track, Tape 1 … Tape 8, Tape Mix, Input	Edit Track
bounce.to	Bounce Target	Bounce-Ziel	Grain, Sampler	Grain
bounce.loop	Endless Loop	Endlosschleife	on / off	on
bounce.len	Input Length	Eingangslänge	1 Bar, 2 Bars, 4 Bars, 8 Bars	4 Bars
bounce.go	Bounce	Bounce	trigger	off

Motion recorder

Appended after the bounce block (`bounce.go`), part by part (`p1.motion.*` … `p4.motion.*`, 20 IDs). Guides: English · Deutsch. The recorded lanes (four per part: target parameter and 8 points per sixteenth, up to 64 sixteenths) are stored in projects and the plug-in state (section `"motion"`); part presets neither carry nor reset the motion block (a new sound keeps the part's lanes).

ID	English	Deutsch	Range	Default
pN.motion.on	Part N Motion Play	Part N Bewegung abspielen	Off / On	Off
pN.motion.rec	Part N Motion Record	Part N Bewegung aufnehmen	Off / On	Off
pN.motion.len	Part N Motion Length	Part N Bewegung Länge	1 … 64 sixteenths	16
pN.motion.amt	Part N Motion Depth	Part N Bewegung Tiefe	0 … 1 (0 = the knob's own value)	1
pN.motion.clear	Part N Motion Clear	Part N Bewegung löschen	trigger (not part of projects)	off

The modulation matrix gains the sources **Motion 1 … Motion 4** (23 – 26): the value of lane 1 … 4 at the play position (0 … 1; 0 where nothing is recorded or the transport stops). The source choice list grows at its end.

Performance pads

Appended after the motion block (`p4.motion.clear`), 11 global IDs. Guides: English · Deutsch. The pads work on the main mix in the master chain (after the master FX rack, before compressor, saturation and limiter). The four pad switches are not part of project presets.

ID	English	Deutsch	Range	Default
`pads.stutter`	Pad Stutter	Pad Stottern	Off / On (held)	Off
`pads.stop`	Pad Tape Stop	Pad Bandstopp	Off / On (held)	Off
`pads.sweep`	Pad Filter Sweep	Pad Filter-Sweep	Off / On (held)	Off
`pads.freeze`	Pad Reverb Freeze	Pad Hall-Freeze	Off / On (held)	Off
`pads.quant`	Pad Quantise	Pad-Quantisierung	Off, 1/16, 1/8, 1/4, 1/2, 1 Bar	1/16
`pads.stutter.div`	Stutter Length	Stotter-Länge	1/4, 1/8, 1/16, 1/32, 1/8T, 1/16T	1/16
`pads.stop.len`	Tape Stop Time	Bandstopp-Dauer	1/8, 1/4, 1/2, 1 Bar, 2 Bars	1/2
`pads.sweep.mode`	Sweep Filter	Sweep-Filter	Low-pass, High-pass	Low-pass
`pads.sweep.len`	Sweep Time	Sweep-Dauer	1/8, 1/4, 1/2, 1 Bar, 2 Bars, 4 Bars	1 Bar
`pads.sweep.res`	Sweep Resonance	Sweep-Resonanz	0 … 100 %	45 %
`pads.freeze.mix`	Freeze Level	Freeze-Pegel	0 … 100 %	60 %

The modulation matrix gains the sources **Pad Sweep** (27, the sweep position 0 … 1) and **Pads Held** (28, 1 while a pad is engaged). The source choice list grows at its end; no new targets.

Harmony brain

Appended after the performance pads (`pads.freeze.mix`): five global IDs, then the follow mode part by part (9 IDs). Guides: English · Deutsch. The memory (key, chord, the last eight chords, the pitch-class weights) is stored in projects and the plug-in state (section `"harmony"`); part presets neither carry nor reset `harmony.pN.follow` (its role in the song, not its sound).

ID	English	Deutsch	Range	Default
`harmony.memory`	Harmony Memory	Harmonie-Gedächtnis	1 … 32 bars (half-life of the key memory)	8
`harmony.hold`	Harmony Hold	Harmonie halten	Off / On	Off
`harmony.take`	Harmony Take Key	Harmonie Tonart übernehmen	trigger (not part of projects)	off
`harmony.clear`	Harmony Clear	Harmonie vergessen	trigger (not part of projects)	off
`harmony.kb.layout`	Grid Layout	Raster-Anordnung	Fourths, Thirds, Whole Tones, Sequential (p59; was In Key, see p59 below)	Fourths
`harmony.pN.follow`	Part N Harmony Follow	Part N Harmonie folgen	Off, Scale, Chord, Root	Off

The modulation matrix gains the sources **Chord Root**, **Chord Change** and **Tension** (29 – 31), the same value in every part. The source choice list grows at its end. No new targets.

In-app updater

No new parameters / keine neuen Parameter: the updater (English · Deutsch) is not a sound or performance control. Its settings (automatic check, channel, last check, skipped version) are stored per user in `Settings/update.json`, not in projects or presets / Seine Einstellungen liegen pro Benutzer in `Settings/update.json`, nicht in Projekten oder Presets.

Pressure glide

Appended after the harmony brain (`harmony.p4.follow`), part by part: `p1.glide.press` … `p4.glide.press` (4 IDs, group `partN`). Guides: English · Deutsch. Part presets carry it as `glide.press`.

ID	English	Deutsch	Range	Default
`pN.glide.press`	Part N Glide Pressure	Part N Glide-Druck	−1 … +1 (−100 … +100 %)	0

Aftertouch (the larger of channel pressure and the newest note's pressure, poly / MPE) scales the glide time (`pN.glide` or the CC 5 portamento time): amount > 0 → time × (1 + 3 × amount × pressure) (up to 4×, slower); amount < 0 → time × (1 + amount × pressure), at least 0 (−1 at full pressure = no glide). 0 or no pressure = unchanged. Not a modulation-matrix target (no new sources or targets). / Aftertouch skaliert die Glide-Zeit; 0 oder kein Druck = unverändert. Kein neues Ziel in der Modulationsmatrix.

SUB layer

Appended after pressure glide (`p4.glide.press`), part by part (`p1.sub.*` … `p4.sub.*`, 16 IDs, group "partN/sub"). Guides: English · Deutsch.

ID	English	Deutsch	Range	Default
`pN.sub.level`	Part N Sub Level	Part N Sub-Pegel	0 … 1 (0 – 100 %; 0 = off)	0
`pN.sub.oct`	Part N Sub Octave	Part N Sub-Oktave	−1 Oct, −2 Oct	−1 Oct
`pN.sub.shape`	Part N Sub Shape	Part N Sub-Form	Sine, Round, Square	Sine
`pN.sub.direct`	Part N Sub Direct	Part N Sub direkt	Off / On	Off

A sub oscillator under every voice of the part's main engine (all eight synth engines; not the layer engine, not the drum kit), phase-locked to the voice's pitch (glide, bend, pitch envelope, matrix pitch, life), band-limited, mono, on the voice's amp envelope. Level 0 leaves the sound bit-identical. Direct off = into the voice filter (Analog, Wavetable); the other engines always add it after their own tone shaping. The modulation matrix gains the target **Sub Level** (85, key `sublevel`, after Wide 84); the target choice list grows at its end. No new sources.

Expressive playing

Guides: English · Deutsch. Twelve parameters per part, appended after `p4.sub.direct` (p31 SUB layer) (`p1.expr.vel.dst1` … `p4.expr.at.curve2`, per part in this order: velocity slot 1 target / amount / curve, velocity slot 2, aftertouch slot 1, aftertouch slot 2). Each slot is a routing of the part's modulation matrix; amount 0 (the default) leaves the sound untouched. Part presets carry them; preset breeding and Discover leave them as they are (how a sound responds to playing).

ID	English	Deutsch	Range	Default
`pN.expr.vel.dst1` / `.dst2`	Part N Velocity 1 / 2 Target	Part N Anschlag 1 / 2 Ziel	Off, Cutoff, Resonance, Level, Timbre, Amp Attack, Amp Decay, Amp Release, Mod Env Amount, LFO Rate, LFO Depth, Vibrato, Glide Time, FM Amount, Position, Grain Spray, Morph X, Morph Y, Drive, FX Mix, Sub Level, Pan, Pitch, Life (append-only)	Cutoff / Amp Attack
`pN.expr.vel.amt1` / `.amt2`	Part N Velocity 1 / 2 Amount	Part N Anschlag 1 / 2 Menge	−100 … +100 %	0 %
`pN.expr.vel.curve1` / `.curve2`	Part N Velocity 1 / 2 Curve	Part N Anschlag 1 / 2 Kurve	Linear, Exponential, Logarithmic, S-Curve, Steps	Linear
`pN.expr.at.dst1` / `.dst2`	Part N Aftertouch 1 / 2 Target	Part N Aftertouch 1 / 2 Ziel	as above	Vibrato / Cutoff

ID	English	Deutsch	Range	Default
`pN.expr.at.amt1` / `.amt2`	Part N Aftertouch 1 / 2 Amount	Part N Aftertouch 1 / 2 Menge	−100 … +100 %	0 %
`pN.expr.at.curve1` / `.curve2`	Part N Aftertouch 1 / 2 Curve	Part N Aftertouch 1 / 2 Kurve	as above	Linear

The modulation matrix gains the source **Vibrato (5.5 Hz)** (32) and the targets **Glide Time**, **Amp Attack**, **Amp Decay**, **Amp Release** (86 – 89); both choice lists grow at their end.

Settings menu / Einstellungsmenü

No new parameters / keine neuen Parameter. The settings (English · Deutsch) are per user (`Settings/settings.json`: language, tooltips, help overlay, skin, display brightness, animations, window size) and not automatable. Their MIDI page shows existing parameters – `pN.midich`, `pN.mpe`, `pN.bend`, `pN.mpe.bend`, `sync.src`, `sync.clockout`, `sync.smooth` – unchanged and still automatable / MIDI-mappable / Die MIDI-Seite zeigt bestehende Parameter, unverändert und weiterhin automatisierbar und MIDI-zuweisbar.

Tape – clear track, clear all

Appended at the end (after the p33 expression block). Triggers, not part of presets.

ID	English	Deutsch	Range	Default
`tape.edit.clear`	Tape Clear Track	Tape Spur löschen	trigger	off
`tape.edit.clearall`	Tape Clear All	Tape alles löschen	trigger	off

Tape Clear Track empties the edit track (`tape.edit.track`) at once (one undo step). Tape Clear All opens the confirmation on the screen (Clear the whole tape?); the tape is cleared only when it is confirmed (undoable).

Metronome, count-in, tap tempo

Guides: English · Deutsch. Six parameters appended after `tape.edit.clearall`, group `metronome` (`metro.tap`: `remote`). Host parameters – saved with the session, automatable, MIDI-mappable – but not part of presets (they belong to the player, like the monitoring switch). Defaults leave Ohpe as before: no click, no count-in. No new modulation sources or targets.

ID	English	Deutsch	Range	Default
`metro.on`	Metronome	Metronom	Off, On	Off
`metro.level`	Click Level	Klick-Lautstärke	0 … 100 % (squared; 100 % = −6 dBFS peak)	60 %
`metro.accent`	Click Accent	Klick-Betonung	Off, On (beat 1 higher and louder)	On
`metro.countin`	Count-In	Vorzähler	Off, 1 Bar, 2 Bars / Aus, 1 Takt, 2 Takte	Off
`metro.plugin`	Click in Plug-in	Klick im Plug-in	Off, On (standalone: always clicks when on)	Off
`metro.tap`	Tap Tempo	Tempo tippen	trigger (a rising edge is one tap)	Off

Tape loop by hand / Schleife von Hand

Guides: English · Deutsch. Seven parameters appended after `metro.tap`. The six triggers (group `remote`) are not part of presets; the grid `tape.loopsnap` (group `tape`) is saved with projects like the loop range. The existing `tape.loop`, `tape.loopin`, `tape.loopout` and `tape.tx.loop` keep their meaning. No new modulation sources or targets.

ID	English	Deutsch	Range	Default
`tape.edit.loopin`	Tape Set Loop In	Tape Loop-Start setzen	trigger: loop in at the playhead (on the grid)	off
`tape.edit.loopout`	Tape Set Loop Out	Tape Loop-Ende setzen	trigger: loop out at the playhead (on the grid), loop on	off
`tape.edit.loopundo`	Tape Loop Undo	Tape Loop rückgängig	trigger: previous loop range	off
`tape.edit.loopredo`	Tape Loop Redo	Tape Loop wiederherstellen	trigger: next loop range	off
`tape.edit.loophalf`	Tape Loop Half	Tape Loop halbieren	trigger: half as long from loop in	off
`tape.edit.loopdouble`	Tape Loop Double	Tape Loop verdoppeln	trigger: twice as long from loop in	off
`tape.loopsnap`	Tape Loop Snap	Tape Loop-Raster	Off, Beat, Bar / Aus, Schlag, Takt	Bar

The 3D switch

Guides: English · Deutsch. Four parameters appended after `tape.loopsnap`, group `partN/space`. Host parameters – automatable, MIDI-mappable, saved with the session and in presets – that are a **front** for the existing `pN.space.on` and `space.out`: those two stay the stored truth, no existing ID changed. The plug-in keeps the three in step on the message thread: a moved switch writes `pN.space.on` (and `space.out` for Headphones / Speakers); a changed `pN.space.on` / `space.out` (preset, session, the Monitoring page, automation of the older IDs) moves the switches; after every preset / session load the switches are set from the stored values (older presets and sessions without the switch load correctly). The monitoring is global, so choosing Headphones or Speakers on one part switches it for every part with 3D on. No new modulation sources or targets.

ID	English	Deutsch	Range	Default
`p1.space.mode` … `p4.space.mode`	Part N 3D Space Mode	Part N 3D-Raum Modus	Off, Headphones, Speakers / Aus, Kopfhörer, Lautsprecher (Off = `space.on` off; Headphones = on + `space.out` Headphones; Speakers = on + `space.out` Speakers)	Off

Scale Lock

Guides: English · Deutsch. Two parameters appended after `p4.space.mode`, group `global` like `key.root` / `key.scale`. Host parameters – saved with the session, automatable, MIDI-mappable. The defaults leave Ohpe as before (lock off). The key itself stays `key.root` / `key.scale` (above); no new modulation sources or targets.

ID	English	Deutsch	Range	Default
`key.lock`	Scale Lock	In Tonart	Off, On	Off
`key.lock.mode`	Scale Lock Mode	In-Tonart-Modus	Nearest, White Keys / Nächster Ton, Weiße Tasten	Nearest

Nearest: every note reaching a part (on-screen keys, MIDI, sequencers, arpeggiator) snaps to the nearest note of the key. White Keys: live notes – the white keys play the key's degrees from the root nearest middle C at C4, black keys are silent (the grid keyboard is not remapped). While on, the transpose of the step sequencer and of the Orbit lanes (`seq.step.transpose`, `seq.orbit.lN.transpose`) counts scale degrees instead of semitones (not when they play the drum kit).

Fingerboard mode (MPE HD 14/21-bit)

Guides: English · Deutsch. Twenty parameters appended after `key.lock.mode`, five per part in the order below (`p1.*` … `p4.*`), group `partN`. Host parameters – saved with the session, automatable, MIDI-mappable; on the MPE page of the pro view. Part presets carry them under their local keys (`mpe.hd` …). The defaults leave Ohpe exactly as before: with Auto, plain MPE without CC 87 / NRPN 100–102 / RPN 0 renders sample-identical to Off. A part in Fingerboard mode (MPE on and onset not Velocity) keeps `mpe`, `mpe.bend`, `mpe.hd` and `mpe.onset` when a part preset loads.

ID	English	Deutsch	Range	Default
`pN.mpe.hd`	Part N MPE HD	Part N MPE HD	Off, Auto, On / Aus, Auto, Ein	Auto
`pN.mpe.onset`	Part N Note Onset	Part N Noten-Einsatz	Velocity, Pressure, Velocity x Pressure / Anschlag, Druck, Anschlag x Druck	Velocity
`pN.mpe.round`	Part N Pitch Rounding	Part N Tonhöhen-Rundung	Off, Initial, Continuous / Aus, Anfang, Fortlaufend	Off
`pN.mpe.round.time`	Part N Rounding Time	Part N Rundungszeit	0.01 … 1 s	0.15 s
`pN.mpe.pcurve`	Part N Pressure Curve	Part N Druckkurve	−1 … +1 (exponent 3^curve on the pressure)	0

Received with MPE HD Auto / On: CC 87 as LSB prefix of the next channel pressure / CC 74 / pitch bend on its channel (14-bit 0x0000 … 0x3F80 = 0 … 1, 21-bit bend), NRPN 0/100, 0/101, 0/102 = low-pass cutoff of X / Y / Z in 2 Hz units (0 = no filter; the voices smooth pitch / note bend, slide and the timbre target, pressure and the pressure target – and the level with the pressure onset – each with its own one-pole low-pass per control block, replacing `pN.mpe.smooth` / `pN.mpe.slide`; defaults 120 / 60 / 250 Hz), null NRPN 127/127, RPN 0 on a member channel = member bend range of the zone (overrides `pN.mpe.bend` until reset or Off), RPN 6 on channel 1 / 16 = lower / upper zone. CC 66 (sostenuto) now holds the keys that were down when it was pressed (all parts, with or without MPE).

The modulation matrix gains the source **Key + Bend** (`keybend`, index 33, bipolar, per voice): key track ((note − 60) / 60) plus the note's own per-note bend in semitones / 60, clamped to −1 … +1 – the continuous position on a fingerboard. No new targets.

Grid keyboard: In Key / Chromatic, Fixed C

Appended at the very end of the list (after `p4.mpe.pcurve`); global, saved in projects, automatable and MIDI-mappable. They act on the on-screen grid keyboard and the Grid-Controller pads alike, in every layout.

ID	English	Deutsch	Range	Default
`harmony.kb.inkey`	Grid In Key	Raster nur Tonart	Off (Chromatic: every note) / On (In Key: only key notes, steps in scale degrees)	Off
`harmony.kb.fixed`	Grid Fixed C	Raster C fest	Off (the bottom-left cell is the key's root) / On (it is C)	Off

`harmony.kb.layout` value 3 is now **Sequential** (+1 right, +8 up). Projects and presets from before that stored 3 (the old In Key layout) and no `harmony.kb.inkey` open as **Fourths + In Key**. No new matrix sources or targets.

Tape memory

Appended at the very end of the list (after `harmony.kb.fixed`), 5 IDs, group "tapemem" (`tapemem.go`: "remote"), not part of presets, all MIDI-mappable. Guides: English · Deutsch. A capture adds no sound parameters: it loads its audio into the part's Sampler (saved inside the session, state section `"tapememory"`). No modulation-matrix changes.

ID	English	Deutsch	Range	Default
tapemem.len	Memory Length	Memory-Länge	5 s, 10 s, 20 s, 30 s	30 s
tapemem.src	Memory Source	Memory-Quelle	Main Out, Part 1 … Part 4 / Hauptausgang, Part 1 … Part 4	Main Out
tapemem.mode	Capture Mode	Merk-Modus	Chromatic, Slices / Chromatisch, Slices	Chromatic
tapemem.to	Capture Target	Merk-Ziel	Edit Part, Part 1 … Part 4 / Bearbeiteter Part, Part 1 … Part 4	Edit Part
tapemem.go	Capture	Merken	trigger	off

Voice Control / Stimmsteuerung

Appended after `tapemem.go`; automatable and MIDI-mappable; the player's setup – **not** stored in part presets or projects. See Voice Control / Stimmsteuerung.

ID	English	Deutsch	Range	Default
voice.on	Voice Control	Stimmsteuerung	Off / On	On
voice.gain	Voice Input Gain	Stimme Eingangspegel	−24 … +24 dB	0 dB
voice.gate	Voice Gate	Stimme Schwelle	−80 … −20 dBFS	−50 dBFS
voice.smooth	Voice Smoothing	Stimme Glättung	1 … 250 ms	20 ms
voice.pitch	Voice Pitch Tracking	Stimme Tonhöhe erkennen	Off / On	Off
voice.monitor	Voice Monitor	Stimme abhören	Off / On (input on the main output)	Off
p1.voice.map … p4.voice.map	Part N Voice Quick Map	Part N Stimme Schnellzuweisung	trigger: Vowel Y → Timbre, Vowel X → Morph X, Vowel Y → Morph Y	Off

New matrix **sources** (appended after Key + Bend 33; Part scope, 0 without input): 34 Voice Level (`voicelevel`), 35 Vowel X (`vowelx`, bipolar), 36 Vowel Y (`vowely`, bipolar), 37 Voice Brightness (`voicebright`), 38 Voice Noisiness (`voicenoise`), 39 Voice Pitch (`voicepitch`), 40 Voice Onset (`voiceonset`). No new targets. Plug-in buses unchanged: the existing optional stereo input bus (off by default) is the voice input; the standalone app's input device is chosen in Settings > Audio (none by default).

Key Gravity per part

Appended after `p4.voice.map`, for each part N = 1 … 4 in this order (group "partN/gravity"). Guides: English · Deutsch.

ID	English	Deutsch	Range	Default
pN.grav.amt	Part N Key Gravity	Part N Tonart-Gravitation	0 … 1 (0 – 100 %; 0 = off, bit-identical, no latency)	0
pN.grav.target	Part N Gravity Target	Part N Gravitation Ziel	Scale / Chord / Chord+Scale (Skala / Akkord / Akkord+Skala)	Chord+Scale
pN.grav.glide	Part N Gravity Glide	Part N Gravitation Glide	0.005 … 2 s	0.15 s
pN.grav.curve	Part N Gravity Curve	Part N Gravitation Kurve	−1 … +1 (gentle … magnetic)	0
pN.grav.formant	Part N Gravity Formant	Part N Gravitation Formant	0 … 1 (formant keep)	0.5
pN.grav.mix	Part N Gravity Mix	Part N Gravitation Mischung	0 … 1	1
pN.grav.quality	Part N Gravity Quality	Part N Gravitation Qualität	Low Latency / Normal / High (Geringe Latenz / Normal / Hoch)	Normal

Chord headroom

Appended after `p4.grav.quality`. Details: FACTORY-SOUNDS / Werksklänge.

ID	English	Deutsch	Range	Default
`pN.chordroom`	Part N Chord Headroom	Part N Akkord-Reserve	0 … 24 dB at 8 sounding notes (log curve: 1 note 0 dB, 2 a third, 4 two thirds; 0 = off)	4.5 dB

3D motion retrigger and 3D amount

Appended after `p4.chordroom`, per part (`p1.space.retrig`, `p1.space.amount`, `p2.space.retrig`, …), group `partN/space`. Only the part's 3D stage has them (the Space 3D rack type keeps its block). Matrix target 95 **3D Amount** (`spaceamount`) moves `pN.space.amount`. Details: SPACE / RAUM.

ID	English	Deutsch	Range	Default
`pN.space.retrig`	Part N 3D Motion Retrigger	Part N 3D Bewegung Retrigger	Free / First / Note (Frei / Erste / Note): the motion restarts from its start position never, on a note after all keys were up, or on every note (mono legato notes do not restart it)	Free
`pN.space.amount`	Part N 3D Amount	Part N 3D Stärke	0 … 1 (0 = plain stereo, 1 = full 3D)	1